

**PLASMA BIOMARKERS OF NEURODEGENERATION ACROSS THE SPECTRUM
OF BRAIN HEALTH**

by

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Abstract

Over the past decade, remarkable progress has been made in the development of plasma biomarkers for neurodegenerative diseases, particularly for diagnosis of Alzheimer's Disease (AD). These biomarkers have considerable potential to transform clinical practice by providing accessible, cost-effective, and minimally invasive tools for early detection, and diagnosis of AD. However, despite rapid international progress, significant gaps remain before these biomarkers can be fully implemented in clinical care within Canada. This thesis addresses several of these gaps through analysis of plasma amyloid beta ($A\beta_{42/40}$), phosphorylated tau-181 (p-tau-181), neurofilament light (NfL), and glial fibrillary acidic protein (GFAP) across four cohorts representing the spectrum of brain health.

Using a large normative population representing Canadians across the lifespan, we address the need for a deeper understanding of how biomarker concentrations vary across normal aging through the establishment of age-specific reference intervals. Here, we found that between the ages of 3-79 years old, all biomarkers exhibited U-shaped curves as a function of age. Next, we explore biomarker signatures in individuals with cognitive resilience, providing insight into biomarker response among individuals who remain cognitively intact despite advanced age. This analysis revealed potentially resilient levels of $A\beta_{42/40}$ and p-tau-181, but levels of NfL and GFAP that were reflective of the normative population. Subsequent chapters link plasma biomarker profiles to autopsy-confirmed neuropathology, addressing a critical knowledge gap in understanding the pathological specificity of these measures for AD compared with other dementias. Here, we were able to identify the domains of AD neuropathology that plasma biomarkers best inform on. Finally, by leveraging a multi-cohort design, this thesis identifies and validates diagnostic cut-offs for AD pathology. This work provides evidence for biomarker application in real-world settings, and assesses concordance between pathology and clinical diagnoses across disease stages. Collectively, the studies presented in this thesis contribute foundational data to guide the clinical translation of neurodegenerative blood biomarkers in Canada, enabling their eventual adoption into diagnostic workflows for dementia.

Lay Summary

Alzheimer's disease affects millions of people worldwide as the leading causes of dementia. Current diagnosis relies on memory tests, which can confirm dementia but not its underlying cause. Accurate detection of the biology hallmarks of Alzheimer's disease will be critical for developing and delivering effective treatments. Existing diagnostic tools such as brain scans and lumbar punctures are invasive and costly, limiting their accessibility and scalability. Recent discoveries of blood-based biomarkers offer a promising alternative. This thesis focuses on transforming these discoveries into practical, reliable blood tests. Using over 4,000 blood samples from Canadian research participants, we established age-specific reference ranges and examined how factors like sex influence results. We also identified how these biomarkers can distinguish Alzheimer's disease from other dementias. These blood tests will provide a low cost, accessible tool to help identify those with Alzheimer's disease for clinical trials and newly approved treatments.

Preface

This thesis is an original intellectual product of the author, J.G. Cooper. No generative Artificial Intelligence (AI) tools were used in the writing of the thesis. AI tools were utilized in the editing of statistical analysis code run on R in chapters 3 and 8 (ChatGPT), and to assist with literature searches alongside traditional methods (Consensus AI). All work in this thesis was performed at the Djavad Mowafaghian Centre for Brain Health and the UBC Statistics Canada Research Data Center. All projects and methods were approved by the University of British Columbia Research Ethics Board.

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Components of Chapter 4 will be utilized in publications regarding each data chapter to provide the context of all quality control. I designed all experiments and cross lot analysis included in this chapter, with supervisory oversight from Wellington CL and Stukas S. I performed all sample and data analysis, prepared figures and wrote the manuscript.

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Chapter 8 is an extension of the work from Chapter 7 to provide an exploratory analysis of biomarker for other forms of dementia. This chapter provides preliminary pilot data that has been used to plan a larger study focused on biomarker discovery. I designed the study with supervisory oversight from Wellington CL and Stukas S. Hsiung GY, Pfulueger S, and Scott I, provided plasma samples from UBC CARD, conducted the original study, and provided study data. Mackenzie IR and Hirsch-Reinshagen V conducted the neuropathological examinations and, with Futhey NC, provided all autopsy data. Agbay A analyzed the plasma specimens on the ARGO platform. I conducted all data analysis, under the guidance of Yanta C, and prepared the manuscript.

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List of Abbreviations

3MS	Modified Mini-Mental State
ACHE	Acetylcholinesterase
AD	Alzheimer's Disease
AD-8	Ascertain Dementia 8
ADNC	Alzheimer's Disease neuropathologic change
ADNC-PPS	Alzheimer's Disease neuropathologic change plasma probability scores
ADRD	Alzheimer's Disease and Related Dementias
AEB	Average enzyme bound
AGD	Argyrophilic grain disease
AICD	Amyloid precursor protein intracellular domain
ALS	Amyotrophic Lateral Sclerosis
ANOVA	analysis of variance
APOE	Apolipoprotein E
APP	Amyloid precursor protein
ARIA	Amyloid-related imaging abnormalities
AUROC	Area under the receiver operator curve analysis
A β	Amyloid Beta
BIA	budget impact analysis
BMI	Body mass index
bvFTD	Behavioural variant FTD
CAA	Cerebral amyloid angiopathy
CADTH	Canadian Agency for Drugs and Technologies in Health
CARD	Clinic for Alzheimer's Disease and Related Disorders

CBD	Corticobasal degeneration
CCCDTD	Canadian Consensus Conference on the Diagnosis and Treatment of Dementia
CCL11	C-C motif chemokine ligand 11
CCNA	Canadian Consortium on Neurodegeneration and Aging
CDR	Clinical Dementia Rating
CDR-SB	Clinical Dementia Rating Sum of Boxes
CERAD	Consortium to Establish a Registry for Alzheimer's Disease
CHI3L1	Chitinase-3-like protein 1
CHMS	Canadian Health Measures Survey
CI	Confidence interval
CIHR	Canadian Institutes of Health Research
CLIA	Clinical Laboratory Improvement Amendments
CLSI	Clinical Laboratory Standards Institute
CNS	Central nervous system
COMPASS-ND	Comprehensive Assessment of Neurodegeneration and Dementia
CSF	Cerebrospinal fluid
CSF2	Colony-stimulating factor 2
CST3	Cystatin C
CTE	Chronic traumatic encephalopathy
CV	Coefficient of variance
CVD	Cardiovascular disease (Ch 5 & 6)
CVD	Cerebrovascular disease
CXCL10	C-X-C motif chemokine ligand
ELISA	enzyme-linked immunosorbent assay

EOAD	Early-Onset Alzheimer’s Disease
FDA	Food and Drug Administration
FDR	false discovery rate
FTD	Frontotemporal dementia
FTLD	Frontotemporal lobar degeneration
fvAD	Frontal variant of AD
GAIC	Generalized Akaike information criterion
GAMLSS	Generalized additive models for location, scale and shape
GDNF	Glial derived neurotrophic factor
GDS	Geriatric Depression Scale
GFAP	Glial fibrillary acidic protein
GGT	Globular glial tauopathy
GPCOG	General Practitioner Assessment of Cognition
HS	Hippocampal sclerosis
HTA	Health technology assessment
IADL	Instrumental Activities of Daily Living Scale
IC	Internal control
ICAM1	intercellular adhesion molecule 1
ICER	incremental cost-effectiveness ratio
IL12p70	Interleukin-12 p70
IPC	Inter-plate controls
IQR	Interquartile range
IVDs	In-vitro diagnostic devices
LASSO	Least Absolute Shrinkage and Selection Operator
LATE	limbic predominant age-related TDP-43 encephalopathy

LBD	Lewy body dementia
LDT	Laboratory-developed tests
LMS	Lambda-Mu-Sigma
LOAD	Late-onset Alzheimer's Disease
LOD	Limit of detection
lvPPA	Logopenic progressive aphasia
MAPT	Microtubule-associated protein tau
MCI	Mild cognitive impairment
MDD	Medical Devices Directorate
MEC	Mobile Examination Center
MIS-CDT	Mini-Cog, Memory Impairment Screen with Clock Drawing
MLR	multivariable linear regression
MMSE	Mini-Mental State Examination
MoCA	Montreal Cognitive Assessment
MRI	Magnetic resonance imaging
MS	Multiple Sclerosis
MSA	multiple systems atrophy
N4PE	Neurology-4-plex E
NC	Negative controls
NfL	Neurofilament light chain
NIA-AA	National Institute on Aging and Alzheimer's Association
NPQ	NULISA Protein Quantification
NPTX2	neuronal pentraxin 2
NPTXR	Neuronal pentraxin receptor
NPV	Negative predictive value

NPY	Neuropeptide Y
NULISA	Nucleic acid Linked Immuno-Sandwich Assay
PART	Primary age related tauopathies
PCA	Posterior cortical atrophy
PD	Parkinson's disease
PDD	Parkinson's disease dementia
PET	Positron emission tomography
PLMS	Provincial Laboratory Medicine Services
POSTN	Periostin
PPA	Primary progressive aphasia
PPS	Plasma probability scores
PPV	Positive predictive value
PSP	Progressive supranuclear palsy
p-tau	Phosphorylated tau
QALYs	Quality adjusted life years
RC-PPS	Reference curve plasma probability scores
REB	Research Ethics Board
REST	RE1-silencing transcription factor
RI	Reference interval
ROC	Receiver operating curve
RUDAS	Rowland Universal Dementia Assessment Scale
RUO	Research use only
S100A12	S100 calcium-binding protein A12
SCI	Spinal cord injury
SCI	Subjective cognitive impairment

SD	Standard deviation
SIMOA	Single Molecule Array
SLIT2	slit guidance ligand 2
SLR	simple linear regression
SNAP25	Synaptosomal-associated protein 25
SQTM1	Sequestosome 1
TAFA5	TAFA chemokine like family member 5
TBI	Traumatic brain injury
TDP-43	Transactive response DNA binding protein 43 kDa
TREM2	Triggering receptor expressed on myeloid cells
TUG	Timed Up and Go test
UBC	University of British Columbia
UCHL1	ubiquitin C-terminal hydrolase L1
VCID	Vascular Contributions to Cognitive Impairment
VSNL1	Visinin-like protein 1
α -syn	Alpha synuclein

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Dedication

For my grandmothers, Grace and Diane, who both battled dementia. Though they are no longer here to see this accomplishment, they were proud supporters earlier in my life and continue to be part of me through namesake. I hope that through this work, we can move one step closer to a world where no one has to watch their loved ones face this disease.

Chapter 1: Introduction

1.1 Summary

Alzheimer's disease (AD) is the most common cause of dementia, affecting millions of people worldwide. As life expectancy increases, the number of people affected continues to rise in Canada and around the world, creating growing social and economic pressures on healthcare systems. A key challenge with AD is diagnosis, which has traditionally been achieved through cognitive testing in symptomatic persons. However, as disease-modifying therapies for AD gain clinical approval, the need for accurate, accessible, and biologically grounded diagnostic tools is more urgent than ever. Recent development of blood-based biomarkers has the potential to transform the AD diagnostic landscape. Using ultrasensitive assays capable of detecting proteins at femtomolar concentrations, key AD pathological hallmarks can now be reliably measured in blood samples, most commonly in plasma. The use of plasma biomarkers offers an accurate, lower-cost, accessible tool to improve diagnosis of AD and related dementias (ADRD). Although many advancements have been made to move diagnostic AD plasma biomarkers towards clinical use, several gaps remain in our knowledge about how to implement them, especially in Canadian populations and clinical landscape. To address these gaps, this thesis aims to gain a deeper understanding of AD plasma biomarkers across the full spectrum of brain health to reveal novel insights and considerations for their interpretation and implementation.

1.2 Alzheimer's Disease

AD is a neurodegenerative disease that primarily affects memory and cognitive functions, with symptoms progressing over time and eventually inhibiting the independence of those affected. AD has both clinical and neuropathological definitions, which for many years has led to debate around the terminology used to define AD. A major reason for this debate is the temporal discordance between the development of neuropathology and the onset of clinical symptoms, which can be up to 30 years apart^{1,2}, raising the question of when we can say a person has AD. In recent years, most scientists and clinicians have moved toward accepting biological definitions for AD, based on the presence of its neuropathological hallmarks, while defining

the clinical symptoms as AD dementia or AD syndrome. One of the main driving factors in this shift is the ability to now detect AD neuropathologic change in living people, without needing to rely on symptom evaluation as the primary diagnostic methodology.

1.2.1 Epidemiology

The growing prevalence of ADRD is a major international concern. Globally, it is estimated that there are over 50 million people worldwide living with dementia, with this number expected to triple, rising to over 150 million by 2050³. As population ageing continues, it is expected that the prevalence and incidence of dementia will rise significantly in coming decades. Mirroring global trends, dementia prevalence has steadily increased in Canada. In 2018, approximately 614,000 Canadians were living with dementia, with projections suggesting that this will nearly double to 937,000 by 2031^{4,5}. The annual incidence among those aged 65 and older is estimated at approximately 20 per 1,000 persons, translating to over 60,000 new cases per year^{6,7}. Dementia prevalence rises sharply with age, especially after 80 years, and is higher among women and those with lower education⁴. The economic burden of dementia in Canada is substantial, with costs projected to rise from \$33 billion in 2011 to \$293 billion by 2040^{4,5}. As AD is the leading cause of dementia, accounting for 60-80% of all dementia cases⁸, improving care through earlier and more accurate diagnosis can reduce these projections and ultimately lessen the burden on healthcare systems.

1.2.2 Clinical Presentation and Variants

AD has a very broad clinical definition with both typical and atypical presentations. The typical, amnesic presentation is dominated by memory impairment, especially episodic memory loss. In the earliest stages, individuals often experience forgetfulness, difficulty recalling recent events, or challenges with word retrieval, while long-term and procedural memory remain relatively preserved⁹⁻¹¹. As the disease progresses, memory impairment worsens, interfering with daily functioning and the ability to perform complex tasks. This is typically accompanied by other cognitive deficits, including language, visuospatial processing, attention, and executive function. Patients may have impaired problem-solving, or difficulty interpreting spatial relationships. Behavioral symptoms, such as apathy, irritability, anxiety,

and depression, often develop in later stages. The most advanced stages are often associated with complete loss of independence, loss of communication abilities, and overall functional decline. This typical amnesic presentation is due to AD neuropathology affecting specific brain regions. Neurodegeneration from AD is often present within the medial temporal lobe, particularly the hippocampus and entorhinal cortex, which are essential for encoding and consolidating new episodic memories¹²⁻¹⁴. While the typical presentation is amnesic, with memory loss being the primary symptom, there are also a variety of atypical AD subtypes. Many of these AD subtypes are non-amnesic due to atypical brain regions where the pathology is localized^{15,16}. Posterior cortical atrophy (PCA) primarily affects the posterior region of the brain, causing visual-spatial difficulties. A frontal variant of AD (fvAD) impacts the frontal lobes, leading to behavioural and executive function problems. Logopenic variant primary progressive aphasia (lvPPA) is another clinical phenotype characterized by language difficulties, particularly with word-finding. Additionally, AD can be defined as early onset AD (EOAD) or late onset AD (LOAD) based on the age of diagnosis, where those under 65 years of age are classified as EOAD^{17,18}. LOAD is the more typical presentation, while EOAD accounts for only about 5-10% of AD cases. EOAD is often linked to genetic mutations in presenilin-1 or amyloid precursor protein that cause a more aggressive progression of disease.

1.2.3 Risk factors

ADRD have an extensive list of risk factors that can be either modifiable, such as lifestyle factors, or non-modifiable, such as genetics. In 2024, the Lancet Commission on Dementia published their report on risk and prevention of dementia, identifying 14 distinct risk factors that together can modify approximately 45% of overall dementia risk¹⁹. AD risk factors include poor education in early life, midlife hypertension, hearing impairment, obesity, depression, diabetes, a traumatic brain injury (TBI), high levels of low-density lipoprotein cholesterol, physical inactivity, smoking, and excessive alcohol consumption, late life social isolation, vision loss, and air pollution. In terms of genetics, apolipoprotein E (*APOE*) genotype is the largest genetic risk factor for AD. *APOE* encodes the apoE lipoprotein that regulates lipid metabolism and cholesterol transport and is the major lipid carrier in the central nervous system (CNS). The *APOE* gene has three major alleles, *APOE2*, *APOE3*, and *APOE4*, which each affect risk and age of onset of AD. *APOE2* is protective, *APOE3* is neutral and *APOE4* is

detrimental, with one copy of an *APOE4* allele approximately doubling AD risk and two *APOE4* copies increasing risk by 10 to 15 times^{20,21}. Many animal and human studies have demonstrated the pleiotropic effects of ApoE4, including promoting amyloid plaque formation, decreasing amyloid beta (A β) clearance from the brain, promoting phosphorylation of tau, increasing microglial mediated neuroinflammation, and increasing cerebrovascular damage²⁰. Biological sex is another non-modifiable risk factor, with females having a higher risk. Epidemiologically, females represent nearly two-thirds of all individuals living with AD, even after adjusting for longer life expectancy²². Clinically, women with AD often present with more pronounced memory deficits, though this can vary by disease stage and comorbidities²³.

1.3 Neuropathology of Alzheimer's Disease

AD neuropathology is defined by three main pathological hallmarks: 1) development of A β plaques; followed by 2) the formation of neurofibrillary tangles; and 3) subsequent neurodegeneration. In order for the neuropathology to be considered AD, A β plaques must be present. While neurofibrillary tangles and neurodegeneration are additional neuropathological hallmarks of AD, they can also occur in the absence of A β plaques where they are considered indicators of other forms of dementia or neurological disorders. This neuropathological description was proposed as a diagnostic framework called the ATN system that describes testing for amyloid beta (A), tau (T), and neurodegeneration (N)²⁴ in living people. The framework was first published in 2018, with an updated framework published in 2024. However, it is essential to note that the gold standard diagnostic tool for AD is still considered to be neuropathological examination on autopsy. Similar to the ATN framework, guidelines specific to neuropathologic assessment of AD were developed by the National Institute on Aging and Alzheimer's Association (NIA-AA), and were updated most recently in 2012^{25,26}. The framework for neuropathological grading outlines a scoring system for AD neuropathological change called the ABC score, standing for Amyloid, Braak, and CERAD. Specifically, the "A" or amyloid score is based on the Thal phase for detection of A β plaques²⁷, which describes amyloid plaque distribution and density throughout the brain. "B" or Braak stage describes the staging of neurofibrillary tangles^{28,29}, which is similar to the Thal phase in that it describes neurofibrillary tangle distribution and density. "C" or CERAD stands for the "Consortium to Establish a Registry for Alzheimer's Disease", which scores the frequency of

neuritic plaques³⁰ and represents a later stage of amyloid plaque development. Each component of the ABC score is scored from 0 to 3 to produce a composite score determining the level of AD neuropathic change from none, low, intermediate, and high. Intermediate to high, which requires a minimum of A1B2C2, is the cut-off of neuropathic change that is sufficient to significantly contribute to a person's cognitive impairment^{25,26}.

1.3.1 Amyloid Pathology

The primary defining feature of AD is the amyloid pathology as defined by the presence of plaques composed of A β peptides. A β peptides are pathogenic proteolytic cleavage products of amyloid precursor protein (APP). APP is a transmembrane protein with a large extracellular domain. In a non-pathologic proteolytic pathway, APP is processed by sequential cleavage of the extracellular domain by α -secretase, to create a soluble extracellular n-terminal fragment called APP alpha (sAPP α), which is then cleared from the brain^{31,32}. The remaining APP fragment, C-terminal fragment-83, is then cleaved by γ -secretase to produce an APP intracellular domain (AICD), and an extracellular P3 peptide^{31,32}. However, in the pathological proteolytic pathway, APP cleavage is first conducted by β -secretase instead of α -secretase, which generates a shorter extracellular section of the n-terminal fragment of APP known as sAPP β ³³. When γ -secretase then cleaves the remaining APP fragment, C-terminal fragment-99, it can cleave at a variety of sites to create the AICD. Importantly, this creates the extracellular A β fragments associated with plaque formation. These A β fragments range in length from approximately 30-50 amino acids, and can undergo further processing in endocytic compartments³⁴. The two most common major variants of the A β protein fragments found extracellularly are 40 and 42 amino acids in length^{31,33,35}. Of these two peptides, A β 42 is more hydrophobic due to the additional amino acids at the C-terminus, and therefore more likely to form aggregates. Individual A β monomers will begin to misfold and aggregate together to form oligomers, followed by larger protofibrils. In these stages the aggregates are still soluble and considered to be neurotoxic, disrupting synaptic function^{36,37}. The protofibrils then elongate into insoluble fibrils which can compact together to form the initial diffuse A β plaque. This is a highly dynamic process, as these early diffuse plaques have the ability to form and dissolve back to earlier A β aggregate forms³². However, over time, diffuse plaques can consolidate and compact to form a dense core plaque and eventually progress to form a senile neuritic

plaque^{32,38}. The presence and density of neuritic plaques can be described using the CERAD score, which can be conducted using a modified Bielschowsky silver stain. The CERAD score is found to correlate more with the severity of cognitive impairment than the amount of diffuse plaques³⁰. The distribution of A β in the brain is described by Thal staging, which is conducted using immunohistochemistry with anti-A β antibodies. Thal phases describe how accumulation begins in the neocortex (phase 1), progresses to the hippocampus (phase 2), then to subcortical structures including the striatum and thalamus (phases 3–4), and finally involves the brainstem and cerebellum (phase 5)²⁷.

1.3.2 Tau Pathology

Tau pathology represents the second major defining neuropathological hallmark of AD. Tau is a microtubule-associated protein predominantly expressed in neurons, where it binds to and stabilizes axonal microtubules. Under normal conditions, tau's binding to microtubules is regulated by phosphorylation to ensure a balance between stability and flexibility³⁹. In AD, tau becomes abnormally hyperphosphorylated at numerous serine and threonine residues, with about 50 of the approximately 80 possible phosphorylation sites on tau being associated with pathological tau⁴⁰. Hyperphosphorylated tau (p-tau) has a reduced affinity for microtubules, causing tau to detach and accumulate inside neurons^{41,42}. This is thought to be triggered in part by A β oligomers creating a cascade due to their neurotoxic effects that causes excessive activity of kinases that are responsible for tau phosphorylation. The detached tau undergoes further misfolding, and tau monomers begin to self-associate into soluble oligomers^{41,43}. Similar to A β monomers, p-tau oligomers are considered neurotoxic due to their ability to impair synaptic function and disrupt intracellular signaling. Tau oligomers further aggregate into filaments and finally into neurofibrillary tangles. Early stage, or pre-tangles, are composed of loosely arranged filaments that are partially soluble in the intracellular space, but, over time, crosslinking of filaments cause them to convert to insoluble mature tangles^{41,44}. These mature tangles disrupt intracellular transport, impair mitochondrial function, and ultimately contribute to cytoskeletal collapse and neuronal death⁴³. The distribution of neurofibrillary tangles in the brain in AD is described by Braak staging. Braak staging was traditionally conducted by detecting neurofibrillary tangles with Gallyas silver stain, but in the past two decades it has become standard to use AT8-immunostaining to detect hyperphosphorylated tau²⁹. Braak

staging begins by assessing the transentorhinal region (stages I–II), followed by limbic regions (stages III–IV), and finally to neocortical regions (stages V–VI)²⁹. This anatomical progression correlates strongly with cognitive decline and clinical disease stage⁴⁵.

1.3.3 Neurodegeneration

Though not unique to AD, neurodegeneration is a key hallmark in the cascade of AD pathology. The accumulation of A β in amyloid plaques and p-tau in neurofibrillary tangles primarily causes synaptic damage and dysfunction, which is recognized as the strongest neuropathological correlate of cognitive impairment^{46,47}. Additionally, plaques and tangles can also disrupt axonal transport and impair mitochondrial energy metabolism, leading to oxidative stress⁴⁸. The combination of these insults triggers apoptotic and necroptotic cell death pathways, resulting in irreversible neuronal loss⁴⁶. In AD, the earliest and most prominent neurodegenerative changes occur within the hippocampus and entorhinal cortex, regions crucial for memory and learning, before extending to the association cortices and, in later stages, the primary sensory and motor areas¹²⁻¹⁴. This progressive pattern of neuronal loss contributes to the disintegration of large-scale brain networks, which underlies many of the characteristic cognitive symptoms of the disease. Overall, neurodegeneration reflects the downstream manifestation of amyloid and tau toxicity and serves as a critical determinant of clinical progression in AD.

1.3.4 Neuroinflammation

In addition to its main neuropathological hallmarks, a variety of other neuropathological sequelae are associated with AD, with one key feature being neuroinflammation. The inflammatory response in AD is primarily mediated by microglia and astrocytes, which become chronically activated in response to A β plaques and degenerating neurons^{49,50}. In early stages, activated microglia play a protective role by phagocytosing amyloid deposits and releasing anti-inflammatory cytokines. However, prolonged exposure can cause a transition to a pro-inflammatory state, with production of cytokines and reactive oxygen species that can exacerbate neuronal injury⁵⁰. Astrocytes similarly undergo reactive gliosis, causing reactive astrocytes to surround amyloid plaques and participate in both the clearance and deposition of

A β ^{51,52}. Due to their reactive state, astrocytes also release inflammatory cytokines that amplify microglial activation. Chronic astrocyte activation contributes to impaired synaptic regulation and breakdown of the blood–brain barrier that further promotes neurotoxicity⁵². Overall, neuroinflammation acts as both a response to and a driver of AD pathology. While initially protective, chronic activation of glial cells causes a cycle of inflammation causing neurodegeneration^{49,50}. This recognition has prompted growing interest in targeting neuroinflammatory pathways as a potential therapeutic strategy to slow or prevent disease progression.

1.4 Treatment of Alzheimer’s Disease

1.4.1 Disease-modifying therapies

In 2021 the USA Food and Drug Administration (FDA) approved the first disease-modifying therapy for AD. Aducanumab is a monoclonal antibody that targets amyloid plaques to break them down and remove them from the brain. Although aducanumab was discontinued in 2024 due to concerns over safety and limited efficacy, two additional anti-amyloid therapies have also been approved by the FDA, Lecanemab in 2023⁵³ and Donanemab in 2024⁵⁴. Both drugs are intended for those in the early stages of dementia, with Donanemab also being approved for use in those with mild cognitive impairment (MCI). Both drugs are approved for clinical use in Japan, China, the United Kingdom (UK), and the European Union (EU). However, both the EU and UK restrict the use of both drugs to those who are not *APOE4* homozygotes, as being a carrier of two copies of *APOE4* is associated with increased risk of adverse effects, such as amyloid-related imaging abnormalities (ARIA)^{55,56}. There are two main categories of ARIA, one involving edema and the other involving hemorrhage in the brain. Although mild ARIA cases can continue treatment with close monitoring, severe ARIA cases can require hospitalization and result in discontinued use of the drugs⁵⁷. Health Canada very recently approved Lecanemab on Oct. 25th, 2025. Similar to the structure in the EU and UK, Canadians will only be prescribed Lecanemab if they are not *APOE4* homozygotes. In addition to the currently approved anti-amyloid therapies, there are a variety of other treatments targeting amyloid and tau pathology in the clinical trial pipeline. At the beginning of 2025, it was reported that 138 drugs were in clinical trials across phases one to three, with 74% of drugs in the pipeline being disease-targeting therapeutics⁵⁸. Of these, 30 target amyloid pathology, 15

target tau pathology, and 25 target inflammatory processes. As these trials conclude and more drugs may be approved, it will be imperative that we have accurate and accessible methods to detect the underlying pathologies that these drugs target. Additionally, methods to assess safety and intervention efficacy are equally important. Blood based biomarkers have potential roles in all of these contexts of use.

1.5 Non-Alzheimer's Disease Dementias

1.5.1 Frontotemporal Dementias

Frontotemporal dementia (FTD) classifies a clinical group of early-onset dementias that often first present with behavioural changes⁵⁹. FTD has two main frontotemporal lobar degeneration (FTLD) subtypes based on the neuropathology that appears in the frontal and/or temporal lobes. The first FTLD pathology is the hyperphosphorylation and accumulation of tau^{60,61}. FTLD-tau is considered a primary tauopathy, while AD is considered a secondary tauopathy. Tauopathies are also classified by the isoforms of tau occurring in the neurofibrillary tangles⁶². The main subtypes are those with three microtubule-binding repeats (3R) and those with four repeats (4R). AD is an example of a condition that has both 3R and 4R isoforms present in filaments, while the FTLD-tau subtypes all have only 3R or 4R. Pick's disease is the subtype that has exclusively 3R tau. Pick's disease was originally referred to as FTD and is often diagnosed in those under 65 years old. It is known to cause two different clinical conditions: Behavioural variant FTD (bvFTD), which is classified by loss of inhibition, empathy and executive functions along with compulsive behaviour and apathy; and primary progressive aphasia (PPA), which results in a loss of language skills. There are a variety of rarer FTLD-tau subtypes that are the result of 4R tau, including corticobasal degeneration (CBD), progressive supranuclear palsy (PSP), and argyrophilic grain disease (AGD). CBD is a condition associated with problems with movement and speech. PSP affects movement and coordination. AGD has a diverse range of symptoms, but most often is associated with psychiatric symptoms and behavioural changes. In addition to these FTLD-tau subtypes, another important tauopathy is primary age related tauopathies (PART). This is a tau subtype similar to AD with a combination of 3R and 4R, however, these individuals lack A β plaques. PART can be regularly found in aging brains, spanning the spectrum from cognitively normal people to those with MCI or dementia.

The second FTLN pathology involves hyperphosphorylation and abnormal cytoplasmic localization of TAR DNA binding protein 43 kDa (TDP-43)^{60,61}. TDP-43 is a protein located in the nucleus responsible for regulating RNA processing. In a pathological state, it can become phosphorylated and aggregate to form inclusions in the cytoplasm of neurons. Similar to FTLN-tau, FTLN-TDP-43 can manifest both as bvFTD and PPA. Another important TDP-43 pathology is Amyotrophic Lateral Sclerosis (ALS), which features similar TDP-43 inclusions, but with a more widespread distribution throughout the brain and spinal cord. As ALS primarily affects motor neurons, it manifests clinically with muscle weakness and eventual paralysis. A final TDP-43 pathology that has been newly classified is limbic predominant age-related TDP-43 encephalopathy (LATE). LATE has a more limited distribution of TDP-43 in the brain than FTLN-TDP-43, and similar to PART, is very commonly found in aging brains.

1.5.2 Dementia with Lewy Bodies

Lewy bodies are inclusion bodies of abnormally aggregated alpha synuclein (α -syn) in the cytoplasm of neurons⁶³. They are associated with two main forms of dementia, Lewy body dementia (LBD) and Parkinson's disease (PD) dementia (PDD). These conditions differ based on the distribution of Lewy bodies in the brain, where PD has α -syn concentrations in the substantia nigra, and LBD has a more diffuse distribution of α -syn across the neocortex. PD is primarily a movement disorder resulting in tremors, slowness and stiffness. There is often a significant amount of time during which patients with PD experience motor symptoms prior to advancing to PDD, where memory problems become prominent. LBD, on the other hand mimics the memory problems seen in AD, but also includes balance and visuospatial problems⁶³, hallucinations, and behavioural and sleep problems. An additional rare Lewy Body disorder is multiple systems atrophy (MSA), where Lewy bodies primarily affect glial cells rather than neurons. It is often clinically confused with PD, however, it has a much more rapid progression than PD or LBD.

1.5.3 Vascular Dementia

Vascular dementia is caused by a range of conditions that can affect cerebrovascular physiology¹⁹ or disrupt blood flow to the brain, such as prior strokes or transient ischemic

attacks⁶⁴. Two major categories are multi-infarct dementia, resulting from multiple large or small strokes that cumulatively damage brain tissue^{65,66}, and small vessel disease, which is associated with chronic ischemic damage. The clinical presentation of vascular dementia is often heterogeneous and depends on the size, location, and number of vascular lesions. Common cognitive features include slowed processing speed, impaired executive function, and difficulties with attention and planning, whereas memory impairment tends to be less severe than in AD^{65,66}. Beyond clinically defined vascular dementia, vascular contributions to cognitive impairment and dementia (VCID) describe a broader spectrum of cognitive decline attributable to vascular pathology. VCID encompasses the broad spectrum of cognitive impairment resulting from acute or chronic reductions in cerebral blood flow. It is a common co-pathology to AD, with up to 80% of those with AD have co-occurring vascular pathology at autopsy⁶⁷. Cardiovascular risk factors have major effects on VCID, with 11 of the 14 recognized modifiable risk factors for AD also being cardiovascular risk factors^{19,68}. These include hypertension, diabetes, smoking, obesity, and elevated LDL cholesterol, and highlight the tight overlap between vascular health and dementia risk. Another important cerebrovascular co-pathology to consider in the context of AD is cerebral amyloid angiopathy (CAA). CAA is characterized by A β , primarily the A β 40 peptide, deposited in the walls of cerebral blood vessels, leading to loss of vascular smooth muscle cells, vessel wall thickening, and vessel fragility^{69,70}. CAA is a highly common co-pathology to AD but can also occur independently⁷¹. It can contribute to cognitive decline causing microbleeds and infarcts, and puts patients at risk of hemorrhages⁷⁰. As a co-pathology in AD, this often exacerbates a patient's cognitive decline⁷¹.

1.5.4 Co-pathologies Across Dementias

Although different forms of dementia are clinically distinct, neuropathology is very often mixed. AD, for example, is the most common clinical diagnosis, yet pure AD neuropathology is surprisingly rare compared to mixed pathologies found at autopsy. Dementia with mixed etiology is estimated to account for over 50% of cases⁷²⁻⁷⁵, yet the true prevalence is unknown. The uncertain nature of a clinical diagnosis is mainly due to the lack of diagnostic tools that accurately detect neuropathological hallmarks in living people, leaving neuropathological examination at autopsy as the gold standard of diagnosis. Through various autopsy-based

studies, more insight has been gained to inform on what pathologies most often co-occur with AD. Vascular co-pathologies are one of the most common entities found in those with AD, being reported in up to 80% of participants⁷⁶⁻⁷⁹. This is due to vascular changes being highly prevalent in aging, sharing risk factors with AD, and directly impacting the clearance of A β . A large portion of this prevalence is due to the common occurrence of CAA, with other vascular conditions such as arteriolosclerosis, atherosclerosis, and infarcts also significantly contributing. When it comes to proteinopathies, Lewy bodies are also a very common co-pathology as they are found in 20–60% of AD cases at autopsy, with higher rates in older populations and those with more advanced AD pathology^{76,77}. TDP-43 pathology occurs least commonly with AD compared to these other pathologies. However, when it does occur, it is often due to LATE-NC as the TDP-43 pathology is localized in the limbic structures rather than the frontotemporal region. LATE-NC has been observed in approximately 40-50% of AD cases⁸⁰⁻⁸².

1.6 Dementia Resilience

Although the high prevalence of dementia leads many to view it as an inevitable part of aging, many people are able to progress through life without developing dementia. These people are often referred to as being cognitively resilient. Cognitive resilience can be maintained either through a resistance to developing the pathology that causes dementia, or resilience to that pathology manifesting as cognitive decline. On the side of being resilient to the clinical manifestations of neuropathologies, previous research has shown that up to 40% of cognitively intact seniors display some AD neuropathology, including amyloid plaques and neurofibrillary tangles⁸³⁻⁸⁶. It is important to note that amyloid pathology has been observed to be more prevalent in cognitively resilient seniors than tau pathology, indicating resilience in mechanisms between these components of the AD pathology cascade⁸⁷. Studies have also shown that some degree of neurodegenerative processes occur during normal aging⁸⁸⁻⁹⁰. However, it has also been shown that those with pathological changes at older ages are still more likely to convert to clinical dementia than those without^{83,91}. Gaining a deeper understanding of the pathological underpinnings of cognitive resilience can provide additional insights to improving diagnostic accuracy and differentiation between normal cognitive aging and prodromal stages of dementia.

1.7 Current Diagnosis of Alzheimer’s Disease and Related Dementias

1.7.1 Cognitive testing

Cognitive testing has been the standard clinical tool to assess dementia for many years and refers to a broad spectrum of tests that assess multiple domains of cognition and memory. In 2020, the Fifth Canadian Consensus Conference on the Diagnosis and Treatment of Dementia (CCCDTD-5) published guidelines for how cognitive testing should be performed in the clinical pipeline in Canada⁹². Cognitive assessments often begin with screening tests such as the Montreal Cognitive Assessment (MoCA) and Mini-Mental State Examination (MMSE). The MoCA is considered to be highly sensitive to MCI and early AD through its assessment of short-term memory recall, visuospatial skills through drawing, executive functioning, concentration, language, and orientation⁹³. The MMSE, however, is recommended for situations where clinicians have more time and are aiming to assess tracking of moderate dementia rather than the earlier stages of MCI⁹⁴. The MMSE primarily focuses on orientation, registration, attention and calculation, recall, and language. The CCCDTD-5 also highlights several alternative screening measures appropriate for different clinical settings and patient populations, including the General Practitioner Assessment of Cognition (GPCOG), Rowland Universal Dementia Assessment Scale (RUDAS), Modified Mini-Mental State (3MS), Mini-Cog, Memory Impairment Screen with Clock Drawing (MIS-CDT), and the Ascertain Dementia 8 (AD-8) informant questionnaire⁹². Beyond screening, domain-specific neuropsychological tests are used in specialty settings to delineate cognitive profiles and guide differential diagnosis across dementia subtypes. These will target domains such as memory, executive function, language, and visuospatial processing. In addition to cognitive testing in clinical practice there are also a wide variety of cognitive tests that are regularly used alongside these measures in research and clinical trials. One of these regularly used tests is the Clinical Dementia Rating (CDR) and its affiliated CDR-Sum of Boxes (CDR-SB), which are validated for use in clinical practice but more widely used in research settings to track disease severity and progression. While cognitive tests remain fundamental for identifying cognitive decline, they do not capture the full complexity of dementia, which requires further understanding of its underlying biology.

1.7.2 Biological definition and the “ATN” Framework

Although cognitive testing has been a key tool for dementia diagnosis for decades, it has been shown that a clinical diagnosis of AD does not correspond with the underlying neuropathology approximately 30% of the time when the brain is examined at autopsy⁹⁵. As previously introduced in section 1.3, the National Institute on Aging and Alzheimer's Association (NIA-AA) developed a research framework in 2018 called the ATN system to define AD through a biological lens²⁴. The ATN framework is based on biomarker assessment of amyloid plaque pathology “A”, tau pathology “T”, and neurodegeneration “N”. Gold standard “A” or amyloid tests include a specialized positron emission tomography (PET) brain scan in which a patient is injected with a radioactive tracer that binds to brain amyloid⁹⁶. It is also possible to infer brain amyloid indirectly by quantifying the levels of A β 42, the most abundant A β peptide found in parenchymal amyloid plaques, in cerebrospinal fluid (CSF) obtained with a lumbar puncture⁹⁷. Importantly, amyloid can accumulate silently for years without causing memory problems⁹⁸. “T” or tau tests also include PET and CSF tests^{99,100} can measure the precise forms of phosphorylated tau (p-tau) found in neurofibrillary tangles, including p-tau-181 and p-tau-217^{101,102}. “N” biomarkers of neurodegeneration include magnetic resonance imaging (MRI¹⁰³⁻¹⁰⁵) as well as CSF tests^{106,107}. In 2024, a revised criteria were published to expand the ATN diagnostic framework and incorporate blood biomarkers as another key testing measure¹⁰⁸. In the updated 2024 framework, core amyloid and tau biomarkers were divided into Core 1 and 2 biomarkers with amyloid and phosphorylated or secreted tau markers being defined as Core 1 and tau proteinopathies being defined as Core 2. Additionally, new marker categories were added to the framework that focused on non-specific neuropathologies or co-pathology indicators. This included inflammation “I”, vascular “V”, and α -synuclein “S” pathologies.

1.7.3 Neuroimaging

Neuroimaging has multiple uses in the context of ADRD diagnosis. As a widely accessible diagnostic modality, MRI is often used as a first-line measure to assess neurodegeneration. MRI can be used to detect regional atrophy and structural abnormalities that can be indicators both of general neurodegeneration and have utility in some differential diagnosis. Distinct atrophy patterns often correspond to specific dementia subtypes, such as medial temporal and

hippocampal atrophy in AD, frontal atrophy in FTD, and posterior thinning in LBD dementia¹⁰⁹⁻¹¹¹. Volumetric analysis can also provide insights into brain volume loss to assist in disease tracking. In addition to informing on neurodegeneration, MRI is very useful for assessing cerebrovascular pathologies, aiding in the diagnosis of vascular dementia or vascular co-pathologies. This can be achieved by assessing white-matter hyperintensities, lacunes, and microbleeds, using T2-weighted or FLAIR (Fluid-Attenuated Inversion Recovery) sequences on MRI^{109,111,112}. For more specific assessment of AD pathology, PET scans are utilized. In PET scans, a radioactive tracer targets either a specific pathology or areas of high metabolism. The uptake of these tracers can then be visualized to detect specific pathological hallmarks. Tracers to indicate pathologies from the entire ATN framework have been developed. Amyloid PET tracers such as [¹¹C]PiB, [¹⁸F]florbetapir, [¹⁸F]florbetaben, and [¹⁸F]flutemetamol bind to A β plaques, with the spatial distribution of amyloid PET signal following Thal amyloid phases¹¹³⁻¹¹⁵. Tau PET tracers such as [¹⁸F]flortaucipir, [¹⁸F]MK-6240, or [¹⁸F]RO-948 target neurofibrillary tangle aggregates, where uptake mirrors Braak stage¹¹⁶⁻¹¹⁸. Neurodegeneration is assessed by PET slightly differently, using [¹⁸F]fluorodeoxyglucose (FDG)-PET, which measures regional cerebral glucose metabolism as a proxy for synaptic function, rather than targeting a specific pathological structure^{119,120}. Although the gold standard for diagnosis of ADRD pathologies remains autopsy confirmation, PET scans are often considered to be the gold standard for detection of AD neuropathology in living people. Because of this, PET scans have also served as a highly useful reference standard for validation of new biomarkers, especially fluid biomarkers like CSF and blood biomarkers^{121,122}. Despite profound contributions to the development of new diagnostic markers, PET scans are highly expensive and not easily accessible tools. Importantly, compared to other countries, Canada has not widely adopted PET in the AD clinical workflow.

1.7.4 Fluid Biomarkers

Fluid biomarkers are newer tools for the diagnosis of ADRD, and are considered an incredibly important advancement due to far lower cost and far greater accessibility compared to PET. CSF biomarkers were developed in a similar framework as the PET markers, in order to assess aspects of the ATN framework. In CSF, amyloid pathology is assessed by reduced concentrations of A β 42 and a lowered A β 42/A β 40 ratio. In AD, both markers decline due to

A β 42 becoming retained in plaques in the brain, causing less A β 42 to be released into CSF¹²³. The A β 42/A β 40 ratio is considered a preferred method as A β 40 acts as a reference to standardize against, improving reliability of the measures^{124,125}. Neurofibrillary tangle pathology is assessed using hyperphosphorylated tau (p-tau) markers, specifically tau phosphorylated at threonine 181, 217, or 231¹²⁶⁻¹²⁸. Total tau (t-tau) however, is used to assess overall neurodegeneration and axonal injury^{129,130}. Additionally, tau markers have also been assessed for use in ratios, such as p-tau/A β 42 and t-tau/A β 42, to indicate AD pathology, as they correlate closely with both amyloid and tau PET measures^{131,132}. Although CSF biomarkers have shown improvements in diagnostic strategy, they still have several caveats, such as the invasive nature of lumbar punctures and the need for specialized trained professionals to perform the procedure, which again limits accessibility and acceptability to patients. The most recent advancement in fluid biomarker diagnostics is the development of blood-based biomarkers. Blood biomarkers have the advantages of being lower cost, less invasive, and more universally accessible than both PET and CSF measures. This has led to rapid advancements in research and clinical implementation for their use in AD and other neurological conditions.

1.8 Neurological Blood-Based Biomarkers

Blood-based biomarkers are the most recent advancement in diagnostic tools for AD and many other neurological conditions. Neurological biomarkers enter the bloodstream through several mechanisms. One major route is through the glymphatic system, which facilitates the clearance of soluble brain-derived proteins through the interstitial fluid of the brain to the CSF surrounding the brain^{133,134}. This fluid then drains along the perivascular spaces of veins and exits the skull through meningeal and cervical lymphatic vessels, which connect with the systemic lymphatic system, enabling the eventual diffusion of brain-derived biomarkers into blood. Additional research has also proposed that proteins released from neurons and glial cells may diffuse directly from the extracellular space, to the CSF, then to blood¹³⁵⁻¹³⁷. Another important pathway is in extracellular vesicles that are released by neurons and glial cells. These vesicles encapsulate protein biomarkers and are capable of crossing the blood-brain barrier while protecting their contents from degradation¹³⁸⁻¹⁴⁰. In conditions involving blood-brain barrier disruption, some brain-derived proteins can leak directly into the blood, with their

concentrations often mirroring the degree of neural damage^{134,136,140}. Collectively, these mechanisms account for the presence of brain derived biomarkers, that represent important pathological hallmarks (**Figure 1.1**) in blood.

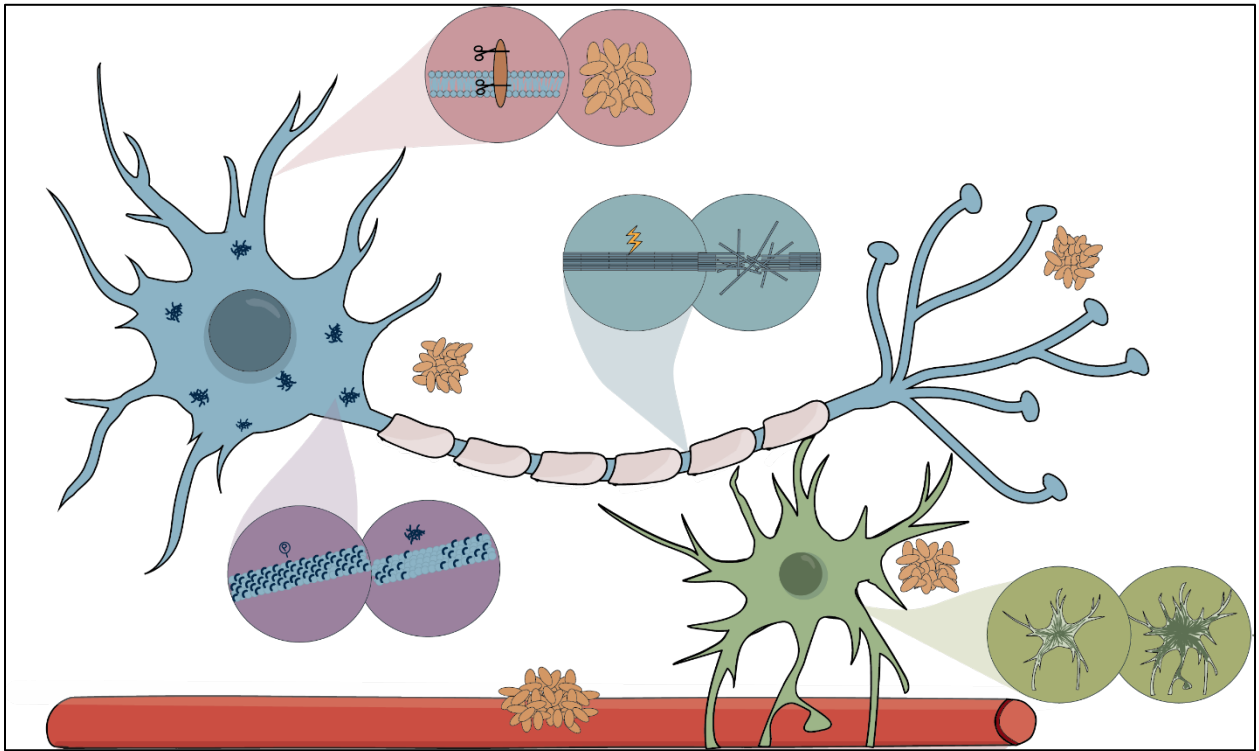


Figure 1.1 Plasma biomarkers of neurodegeneration and the pathologies they are derived from. These pathologies include mis-cleavage of amyloid precursor protein to form amyloid plaques (pink), phosphorylation of tau to form neurofibrillary tangles (purple), release of axonal proteins due to neurodegeneration (blue), and activation of astrocytes (green).

1.8.1 Amyloid Beta

Similar to CSF, the top amyloid beta derived plasma biomarkers are A β 40 and A β 42, which are produced as a result of APP being cleaved by β and γ -secretase to produce two different lengths of A β protein^{34,141}. Due to its direct link to amyloid pathology, studies on A β 42 and A β 40 have been a key focus in AD biomarker research, with a multitude of studies investigating use of A β markers for diagnosis and staging of disease and use in clinical trials. Though both biomarkers have been assessed for their utility individually, studies have shown

that the ratio of A β 42/40 is the superior plasma marker, as normalizing the reduction of A β 42 with A β 40 improves consistency of results as well as test accuracy^{142,143}. The primary area of research for A β 42/40 has been as a diagnostic marker for amyloid pathology. Like CSF A β , plasma A β 42/40 decreases in response to the development of amyloid plaque pathology. This is again due to the A β 42 isoform aggregating into plaques, with less being released into circulation¹²³. A β 42/40 has been shown to be significantly lower in those with confirmed A β pathology across MCI and dementia stages¹⁴⁴⁻¹⁴⁸. It has also been shown to discriminate A β -positive from A β -negative cognitively normal individuals, supporting its potential for early detection and screening¹⁴⁹⁻¹⁵¹. With respect to clinical disease staging, plasma A β 42/40 decreases progressively from preclinical to prodromal and dementia stages^{150,152-155}, with the best diagnostic performance in distinguishing AD dementia from cognitively normal controls^{150,155}. It is also important to note that, compared to some other diagnostic biomarkers, the effect size of A β 42/40 between those with and without AD is considered relatively small, which has limited its adoption in clinical settings as an individual marker^{156,157}. A β is often used in combination with other markers, or modeled alongside variables like age and *APOE* genotype, or other biomarkers, for diagnostic testing^{150,152,158}.

1.8.2 Phosphorylated Tau

Plasma phosphorylated tau is derived from the microtubule-associated tau protein that has been hyperphosphorylated and detached from the microtubule. Different plasma p-tau isoforms are named based on the site at which they are phosphorylated, with different conditions having different hallmark phosphorylation sites. In AD, p-tau-181 and p-tau-217 have emerged as the most promising biomarkers for diagnosing, prognosticating, and monitoring AD. Growing evidence supports their specificity and sensitivity for AD pathology compared to other neurodegenerative diseases. Initially, p-tau-181 and p-tau-217 were investigated to detect the presence of tau pathology and showed strong correlations with tau PET measures¹⁵⁹⁻¹⁶¹. However, despite their ability to detect tau pathology, many studies have also found they have high utility to detect the presence of amyloid pathology¹⁶²⁻¹⁶⁴. In fact, many studies have proposed that based on stronger correlations with amyloid PET measures than tau PET measures, plasma p-tau markers including p-tau-181 and p-tau-217 serve as better markers of amyloid pathology rather than tau^{164,165}. This, in combination with the lower effect size of

plasma A β 42/40 between A β positive and negative individuals, has led to plasma p-tau biomarkers coming to the forefront of AD diagnostic markers. Plasma p-tau-181 and p-tau-217 have consistently shown high utility for differential diagnosis of dementia, due to their specificity to AD^{159,166-168}. Although both the 181 and 217 isoforms have high diagnostic utility, p-tau-217 has been shown to consistently outperform p-tau-181 in its ability to detect amyloid positivity¹⁶⁸⁻¹⁷¹. However, this finding does not erase the utility of p-tau-181. Multiple studies continue to cite p-tau-181 as an excellent diagnostic marker¹⁷²⁻¹⁷⁴, especially as a highly sensitive marker to rule out the presence of AD pathology when p-tau-181 levels are not elevated¹⁷⁵. Additionally, p-tau-181 has been incorporated into large clinical trials as a pre-screening tool to reduce the need for PET imaging⁵⁴, and as a secondary endpoint to look at the efficacy of anti-amyloid therapy on pathology⁵³. As the studies in this thesis were initiated prior to the commercial release of validated p-tau-217 assays, the following chapters report on our findings for p-tau-181. We recognize that future studies on p-tau-217 are exceptionally important but are beyond the scope of the body of work of this thesis.

1.8.3 Neurofilament Light

Neurofilament light (NfL) chain is one of the main subunits of neurofilaments, which are intermediate filaments that form the cytoskeleton of neurons. NfL, together with neurofilament medium and heavy chains, provides structural stability, maintains axonal caliber, and supports nerve conduction by forming a scaffold within axons^{176,177}. NfL is most abundant in large, myelinated axons. Any time the axonal membrane and cytoskeleton are disrupted, NfL is passively released into the extracellular space and is cleared into the bloodstream^{178,179}. This makes blood NfL a very sensitive but non-specific marker of a wide variety of conditions including neurodegeneration and neurotrauma. Plasma and serum NfL have been extensively researched in the field of neurotrauma. For example, following traumatic brain injury (TBI), NfL levels correlate with injury severity, imaging evidence of axonal damage, and long-term functional outcomes¹⁸⁰⁻¹⁸³. As NfL can remain elevated for months to years after injury, it has high utility as a prognostic marker, with persistent elevations indicating persistent symptoms^{180,181}. NfL has also shown utility to inform on outcomes after stroke, and correlates with infarct volume, white matter hyperintensities, and functional outcomes¹⁸⁴⁻¹⁸⁶. With respect to neurodegeneration, NfL has been extensively researched in Multiple Sclerosis (MS), where

elevated plasma NfL has been used as a diagnostic marker¹⁸⁷, but is mainly used for prognosis and monitoring^{188,189}. Specifically, elevated NfL levels are used to indicate flare ups in relapsing-remitting MS, and lowered NfL levels after treatment are used as an indicator of treatment efficacy^{187,189,190}. NfL is elevated across all forms of dementia^{191,192}. Because of this, NfL may be considered a good general marker for dementia to indicate the degree of neuronal loss/neurodegeneration. In terms of diagnostic utility, NfL has been shown to be more elevated in FTD than in other forms of dementia, showing potential for differential diagnosis^{193,194}. As NfL is a marker of axonal damage, it can serve as an excellent marker of disease progression across multiple forms of dementia, with higher elevations showing increasing severity^{192,195,196}. NfL elevations are considered a good predictor of future cognitive decline^{196,197}.

1.8.4 Glial Fibrillary Acidic Protein

Glial fibrillary acidic protein (GFAP) is a type III intermediate filament protein, forming part of the astrocyte cytoskeleton. GFAP provides structural support, maintains cell shape, and contributes to the mechanical strength and stability of astrocytes¹⁹⁸⁻²⁰⁰. GFAP is also involved in the organization of the cytoskeleton, helping astrocytes support neurons and the blood-brain barrier. When astrocytes become activated, or undergo astrogliosis, astrocytes undergo morphological changes and significantly upregulate the expression of GFAP to help extend the cytoskeletal network^{198,199,201}. As neuroinflammatory signalling is one of the key drivers of astrogliosis, increased GFAP is often considered a marker of neuroinflammation^{200,202,203}. However, it is important to note that other mechanisms, such as cerebrovascular or neuronal damage, can trigger astrogliosis as well, indicating that GFAP is not exclusively a marker of inflammation²⁰⁰⁻²⁰². Because of this, similar to NfL, GFAP is considered a highly sensitive but non-specific marker in blood. GFAP has been highly investigated across multiple forms of neurotrauma to inform both on diagnosis and prognosis. For example, plasma GFAP is highly sensitive to detecting intracranial lesions in TBI, where levels rise rapidly after injury, peak within 20–24 hours and remain elevated for up to 72 hours²⁰⁴⁻²⁰⁶. Meta-analyses and large cohort studies confirm GFAP's strong diagnostic performance for identifying CT-positivity after TBI and its ability to stratify injury severity²⁰⁴⁻²⁰⁶. As elevated GFAP correlates with injury severity, GFAP has also shown high prognostic value for both TBI and spinal cord injury (SCI), with higher elevations indicating poorer outcomes^{207,208}. GFAP has also been assessed

across multiple neurodegenerative conditions, with a special interest in AD. Plasma GFAP is consistently elevated in AD, rising through preclinical stages to MCI and dementia. It has utility to detect amyloid pathology, distinguishing those with AD from healthy controls and other dementias with high accuracy²⁰⁹⁻²¹². In other forms of dementia, such as FTD and LBD, GFAP is elevated when compared to controls; however, not to the same extent as it is in AD²¹³⁻²¹⁷. This is likely due to A β plaques triggering the activation of astrocytes in attempts to clear the plaques^{51,52}. Studies have also shown that GFAP levels correlate with amyloid and tau pathology and can predict cognitive decline as levels rise before the onset of clinical symptoms, making it valuable for early detection and prognosis^{209,215,218}. It is important to note that due to the non-specific nature of GFAP, many studies often investigate it alongside multiple other markers for AD diagnosis, where GFAP can add additional value to markers like p-tau and A β ^{214,219,220}.

1.9 Blood Based Biomarker Analytical Technologies

Blood-based biomarkers for neurology are a fairly new concept, only emerging at a larger scale in the past 10 years. The reason for this is that proteins derived from the CNS are in such low abundance in blood that analytical methods to accurately detect them were not developed until very recently. Standard analytical platforms simply lacked the sensitivity to detect these low-abundance analytes in blood, which exist at attomolar to picomolar concentrations in blood. However, several technological advancements have since been developed that not only allow for greatly increased sensitivity, but also improved robustness and reproducibility.

1.9.1 Quanterix Simoa

One of the first major milestones in the advancement of neurological plasma biomarker testing was the development of the Simoa HD-1, and successor HD-X, analyzers from Quanterix. These analyzers use Single Molecule Array (Simoa) technology to achieve its advanced sensitivity. This technology is based on the principles of a sandwich enzyme-linked immunosorbent assay (ELISA)²²¹, where two antibodies bind to the analyte of interest, with a detector bound to one antibody to indicate the presence of the analyte. In the Simoa technology, the primary antibodies are bound to microscopic beads in suspension. Each bead has the

capacity to capture a single molecule of the analyte of interest. Biotinylated detection antibodies are then added to form a complex with the analyte of interest and bead-bound primary antibody, followed by the addition of streptavidin- β -galactosidase, causing streptavidin to bind the biotin on the detection antibodies. Resorufin β -D-galactopyranoside, the substrate for the β -galactosidase enzyme, is finally added before the resulting solution with beads is loaded into arrays consisting of approximately 200,000 microwells that can each hold up to one single bead, and then sealed with oil. Within the femtoliter-sized wells the enzymatic reaction takes place to create a fluorescent product. The arrays are then digitally imaged to quantify the number of wells containing the analyte of interest to determine the concentration of the original sample. The Simoa technology was the first of its kind with the capability of detecting analytes at femtomolar concentration, and became the first widely adopted platform for analyzing neurological biomarkers in serum and plasma due to its improved sensitivity over older platforms. In the initial years of the expansion of blood biomarker work, Simoa technology was at the forefront and also responsible for the development of some of the most widely-used assays in plasma biomarker research to date.

1.9.2 Almar ARGO

Though the Simoa technology laid the foundation for blood biomarker research, it is limited in its ability to multiplex more than four analytes at one time. While small multiplexed assays work well for clinical validation studies, biomarker discovery projects require higher multiplexing capacity to accelerate assessment of tens or hundreds of potential biomarkers. One of the emerging platforms focused on neurological biomarkers is the Alamar ARGO HT analyzer, which utilizes Nucleic acid Linked Immuno-Sandwich Assay (NULISA) technology²²². Similar to Simoa, NULISA leverages the basic principles of sandwich ELISAs and utilizes beads for antibody binding. In the NULISA technology, a pair of antibodies are added to solution with the sample to bind the analyte of interest. Each antibody is conjugated to a unique nucleic acid tail that allows them to bind to beads for two sequential wash steps. One of the antibodies has a polyA-tail that binds a polyT sequence on a bead which is pulled down and washed in the first step, while the other antibody has a biotin tail that binds to a streptavidin bead which is then pulled down and washed in the second step. The DNA tails conjugated to both antibodies also include half a barcode, which becomes ligated in the final

reaction step. This DNA barcode is specific to each antibody pair and reaction well, and is quantified through next-generation sequencing. This technology therefore allows not only high sensitivity detection, but also facilitates multiplexing with up to 250 different targets detectable within one sample.

1.9.3 Clinically used biomarker platforms

Although the Quanterix Simoa technology has been a leader in the field of neurological blood biomarker research, the HD-X platform has several limitations, including its designation as a research use only (RUO) platform. However, in more recent years, several major diagnostic companies have started to develop neurology assays on immunoassay platforms that are already approved for clinical use. For example, the Roche Cobas and Fujirebio Lumipulse are two immunoassay platforms leading translation of AD fluid biomarkers into the clinical space. Both Roche Cobas and Fujirebio Lumipulse are considered fully automated, high-throughput analytical platforms for immunoassays, and both currently have plasma tests approved for clinical use in AD diagnosis. Other major vendors of clinical immunoassay systems such as Siemens, Abbott, and Beckman are all developing commercial assays for AD and other contexts of use with some assays also receiving regulatory approval in recent years. Importantly, the availability and release of clinically validated assays and analytical platforms represent significant steps forward in bridging the gap between high-sensitivity research platforms such as Quanterix Simoa HD-X or discovery-focused systems like the Alamar ARGO HT, with clinical laboratories expected to become responsible for large-scale diagnostic implementation.

1.10 Current Use and Implementation of Blood Based Biomarkers

1.10.1 Alzheimer's Disease Blood Biomarkers

As previously introduced, revised criteria for the ATN system were published in 2024 to expand the diagnostic framework and incorporate blood biomarkers as a diagnostic tool for AD¹⁰⁸. In this updated ATN framework, plasma and CSF biomarkers are considered equivalent markers of AD pathologies. Biomarkers are divided into categories, with the first being core AD biomarkers, which are further divided into core 1 and 2 categories. In Core 1, A β 42 is

listed as an indicator of A β proteinopathies, while p-tau-217, p-tau-181, and p-tau-231 are listed as indicators of phosphorylated and secreted AD tau. However, it is important to note that these p-tau isoforms have been recognized in multiple studies as being excellent indicators of A β pathology. Core 2 biomarkers represent AD tau proteinopathies and are indicated by mostly non-phosphorylated tau forms, such as brain-derived or microtubule-binding region tau. Markers of non-specific neurodegenerative processes include NfL as a marker for neurodegeneration and GFAP as a marker of inflammation due to astrocytic activation. The most recent ATN framework also highlights plasma p-tau-217 as having the highest utility for AD diagnosis, with other biomarkers aiding more in staging AD pathology. Some AD blood biomarkers have also succeeded in receiving clinical approval for use in AD diagnosis. In 2025, the US FDA granted approval for the Fujirebio Lumipulse Plasma p-tau217/A β 42 assay, indicated for use in individuals aged ≥ 50 years presenting with cognitive decline to determine the presence or absence of amyloid pathology²²³. This assay is designed for use within specialized or memory-care settings as part of the diagnostic work-up, providing a minimally invasive alternative to PET or CSF testing. The FDA also approved the Roche Elecsys Plasma p-tau-181 assay in 2025, intended for use in primary care or general neurology settings as a rule-out test for AD pathology¹⁷⁵. When used in combination with clinical evaluation and cognitive testing, low p-tau-181 concentrations can effectively exclude significant amyloid or tau pathology, allowing clinicians to focus on further investigation of alternative causes of cognitive symptoms. Additional approvals for multiple biomarkers and tests are likely to come soon, as many other clinical practice guidelines are being published²²⁴⁻²²⁶.

1.10.2 Blood Biomarkers in Other Dementias

Research on plasma biomarkers for other forms of dementia lag far behind the progress made with AD. A major reason for this is the prior advancements in AD to detect A β pathology in living persons using PET and CSF measures as similar tools are not readily available to detect pathologies such as α -syn and TDP-43. Some advancements have been made in recent years with the development of tau PET that may aid in diagnosis of FTLT-tau²²⁷, and a CSF assay to detect α -syn²²⁸. However, these tests are newly developed or under development, and therefore not widely implemented. As biomarker advances in other dementias have been incremental to date, the diagnostic gold standard for most of these pathologies continue to be

neuropathological examination at autopsy. Importantly, most research on dementia biomarkers for both AD and non-AD dementia has been conducted in clinically-diagnosed living participants who lack a neuropathological diagnosis, as opposed to research on banked specimens collected from participants who subsequently undergo neuropathological classification at autopsy. For a condition like FTD, this approach can result in limitations as multiple pathologies can lead to the same clinical diagnosis^{60,61}. Recent research has shown that NfL may differentiate FTD from other dementias due to its extreme elevations in FTD compared to more subtle elevations in AD¹⁹³. Although this may aid in a clinical differential diagnosis, markers to detect specific FTD pathologies, like TDP-43, will likely be required to assist in the development of FTD disease modifying therapies.

1.10.3 Blood Biomarkers in Other Contexts of Use

Beyond AD and other dementias, blood biomarkers have achieved clinical implementation in other contexts, providing valuable precedents for their translation into routine clinical care. NfL has emerged as a general marker of axonal injury and neurodegeneration across a range of neurological conditions. In 2023, the US FDA approved the use of plasma NfL to monitor disease activity in patients with relapsing-remitting or secondary progressive MS using the Roche Cobas analyzer. Health Canada subsequently approved an NfL assay on the Siemens Atellica platform in 2024. In MS, elevated NfL levels correlate strongly with new or enlarging MRI lesions, active inflammation, and long-term disability progression, establishing its role as an objective marker of disease activity and treatment response^{188,229}. Similarly, GFAP has been validated as a marker of acute brain trauma. In 2021, the FDA authorized a rapid whole-blood GFAP test on the Abbott i-STAT analyzer to aid in determining the need for head CT scanning following mild TBI or concussion. Health Canada approved the same assay in 2025, marking one of the first blood-based neurotrauma diagnostics available for point-of-care use. After TBI, astrocytic activation leads to the rapid release of GFAP into the bloodstream, where its concentration rises proportionally to injury severity^{215,230}. This acute response enables sensitive detection of brain injury and supports early diagnostic triage. Beyond clinical implementation, these biomarkers are being actively investigated across a range of other neurological conditions for their utility in prognosis, monitoring disease progression, and evaluating treatment response. Ongoing research continues to expand the contexts in which these markers may

prove to be clinically valuable, providing critical insights for the broader development and implementation of blood-based biomarkers.

1.11 Pathway towards clinical implementation of blood-based biomarkers

1.11.1 Implementation of laboratory tests

There are many steps in the process of developing a biomarker into a clinically useful blood test. The overall pipeline involves discovery of the biomarker, analytical validation of the assay, clinical validation of biomarker utility, and then regulatory approval. Once a biomarker has been discovered, assay development and analytical validation involves optimizing the reliability of the assay, which includes evaluation of key performance parameters. These parameters include robustness (the assay's resilience to small procedural variations), precision (the consistency of repeated measurements), trueness (the closeness of results to a reference value), measurement uncertainty (the estimated range within which the true value lies), limits of quantification (the highest and lowest reliably measurable analyte concentrations), dilutional linearity (the ability to maintain accuracy when diluting samples above the assay range), parallelism (the agreement between diluted biological samples and calibrators), recovery (the proportion of analyte that can be accurately extracted and detected from the biological matrix), selectivity (the ability to specifically measure the analyte in the presence of other components), and sample stability (the degree to which the analyte remains chemically intact under defined conditions)²³¹. When conducting discovery and analytical validation stages, a larger number of targets are often assessed using smaller sample sizes. The previously highlighted blood-based biomarkers of this thesis currently lie mostly in the clinical validation or regulatory approval phases. Here, more specific targets are assessed using larger samples sizes that are meant to represent the expected patient population at large. Clinical validation evaluates the biomarker's ability to accurately and reliably reflect disease presence, stage, or prognosis in real-world populations according to pre-specified contexts of use, as test requirements differ depending on whether the test is used for screening, diagnostic, and prognostic purposes. For example, a diagnostic marker must have high accuracy in its ability to detect the presence of disease, a screening test is intended to identify those who are at risk of developing a condition, and a prognostic test is intended to predict clinical disease progression. In AD, a diagnostic test would require high accuracy for amyloid pathology, a

screening test would require the ability to assess risk of developing AD pathology, and a prognostic test would be required to predict conversion through clinical AD stages as well as predicting the rate of cognitive decline. The current focus on AD biomarkers is primarily towards diagnosis, where clinical validation studies aim to define the test's ability to accurately determine presence or absence of A β pathology. In this stage, there are also many considerations made for test interpretation. Once sufficient analytical and clinical evidence is established, regulatory approval is conducted. This involves assessments to ensure that the assay meets standards for safety, effectiveness, and quality, which are set by each country. The process also includes assessments that evaluate clinical impact, cost-effectiveness, value added, and health equity of the test.

1.11.2 Considerations for clinical validation

An essential component of clinical validation is the determination of appropriate cut-off values that distinguish normal from abnormal biomarker concentrations. Establishing thresholds is complex and must account for sources of biological variability. Cut-offs are typically derived from receiver operating curve (ROC) analyses in large, well-characterised cohorts, balancing sensitivity and specificity to optimize diagnostic performance. However, cut-offs must be tailored to the intended clinical use of the biomarker, shifting the test's sensitivity, specificity, and positive and negative predictive values for the desired outcomes. Sensitivity refers to a test's ability to correctly identify true positives, giving the percentage of patients that will test positive and truly have the disease based on an external reference standard, preferably the gold standard. For AD, a test with high sensitivity would have a large percentage of people who have a positive plasma test and have A β pathology confirmed by autopsy (gold standard), and PET or CSF (reference standards). Specificity, on the other hand, refers to a test's ability to correctly identify true negatives, giving the percentage of patients that will test negative and do not have the disease based on the reference standard. A test with high sensitivity minimizes false negatives while a test with high specificity minimizes false positives. Although we want balance between the two, one is often prioritized depending on the intended use. In a condition where treatment has potential adverse effects, it is important to have a highly specific test that will ensure minimal false positives so that treatment is provided for those almost certain to have the pathology. This is considered a good “rule in” test, and is highly relevant in AD, where

anti-amyloid therapy can carry high risks for adverse complications like ARIA. In a condition where the purpose of the test is to “rule out” a condition, sensitivity becomes more important than specificity. In AD, a highly sensitive test would catch all those who may be positive, but would likely require further testing for final confirmation of pathology prior to treatment with disease modifying therapy. While sensitivity and specificity provide useful information about a test in research settings, positive predictive value (PPV) and negative predictive value (NPV) are crucial metrics for interpreting test results in real-world clinical practice. Calculating these metrics relies on the study population having an accurate prevalence of the disease. For detecting A β pathology, the prevalence would change across the clinical spectrum. For example, detecting amyloid in cognitively normal populations would require a cohort with ~10-20% A β positivity²³²⁻²³⁵, in MCI ~25-50% A β positivity²³²⁻²⁴⁰, and in those with clinical suspicion of AD dementia ~70-80% A β positivity^{232,237,241,242}. If the cohort is representative of this prevalence, we can calculate PPV to indicate the probability that, following a positive test result, an individual will truly have the disease. Whereas NPV will indicate the probability that, following a negative test result, an individual will truly not have the disease. It is important to note that prevalence of the disease will impact expected results of these metrics as lower disease prevalence decreases PPV and increases NPV. For example, when comparing testing in cognitively unimpaired and dementia groups, the test will have a high PPV in dementia and higher NPV in cognitively unimpaired individuals.

The recently published Alzheimer’s Association guideline recommends that AD blood tests for detection of amyloid pathology should meet a performance benchmark of $\geq 90\%$ sensitivity and $\geq 75\%$ specificity when used as a triage tool, and $\geq 90\%$ sensitivity and specificity if intended to substitute for amyloid PET or CSF testing²⁴³. When determining cut-offs, it is also critical to consider population heterogeneity and comorbid medical conditions that can influence biomarker concentrations. Factors such as age, sex, body mass index (BMI), renal function and genetics, as well as vascular comorbidities, can modify plasma biomarker levels and potentially confound interpretation^{188,244-250}. Therefore, assessment of these variables in large, diverse cohorts is an important part of establishing clinical implementation and interpretation. Finally, assay harmonization and standardization are necessary to ensure generalizable cut-offs. Variability between assay manufacturers, calibration protocols, and

sample-handling procedures can result in disparate cut-off values and reduce comparability across studies or clinical sites. Collaborative standardization efforts and development of well-characterized, stable samples or substances for use as reference materials to calibrate assays and verify analytical performance, are strongly encouraged by the guideline as a way to improve reliability and applicability of biomarker thresholds.

1.11.3 Canadian-specific requirements for blood tests approval

Approval and implementation of new blood tests in Canada involve both regulatory oversight and evidence-based evaluation. Laboratory tests in Canada are approved through two main pathways, as in-vitro diagnostic devices (IVDs) or as laboratory-developed tests (LDTs). IVDs require approval from Health Canada, which assesses the safety, effectiveness, and quality of both the testing platform and assay²⁵¹. Following approval, the Canadian Agency for Drugs and Technologies in Health (CADTH) may conduct a health technology assessment (HTA) to evaluate the clinical, ethical, and economic implications of the test. Provincial health authorities then use this evidence to determine implementation and public funding²⁵². Research into plasma biomarkers will play a critical role throughout this process, as data generated from analytical and clinical studies provide the foundation for both regulatory approval and HTA review. Evidence of accuracy, reliability, and clinical utility are required to support the case for adoption into clinical practice. Economic and public health evaluations further guide implementation by assessing cost-effectiveness, health equity, and system impact²⁵³. The research evidence, regulatory review, and economic evaluation ensure that new blood tests introduced in Canada are safe, effective, and deliver value to the healthcare system. LDTs, in contrast, are regulated at the provincial level and are approved for use within a single accredited laboratory, limiting large-scale application. Because of this, for widely needed diagnostic tools such as blood tests for Alzheimer's disease, the IVD route is more appropriate.

1.12 Summary, research hypothesis, and specific objectives of this thesis

1.12.1 Summary

Over the past decade, remarkable progress has been made in the development of blood-based biomarkers for neurodegenerative diseases, particularly for AD diagnosis. Advances in

ultrasensitive assay technologies, capable of measuring down to femtomolar concentrations, have enabled the reliable quantification of CNS enriched proteins such as A β 42/40, p-tau-181, NfL, and GFAP in plasma. These biomarkers have considerable potential to greatly advance clinical practice by providing accessible, cost-effective, and minimally invasive tools for early detection, diagnosis, and monitoring of AD. However, despite rapid international progress, significant gaps remain before these biomarkers can be fully implemented in clinical care within Canada. This thesis addresses several of these gaps through analysis of plasma A β 42/40, p-tau-181, NfL, and GFAP across four cohorts representing components of the brain health spectrum as displayed in **Figure 1.2**. Using a large normative population representing Canadians across the lifespan, we address the need for a deeper understanding of how biomarker concentrations vary across the spectrum of normal aging through the establishment of reference intervals. This information helps to clarify how biomarker concentrations vary across age, sex, and other demographic factors, which are essential prerequisites for clinical implementation. Next, we explore biomarker signatures in individuals with cognitive resilience, providing insight into biomarker behavior among individuals who remain cognitively intact despite advanced age. Subsequent chapters then link plasma biomarker profiles to autopsy-confirmed neuropathology, addressing a critical knowledge gap in understanding the pathological specificity of these measures for AD compared with other dementias. Overall, by leveraging a multi-cohort design, this thesis identifies and validates diagnostic cut-offs for AD-specific pathology and assesses their concordance with clinical diagnoses across disease stages, providing evidence for their application in real-world settings. Collectively, the studies presented in this thesis contributes foundational data to guide the clinical translation of neurodegenerative blood biomarkers in Canada, enabling their eventual adoption into diagnostic workflows, regulatory frameworks, and precision medicine approaches for dementia care.

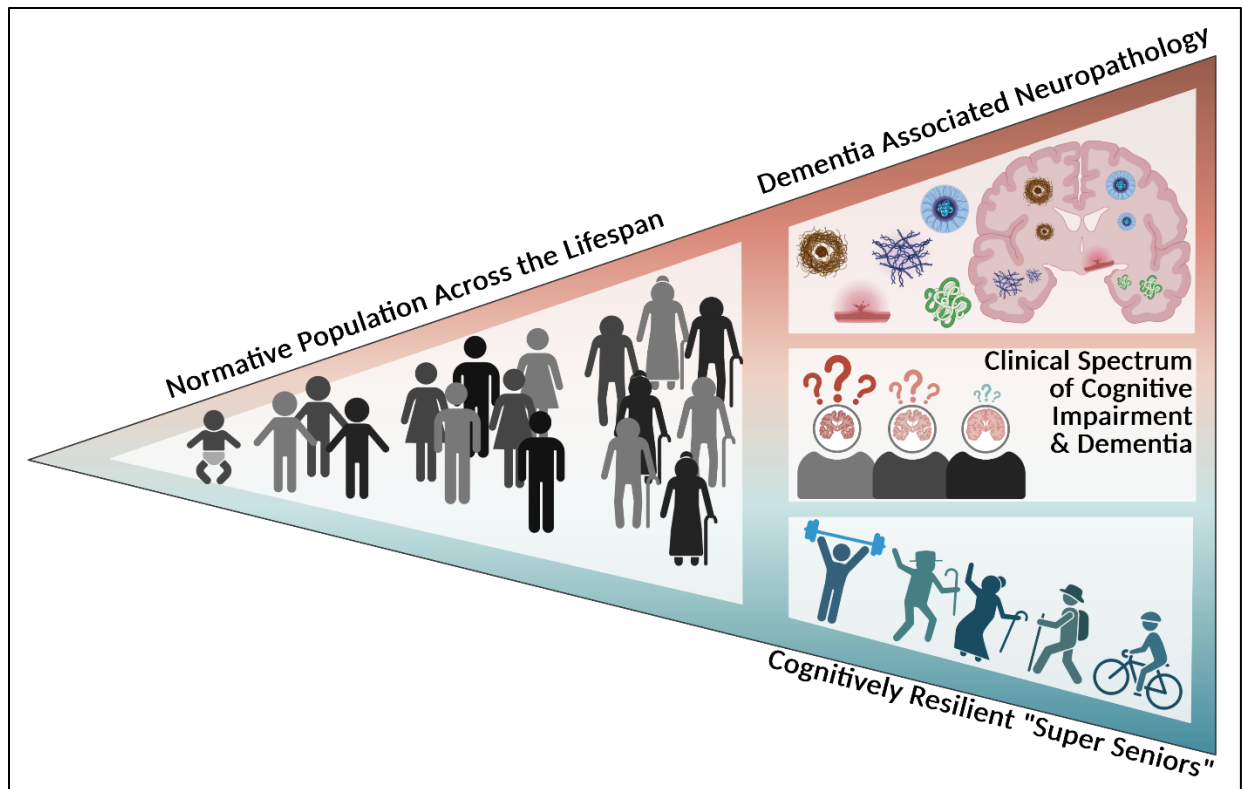


Figure 1.2 Cohort overview for assessing plasma biomarkers across the spectrum of brain health.

1.12.2 Research hypothesis

Blood-based biomarkers of neurodegeneration will significantly differ across the spectrum of health. Across the normative population, we hypothesize that biomarkers will display age-related changes resulting in the need for age-specific reference intervals. We hypothesize that in cognitively resilient populations, plasma biomarkers will display more condensed concentrations than normative populations that trend in the opposite directions anticipated from neurodegenerative diseases. We expect findings from these populations will provide insights into important factors to consider in the clinical interpretations of these biomarkers. We hypothesize that all plasma biomarkers will associate with measures of autopsy confirmed AD neuropathology, allowing us to derive accurate cut-offs for plasma biomarkers from this population. We hypothesize that, in a large cohort, these biomarkers will serve as good proxy markers for the indication of AD pathology across the clinical spectrum of dementia.

1.12.3 Specific objectives

1. Create valid age-specific reference intervals for the following blood-based biomarkers of neurodegenerative disease: A β 42/40, p-tau-181, NfL, and GFAP.
2. Quantify markers from objective 1 in a group of “super seniors” over 85 years old to observe levels in people considered to be extremely healthy and potentially resilient to neurodegeneration.
3. Investigate associations between plasma biomarkers and autopsy-confirmed neuropathology to evaluate the specificity of A β 42/40, p-tau-181, NfL, and GFAP for the detection of AD pathology.
4. Identify which markers from objective 1 have utility in the diagnosis of AD-specific pathology by using a multi-cohort approach to establish and validate cut-offs. Apply cut-offs across the clinical diagnosis spectrum and examine concordance between clinical diagnosis and underlying pathology.

Chapter 2: Age Specific Reference Intervals for Plasma Biomarkers of Neurodegeneration and Neurotrauma in a Canadian Population

2.1 Introduction

Blood-based biomarkers, predominantly derived from the central nervous system (CNS), are promising tools to screen, diagnose, prognosticate and manage a variety of neurological disorders. These biomarkers are considered to be a low cost, minimally invasive proxy measurement of CNS pathologies. Many prominently investigated biomarkers focus on Alzheimer's disease (AD), the most common form of dementia. AD is neuropathologically defined by amyloid plaques of aggregated β -amyloid ($A\beta$) peptides and neurofibrillary tangles of hyperphosphorylated tau. In plasma, the ratio of $A\beta_{42/40}$ can be measured as an indicator of amyloid plaque pathology^{144,254}. Phosphorylated tau at residue 181 (p-tau-181), a neuronal microtubule-associated protein phosphorylated at amino acid position 181, has recently emerged as a promising plasma biomarker that closely tracks with amyloid burden in AD patients^{162,255}. Neurofilament light (NfL) is an axonal intermediate filament protein that, when measured in plasma, is elevated in AD²⁵⁶, other neurological conditions such as multiple sclerosis (MS)¹⁸⁸, frontotemporal dementia¹⁹³, and neurotrauma²⁵⁷. Glial fibrillary acidic protein (GFAP) is an astrocytic intermediate filament protein that is presumed to reflect astrocyte activation when measured in plasma. Similar to NfL, plasma GFAP levels are increased in several neurodegenerative conditions as well as after in neurotrauma and acute cerebrovascular insults^{215,258}.

These four plasma biomarkers are at various stages of analytical and clinical validation depending on the specific context of use being examined. An important step in the clinical validation process is to establish cut-offs to discriminate normal from abnormal values. This can be done by creating reference intervals (RI) to define normal concentrations for a healthy or normative population based on a person's age or sex. As many of the above biomarkers have been found to associate with age^{154,249}, it is important develop age-dependent RIs across the lifespan to facilitate interpretation of their plasma levels. In this study, we developed discrete RIs using methodology recommended by the Clinical Laboratory Standards Institute guidelines (CLSI) (EP28-A3c)²⁵⁹ and continuous RIs using smoothed quantile regression, to

provide age specific cut-offs for plasma A β 42/40, p-tau-181, NfL, and GFAP reflective of the Canadian population.

2.2 Methods

2.2.1 Ethical approval

This study was reviewed and approved by The University of British Columbia Clinical Research Ethics Board, Vancouver, Canada (H19-01445). The Canadian Health Measures Survey (CHMS), which provided plasma specimens for this study, was reviewed and approved by the Health Canada and the Public Health Agency of Canada Research Ethics Board, Ottawa, Canada.

2.2.2 Participant and specimen selection

Specimens were acquired from the Statistics Canada Biobank which holds specimens from CHMS²⁶⁰. CHMS is a population-based national survey that collects information on the health of Canadians²⁶¹. Specimens were from CHMS cycle 5 where plasma specimens were collected in 2016 and 2017 from Canadians aged 3 to 79 years old. The CHMS study excluded individuals who live in the three Canadian territories (Northwest Territories, Yukon and Nunavut), live on reserves and other Indigenous settlements in Canadian provinces, are full-time members of the Canadian Forces, are institutionalized, or are residents of remote regions with under 10,000 people in a 75km radius. These exclusions represent approximately 4% of the targeted Canadian population²⁶². No further exclusion criteria for this study were applied beyond CHMS eligibility criteria. Blood was collected in EDTA vacutainers and centrifuged at 8° Celsius for 15 minutes at 1,800 g-force. Plasma was separated and stored directly into a -80° Celsius freezer within a maximum of 4 hours from collection. N=5,232 specimens were obtained from the Statistics Canada Biobank, and N=900 participant specimens across the full age range were randomly selected for analysis by a Statistics Canada methodologist to ensure CHMS participant information was protected and anonymized. A subset of N=384 specimens was initially selected to determine how many additional specimens would need to be analyzed to establish RIs, which were selected by Statistics Canada methodologists based on an even distribution across CHMS age groupings (3-5, 6-11, 12-19, 20-39, 40-59, 50-79 years). For

each age group, 4 participants (2 males, 2 females) were selected from each of the 16 collection sites across Canada, for a total of 64 subjects per age group. A Statistics Canada representative then selected an additional 516 specimens across age bins with an even distribution of sex and geographical region of collection to ensure a large enough sample size to comply with CLSI EP28-A3c guidance²⁵⁹. Although our results are based on unweighted population data and are therefore not fully representative of the Canadian population, our data exceed CLSI recommendations for RI generation.

2.2.3 Biomarker Analysis

Biomarker concentrations were quantified using the single-molecule array enzyme linked immunoassay (Simoa) HD-X platform from Quanterix (Billerica, MA) following the manufacturer's protocol. Plasma GFAP, NfL, A β 40, and A β 42 were measured using the Neurology-4-plex E advantage assay (catalogue #103670, lot 503105), and plasma p-tau-181 was measured using the ptau-181 V2 advantage assay (catalogue #103714, lot 502923). Each assay included an 8-point calibrator curve, two internal kit controls, and three plasma controls. All specimens were assayed in duplicate and the mean value was reported as the result. Analysis was completed using a single lot of each assay to reduce analytical variability that can be introduced by harmonizing lots. For all analytes, quality control information was documented across all runs. Standard curve concentrations had an average absolute error of 2.7-5.8% and an average recovery of 96.5-100%. Internal kit control concentrations had an average absolute error of 7.6-24.7% and an average recovery of 97.3-124%. The inter-plate CVs, calculated using plasma control concentrations analyzed in duplicate across every plate, were between 8.4-14.4%. The average intra-plate CVs, calculated using participant specimen, were between 2.0-7.0%. Analysis was conducted approximately 5 years after sample collection. It has been shown that A β 40 and A β 42 are stable over this time period²⁶³. While NfL, GFAP and p-tau-181 have not been investigated for their stability in plasma over 5 years, they have been shown to be robust, stable analytes in plasma that are resilient to varied storage conditions and freeze thaw cycles²⁶⁴.

2.2.4 Cross Lot Analysis

Although all specimens were analyzed using the same HD-X instrument and assay lot, a cross lot analysis was conducted to interpret the generalizability of the results. N=80 CHMS specimens were randomly selected across the dynamic range of the original N=900 to be re-analyzed on two separate lots of reagents. N=40 of the cross lot specimens were re-analyzed on a different lot (N4PE: 503212, p-tau-181: 503199) using the same HD-X instrument as the original study. Another N=40 specimens were re-analyzed on a different lot (N4PE: 503811, p-tau-181: 503545) and a different HD-X instrument from the original study. In addition to re-analyzing CHMS specimens across lots and instruments, we also tracked 3 plasma controls that were used in the original study to monitor inter-plate variability. These controls were generated from healthy plasma, with a low concentration control being unaltered plasma, and medium and high concentration controls created by spiking the plasma with the top calibrators provided with the assay kits. These plasma controls were carried over onto 10 additional plates, 5 plates using a different lot (N4PE: 503212, p-tau-181: 503199) and the same HD-X instrument as the original study, and another 5 plates using a different lot (N4PE: 503811, p-tau-181: 503545) and a different HD-X instrument from the original study. We then determined if each control value was within the acceptable ranges, which were determined by taking the 95% CI of the results obtained from each of the 26 plates analyzed in the original study.

2.2.5 Statistical Analysis

Discrete RIs were produced according to CLSI EP28-A3c guidelines²⁵⁹. Scatterplots for each biomarker and age were used to visually estimate break points for partitioned RIs. Breakpoints were established using the Harris and Boyd method²⁶⁵. Within each age partition, a Shapiro-Wilk test for normality was performed; all partitions contained non-normally distributed data. A Mann-Whitney U-test and Harris and Boyd method²⁶⁵ was used to test for a statistical difference between sexes within each partition. For all biomarkers, significant partitions were required for age but not sex. Within partitions, RIs were created using the *referenceIntervals* package²⁶⁶ for the R Statistical Programming Language. Outliers were removed from the dataset using Tukey's test for outliers. Upper and lower limits of the RIs were determined using the non-parametric method²⁵⁹, where the upper limit is denoted by the 97.5th percentile of the

population and the lower limit by the 2.5th percentile of the population. 90% confidence intervals (CIs) of upper and lower limits were also generated using the non-parametric method.

Continuous RIs were created for each analyte with the *quantregGrowth* package^{267,268} for the R Statistical Programming Language. No outliers were removed for this analysis. Smoothed regression curves were generated at the 95th and 5th percentiles to represent the upper and lower limits of the RIs, respectively. A regression curve was also generated at the 50th percentile to indicate the population median. The smoothing factor lambda for each regression function was determined by cross validation, and visually adjusted to determine fit based on biological expectations. 95% CIs were also created for both upper and lower limit regression curves. Point intervals for each age from 3-79 were then created using the predict function.

Bland-Altman analysis and plots were used to determine agreement between lots for the cross lot analysis. Agreement was determined as the and percentage bias and the 95% limits of agreement. All statistical analyses were performed using RStudio version 4.1.2 and GraphPad Prism version 9.5.1.

2.3. Results

2.3.1 Reference Intervals

A total of N=900 specimens from CHMS participants aged 3 to 79 years were analyzed to create discrete and continuous RIs. The sex distribution of participants was equal, with N=450 (50%) males and females. N=369 participants were under 20 years of age, N=342 participants were between 20 to 60 years of age, and 189 participants were over 60 years of age. The distribution of participants by age and sex is shown in **Figure 2.1**.

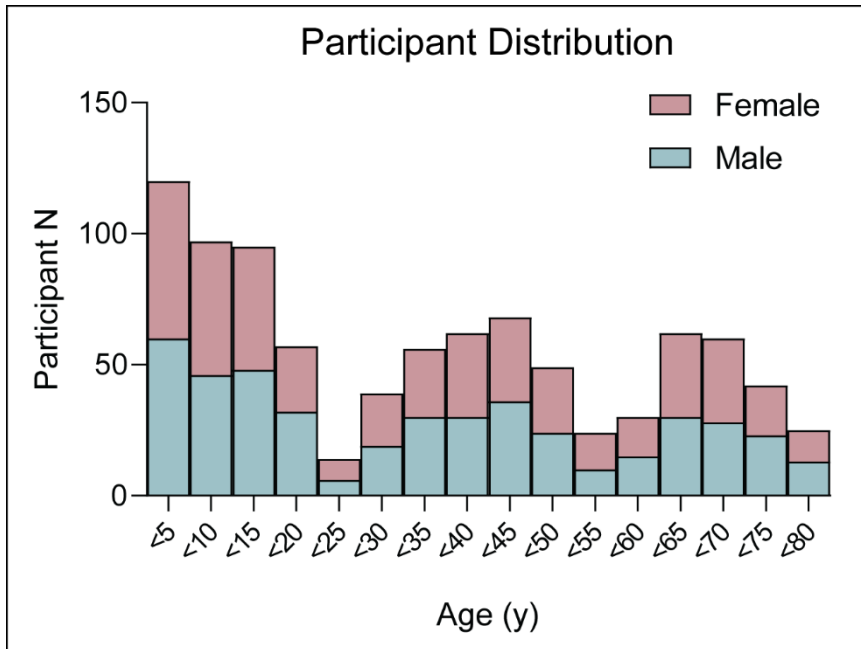


Figure 2.1 Distribution of participants by age and sex. N=900 participants from CHMS, cycle 5 collected in 2016/2017.

For discrete RIs, significant partitions were determined to create age bins for each biomarker. The age bins for each analyte were: A β 42/40: 3 to <55 years, and 55 to <80 years; p-tau-181: 3 to <12 years, 12 to <60 years, and 60 to <80 years; NfL: 3 to <40 years, 40 to <60 years, and 60 to <80 years; GFAP: 3 to <10 years, 10 to <60 years, and 60 to <80 years. No significant sex differences were determined, allowing the same RIs to be used for both sexes. The values for the upper (97.5th) and lower (2.5th) limits for the RIs with associated 90% CIs for each age bin are displayed in **Table 2.1** and **Figure 2.2**.

Table 2.1 Discrete reference intervals for plasma A β 42/40, p-tau-181, NfL, and GFAP. Significant age partitions were determined by Harris and Boyd method, no significant sex partitions were detected. Tables display N per each age bin after outliers were removed using Tukey test, values for upper (97.5th) and lower (2.5th) limits with corresponding 90% confidence.

A β 42/40					
Age Bin (y)	N	Lower Limit (pg/ml)	90% CI (pg/ml)	Upper Limit (pg/ml)	90% CI (pg/ml)
3 to <55	661	0.053	0.050, 0.057	0.098	0.096, 0.099

55 to <80	221	0.04	0.028, 0.045	0.09	0.088, 0.095
p-tau-181					
Age Bin (y)	N	Lower Limit (pg/ml)	90% CI (pg/ml)	Upper Limit (pg/ml)	90% CI (pg/ml)
3 to <12	239	1.4	1.4, 1.5	5.6	5.0, 6.6
12 to <60	444	0.8	0.7, 0.8	3.1	2.9, 3.6
60 to <80	202	0.9	0.9, 1.0	4	3.6, 5.7
NfL					
Age Bin (y)	N	Lower Limit (pg/ml)	90% CI (pg/ml)	Upper Limit (pg/ml)	90% CI (pg/ml)
3 to <40	522	2.6	2.5, 2.7	11.3	10.7, 12.8
40 to <60	165	4.6	4.3, 4.9	17.1	15.5, 18.0
60 to <80	202	8.1	7.5, 9.1	47.1	37.5, 56.6
GFAP					
Age Bin (y)	N	Lower Limit (pg/ml)	90% CI (pg/ml)	Upper Limit (pg/ml)	90% CI (pg/ml)
3 to <10	200	47	36.5, 55.2	226.3	190.0, 231.9
10 to <60	491	21.2	19.7, 22.7	91.9	86.2, 96.5
60 to <80	199	40.7	38.0, 44.0	227.5	184.4, 255.8

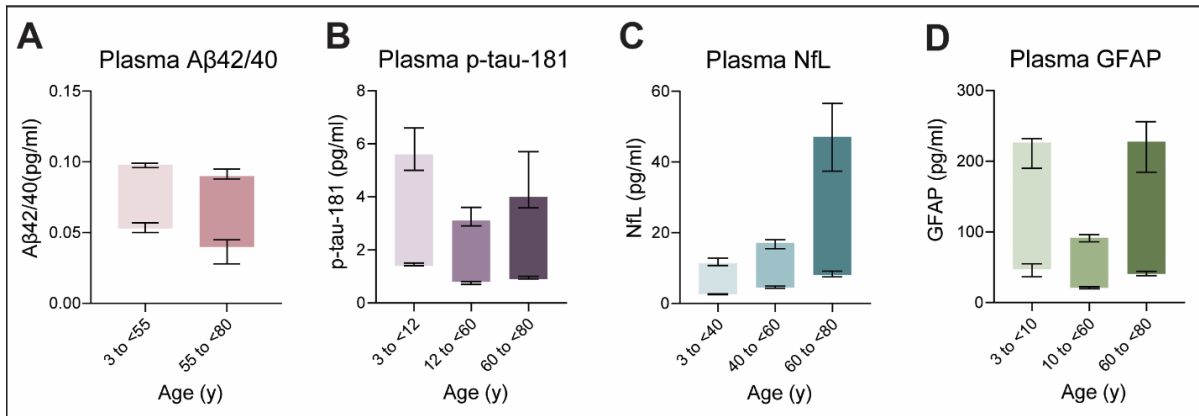


Figure 2.2 Discrete reference intervals for plasma A β 42/40 (A), p-tau-181 (B), NfL (C), and GFAP (D). Significant age partitions were determined by Harris and Boyd method, no significant sex partitions were detected. The shaded regions display the reference interval with upper (97.5th percentile) and lower (2.5th percentile) limits of reference range, error bars indicate the 90% CI of the limits.

Continuous RIs were created to determine the upper and lower limits for each year of age. As no sex differences were determined, data were combined to create continuous RIs to be used

for both sexes. Graphs displaying continuous RIs are displayed in **Figure 2.3**. The upper and lower limits are displayed as smoothed curves across the 95th and 5th percentiles of the population. The 50th percentile is also displayed to show the population median. Exact values for the upper and lower limits of the RI at every year of age with associated 95% CIs can be found in **Appendix 1**.

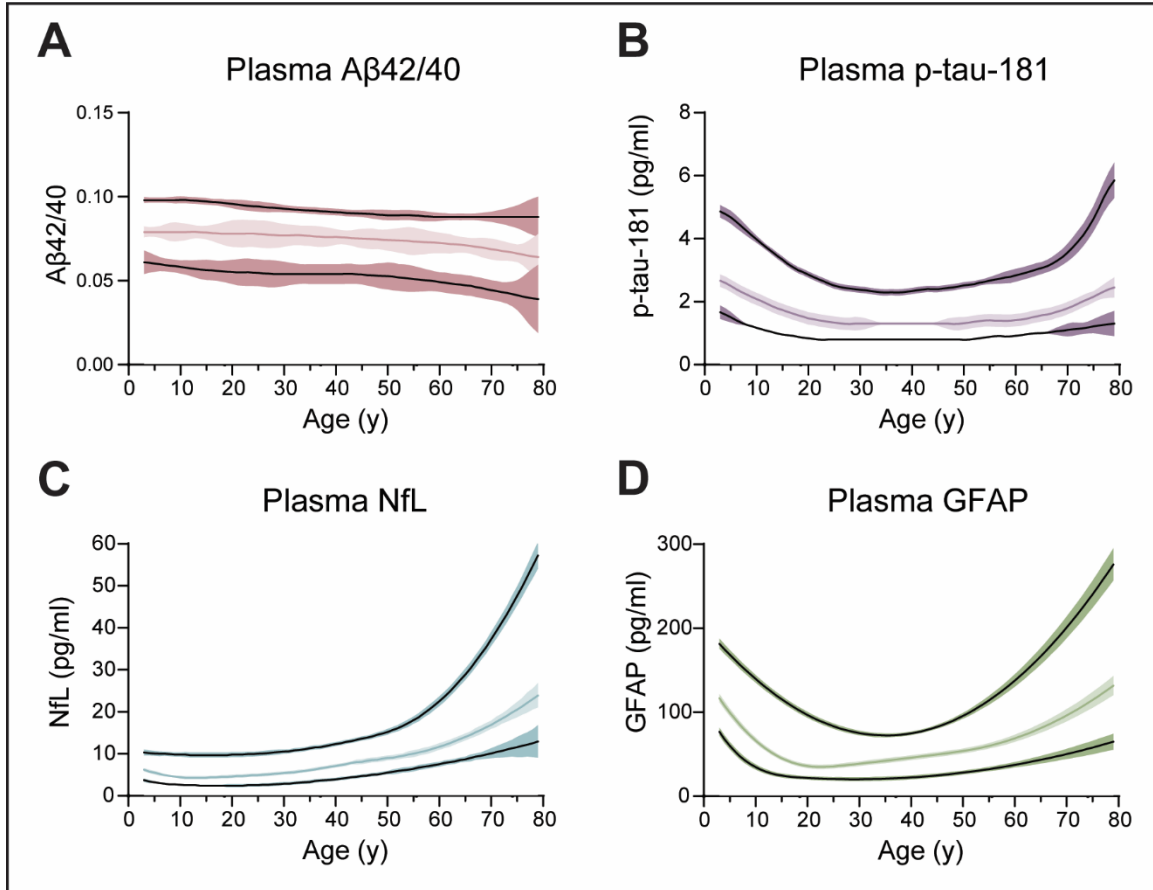


Figure 2.3 Continuous reference intervals for plasma Aβ 42/40 (A), p-tau-181 (B), NfL (C), and GFAP (D). N=900 for all figures. Black lines are 95th (upper limit) and 5th (lower limit) percentiles. Lighter coloured line is the 50th percentile. Shaded area around lines are the 95% CIs.

2.3.2 Assay Lot Agreement

To determine the generalizability of our results, we conducted cross lot investigations by reanalyzing a subset of N=80 CHMS specimens to compare results across another lot of

reagents and a separate HD-X instrument. The first cross lot analysis involved re-analyzing N=40 specimens on the same HD-X instrument as the original study with different assay lots. For the Neurology 4-Plex E assay there was an average bias of 8.4%, and for the p-tau-181 assay there was a bias of 22.1%. The second cross lot analysis involved re-analyzing a separate set of N=40 specimens on a different HD-X instrument than the original study with different assay lots. For the Neurology 4-Plex E assay there was an average bias of 9.7%, and for the p-tau-181 assay there was a bias of 13.7%. The bias for each individual analyte can be found in **Table 2.2**.

Table 2.2 Cross lot comparison for Quanterix Neurology-4-plex E (N4PE) advantage and ptau-181 V2 advantage assays. N=80 samples were randomly selected across the dynamic range of the sample set to be re-analyzed on two separate lots of reagents. One lot was analyzed on the original HD-X that this study was conducted on, and the other lot was analyzed on a different HD-X instrument. Bland-Altman analysis was conducted to determine agreement between lots and reported here as the percentage bias.

Compared to original study analysis:	Different Lot	Different Lot
	Same HD-X Instrument	Different HD-X Instrument
Cross-Lot N	40	40
Analyte	Mean Bias, %, (SD)	Mean Bias, %, (SD)
A β 40	9.8 (12.9)	4.4 (14.7)
A β 42	3.7 (11.9)	7.4 (5.4)
A β 42/40	-6.2 (13.4)	-2.9 (14.5)
NfL	11.0 (32.4)	-21.8 (20.0)
GFAP	9.4 (24.3)	-5.3 (18.4)
p-tau-181	-22.1 (17.6)	-13.8 (11.8)

Plasma controls were also tracked onto different lots to determine their agreement with the ranges generated on the original lot. Acceptable ranges were determined by calculating the 95% CI of plasma control results across 26 plates analyzed on the original lot. When analyzed on the same HD-X instrument as the original study using a different assay lot, plasma controls were within range for 80% of A β 40 data points and 93% of GFAP data points, with 100% of the data points for A β 42, A β 42/40, p-tau-181 and NfL falling in acceptable ranges. When analyzed on a different HD-X instrument and using a different assay lot than the original study, plasma controls were within acceptable ranges for 67% of A β 42 data points and 80% of p-tau-181 data points, and 100% of the data points for A β 40, A β 42/40, NfL, and GFAP fell within

acceptable ranges. Thus, an average of 92% of plasma control data points per analyte fell within acceptable ranges.

2.4. Discussion

2.4.1 Interpretation of Reference Intervals

We created RIs for plasma A β 42/40, p-tau-181, NfL and GFAP that included cut-offs for age bins and for each year of age. Sex was not identified as a modifying variable for any biomarker. At least one significant age partition was required for each biomarker. For A β 42/40, only one partition point was determined at age 55. A β 42/40 continuous RI showed little change over the lifespan, with an average yearly change of <1%. To our knowledge, no other study has created RIs for plasma A β 42/40, though RIs for the individual markers have been investigated^{269,270}. p-tau-181 had two partition points at the ages of 12 and 60 years. Examination of continuous RI reveals that median p-tau-181 concentrations decreased by an average of 3% per year for participants under 20 years of age, experienced no significant change during middle age, and increased by an average of 3% per year in participants over 60 years old. The increase in p-tau-181 aligns with previous findings showing a positive association of p-tau-181 levels with age in controls without AD from an American cohort¹⁶³. However, to our knowledge, no study has produced age specific reference intervals for p-tau-181. To our knowledge, this is the first study characterizing normative levels of p-tau-181 in pediatric participants.

For NfL, two partition points were determined at the ages of 40 and 60 years. The continuous RI indicated median NfL concentration increased by an average of 3% per year from age 13 to 60 years, increasing to a 4% per year increase after the age of 60 years. NfL RIs have been the most extensively investigated compared to the other biomarkers this study^{188,250,271,272}. Our results align with previous findings that show median levels of NfL increase by approximately 3% per year of age in American adults²⁵⁰, and that this increase becomes steeper after the age of 50 years¹⁸⁸. Similar to a study by Benkert et al., which created continuous cut offs for serum NfL in a European and American cohort¹⁸⁸, we also observed a general increase in the 50th and 95th percentiles and increasing separation between the 50th and 95th percentiles as age increases. Our discrete RI data has similar age cut points to a study by Simren et al. that utilized data

from multiple European cohorts to create cut-offs for plasma NfL²⁷²; our upper limits are quite similar. Our discrete RI upper limits are within an average of 4 pg/ml from each other under the age of 60, however, they do differ by up to 27 pg/ml in age bins over 60 years old.

For GFAP, two partition points were determined at the ages of 10 and 60 years. For participants under the age of 20 years, continuous RI indicated that median GFAP concentrations decreased by an average of 7% per year. Above 20 years old, median GFAP concentrations increased by an average of 2% per year, with a 3% yearly increase over 60 years old. Tybirk et al. previously published two studies on discrete and continuous RIs for serum GFAP in Danish pediatric and adult populations^{248,273}. Compared to their pediatric cohort, although our raw GFAP concentrations are different in terms of absolute levels, the overall trajectory is similar with an approximately 7% decrease per year of age. For adults, absolute GFAP levels differs between our results and those of Tybirk, yet a similar trajectory and percent change with age for both the median and upper RI are observed.

The results of our study for both NfL and GFAP are also consistent with the findings of another study conducted by our group in which Stukas et al. created pediatric RIs for serum NfL and GFAP using a separate Canadian cohort²⁷⁴. Although the studies were conducted using a different matrix, cohort, and assay formulation, our results largely overlap in terms of cut-offs and changes in median concentrations. For GFAP, both studies observed a decrease in GFAP of 7-8% per year between ages 3 to 18 years old. The lower limits of the RI differed by an average of 4 pg/ml and the upper RI differed by an average of 33 pg/ml, with results being closest in the older ages. For NfL, both studies observed an initial yearly decrease of median NfL by 4-6% from 3 to 12 years old, followed by a yearly increase of 1-3% from 13 to 18 years old. The lower limits of the RIs differed by less than 1 pg/ml at every age and the upper limits differed by an average of 1.5 pg/ml.

Some patterns were observed in biomarker trajectories across continuous RIs. P-tau-181 and GFAP had U-shaped curves with high concentrations in pediatric specimens, lowest levels in early adulthood and increasing concentrations in later adulthood. Although higher concentrations in pediatric age ranges were not observed for NfL or the A β 42/40 ratio, this

pattern was observed for A β 42 and A β 40 as individual markers, and for NfL in the separate pediatric study conducted by our group²⁷⁴, where serum NfL levels were highest in 1-year olds, with an average decrease in concentration of 10% per year up to 3 years old. A pattern of high biomarker levels in young children likely reflects neurodevelopment, with brain growth, synaptic pruning and maturing CNS clearance pathways²⁷⁵ all potentially contributing to high biomarker levels in blood. Although A β 42, A β 40, and p-tau-181 are often reported as AD-specific markers, their high concentrations in pediatric specimens indicate that other factors can influence their plasma concentrations. Except for A β 42/40, all biomarkers increased with age in adulthood. This association with age has been previously reported, however, most previous studies determined only a single cutoff for these biomarkers for all ages^{254,276-278}. Because of the dynamic changes noted in biomarker concentrations, especially under 20 and over 60 years old, it is recommended to use continuous rather than discrete RIs for improved interpretation of biomarker concentrations. We provide yearly cut points for each biomarker in **appendix 1** that were derived from the continuous RI regression curve.

2.4.2 Current and Future Utilities of Biomarkers

Age-specific RIs can help refine reporting and interpreting cut-off values to improve interpretation of laboratory tests in a variety of contexts. Currently, the United States Food and Drug Administration has approved all four biomarkers as breakthrough devices for distinct contexts of use. Though the approvals are on different devices, both plasma p-tau-181 and cerebrospinal fluid A β 42/40 have been approved to aid in the diagnostic evaluation of AD^{255,279-281}. Plasma NfL has been clinically validated to monitor disease progression and therapeutic response in patients with multiple sclerosis^{188,282}. Plasma GFAP is approved for use in adults with mild traumatic brain injury or concussion to determine the need for head computed tomography (CT)²¹⁵. In terms of future utility of these biomarkers, A β 42/40, p-tau-181, NfL and GFAP are all of interest for diagnostic and prognostic utility in neurodegenerative diseases^{283,284}, and NfL and GFAP are also of interest for neurotrauma and other acute neurological insults^{207,215,257}. In 2022, the Alzheimer's Association published recommendations for appropriate use of blood biomarkers in AD, which outlined how these biomarkers are currently being used for screening in clinical trials and may be used in memory clinics to assist in diagnosis when confirmed by secondary testing²²⁶. One of the research

priorities identified by the group was to establish cut points for different contexts of use²²⁶, a gap which this study helps fill.

2.4.3 Generalizability of Reference Intervals

The RIs generated here can only be used for interpretation of data analyzed using the Simoa HD-X platform. Importantly, compared to other studies mentioned in section 2.4.1 of this manuscript, our results align well with those of others despite analyses being performed on different cohorts from different countries using different assay lots and instruments. When investigating the applicability of our own results to other HD-X users, we conducted cross lot experiments that showed a bias between 3% to 22% for all analytes investigated in this study. We also found that the use of a different HD-X did not increase the overall bias across the assays used. Over both lots A β markers and GFAP tended to have a lower bias <10%, while NfL and p-tau-181 had larger bias. NfL had a bias of 11% and 21% and p-tau-181 had a bias of 22% and 14% on each lot. The results of this study suggest that there may be a mild bias of 10-20% between results produced on different assay lots for certain analytes. However, use of a different instrument will likely not increase this bias. Although a bias over 20% may be cause for concern, there are ways to mitigate this bias such as harmonizing the data through transformation if lots have a large bias with small limits of agreement. When tracking plasma controls over different lots, we observed that an average of 92% of data points fell in the acceptable ranges defined on the original lot. The only analyte that was found to have less than 80% of data points in the acceptable range was A β 42. Though this also may be some cause for concern, the corresponding A β 42/40 ratio had 100% of data points deemed acceptable, raising confidence in the translation of A β 42/40 over lots. As a mild bias between assay lots is not uncommon in laboratory medicine, the RIs shown in this study be used as a guideline for research-use-only interpretation of biomarker levels, and values that fall within the 95% CI of the RI should still be interpreted with caution.

Future studies will be conducted to incorporate a larger number of samples on multiple lots, and to develop widely available reference materials to facilitate calibration of results by other investigators to increase the utility of using RI to improve biomarker interpretation across the lifespan. Prior to the creation of reference materials for calibration, we recommend

transference studies are conducted by other laboratories to improve confidence in use of these RIs. When utilizing the same analyzer and assay type, CLSI EP28-A3c guidelines recommend validation of a reference population to accomplish transference²⁵⁹. This may be conducted on a small group of N=20, that represent the laboratories normative population, where it is expected that 90% of specimens fall within the reference limits. More robust validation can also be conducted with a larger reference population.

2.4.4 Limitations

One limitation of this study is its reliance on a population-based sample, making these RI representative of normative biomarker concentrations rather than concentrations in healthy persons rigorously screened to exclude neurological disease. Participation in the CHMS is voluntary. CHMS participants are not excluded from the CHMS study based on any neurological indications or cognitive impairment. Nevertheless, participants with cognitive impairment severe enough to limit their ability to provide informed consent would have been ineligible, and participants from whom blood could not safely be drawn or who did not provide consent for sample storage would not have their samples stored in the Statistics Canada Biobank. Thus, it is possible that underlying neurological disorders in study participants may underlie some of the increased data spread between the median and 95th percentile especially in those over 60 years old for p-tau-181, NfL and GFAP. However, using a normative population also has its advantages, particularly for biomarker studies on acute neurological insults that can happen at any age and can occur with co-morbid neurodegeneration in the elderly. As the biomarkers investigated here have multiple potential contexts of use, having cut-offs that represent normative population levels can improve diagnostic precision in a broader population. A second limitation is that data is not adjusted for survey weights of CHMS, and thus our results are not fully representative of the Canadian population. Previous studies have shown that A β 42/40, p-tau-181, NfL and GFAP levels in blood are modified by kidney function and body mass index, and diabetes and cardiovascular disease may be additional modifiers^{163,249}. Future studies will identify variables that modify the concentration of these biomarkers and how to account for them when interpreting RIs. A third limitation is that the Quanterix instruments are not clinically approved, and the RI data presented here cannot be used to aid biomarker interpretation for assays performed on other instruments.

Finally, reference materials will be required to calibrate future results to the RIs developed here, a limitation that also exists for all published RI studies on these biomarkers.

2.5. Conclusion

Discrete and continuous RIs for neurological plasma biomarkers will help refine normative cut-offs that can be used for interpretation in research settings to improve the precision of interpreting biomarker levels. These RIs also help us to understand the dynamic changes these biomarkers display across the lifespan.

Chapter 3: Age-Specific Reference Curves for Plasma Biomarkers of Neurodegeneration

3.1. Introduction

In recent years, research on plasma biomarkers for a variety of neurological conditions has surged, and several biomarkers have received regulatory clearance and are advancing into clinical use. Some of the most thoroughly investigated biomarkers include amyloid beta 42 to 40 ratio ($A\beta_{42/40}$), phosphorylated tau at residue 181 (p-tau-181), neurofilament light (NfL), and glial fibrillary acidic protein (GFAP). $A\beta_{42/40}$ and p-tau-181 are biomarkers of amyloid plaque pathology in Alzheimer's disease (AD)^{144,162,254}, and p-tau-181 also serves as a biomarker of neurofibrillary tangles^{255,278}. NfL is a highly sensitive, albeit non-specific, biomarker of neurodegeneration that is elevated in many neurological conditions where axonal damage is present. NfL can be used to monitor intervention efficacy in the context of multiple sclerosis (MS)¹⁸⁸, and shows promise as a diagnostic biomarker of frontotemporal dementia^{193,285}. GFAP is a biomarker of astrocytic activation that is of interest in AD²¹⁷ and in various forms of neurotrauma^{215,258} where neuroinflammation is pronounced.

As these biomarkers can be useful for multiple purposes across many neurological conditions, interpretation and cut-off values will depend on the specific context of use. Previous work has demonstrated that age and other demographic and health variables, such as sex, body mass index (BMI), and kidney function, can modify biomarker levels in selective populations^{188,248,274,286}. Therefore, having a large representative reference population to generate normative values of these biomarkers is extremely useful, as it would yield insights into how age, biological sex, and other demographic and health variables may affect normal biological variability in a biomarker across the entire lifespan. Utilizing a normative population as the basis of age-specific cut-off generation aligns with standard methods for reference interval generation²⁸⁷. However, a single reference interval may be limited in application, whereas normative reference populations provide more granular data. Detailed percentiles and z-scores, for example, allow for the testing of multiple percentiles as cut-offs for different contexts of use, and the ability to track a patient's percentile change over time.

In this study, we expanded on our previous work that produced age-specific reference intervals (RI) from n=900 Canadians drawn from the Canadian Health Measures Survey (CHMS)²⁸⁶, to now produce age-specific reference curves that provide normative distribution of neurological plasma biomarkers. While an RI can provide an upper and lower limit of normative biomarker concentrations, reference curves characterize the entire age-dependent distribution, enabling assessment of where an individual's biomarker value lies relative to the population. In this expanded analysis of n=2,500 CHMS participants from 3 to 79 years old, we report the 1st, 3rd, 5th, 15th, 25th, 50th, 75th, 85th, 95, 97th, and 99th percentiles of plasma A β 42/40, p-tau-181, NfL, and GFAP to provide a distribution representative of the Canadian population. We also developed an application that enables externally produced biomarker results to be directly compared to this reference curve database, including both a percentile and z-score output. Finally, we conducted an exploratory analysis on selected demographic and health variables, such as sex, body mass index (BMI), and kidney function, to assess their impact as potential modifier variables that may be important to consider when interpreting plasma biomarker results.

3.2. Methods

3.2.1. Ethical approval

This study was reviewed and approved by the University of British Columbia Clinical Research Ethics Board, Vancouver, Canada (H19-01445). The CHMS, which provided plasma specimens for this study, was reviewed and approved by the Health Canada and the Public Health Agency of Canada Research Ethics Board, Ottawa, Canada.

3.2.2. Participant and specimen selection

Specimens were acquired from the Statistics Canada Biobank, which holds specimens from CHMS²⁶⁰. CHMS is a population-based national survey that collects information on the health of Canadians²⁶¹. Specimens were from CHMS cycle 5, wherein plasma specimens were collected between 2016 to 2017 from Canadians aged 3 to 79 years old. The CHMS study excluded individuals who live in the three Canadian territories, live on reserves and other Indigenous settlements in Canadian provinces, are full-time members of the Canadian Forces,

are institutionalized, or are residents of certain remote regions²⁶². No further exclusion criteria for this study were applied beyond CHMS eligibility criteria. At CHMS collection sites, blood was collected into EDTA vacutainers and centrifuged at 8° Celsius for 15 minutes at 3,400 revolutions per minute. Plasma was separated and stored directly into a -80° Celsius freezer within 4 hours of collection. N=5,232 specimens were obtained from the Statistics Canada Biobank, and specimen selection for analysis was performed by a Statistics Canada methodologist to ensure CHMS participant information was protected and anonymized. Sample selection was performed in 3 parts. Initially, a subset of N=384 specimens was selected to determine how many additional specimens would need to be analyzed to establish RIs. These were selected by the Statistics Canada methodologists based on an even distribution across CHMS age groups (3-5, 6-11, 12-19, 20-39, 40-59, 50-79 years). For each age group, 4 participants (2 males, 2 females) were selected from each of the 16 collection sites across Canada, for a total of 64 participants per age group. The Statistics Canada methodologist then selected an additional 516 specimens across age bins with an even distribution of sex and geographical region of collection to ensure a large enough sample size to comply with Clinical and Laboratory Standards Institute (CLSI) EP20-A3c guidance²⁵⁹ to create discrete RIs. Although these data exceed CLSI recommendations for RI generation, the sample size was insufficient to meet Statistics Canada criteria to include population weights, making our previous results not fully representative of the Canadian population. To enable continuous reference curves representative of the Canadian population, an additional 1,600 participants were selected by Statistics Canada methodologists to enable the creation of a weighted subset of the CHMS database.

The subsample of n=2,500 was selected among CHMS Cycle 5 Mobile Examination Center (MEC) respondents with biobank measures. As such, the weighting follows typical CHMS weight adjustment steps for CHMS MEC respondents: site selection weight, dwelling selection, removal of out-of-scope units, household nonresponse, person selection, questionnaire-level nonresponse and MEC-level nonresponse. After MEC-level nonresponse adjustment, two additional adjustments were applied to the weights to account for this subsample: adjustment for MEC respondents that were not eligible for the subsample (MEC respondents with no biobank measures), and adjustment for the subsample. The former resulted

in redistribution of weights of MEC respondents not eligible for the subsample to those who were eligible; this was done by collection site, CHMS age group, and sex. The latter resulted in redistribution of weights of MEC respondents eligible for the subsample but who were not selected versus those who were selected for the subsample; this was done by collection site, age groups subsampled for this project, and sex. The subsample weights were then winsorized and calibrated and the bootstrap weights were generated following the typical CHMS steps. The resulting weights are representative of the Canadian population age 3-79 years.

3.2.3. Biomarker Analysis

Biomarker concentrations were quantified using the single-molecule array enzyme-linked immunoassay (Simoa) HD-X platform from Quanterix (Billerica, MA) following the manufacturer's protocol. Plasma GFAP, NfL, A β 40, and A β 42 were measured using the Neurology-4-plex E advantage assay (catalogue #103670), and plasma p-tau-181 was measured using the p-tau-181 V2 advantage assay (catalogue #103714). Each assay included an 8-point calibrator curve, two internal kit controls, and three plasma controls. All specimens were assayed in duplicate, and the mean value was reported as the result. For all analytes, quality control information was documented across all runs. Standard curves had an average absolute error of 2.5-7.8% and an average intra-run CV of 4.5-7.6. Internal kit controls had an average absolute error of 6.1-11.8%, an average intra-run CV of 1.8-5.3%, and an average inter-run CV of 3.0-23.8%. For the plasma controls, the average intra-plate CVs were between 2.6-6.7% and average inter-plate CVs were between 4.6-14.4%. The initial n=900 (36%) specimens were analyzed on lot 503105 for N4PE and lot 502923 for p-tau-181. The remaining n=1,600 (64%) specimens were analyzed on lot 503811 for N4PE and lot 503545 for p-tau-181. A cross-lot analysis was performed by selecting n=40 specimens from the first subset across the dynamic range of NfL for N4PE and p-tau-181 for p-tau-181. A Spearman's correlation was performed between the two lots, and for all analytes, the rho value was between 0.922-0.985, showing excellent correlation between lots. A Bland-Altman analysis was conducted to examine bias between lots. A β 40 (4.4%), A β 42 (7.4%), and GFAP (5.3%) all had a very low absolute bias between lots, while NfL (21.8%) and p-tau-181 (21.1%) had slightly higher bias. No correction was performed across lots. Incorporating raw data from two independent reagent lots captures the lot-to-lot variability, improving the generalizability of

results across lots of N4PE and p-tau-181, rather than creating the need to harmonize to a single lot. Detailed quality control metrics and cross-lot comparison methodology across all thesis chapters are described in Chapter 4.

3.2.4. Reference curve creation

Reference curves were created to report age-specific percentiles and z-scores using generalized additive models for location, scale, and shape (GAMLSS)²⁸⁸. Using the *GAMLSS* package in R, we derived a Lambda-Mu-Sigma (LMS) model with the biomarker as the response variable and age as the predictor. An age transformation was applied to account for rapid changes seen on the tail ends of age experienced by the biomarkers. From the LMS model, we extracted the age transformation and best fitting family for the distribution. All biomarkers used a Box-Cox t distribution, which allows for heavier tails than a standard Box-Cox distribution. Using the age transformation and selected family, GAMLSS models were built that incorporated normalized survey weights. Four smoothing parameters (μ , σ , ν , and τ) were optimized using a generalized Akaike information criterion (GAIC) with k penalties for fitting. Q stats and worm plots were utilized to assess the model fit and adjustments needed for each smoothing parameter. From the final models, percentiles and z-scores were extracted at each age, where the midpoint of an age is used to represent the range of that year of age. For percentiles, we reported the percentiles: 1, 3, 5, 15, 25, 50, 75, 85, 95, 97, 99 in accordance with the World Health Organization standard growth curves²⁸⁹. The 95th and 5th percentiles were selected to represent the upper and lower limits of the RIs, respectively²⁸⁶. For z-scores, we report -2, -1.5, -1, -0.5, 0, 0.5, 1, 1.5, 2.

3.2.5. Statistical analysis of associations between biomarkers and health variables

As an exploratory analysis, a set of predetermined CHMS variables were assessed for their potential associations with plasma biomarkers. Variables were selected based on known associations with biomarkers or neurological conditions, such as age²⁸⁶, BMI²⁹⁰, and kidney function²⁹¹, and variables regularly used in the interpretation of blood tests, such as sex, ethnicity, and fasting status. Analysis was performed using the *survey* package in R²⁹², which allows for the integration of survey weights. As many of the predetermined variables chosen

are also highly associated with age, we created age bins to conduct separate analyses for participants 3-19 years old, 20-39 years old, 40-59 years old, and 60-79 years old. Within each age bin, outliers were removed from biomarkers and survey variables using a Tukey test with a fence of 3. Univariate regression analysis with biomarker as the response variable and health variables as the predictors was performed using the *svyglm* function in each age bin. Results of each regression were reported using standardized betas and confidence intervals. **Appendix 2** contains a list of all CHMS variables analyzed, with the original CHMS questionnaire description of how questions were asked and coded. For categorical variables, any variable that was not binary was converted to binary by defining a new threshold. The details of which variables were modified from the original CHMS questionnaire can be found in **Appendix 2**. In alignment with Statistics Canada data privacy protocols, regressions were not completed for variables with less than n=30 respondents. For additional statistical robustness, regressions were only performed on variables where a minimum of 25% of the age bin had responded to the question. For binary variables, regressions were only performed when both categories contained >10% of the respondents for that variable. All analyses and reporting followed Statistics Canada's confidentiality protection policies. No raw microdata or individual-level information left the secure research environment. To protect participant confidentiality, all analytic outputs are subject to strict disclosure controls: result tables were reviewed to ensure that low respondent counts do not display exact values, and estimates were rounded in accordance with Statistics Canada guidelines. All findings presented represent aggregate data, and direct or indirect identification of individual study participants is not possible. All results contained in this publication were independently reviewed by Statistics Canada agents to ensure any rounded or modified values presented here accurately reflect the underlying secured data.

3.3 Results

3.3.1 Continuous reference intervals

A total of n=2,500 plasma specimens from CHMS participants aged 3 to 79 years were analyzed to create the reference curves. The sex distribution of participants was equal, with n=1,250 (50%) males and females. N=695 participants were under 20 years of age, n=1,090

participants were between 20 to 60 years of age, and n=715 participants were over 60 years of age. The distribution of participants by age and sex is shown in **Figure 3.1**.

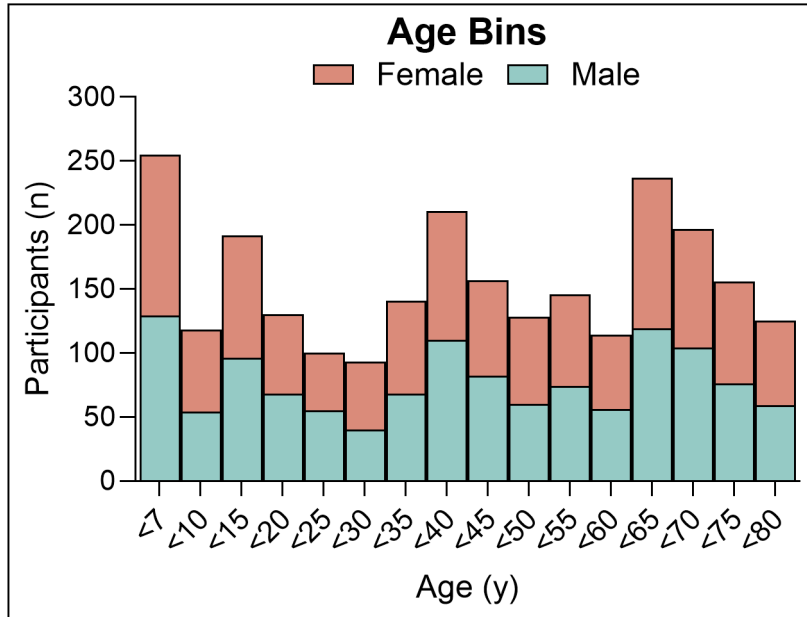


Figure 3.1 Distribution of CHMS participants by age and sex.
N=2,500 participants from cycle 5 collected in 2016/2017.

Continuous RIs were created to determine the upper and lower limits for each year of age, defined as the 5th and 95th percentiles of each biomarker. Additional percentiles are reported as reference curves to characterize the normative distribution across ages to provide reference curves for how biomarkers change with age. We determined the following percentiles for each biomarker: 1st, 3rd, 5th, 15th, 25th, 50th, 75th, 85th, 95, 97th, and 99th. A visualization of the reference curves can be found in **Figure 3.2**, and the detailed concentrations that correspond to each percentile at each age are reported in **Appendix 3**. In addition to providing reference curves that display the change in biomarker concentrations with age, we also report the z-score distribution for all biomarkers to improve the utility of the reference curves for other populations. The -2, -1.5, -1, -0.5, 0, 0.5, 1, 1.5, 2 z-scores are displayed in **Figure 3.3**.

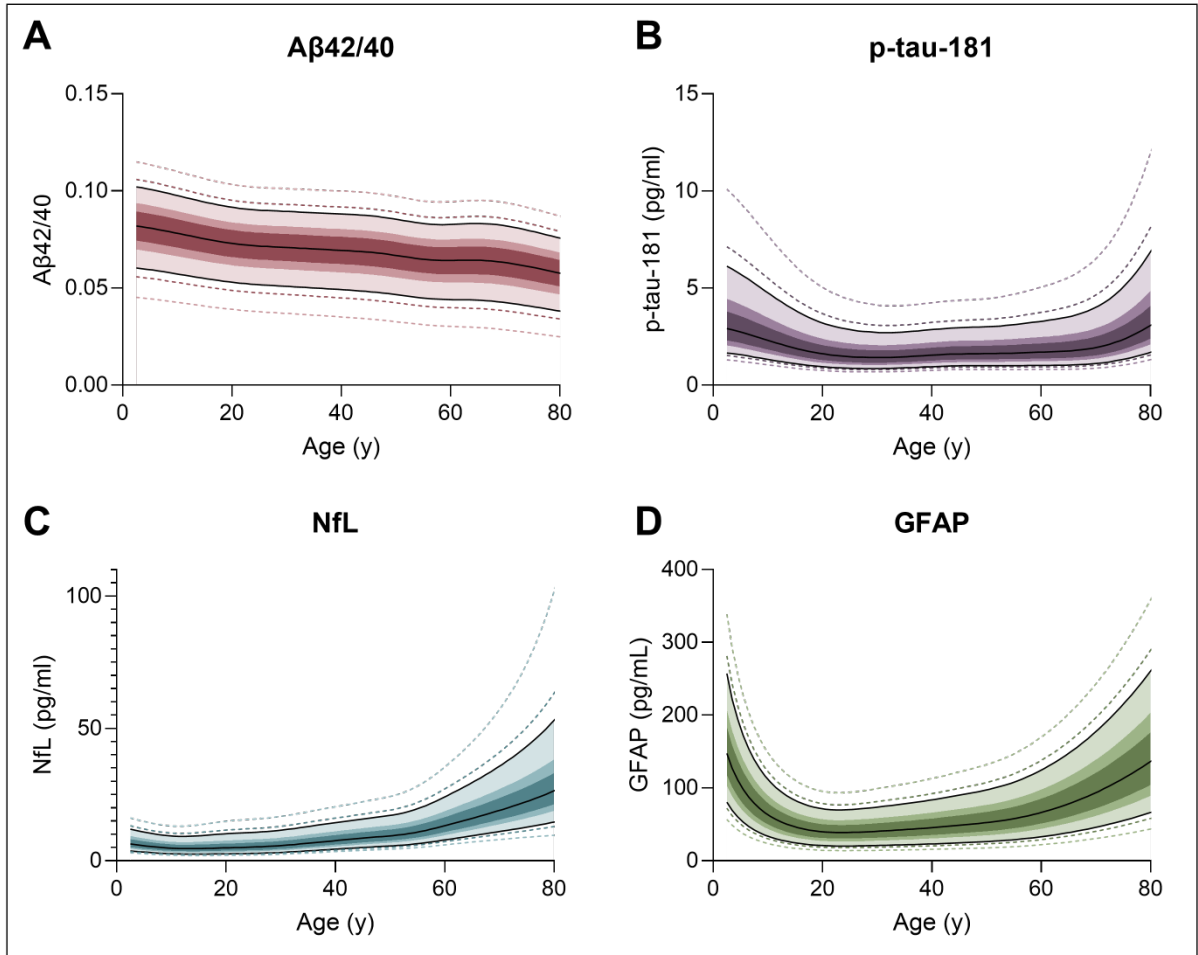


Figure 3.2 Reference Curves for plasma Aβ42/40 (A), p-tau-181 (B), NfL (C), and GFAP (D). Produced using n=2,500 CHMS participants and GAMLSS regression to determine smoothed percentiles. Displayed percentiles include 1st (lowest dotted line), 3rd (second lowest dotted line), 5th (lowest solid line and first shaded gradation from bottom), 15th (second shaded gradation from bottom), 25th (third shaded gradation from bottom), 50th (central solid line), 75th (fourth shaded gradation from bottom), 85th (fifth shaded gradation from bottom), 95th (upper solid line and end of of shaded gradations), 97th (second highest dotted line), and 99th (highest dotted line), with the shaded region indicating the reference interval from the 5th to the 95th percentile.

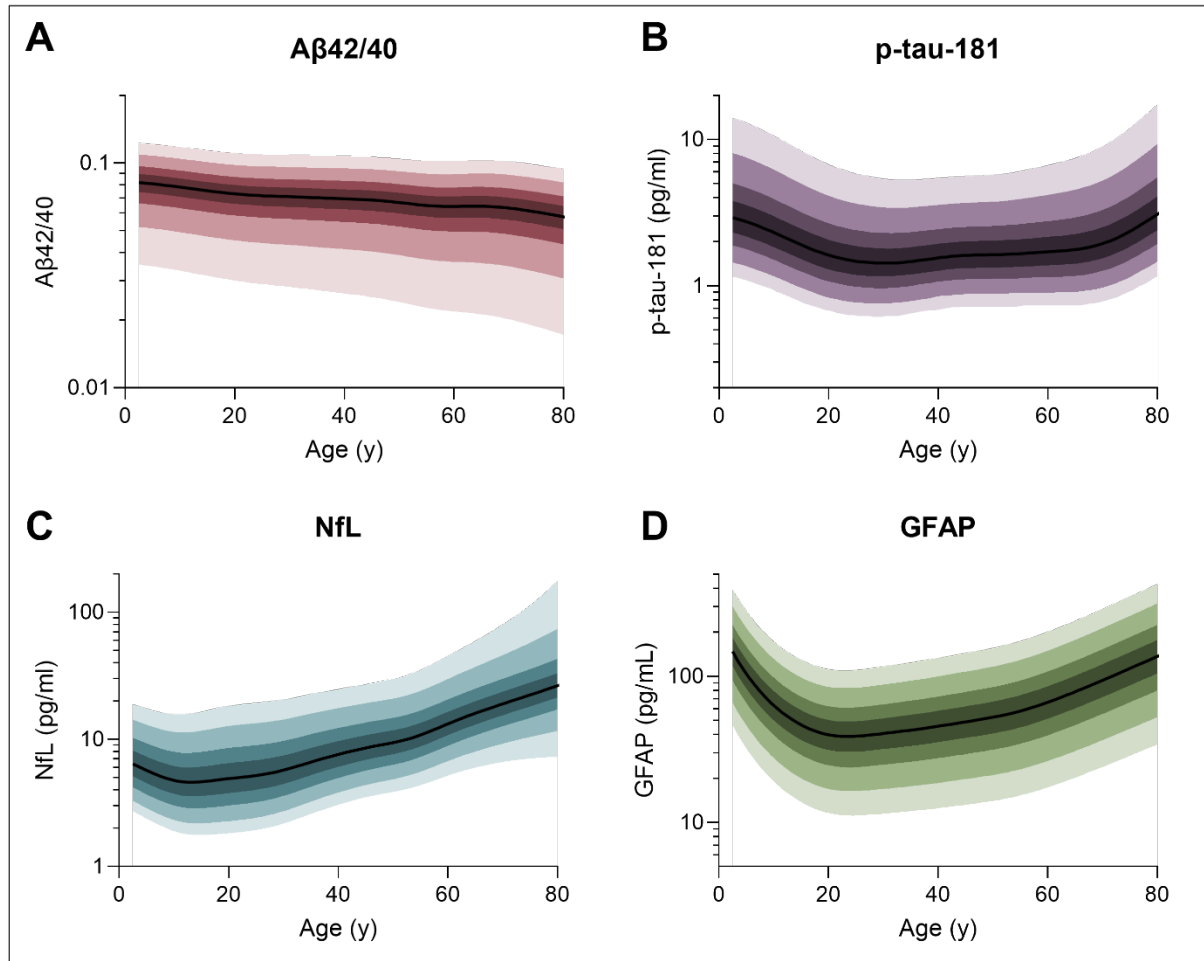


Figure 3.3 Z-score curves for plasma Aβ42/40 (A), p-tau-181 (B), NfL (C), and GFAP (D). Curves were produced using n=2,500 CHMS participants and GAMLSS regression to determine smoothed percentiles. Displayed z-scores include -2 (bottom of lowest gradation), -1.5 (bottom of second gradation), -1 (bottom of third gradation), -0.5 (bottom of fourth gradation), 0 (black line), 0.5 (top of fourth gradation), 1 (top of fifth gradation), 1.5 (top of sixth gradation), and 2 (top of final gradation).

We next investigated if any biomarker required sex specific RIs. Sex effects were assessed in evenly distributed age bins for all biomarkers (3-19, 20-39, 40-59, and 60-79 years old) using the Harris and Boyd method²⁶⁵, followed by validation with simple linear regression. Neither Aβ42/40 nor NfL displayed any association with sex and therefore did not require sex specific RIs. P-tau-181 was associated with sex in age bins from 20 to 79 years, where males had higher p-tau-181 levels than females. GFAP was also found to associate with sex, with females in age bins 40 to 79 years having higher GFAP levels than males. **Figure 3.4** displays the 5th, 50th and

95th percentiles of p-tau-181 (A) and GFAP (B), indicating that the sex effects are specific to the upper range of biomarker concentrations at the 95th percentile, while the 5th percentiles greatly overlap and the 50th percentiles show minimal variation.

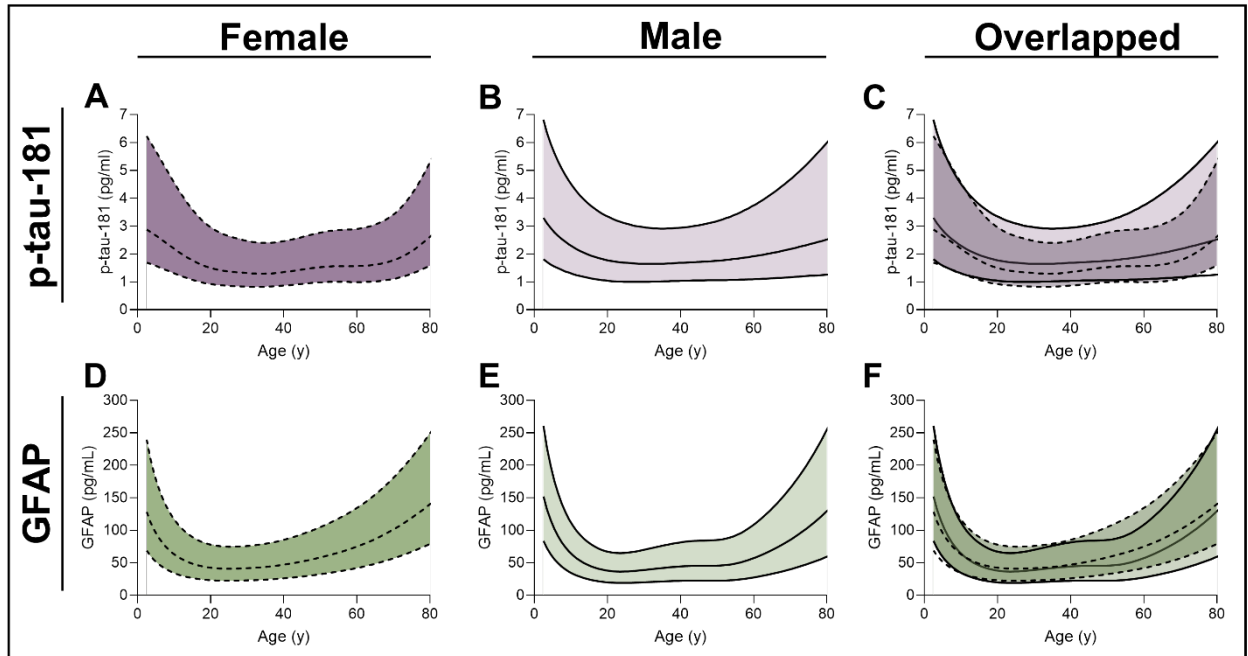


Figure 3.4 Sex specific continuous reference intervals for plasma p-tau-181 (A-C) and GFAP (D-F). Produced using n=2,500 CHMS participants and GAMLSS regression to determine smoothed percentiles. Females are displayed with dashed lines (n=1,250) and males are displayed with solid lines (n=1,250), the 5th, 50th, and 95th percentiles are displayed.

3.3.2 Associations between biomarkers and health variables

Analysis of associations between plasma biomarkers and health variables required consideration of plasma biomarker trajectories and distinct prevalences of various health conditions across the lifespan. We thus generated four even age bins: 3-19 years, 20-39 years, 40-59 years, and 60-79 years. Each biomarker and variable were assessed for outliers to be removed in each age bin using a Tukey test with a fence of 3 to ensure only extreme outliers were removed. Age, sex, ethnicity, fasting, and urine creatinine did not have outliers removed; however, BMI had <10 outliers removed from age 3-19, and serum creatinine had <10 outliers removed from age 60-79. All plasma biomarkers had ≤10 outliers removed from the younger age bins; NfL and p-tau-181 had an additional n=18 and n=11 outliers removed from the 60-

79 year age bin, respectively. A full description of outliers removed can be found in **Table 3.1**. The final sample sizes used in the analysis for each health variable can be found in **Appendix 1**.

Table 3.1 Outliers removed from association analysis. Outliers removed by age bin using a Tukey test with a fence of 3 for removing only extreme outliers. For bins where outliers $n < 10$, sample size has been rounded to comply with the Statistics Canada policies. Abbreviations: BMI, body mass index.

Variable	Outliers (n)			
	Age 3-19	Age 20-39	Age 40-59	Age 60-80
AB42/40	<10	<10	0	0
NfL	<10	10	<10	18
GFAP	<10	<10	<10	10
p-tau-181	<10	<10	10	11
BMI	<10	0	0	0
Creatinine	0	0	0	<10

Biomarkers were assessed for their association with health variables by linear regression analysis and reporting of a standardized beta and confidence interval (**Figure 3.5**), with detailed numeric results available in **Appendix 4**. With respect to demographic variables, all biomarkers were found to associate with age in multiple age bins, following the expected associations identified by the RIs. P-tau-181 was significantly associated with sex in all age bins from 20-79 years, with males having higher concentrations. GFAP was significantly higher in females from 40 to 79 years. Associations with ethnicity were determined by assessing ethnic minorities compared to those who identified as Caucasian, in order to have a large enough sample size for analysis. Being a non-Caucasian was associated with lower A β 42/40 in those aged 3 to 19, and higher NfL and lower p-tau-181 in ages 20 to 39 years. NfL and GFAP consistently showed a negative association with BMI in all age bins, while this association was only seen with A β 42/40 in ages 3 to 19 and 40 to 59 years, and with p-tau181 in ages 3 to 19 years. Kidney function was assessed by both blood and urine creatinine. A β 42/40 was negatively associated with both blood and urine creatinine from ages 3 to 19; however, it was negatively associated with urine creatinine in ages 40 to 59 but positively associated with blood levels in ages 20 to 39. P-tau-181 was negatively associated with both blood and urine creatinine in ages 3 to 19, but positively associated with blood creatinine in

ages 40 to 79. NfL was positively associated with blood creatinine in ages 20 to 39 and 60 to 79, but showed no association with urine levels. GFAP was negatively associated with both blood and urine creatinine levels in ages 3 to 39 and only in urine creatinine for ages 60 to 79. Finally, we assessed if fasting prior to blood sampling affected biomarker levels; we observed that fasting was associated with lower GFAP in ages 3 to 19 and higher NfL in ages 40 to 79 years.

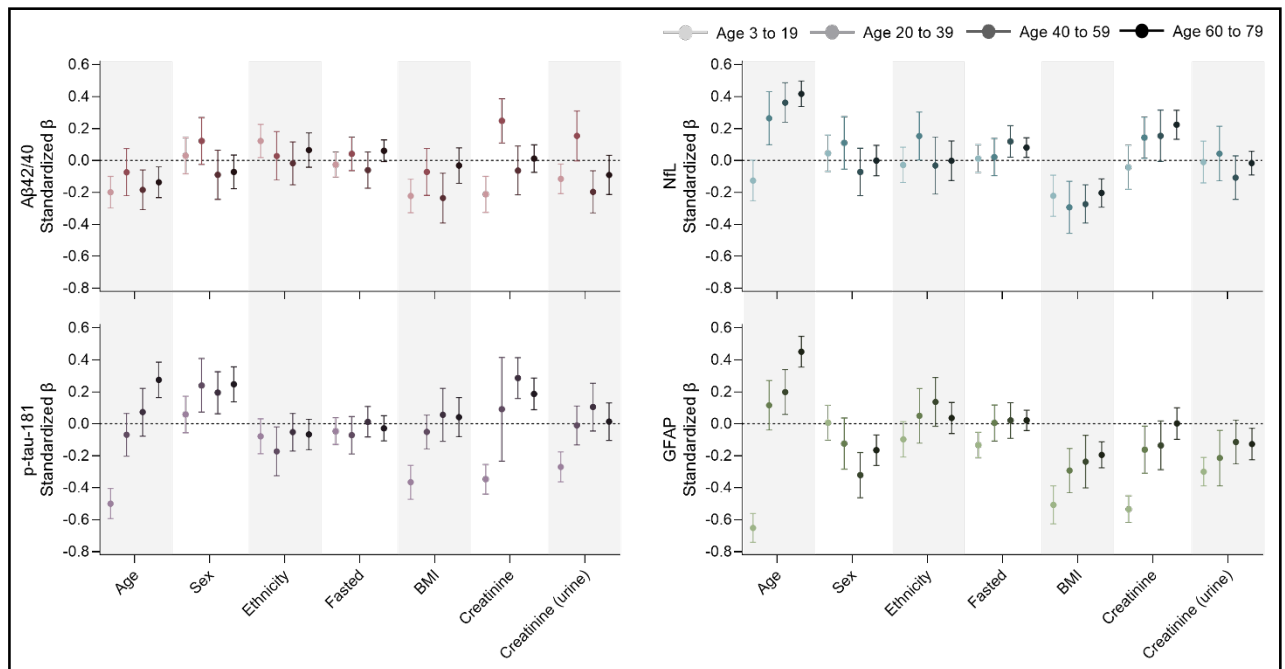


Figure 3.5 Association of plasma biomarkers with health survey variables. Associations were assessed by linear regression with the plasma biomarkers as the response variable. Displayed results show standardized beta and 95% CI. For binary variables the reference group was those who responded no to the survey question, with the exception of sex, where females were used as the reference group, and ethnicity, where white was the reference group. Abbreviations: BMI, body mass index.

3.4 Discussion

3.4.1 Age-related biomarker trends

This study extends our previous work in Chapter 2, which developed age-specific RIs for A β 42/40, p-tau-181, NfL, and GFAP in n=900 Canadians from age 3 to 79²⁸⁶. Here, we expanded the sample size to n=2,500, enabling the creation of reference curves to increase reporting of percentiles to display more granular biomarker distributions by age, and to assess the influence of demographic and health variables on biomarker concentrations. With this increased sample size, we replicated our previous findings of age as a variable that modifies biomarker concentrations across the lifespan. Plasma p-tau-181, NfL, and GFAP displayed U-shaped curves where concentrations decreased in childhood, stabilized in adulthood, and significantly increased with aging. A β 42/40 displayed a different trajectory, slowly decreasing over the lifespan. Not only were the patterns replicated, but the biomarker concentrations at the 5th, 50th, and 95th percentiles were highly reproducible, with the average percent difference for biomarker concentrations ranging from 4-13%. For A β 42/40 and NfL, concentrations for all percentiles were never greater than 15% different from the previously defined RIs. P-tau-181 and GFAP generally fell within this parameter as well, with the exceptions of slightly higher p-tau-181 values for all percentiles between the ages of 38 to 55 years, and a dip in GFAP's 95th percentile between the ages of 10 to 25 years.

We had previously shown that these age-related trajectories were aligned with independent cohort generated RIs for these biomarkers^{188,248,250,272-274,286}. However, since 2023, additional RI studies have been published for these biomarkers. To date, very little work has been done to further investigate age-specific RIs for plasma A β 42/40 and p-tau-181, with only one new study examining n=193 participants over the age of 50 years²⁹³. Although this study only reported discrete RI, their findings are consistent with ours, with an increase in the p-tau-181 95th percentile becoming more pronounced at older age, and a steady decrease of A β 42/40 levels with aging. NfL and GFAP RIs have been more thoroughly investigated, with multiple RI studies published since 2023²⁹⁴⁻²⁹⁷. For NfL, one study of n=223 healthy participants from age 20 to 60, displayed the same increase with age that we observed²⁹⁴. However, this study reported values of the 97th percentile that were consistently 3-4 pg/mL lower than our values, which is likely attributed to use of a different assay formulation. Another study examined NfL

and GFAP in n=116 controls aged 18 to 69 years where upper 95th or 97th percentiles were not reported, but similar median changes for both biomarkers were observed when compared to our cohort²⁹⁵. Two RI publications for GFAP utilized different platforms, which expectedly produce different absolute concentrations. However, both studies still displayed the same age-related trend we report, showing an almost exponential increase into the older ages^{296,297}. There is now substantial evidence to verify the age-related trajectories for NfL and GFAP, with considerable evidence backing similar absolute concentrations when using the same analytical platform.

3.4.2. Multiple utilities of reference curves

One of the enhanced features of the reference curves is the ability to report more granular details such as an increased number of percentiles across the lifespan. Here we report the 1st, 3rd, 5th, 15th, 25th, 50th, 75th, 85th, 95th, 97th, and 99th percentiles of the normative population, allowing for expanded utility beyond use as standard RIs. When utilizing age-specific RIs, a common practice is to use the 97th or 95th percentiles of the normative population as a cut-off to discriminate normal from abnormal values^{286,298}. In contrast, for a biomarker like A β 42/40 that decreases in AD, the 3rd or 5th percentile would be used as the cut-off.

Providing additional percentiles allows for more opportunities for future testing of other percentiles as cut-offs, recognizing that cut-off values will be context specific. For example, in AD, pathologic change appears well before symptom onset, appearing in cognitively normal people over the age of 50 and increasing in prevalence with age^{83,234}. Although our present study lacks measures such as PET or CSF biomarkers to confirm the presence of AD pathology in CHMS participants, our results raise the potential to determine if percentiles such as the 75th or 85th may serve as more sensitive cut-offs for use as a potential screening test. Testing this hypothesis will require additional validation against a reference standard to assess cut-off performance, which we investigate in Chapter 8.

The reference curves presented in this study also have many utilities beyond serving as normative cut-offs. For example, knowing a patient's or participant's baseline percentile can be

extremely valuable. In the context of screening for AD pathologic change, samples measured at regular intervals over time may indicate regular age-related trends if an individual's percentile remains stable. Conversely, it could be indicative of pathogenic progression if an individual's percentile changes over time. Another key advantage of reference curves is to serve as a proxy reference population in a research context. Here, researchers can compare how their population of interest compares to this reference population without the need for a large control group. One way this can be done is by comparing the observed distribution of the study population across the reference percentiles compared to the expected distribution of the normative population, as we have explored in Chapter 5. However, one of the most advantageous ways of comparing to this reference database is through the use of z-scores that can be generated from our reference curve models. To enable researchers to do this, we developed an application that allows researchers to input biomarker concentrations with participant ages to output their exact percentile and z-score (**Appendix 5**). Such an approach is expected to be very useful for acute neurological conditions such as traumatic brain injury or stroke, where pre-injury or baseline measures are not normally available.

3.4.3. Reproducibility and translatability of reference curves

One important consideration when producing reference curves or intervals is to ensure their translatability and generalizability. A key aspect of this is reproducibility, which we have achieved both through replication of our previous RI and showing strong alignment of results from other groups discussed previously and in section 3.4.1. This speaks to the robustness of our results, as other studies investigated different cohorts from different countries using different assays and analytical instruments. We have also previously shown that there is a good lot-to-lot translatability of the N4PE and p-tau-181 assays utilized in this study, through low bias in cross-lot analysis²⁸⁶. To mitigate any minor biases that do exist between lots we also incorporated two separate lots into the creation of these reference curves to capture any inherent analytical variability that may occur with these assays. To assess the translatability of these reference curves to other assay formulations, we conducted a separate study crossing CHMS plasma samples to other assay formulations. Here, we found that the bias between assays varied greatly, with some producing values less than 5% different, and others having a bias of 30-40% from our reference assays²⁹⁹. This finding illustrates that despite alternative

assays following consistent trends, there may still be a need to harmonize results when utilizing reference curves to compare results produced using alternate assays. Development and dissemination of reference materials is a critically important step to harmonize results across laboratories and studies. Another method to calibrate data against reference curves is a transference study in accordance with CLSI guidelines. When utilizing the same analyzer and assay type, CLSI EP28-A3c guidelines recommend validation of a reference population to accomplish transference²⁵⁹. This may be conducted on a small group of n=20, that represent the laboratories normative population, where it is expected that 90% of specimens fall within the reference limits. More robust validation can also be conducted with a larger reference population.

3.4.4. Additional considerations for biomarker interpretation

One of the variables that is often used to partition separate RIs is sex. Our analysis demonstrated that neither A β 42/40 nor NfL vary with sex in any of the age bins analyzed. However, significant associations with sex were found for p-tau-181 and GFAP, consistent with previous reports^{293,296}. As these sex specific associations were only found in specific age bins, we created both age and age and sex specific RIs for both biomarkers. Both p-tau-181 and GFAP displayed sex-specific associations only in adulthood, with p-tau-181 being significantly higher in males over 20 years and GFAP being significantly higher in females over 40 years. The reason our previous study did not detect sex differences²⁸⁶, is likely due to the smaller sample size. Sex-specific differences in p-tau-181 and GFAP levels were identified in a limited set of previous RI studies, although both prioritized the use of age-specific RIs alone^{293,296}. Thus, we produced distinct sex-specific reference curves for both p-tau-181 and GFAP, reporting the 5th, 50th, and 95th percentiles for each. We observed that the significant differences were highly driven by the 95th percentile, with both the 5th and 50th percentiles majorly overlapping between sexes. As the 95th percentile is the limit typically used as a cut-off, the decision on whether to use sex specific cut-offs would depend on a variety of factors. For one, they should only be used at ages where sex associations are significant. Also, they should be considered if disease prevalence or the biomarker–disease relationship differs by sex, such that sex-specific thresholds could improve diagnostic performance.

Another factor often considered in the creation of RI is ethnicity. Although we did not have sufficient sample size to investigate non-Caucasian ethnic groups, after dichotomizing to Caucasian vs non-Caucasian groups we observed mild associations with A β 42/40, p-tau-181, and NfL, only in one age bin each. Specifically, being non-Caucasian trended A β 42/40 under age 20 and p-tau-181 between 20 and 39 years in the direction of healthier biomarker levels, whereas NfL was elevated in non-Caucasians age 20 to 39 years. These observations underscore the importance of future studies that are powered to detect significant associations in specific ethnic groups, as Caucasians are overrepresented in many studies of plasma biomarkers used to derive cut-offs, which may limit the applicability to other diverse populations^{300,301}.

In addition to demographics utilized to create separate reference ranges, an additional consideration in laboratory medicine is the potential impact of fasting on blood tests. Here, we found that fasting status was not associated with A β 42/40 or p-tau-181 levels. However, we observed higher NfL levels in fasted individuals over the age of 40 years, and lower GFAP levels in fasted individuals under the age of 20 years. Previous literature has been limited, but a few studies have shown increases of NfL and GFAP after fasting^{302,303}, although these findings were dependent on the length of fasting, which may also influence our results. As these associations are mild, they are unlikely to alter biomarker results enough to warrant the need to fast. However, when comparing results across studies, fasting may be a factor of importance.

In addition to the regular measures assessed in laboratory medicine, we also sought to validate recent findings that show the modifying effects of BMI and kidney function on biomarkers^{188,244,247,290,291,304,305}. BMI showed a strong negative association with NfL and GFAP at all ages, with p-tau-181 under age 20 years, and with A β 42/40 in ages 3-20 and 40 to 59 years. This association with BMI and these biomarkers has been previously reported by multiple other investigators, and is believed to indicate a dilution effect, where higher blood volume in those with higher BMI dilutes biomarker concentrations^{188,244,290}. One study previously created NfL RI with different cut-offs based on BMI for the prognosis of MS¹⁸⁸. However, it has not been assessed if BMI alters the utility of these biomarkers for use in other

contexts and disease states. Kidney function was assessed both through serum and urine creatinine, both of which showed similar trends with biomarker concentrations. However, stronger associations were often observed with serum creatinine, indicating neurology biomarkers are more likely to be associated with long-term kidney health and function rather than short-term filtration. We found that serum creatinine was negatively associated with A β 42/40, p-tau-181 and GFAP in those under 20 years. That these associations were strongly replicated by urine creatinine suggest that they may be more indicative of normative levels, rather than a relationship to pediatric kidney function. While we have shown that neurological biomarkers are often higher in pediatric than adult plasma specimens, creatinine shows the opposite trajectory with aging³⁰⁶ being low in childhood and increasing in adults. In older age groups, we observed a reverse association, with higher levels of neurological biomarkers associated with markers of worse kidney function. This is an established association seen across multiple studies^{247,291,304,305}, and it is thought to be due to reduced clearance of biomarkers due to impaired kidney function, or the direct impact of renal failure on the nervous system^{291,307}. Chronic kidney disease can lead to inflammation and vascular damage, which may impact the blood-brain barrier, resulting in a greater release of these biomarkers into circulation^{308,309}. However, it is important to note that the effect of kidney function has not been shown to have an impact on the diagnostic utility of these biomarkers^{307,310,311}.

3.4.5. Limitations

One limitation of this study is its reliance on a population-based cohort, reflecting a normative cohort rather than a uniformly healthy one. Although no specific exemptions were made against those with cognitive impairment, fewer than 30 (1%) CHMS participants reported any memory or cognitive complaints in their survey responses. However, undiagnosed neurological conditions or pre-clinical neuropathology among participants may contribute to wider variation between the median and 95th percentile for p-tau-181, NfL, and GFAP, particularly in participants over 60 years. However, utilizing the full spectrum of the normative population allows us to propose cut-offs at various percentiles of the population rather than just outer limits, although these will require validation against reference standards for specific contexts of use. We also acknowledge that the preliminary investigation of associations between plasma biomarkers and other CHMS health variables is highly exploratory. Our aim is to report broad

associations in a large sample size across the lifespan in order to inform on what variables could be investigated further.

3.5. Conclusions

We generated age-specific reference curves that describe the 1st, 3rd, 5th, 15th, 25th, 50th, 75th, 85th, 95, 97th, and 99th percentiles of plasma A β 42/40, p-tau-181, NfL, and GFAP in a normative Canadian population from age 3-79 years. These reference curves have utility as age-specific RIs and as an age-matched reference database. Additionally, we describe the importance of interpreting biomarkers alongside important demographic and health variables, such as sex, BMI, and kidney function.

Chapter 4. Investigating Reproducibility of Plasma Biomarker Results Through Cross Lot Analysis of Simoa Assays

4.1 Introduction

Throughout this thesis, analysis is performed consistently using the same platform, the Simoa HD-X, and same assay kits, Neurology-4-plex E and ptau-181 V2 advantage assays. However, different cohorts were assessed using different lots of these assays. The ability to compare key assay characteristics across these lots is essential for making meaningful comparisons across studies. Simoa principles are based on using capture beads to bind analytes in fluid matrices such as plasma and isolating immunocomplexes in femtoliter-sized array wells. The array well volume accommodates only a single bead-analyte complex and provides a close-quarters environment for the assay enzymes to react with substrate that enables highly sensitive detection. If successfully bound to the bead-analyte complex, the detector antibody generates a unique fluorescent signal depending on the analyte detected³¹². The Quanterix platform has demonstrated robust analytical performance, characterized by high reproducibility, low intra- and inter-assay variability, and excellent sensitivity across repeated runs and operators. Despite this platform demonstrating robust analytical performance, characterized by high reproducibility, low intra- and inter-lot variability, and excellent sensitivity across repeated runs, it is still a research-use-only platform. Therefore, there are no tight regulations around exactly how comparable results are across lots. To ensure the robustness of all results reported in this thesis, we have characterized each lot utilized based on a set of quality control metrics. Importantly, we have performed a set of cross-lot analyses to demonstrate the validity of comparing results across cohorts that have been analyzed using different assay lots.

4.2 Methods

4.2.1 Biomarker analysis and quality control

All cohorts included in this thesis had samples analyzed using the Simoa HD-X platform from Quanterix. Plasma GFAP, NfL, A β 40, and A β 42 were measured using the Neurology-4-plex E advantage assay (catalogue #103670), and plasma p-tau181 was measured using the ptau-181 V2 advantage assay (catalogue #103714) following the manufacturer's protocol. Each assay included an 8-point calibrator curve, two internal kit controls, and three plasma controls. All

specimens were assayed in duplicate, and the mean value was reported as the result. For all analytes, quality control information was documented across all runs on lots. The average % error is calculated for calibrators and kit provided controls, and calculated by taking the absolute value of the $((\text{measured value} - \text{known value}) / \text{known value})$ multiplied by 100%. This is averaged across calibrators and controls within a run, and then the average values of all runs are averaged to report the average % error per lot. Average intra-run % CVs were calculated for calibrator average enzyme bound (AEB) values, kit and plasma control concentrations, and plasma sample concentrations using the replicate measurements performed by the instrument. This is calculated by dividing the standard deviation of the sample replicates by the average AEB or concentration and multiplying by 100%. This is averaged across calibrators, controls, and samples within a run, and then the average values of all runs are averaged to report the average % CV per lot. Average inter-run % CVs are calculated for kit and plasma controls analyzed across multiple runs

4.2.2 Cross lot analysis

Each time a new lot was utilized, a cross-lot analysis was performed. When starting with the new lot, samples are selected that were run on the previously used lots to perform a cross-lot analysis. For each cross, n=40 samples were selected to re-analyze, in accordance with CLSI guidelines³¹³. The n=40 samples were selected evenly across the dynamic range of previously run analytes, ensuring there was good representation of both the low and high ends of the dynamic range of the cohort. For N4PE, GFAP was utilized as the primary marker to base the dynamic range selection off of. To assess associations between assay results, Spearman correlation was performed to report rho (ρ) and simple linear regression was used to determine the line of best fit for correlation plots. To assess the difference between assays, Bland-Altman analysis was performed to report the average bias, standard deviation, and the 95% limits of agreement across assays.

4.2.4 Reference sample analysis

At the initiation of this study, a set of spiked plasma controls were created to provide a low, medium, and high concentration controls that could be tracked across all lots. This was done

by first creating a pooled concentrate using the top 2 calibrators and the highest kit control of N4PE lot 503105 and p-tau-181 lot 502923. The low control was un-spiked plasma, the medium control was spiked with enough concentrate to aim for values to fall between calibrator E and F, and the high control was spiked with enough concentrate to aim for values between calibrator F and G. The spiked plasma controls were analyzed across all 42 runs of the first lot set utilized in this thesis. The averaged concentrations and standard deviations across runs were determined for each analyte for each control. This was used to determine the accepted range of the control values by adding or subtracting the average concentration by two times the standard deviation to determine the upper and lower limits, respectively. Following this, when a new lot set was introduced, the spiked controls were analyzed on 5 separate runs of the new lot to ensure the controls still fell within the expected range. We considered 80% of measures per analyte falling in the expected limits per lot to be acceptable performance to consider the lots comparable.

4.3 Results

4.3.1 Quality control

In this thesis, three separate lots of N4PE and p-tau-181 Simoa assays were utilized. These were grouped into sets based on the cohorts that were analyzed on each lot. Lot Set A used N4PE lot 503105 and p-tau-181 lot 502923 to analyze the CHMS and Super Seniors cohorts. Lot Set B used N4PE lot 503212 and p-tau-181 lot 503199 to analyze the COMPASS-ND cohort. Lot Set C used N4PE lot 503811 and p-tau-181 lot 503545 to analyze the CHMS and CARD cohorts. For all assay runs, quality control measures were documented to ensure quality of results. All quality control information was compiled by lot to provide the average % error, and intra and inter run CVs for all lots used (**Table 4.1**). Across all lots, the average % error for calibrators never exceeded more than 7.8%, and for kit controls it never exceeded 13.4%. The average intra-run CVs across all measures never exceeded 7.6%. the inter-run CVs for kit and plasma controls never exceeded 14.4%, with the exception of analytes on N4PE lot 503811, averaging 23%.

Table 4.1 Quality control information from all lots of N4PE and p-tau-181 assays per analyte. All quality control metrics have been averaged across all runs per lot.

	Calibrators		Kit Controls			Plasma Controls		Samples
Analyte	Error (%)	Intra-run CV (%)	Error (%)	Intra-run CV (%)	Inter-run CV (%)	Intra-run CV (%)	Inter-run CV (%)	Intra-run CV (%)
Lot Set (A) - N4PE lot 503105; p-tau-181 lot 502923; Cohorts: CHMS, Super Seniors								
A β 40	4.6	7.6	8.8	2.6	5.5	4.8	8.6	2.0
A β 42	2.7	5.2	10.2	2.4	3.0	4.2	9.4	2.4
NfL	5.7	5.7	11.8	4.1	13.9	6.0	14.4	6.6
GFAP	4.9	6.3	6.8	4.8	5.5	6.7	8.4	6.2
p-tau-181	5.5	4.8	7.9	4.2	10.8	5.3	8.1	7.0
Lot Set (B) - N4PE lot 503212; p-tau-181 lot 503199; Cohorts: COMPASS-ND								
A β 40	6.5	7.3	13.2	4.3	8.3	5.9	7.6	3.1
A β 42	3.3	5.0	7.8	2.1	7.3	5.7	6.5	2.7
NfL	7.0	4.4	13.4	3.7	6.9	4.1	8.8	3.6
GFAP	3.6	5.4	4.8	3.6	7.5	5.5	10.2	4.2
p-tau-181	6.4	4.9	7.2	3.7	8.6	5.0	11.3	6.2
Lot Set (C) - N4PE lot 503811; p-tau-181 lot 503545; Cohorts: CARD, CHMS								
A β 40	2.7	5.4	6.1	2.7	22.6	2.6	4.8	1.8
A β 42	2.5	4.8	8.4	1.8	22.5	3.4	4.6	2.7
NfL	7.8	4.5	11.1	5.3	23.8	5.3	6.0	5.4
GFAP	5.6	5.4	7.0	4.8	23.0	4.8	5.8	5.5
p-tau-181	4.8	4.7	10.5	4.9	7.5	4.8	7.2	5.4

4.3.2 Cross lot analysis

For each new lot set used a cross-lot analysis was performed by analyzing n=40 samples that were run on a previous lot set. Samples from lot set A were crossed onto both lot set B and C, and samples from lot set B were crossed onto lot set C. All analytes showed excellent correlations across lots with a rho=0.9 or greater, and linear regression analysis again showed quite tight associations between the concentration produced across lots, with the majority of β values between 0.8 and 1.2 (**Table 4.2**). Bland-Altman analysis was used to indicate how

close the raw concentration values were (**Table 4.2, Figure 4.1**). Across N4PE lot crosses, the average bias between lots was only 9.1%, and other than NfL crossing lot set A to C all biases were under 14%. Limits of agreement were very tight for A β 40 and 42, but spanned a much larger range for NfL and GFAP, indicating a less consistent bias for these analytes across samples. For A β 40 and 42, lot set A produced the highest concentrations, followed by B, and then C, with the lowest concentrations. Interestingly, NfL and GFAP followed a different trend, with lot set C producing the highest concentrations, followed by A, and then B with the lowest concentrations. For p-tau-181, only lot set B to C gave a low bias of 13.5%, while lot set A to B and C had biases over 20%. The limits of agreement were also quite tight. For p-tau-181 lot set C produced the highest concentrations, followed by B, and then A with the lowest.

Table 4.2 Cross-lot analysis of N4PE and p-tau-181 assays. N=40 specimens used for each cross were selected from a previously analyzed lot. Rho was calculated from Spearman correlation, β was calculated from simple linear regression, and bias and limits of agreement (LoA) were generated from Bland-Altman analysis.

N4PE Advantage									
	Crossed Lot	503212 (B)				503811 (C)			
Original lot	Analyte	rho	β	Bias	LoA	rho	β	Bias	LoA
503105 (A)	Aβ40	0.957	0.690	9.8	-15.5-35.1	0.922	0.493	4.4	-24.5-33.3
	Aβ42	0.938	0.896	3.7	-19.7-27.0	0.980	0.894	7.4	-3.2-18.1
	NfL	0.903	3.276	10.95	-52.5-74.4	0.985	0.792	-21.8	-61.0-17.4
	GFAP	0.962	1.569	9.4	-38.1-57.0	0.975	0.987	-5.3	-41.3-30.8
503212 (B)	Aβ40					0.900	1.250	0.1	-33.7-33.9
	Aβ42					0.960	0.881	11.8	-15.3-39.0
	NfL					0.959	1.053	-13.7	-64.1-36.7
	GFAP					0.951	0.410	-11.2	-62.5-40.2
p-tau-181 Advantage V2.0									
	Crossed Lot	503199 (B)				503545 (C)			
Original lot	Analyte	rho	β	Bias	LoA	rho	β	Bias	LoA
502923 (A)	p-tau-181	0.970	1.610	-22.1	-56.5-12.4	0.976	1.206	-21.1	-42.0-0.1
503199 (B)	p-tau-181					0.967	1.017	-13.5	-36.5-9.5

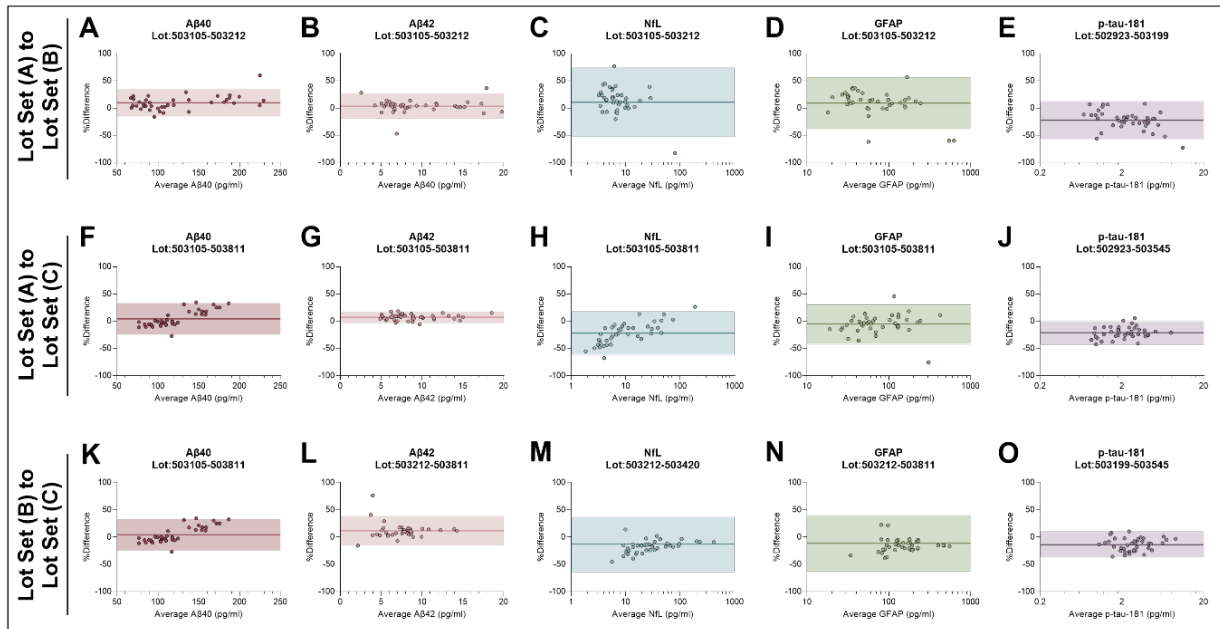


Figure 4.1 Bland-Altman's of cross lot analysis. Each cross consisted of n=40 samples selected from the previously lot to re-analyze on the new lot. Biases are reported in relation to the original lot analyzed.

4.2.3 Analysis of reference samples

A set of controls were generated when using lot set A to serve as low, high and medium controls. These were created using health plasma and spiking a pooled concentrate of calibrator and kit controls into the medium and high controls. These controls were analyzed across all runs of lot set A to determine an expected concentration range, displayed as the shaded regions in figure 6.2. For Aβ40, the expected values of all 3 controls greatly overlapped, and the low and medium controls for Aβ42 also overlapped significantly. For NFL, the low and medium controls mildly overlapped, but none of the expected ranges for GFAP or p-tau-181 overlapped. In lot sets B and C, these controls were analyzed on 5 different runs to determine if they produced concentrations in the expected ranges (**Figure 4.2**). For lot set B, n=2 Aβ40 and n=1 GFAP measurements were outside the expected range. Aβ42 had a low control under the limit and a high control over the limit, both on separate runs. The GFAP control was a high control, being under expected limits. For lot set B, all analytes had between 87 to 100% in the expected ranges. For lot set C, n=3 Aβ42 and n=3 p-tau 181 measures were outside the expected range.

A β 42 had 1 low and 2 high controls that were all under the lower limits, while p-tau-181 had 2 medium controls and 1 high control that were all above the upper limits. These analytes still maintained 80% of measures in the expected ranges, and A β 40, NfL and GFAP all had 100% of measures in the expected ranges.

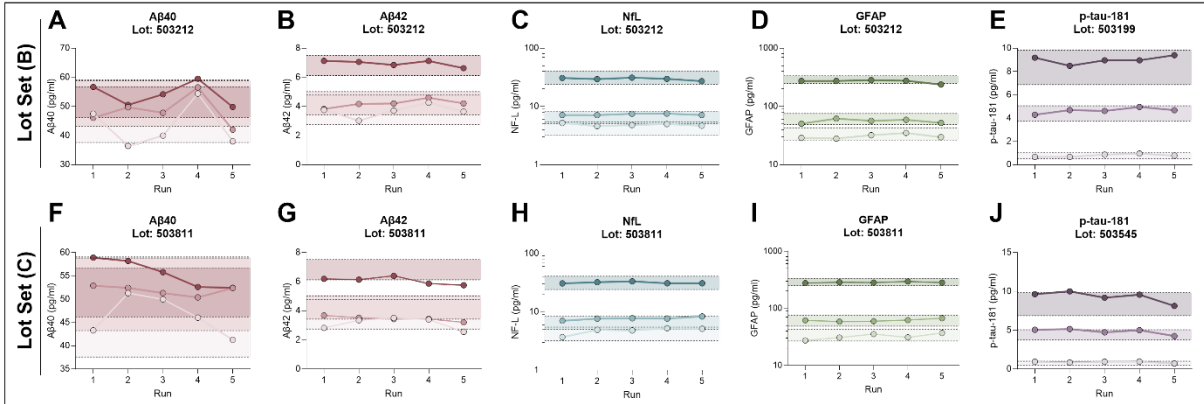


Figure 4.2 Tracking of plasma control samples onto new lots. Expected concentrations of each control were determined from runs on lot set A and indicated by the shaded regions with dashed lines. Controls were tracked across 5 runs with the lighted shade being the low control and the darkest shade being the high control.

4.4 Discussion

4.4.1 Robustness of Simoa Assays

In this study, we aimed to prove the robustness and translatability of Simoa-generated biomarker concentrations to ensure the validity of results produced across studies in this thesis. We first aimed to prove the robustness across a variety of quality control measures over the three different lots of N4PE and p-tau-181 utilized. We found that across all 3 lots, the percent error for known values of kit provided calibrators and controls was very low, with calibrators never having an average error over 8% and controls never having an average error over 14%. Acceptable error can vary across analytes in clinical chemistry, and can be calculated in multiple ways. One way is to assess the biological variation based on within-subject and between-subject variation³¹⁴. Previous literature has established the expected within-individual biological variation, which ranges across analytes³¹⁵⁻³¹⁸, with studies reporting consistent numbers ranging from ~3-10% variability depending on the analyte. These studies also

investigated expected variability between subjects, however these results varied. A β 42/40 had the lowest variability around 7-15%, followed by p-tau-181 with around 20-25%, NfL ranged from 20-35% and GFAP varied greatly from 20-40%. In Chapters 1²⁸⁶ and 2, we previously established the dynamic nature of biological variability between subjects and its increase with aging, which may explain some of these varied results based on the different populations investigated. Rough calculations for the average allowable total error³¹⁴ of these analytes has the optimal error under 10% and desirable error under 30%, which our metrics fall well within. These values are also similar to other acceptable performance guidelines outlined by the Clinical Laboratory Improvement Amendments (CLIA) performance criteria^{314,319}, backing the accuracy of our results.

The other major quality control metric we assessed was the analytical variability of sample replicates (intra-run CV), and running controls across multiple assay runs (inter-run CV). This was done by assessing the kit provided controls, internally collected plasma controls, and the plasma samples from all projects. Average intra-run CVs for all controls and samples never exceeded 7% across all lots, indicating excellent analytical performance for sample replicate measurements. Average Inter-run CVs were only calculated for kit and plasma controls where multiple aliquots were utilized across multiple assay runs. Here, the plasma controls ranged between average CVs of 4-15%, and for two of three lots, the kit controls ranged between average CVs of 3-14%. However, for one lot of the N4PE assay, the average CVs were approximately 23%. Despite these outlier values, the controls from this lot performed well on the other metrics, having consistently low bias and low CVs from sample replicates. Overall, we can conclude that the assays utilized in this thesis produced highly robust results due to their high performing quality control metrics.

4.4.2 Translatability of results across assay lots

In addition to the general robustness of these assays, we aimed to ensure results would be translatable across the three different sets of lots utilized in the different cohorts analyzed. To do this, a cross-lot analysis was performed based on CLSI guidelines for internal method comparison studies³¹³. N=40 samples were selected from the dynamic range of those analyzed on a previous lot to be re-analyzed on the new lot of reagents. Three sets of lots were used and

labelled set A, B, and C. Samples analyzed originally on lot A were crossed to lots B and C and samples from lot B were crossed to lot C. When assessing the translatability of results, we first assessed that results were tightly correlated between lots, with rho values >0.9 for all analytes, and regression coefficients close to 1. This indicates that results can be compared across lots, and that corrections can be performed between the lots if the raw concentrations differ. To assess this, we next looked at the bias and limits of agreement between lots. We found generally low absolute biases between lots, with the majority of crosses having a bias between 0.1 to 14%. In general, A β 40 and 42 had the lowest bias between lots, and p-tau-181 had the highest bias. We deemed a bias under 15% was low enough to not require any harmonization of the data between lots. This number is based on the acceptable bias based on the biological variability of the analyte, similar to the calculations performed for acceptable error³¹⁴. Three crosses did not meet this threshold, NfL lot set A to B and p-tau-181 lot set A to B and A to C. Fortunately, due to how lot A was utilized, harmonization was not required. Lot A was primarily used to create the Reference Intervals generated in Chapter 1. However, in Chapter 2, we introduced updated reference curves that incorporate samples from lots A and C, therefore incorporating this natural bias between lots into the regression models used to create the reference curves. Any comparisons to the reference intervals/curves conducted in the following chapters will utilize the reference intervals with combined lots. This allows us to set a secondary requirement for harmonization, where results did not need to be harmonized unless the bias between the comparison lot and both reference interval lots was over 15%. As not of the analytes crossed this threshold, no data manipulation was performed to compare between cohort results.

In assessing the differences across lots, we also assessed consistency. We found that the trend in biases told a cohesive story of ranking lots from the highest to lowest concentrations reported. However, the ranking was not the same across analytes. A β 40 and 42 both had the same ranking, with lot A producing the highest results and lot C producing the lowest. NfL and GFAP also tracked together but with lot C producing the highest results and lot B producing the lowest. P-tau-181 had the highest results produced by lot C and the lowest by lot A. It was interesting that within the N4PE assay, the different analytes had different relationships across lots. This indicates the importance of understanding how different analytes may perform within

a single assay. Another measure we used to assess consistency across lots was utilizing spiked control samples that were measured across all 3 lots. These samples were first analyzed on lot A to determine the acceptable range of concentrations for each sample. These samples were then analyzed on 5 different runs from lots B and C. For each analyte cross, control values fell within expected ranges 80 to 100% of the time. This further indicates the validity of comparing results across these multiple lots.

4.4.3 Limitations

This study was intended to present the robustness and validity of our biomarker analysis across multiple cohorts and assay lots. However, there are some limitations in the wider applicability of these results. First, these findings are only applicable to the specific assays analyzed, and the results may not be representative of other assay formulations that measure the same analytes. Second, as the initial lot of reagents was used to analyze our normative population, the dynamic range of the samples does not cover the full dynamic range of the assays. Therefore, we cannot speak to the translatability of results in the higher concentration ranges, or whether the biases reported here are consistent or concentration dependent. Though we were able to represent concentrations of a standard dementia population in our crosses between lot sets B and C, we do not reach the upper limits of the assay ranges that can be observed in populations with other neuropathologies, such as traumatic brain or spinal cord injuries.

4.5 Conclusions

In this chapter we demonstrated that all lots utilized in this thesis performed well with low error and CVs, a minimal bias was observed between lots, and controls fell within the ranges expected from previous lots. This indicates the validity of comparing plasma biomarker measurements across the multiple cohorts that were analyzed on different assay lots.

Chapter 5. Cognitively Healthy Seniors Have Resistant Alzheimer's Disease Plasma Biomarker Profiles

5.1 Introduction

Maintenance of good cognitive health into older ages is thought to reflect cognitive resilience or resistance. Cognitive resilience refers to a person not developing cognitive impairment in the presence of underlying pathology, such as in Alzheimer's disease (AD) and related dementias (ADRD), whereas resistance refers to not developing the underlying pathology in the first place. Many studies have shown that some degree of neurodegenerative processes occur during normal aging⁸⁸⁻⁹⁰, and previous research has demonstrated that up to 40% of cognitively intact seniors display some AD neuropathology, specifically, amyloid plaques and neurofibrillary tangles⁸³. Although this observation suggests there are some individuals who may be more resilient to functional decline as a result of AD pathological changes, other studies have shown that those with pathological changes at older ages are still more likely to convert to clinical dementia than those without^{83,91}. These findings underscore the importance of developing approaches to characterize the pathological profiles of individuals who remain cognitively healthy into the oldest ages, as doing so may reveal key mechanisms that protect against cognitive decline.

Plasma biomarkers are easily accessible tools to assess the presence of AD neuropathology. Plasma A β 42/40 and p-tau-181 are both considered markers specific to AD pathology, reflecting the formation of amyloid plaques and neurofibrillary tangles, respectively^{144,162,254,255}. P-tau-181 is also a robust marker of amyloid positivity and is believed to represent an early change in tau upon amyloid deposition^{162,320}. In contrast, plasma NfL and GFAP are general markers of neurodegenerative processes that can result from AD or other causalities. NfL is associated with axonal damage and is used as a non-specific marker of neurodegeneration^{193,256}, and increased GFAP produced in activated astrocytes is used as a general marker of neuroinflammation. Together, these four biomarkers are amongst the most well-studied plasma biomarkers in both clinical and in research contexts¹⁰⁸.

This study was designed to test whether plasma levels of A β 42/40, p-tau-181, NfL and GFAP in cognitively healthy seniors reflect representative population norms. Specifically, we aimed to determine whether these markers display standard age-related trajectories or deviate from a normative population's distribution. Here, we utilized previously established reference curves from n=2,500 plasma samples from the Canadian Health Measures Survey (CHMS; Chapter 3), as a novel approach to test whether cognitively resilient seniors follow the expected distributions previously observed with normative aging. In normative aging, both the median plasma concentration and concentration range of these biomarkers change with age. Therefore, we hypothesize that plasma biomarker concentrations in cognitively healthy Super Seniors would be skewed to more favourable profiles compared to the normative population.

5.2 Methods

5.2.1 Ethical approval

This study was reviewed and approved by the University of British Columbia (UBC) Clinical Research Ethics Board (REB), Vancouver, Canada (H21-01783). The Super-Seniors Study, which provided banked plasma specimens for this study, was reviewed and approved by the joint Clinical REB of BC Cancer and UBC, with harmonized review by the REB of Simon Fraser University (H20-03117).

5.2.2 Participant selection

Specimens for this secondary analysis study were acquired from phase 1 (collected 2004–2007) of the Super Seniors Study, which recruited healthy individuals age ≥ 80 ³²¹, who self-reportedly had never been diagnosed with cancer, CVD, diabetes, dementia, or major pulmonary disease. Health data and prescribed medications of enrolled participants were reviewed and participants were excluded if they had any of the above diseases or were prescribed medications to treat one or more of these conditions. Demographic data included age, biological sex, height, and weight. Functional tests included the Mini-Mental State Examination (MMSE), Instrumental Activities of Daily Living Scale (IADL), Timed Up and Go test (TUG), and Geriatric Depression Scale (GDS). Non-fasting blood samples were

collected in EDTA vacutainers and centrifuged at 4°C for 15 minutes at 2500 g. Plasma was separated and stored in a –80°C freezer within a maximum of 24 hours from collection.

5.2.1 Biomarker analysis

Biomarker concentrations were quantified using the single-molecule array enzyme linked immunoassay (Simoa) HD-X platform from Quanterix following the manufacturer's protocol. Plasma GFAP, NfL, A β 40, and A β 42 were measured using the Neurology-4-plex E advantage assay (catalogue #103670, lot #503105), and plasma p-tau-181 was measured using the p-tau-181 V2 advantage assay (catalogue #103714, lot #502923). As the same assay lots were used to generate reference curves, no data harmonization was required. Each assay included an 8-point calibrator curve, two internal kit controls, and three plasma controls. All specimens were assayed in duplicate and the mean value was reported as the result. For all analytes, quality control information was documented across all runs. Standard curves had an average absolute error of 2.2% to 6.8% and an average recovery of 94% to 106%. Internal kit controls had an average absolute error of 8.0% to 12.0% and an average recovery of 106% to 110%. For plasma controls analyzed in duplicate across every plate, mean values had an average inter-plate coefficient of variation (CV) between 3.7% and 18%. Average intraplate CVs were between 2.3% and 4.3%.

5.2.3 Reference Curves

Reference curves were previously generated in Chapter 2 to encapsulate the normative range of plasma biomarkers from age 3 to 79 years old using $n=2,500$ CHMS samples²⁶². Here, we report the 1st, 3rd, 5th, 15th, 25th, 50th, 75th, 85th, 95, 97th, and 99th percentiles of the population by performing a GAMLSS regression of biomarker concentration across the lifespan. As this study aimed to compare a group of seniors over the age of 80 to the normative population, we first extended the trajectory of the reference curves through interpolation. A simple linear regression was performed using the concentrations that corresponded to each percentile at each age from 70 to 79. For each percentile, this regression formula was used to interpolate an extension of the curves to report the expected concentration percentiles at each age from 80 to 105 years old. Following extension of the reference intervals, plasma biomarker results from

the Super-Seniors were compared to the extended curves. Each participant had their biomarker results coded based on the percentile bin they fell into.

5.2.3 Statistical analysis

Biomarker concentrations and study data were determined to be non-parametric by the Shapiro–Wilk test, therefore all group data are reported as the median with interquartile range. We compared the observed proportion of Super Seniors to the distribution of participants expected to fall in each of the reference curve percentile bins using Mee’s proportion differences, where associated 95% confidence intervals (CI) were calculated and converted into a percent difference by dividing the proportional difference by the expected proportion and multiplying by 100³²². Super Seniors data were then dichotomized based on the quartile bins their biomarkers were determined to be, and deemed to be “clinically” or “biomarker” super. Group comparisons between *APOE4* carriers and non-carriers for demographics, study measures, and raw biomarker concentrations were performed using a Mann-Whitney U test for continuous variables or a Fisher's exact test for categorical variables. All statistical analyses were performed using RStudio with R version 4.4.2 and GraphPad Prism version 10.4.1.

5.3 Results

5.3.1 Cohort description

A total n=480 Super Seniors were included in this study. The cohort had a median age of 88 (87-90) years old, with n=325 (68%) of the cohort being female, and a median BMI of 24 (22-27). In terms of study measures collected, the group had a median MMSE score of 29 (28-30), IADL of 23 (21-23), TUG of 11 (10-14), and GDS of 1 (0-2). The median plasma biomarkers for the full cohort were: A β 42/40; 0.0616 (0.0533-0.0692) pg/mL, p-tau-181; 2.80 (2.05-3.81) pg/mL, NfL; 30.9 (25.1-41.0) pg/mL, and GFAP; 175.8 (137.1-221.7) pg/mL. Of the n=369 Super Seniors genotyped for *APOE*, n=80 (22%) were *APOE4* carriers.

5.3.2 Comparison to population reference intervals

Super Seniors were compared to population-based normative levels utilizing reference curves that describe the 1st, 3rd, 5th, 15th, 25th, 50th, 75th, 85th, 95, 97th, and 99th percentiles of the

Canadian population by age. **Figure 5.1** displays Super-Seniors biomarker data overlayed on top of the interpolated reference curves, where it is visually evident that A β 42/40 concentrations are skewed to the upper range of normal and p-tau-181 concentrations are skewed to the lower range of normal, whereas their NfL and GFAP concentrations are largely similar to those of a normative population. These distributions were confirmed upon coding of Super Seniors biomarker concentrations based on the reference curve percentile bins they fell into (**Table 5.1**).

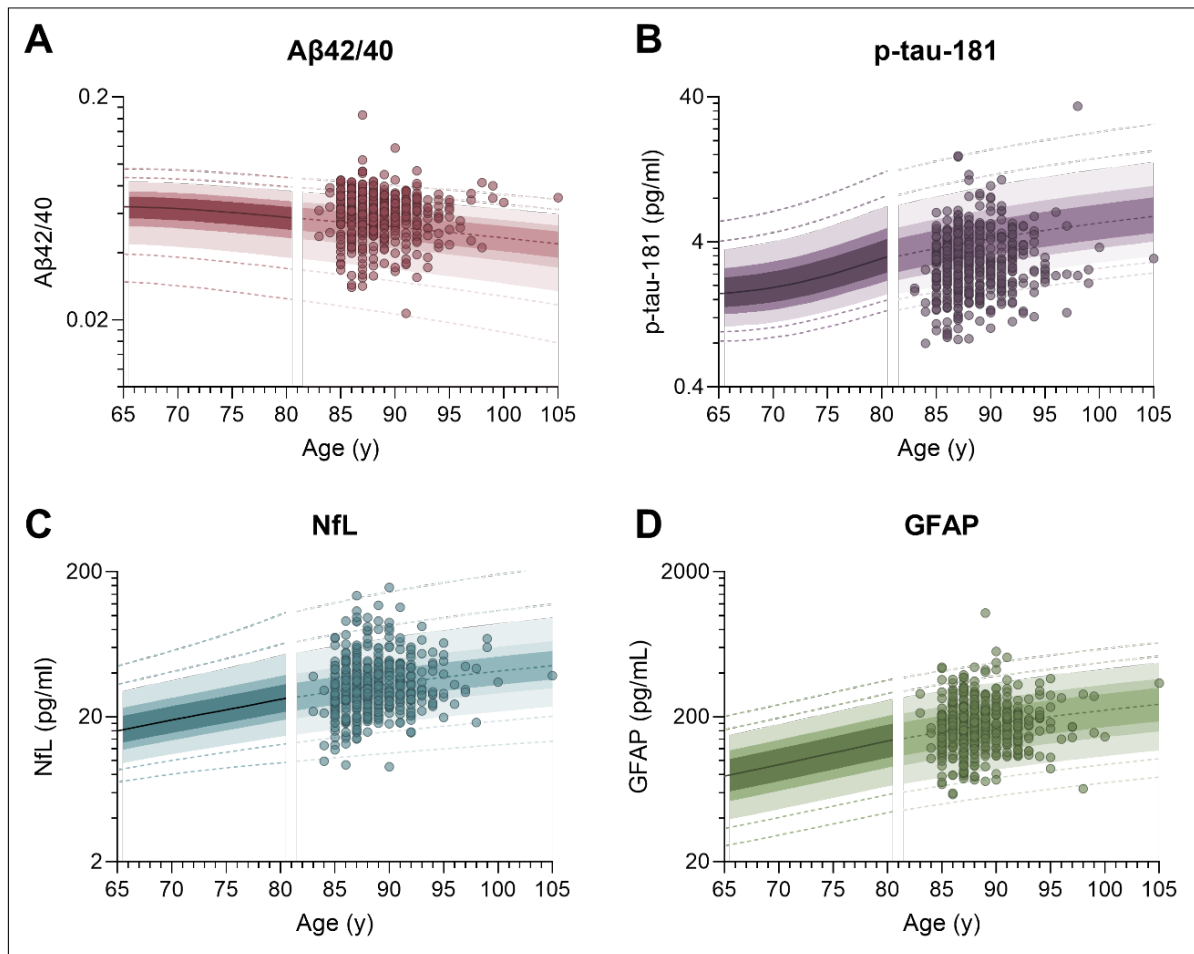


Figure 5.1 Plasma biomarker concentrations of Super-Seniors compared to population-based reference curves. Reference curves were generated using n=2,500 plasma samples from the Canadian Health Measures Survey participants aged 3-79. Reference curves were extended to age 105 using linear regression interpolation. Percentiles displayed include the 1st, 3rd, 5th, 15th, 25th, 50th, 75th, 85th, 95, 97th, and 99th percentiles. N=480 Super Seniors are indicated by shaded dots.

Table 5.1 Proportion of Super Seniors in each reference curve percentile bin of plasma biomarkers.

Percentile Bins	<i>Expected</i>	Aβ42/40	p-tau-181	NfL	GFAP
Total	480 (100%)	480 (100%)	480 (100%)	480 (100%)	480 (100%)
Below 1st	4.8 (1%)	0 (0.0%)	65 (13.5%)	3 (0.6%)	1 (0.2%)
Between 1st and 3rd	9.6 (2%)	5 (1.0%)	59 (12.3%)	12 (2.5%)	4 (0.8%)
Between 3rd and 5th	9.6 (2%)	8 (1.7%)	22 (4.6%)	12 (2.5%)	12 (2.5%)
Between 5th and 15th	48 (10%)	29 (6.0%)	92 (19.2%)	66 (13.8%)	44 (9.2%)
Between 15th and 25th	48 (10%)	17 (3.5%)	46 (9.6%)	60 (12.5%)	56 (11.7%)
Between 25th and 50th	120 (25%)	58 (12.1%)	92 (19.2%)	116 (24.2%)	124 (25.8%)
Between 50th and 75th	120 (25%)	79 (16.5%)	71 (14.8%)	90 (18.8%)	122 (25.4%)
Between 75th and 85th	48 (10%)	68 (14.2%)	15 (3.1%)	51 (10.6%)	49 (10.2%)
Between 85th and 95th	48 (10%)	108 (22.5%)	12 (2.5%)	45 (9.4%)	47 (9.8%)
Between 95th to 97th	9.6 (2%)	40 (8.3%)	2 (0.4%)	12 (2.5%)	6 (1.2%)
Between 97th to 99th	9.6 (2%)	48 (10.0%)	3 (0.6%)	11 (2.3%)	10 (2.1%)
Above 99th	4.8 (1%)	20 (4.2%)	1 (0.2%)	2 (0.4%)	5 (1.0%)

We then compared the observed proportion of Super Seniors biomarker data in each percentile bin to the expected proportions that would exist in a normative population using the interpolated reference curves (**Figure 5.2**). This was calculated as the Mee's proportional difference, with 95% CI, and reported by converting to percent difference. We assessed this for all reported percentile bins, as well as for the summary of the top and bottom quartiles of the reference curves. A β 42/40 and p-tau-181 distributions showed the biggest deviation from the normative population. Super Seniors A β 42/40 data had slightly lower than expected proportions in the bottom 3 quartiles of the reference curve, but a much greater proportion above the 75th percentile, with a 136.7% (112.8 to 156.6%) increase over the expected normative distribution. This was largely driven by the number of participants falling over the 95th percentile, where only 5% of the population should fall. Remarkably, 22.5% of Super Seniors had A β 42/40 data over the 95th percentile. The distribution of p-tau-181 concentrations in Super Seniors trended in the opposite direction, having a larger enrichment at the lower end of the reference curve. We found slightly lower proportions than expected for Super Seniors p-

tau-181 data in the upper three quartiles of the reference curves, and a much larger proportion below the 25th percentile, with a 136.7% (112.8 to 156.6%) increase over the expected normative distribution. For p-tau-181, this was highly driven by participants whose data fell below the 3rd percentile. While only 3% of population data are expected, 25.8% of Super Seniors p-tau-181 data were below the population 3rd percentile. Concentrations of NfL largely followed the distribution of normative populations. However, there was a slight enrichment in the proportion of Super Seniors NfL data under the 25th percentile (27.5%, 4.7 to 50.2%) and a slightly lower than expected proportion of NfL levels between the 50th and 75th percentiles (-25.0%, -45.8 to -4.1%). The distribution of Super Seniors GFAP concentrations did not differ from the normative distribution across the entire reference curve.

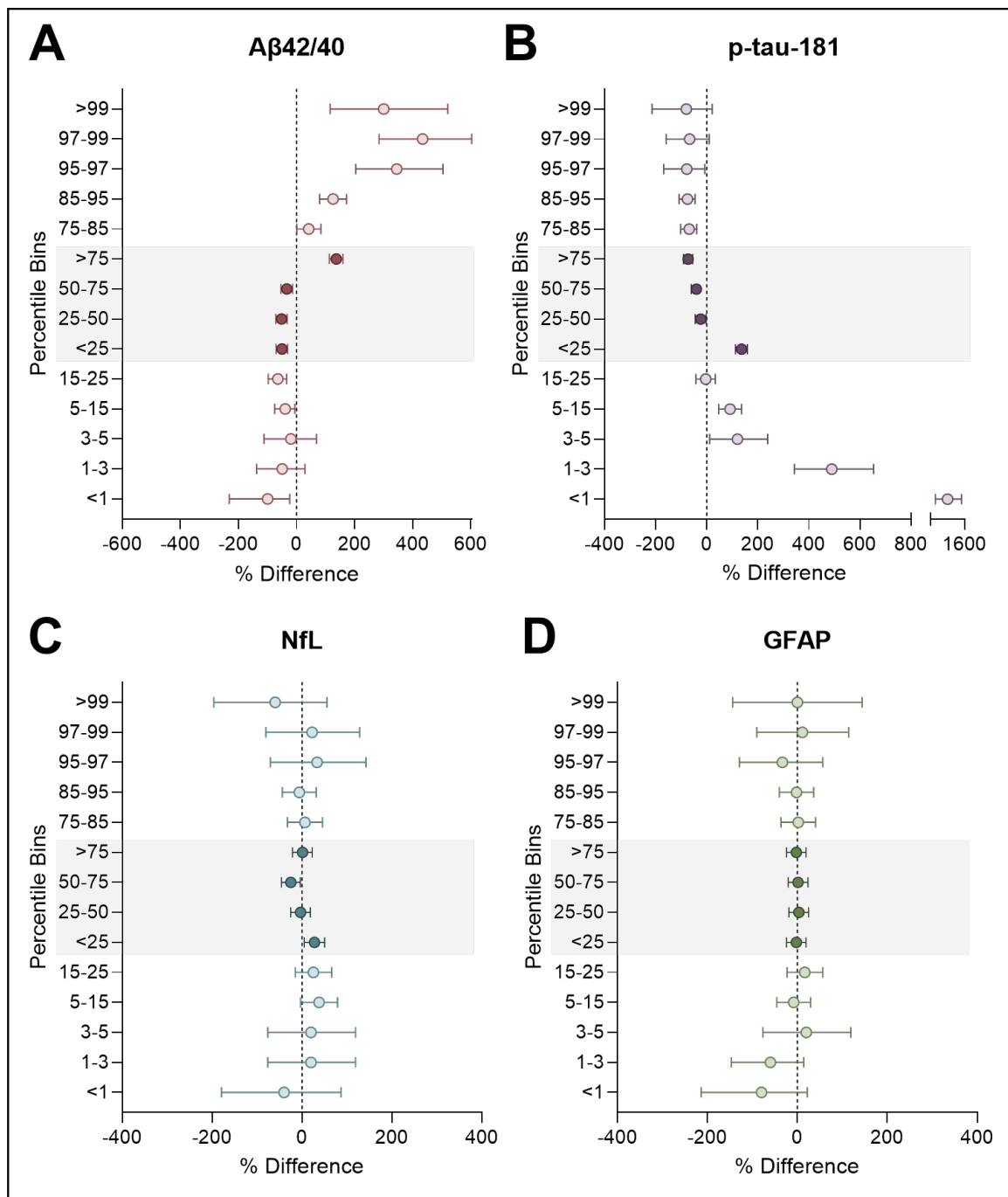


Figure 5.2 Percent difference between expected and observed proportion of Super Seniors in reference curve percentile bins. The proportion difference and 95% confidence interval were calculated using Mee's difference in proportion test. The shaded region of the graphs indicates the quartiles of the reference curves, and the unshaded regions display all detailed percentile bins.

5.3.3 Biomarker distribution

As a significant proportion of Super Seniors had plasma A β 42/40 levels over the 75th percentile and p-tau-181 under the 25th percentile, we investigated whether demographic variables might be driving this skew. We considered the participants who fell in the “healthier” ranges of these biomarker distributions to be resilient based on biomarker standards, while the remaining seniors were considered to be super only by clinical standards. Participants were deemed “biomarker super” if their A β 42/40 was over the 75th percentile and p-tau-181 was under the 25th percentile (n=183; 38% of total Super Seniors), and all other Super-Seniors were deemed “clinically super” (n=297; 62% of total Super Seniors). We compared these two groups to determine what variables may be affecting these participants’ biomarker levels (**Table 5.2**). We found no significant difference in age or BMI, nor in performance of any Super Senior study measure. However, the biomarker super and clinically super groups did significantly differ in their distribution of sex (p=0.0013) and *APOE4* carrier status (0.0006). The biomarker super group had a greater proportion of females (77% female) compared to the clinically super group (64% female). Furthermore, only 12% of the biomarker super seniors were *APOE4* carriers, while 27% of clinically super seniors were *APOE4* carriers.

Table 5.2 Demographics of the Super Seniors Cohort dichotomized by plasma A β 42/40 and p-tau-181 status. Participants were deemed “biomarker super” if their A β 42/40 was over the 75th percentile and p-tau-181 was under the 25th percentile. All other Super Seniors were deemed “clinically super”. Comparisons were performed using a Mann-Whitney test for continuous variables and a Fisher's exact test for categorical variables. *APOE analysis conducted on n=369 participants. Abbreviations: BMI=body mass index, TUG=timed up and go, MMSE=mini mental state exam, IADL=instrumental activities of daily living, GDS=geriatric depression score.

	Clinically Super	Biomarker Super	p-value
Demographics			
N (%)	297	183	
Age, y (median, IQR)	88 (86-90)	88 (87-91)	0.1039
Male Sex (N, %)	112 (38)	43 (24)	0.0013
Female Sex (N, %)	185 (62)	140 (77)	
BMI (median, IQR)	23.8 (21.9-26.1)	24.4 (21.6-27.1)	0.3115
APOE4 Carrier (N, %)*	63 (27)	17 (12)	0.0006

Study Measures			
MMSE (median, IQR)	29 (27-30)	29 (27-29)	0.4336
TUG (median, IQR)	11 (10-13)	12 (10-14)	0.2137
IADL (median, IQR)	23 (22-23)	23 (21-23)	0.1134
GDS (median, IQR)	1 (0-2)	1 (0-3)	0.4384

5.4 Discussion

In this study, we assessed how plasma levels of A β 42/40, p-tau-181, NfL, and GFAP in cognitively healthy Super Seniors compared to the plasma biomarker distribution expected in a normative population. The Super Seniors cohort consists of 480 seniors over the age of 80, who had never been diagnosed with clinical dementia, cancer, diabetes, CVD, and major pulmonary disease. We hypothesized that Super Seniors would have neurodegenerative plasma biomarker levels that are less variable and concentrated farther from pathological levels than found in a representative normative population. This is due not only to the relationship between these biomarkers and the pathology of dementia, but also because these plasma biomarkers are affected by conditions excluded from the Super Seniors cohort, like diabetes and CVD^{249,271,323-325}. This hypothesis was true for plasma A β 42/40 and p-tau-181, two biomarkers used to indicate the presence of AD pathology¹⁰⁸. A larger than expected proportion of Super Seniors had A β 42/40 levels above the 75th percentile, and p-tau-181 levels below the 25th percentile of the normative population. However, for markers associated with general neurodegenerative processes, this was not true, with plasma NfL and GFAP displaying profiles highly representative of a normative population.

5.4.1. Super Seniors may be resistant to Alzheimer's pathology

A β 42/40 is a marker of amyloid plaque formation that decreases in plasma as the pathological load increases^{144,326,327}. A β 42/40 is now recommended for clinical use in both cerebrospinal fluid (CSF) and plasma to indicate the presence of A β pathology in cognitively impaired individuals¹⁰⁸. Plasma p-tau-181 is a marker of neurofibrillary tangles that increases in response to AD pathology^{320,328} and correlates highly with amyloid burden^{162,329}, indicating it can be used as a marker of early AD pathological change. Together, these two markers are

highly indicative of the presence of AD pathology. In this study, we found that Super Seniors had plasma A β 42/40 concentrations clustered toward the upper portion of the normative reference curve and p-tau-181 concentrations clustered to the lower portion. Combined, these data indicate that Super Seniors display resistance to the formation of AD pathology compared to the normative population, and that their healthy cognitive status likely stems as a result of this. It has been shown that AD pathology, specifically A β plaque accumulation, can develop up to 30 years before clinical manifestations of dementia and that up to 40% of those over 80 years old have silent AD pathology⁸³⁻⁸⁶. However, our results show that Super Seniors are more likely to have less underlying AD pathology than extrapolated norms. The drivers of this pathological resistance remain to be determined. Candidate variables include diabetes and other CVD mechanisms that are considered major risk factors for AD but are of low prevalence in Super Seniors^{19,330,331}.

5.4.2. General neurodegenerative plasma biomarkers in Super Seniors

NfL is an axonal protein that is released in response to axonal damage. It is considered a highly sensitive marker of neurodegeneration and is increased across multiple forms of dementia³³²⁻³³⁴. GFAP is an intermediate filament in astrocytes, which is increased when astrocytes are activated. It is considered a marker of neuroinflammation, which is an underlying process in multiple forms of dementia³³⁵⁻³³⁷, and to a lesser extent in normal aging^{88,338,339}. NfL and GFAP are known to increase with normal aging across the full range of the population^{188,248,286}. In contrast to our expectations of a greater proportion of Super Seniors having NfL and GFAP levels in the lower ranges of the normative reference curves, we observed that the levels of both NfL and GFAP in Super Seniors largely reflect the distribution expected in a normative population. This observation indicates that cognitive health does not always indicate resistance to the normal accumulation of neurodegeneration and inflammation seen with normal aging. Our findings add further weight to the hypothesis that there is a certain level of these underlying neurodegenerative processes that can be compensated for to produce cognitive resilience^{340,341}. In Super Seniors, this may be due to healthier lifestyle choices or a reduction in multiple morbidities compounding the risk of progression to a pathological state^{321,342}.

5.4.3. Drivers of biomarker distribution

We further investigated why some Super Seniors appear resistant to AD pathology based on the higher levels of A β 42/40 and lower p-tau-181. To do this, we defined a subgroup of Super Seniors who had concentrations of A β 42/40 over the 75th percentile and p-tau-181 under the 25th percentile of the normative population. We deemed this subgroup to have a “biomarker super” profile, while the remaining participants were considered to be “clinically super”. N=183 (38%) of the Super Seniors cohort met criteria to be deemed biomarker super. When comparing these subgroups, we found only two significant differences between the biomarker vs clinically super subgroups. Specifically, the biomarker resilient subgroup had a higher proportion of females and a lower proportion of *APOE4* carriers. The significantly different sex distribution is aligned with our previous finding from the CHMS cohort in Chapter 3, where we found that females had lower levels of p-tau-181 at older ages. However, we had found that females displayed a lower 95th percentile than males, without the lower percentiles of the reference curve displaying sex specific differences. Therefore, this likely indicates a higher resistance in females, an interesting finding as females are thought to be at greater risk for AD development^{343,344}. Previous studies have shown that females are more susceptible to neurofibrillary tangle pathology^{23,345,346}, which is most closely correlated to the onset of cognitive decline³⁴⁷⁻³⁴⁹. Therefore, females who remain cognitively healthy into older ages may better resist development of AD pathology rather than be resilient to the clinical manifestation of pathology. This resistance in females may be mediated by a variety of factors, such as genetics^{23,350}, leading to this subset of Super Seniors.

The *APOE* gene encodes apolipoprotein E (apoE), a protein that regulates lipid metabolism and cholesterol transport. *APOE* genotype is the largest risk factor for AD, with different alleles affecting the risk and age of onset of AD. The *APOE2* allele is considered protective, *APOE3* is neutral and *APOE4* is detrimental, with one copy increasing risk of AD by 2-4 times, and two copies increasing risk 8-10 times^{20,21}. ApoE4 has been shown to influence the progression of AD through such mechanisms as promoting amyloid plaque formation, decreasing A β clearance from the brain, and promoting phosphorylation of tau^{20,21,351}. It has been previously reported that Super Seniors in general have a lower prevalence of *APOE4* than the general population³⁵². Here, we link that finding to the downstream effects of non-carriers being more

likely to have healthier AD biomarker profiles. It is expected that those with healthier AD biomarker profiles would have the lowest prevalence of *APOE4* carriership. This points to the association of *APOE4* carriers being more likely to develop AD pathology regardless of whether it results in cognitive decline³⁵³⁻³⁵⁵. In Chapter 6, we performed a deeper investigation into the relationship of *APOE* genotype and plasma biomarker concentrations in Super Seniors, finding that *APOE4* carriers have significantly higher p-tau-181 levels, even when accounting for sex³⁵⁶. This work and others indicate there may be a moderating effect between sex and the impact of *APOE* genotype on the development of AD pathology and cognitive decline, where females are more susceptible to the detrimental effects of *APOE4*^{23,350,357-359}.

It is also important to note that within the subgroup of Super Seniors who were deemed clinically super, a subset does have A β 42/40 or p-tau-181 levels that trend in the direction indicative of AD pathology. This indicates that though there are Super Seniors who are resistant to the formation of AD pathology, there are additional Super Seniors who display cognitively resilience despite AD pathologic change. More research will be needed to determine if clinically super seniors developed pathology slightly later in life and will eventually experience cognitive decline, or if there are other factors that allow them to remain cognitively resilient in the presence of AD pathologic change.

5.4.4. Limitations

In this study, we compared Super Seniors biomarker concentrations to previously generated reference curves from a normative population. We acknowledge that the reference curves used are based on a population that only extends to the age of 79 years old. We thus used interpolated curves to extend normative data to compare against Super Seniors data. Although this is a significant limitation, few normative population studies extend to age 80 and over, as the prevalence of dementia and other age-related morbidities greatly increases. Therefore, using an interpolation to assess this age group is justified and represents a conservative estimate of what the general population over age 80 might look like. Another limitation is use of plasma biomarkers as proxy measures of pathological indications without confirmatory autopsy or imaging confirmation. Despite lack of pathology data, the AD biomarkers in this study are now well-established markers to correlate with their representative neuropathologies¹⁰⁸. Finally, an

additional limitation is that not all participants were genotyped for *APOE*, with 77% completed data in this cohort analyzed. However, the proportion of Super Seniors genotyped in each subgroup was equivalent.

5.5. Conclusions

Cognitively healthy older adults aged 80 plus years old displayed biomarker levels indicative of resistance to AD pathology despite normative neurodegenerative biomarker profiles. Super Seniors displayed healthier than normal plasma levels of A β 42/40 and p-tau-181 and highly normative levels of plasma NfL and GFAP consistent with aging. Those with the healthiest biomarker profiles had a decreased prevalence of being an *APOE4* carrier and an increased probability of being female.

Chapter 6. *APOE4* Carrier Status Modifies Plasma p-tau-181 Concentrations in Cognitively Healthy Super-Seniors

6.1 Introduction

Several plasma biomarkers have been investigated for their use as diagnostic or prognostic markers of Alzheimer's disease (AD) and related dementias (ADRD). Promising candidate biomarkers include the ratio of amyloid-beta 42 over 40 ($A\beta_{42}/40$), an indicator of amyloid plaque pathology^{144,254}; phosphorylated tau at residue 181 (p-tau-181), a phosphorylated neuronal microtubule-associated protein that can indicate neurofibrillary tangles and that tracks closely with amyloid burden in AD patients^{162,255}; neurofilament light (NfL), an axonal intermediate filament protein that is elevated in AD²⁵⁶ and other neurological conditions¹⁹³; and glial fibrillary acidic protein (GFAP), an astrocytic intermediate filament protein that reflects astrocyte activation, induced by the presence of amyloid plaques and other factors such as inflammation³⁶⁰.

Although these biomarkers have potential clinical utility, much remains to be learned about the demographic and biological variables that modify plasma biomarker levels. Large scale efforts are underway to investigate potential variables, with studies demonstrating associations between plasma biomarker concentrations and age, body mass index (BMI), and kidney function^{246,271,361}. Common age-related comorbidities such as diabetes and cardiovascular disease (CVD) are also associated with elevated biomarker levels^{249,271,323-325}. To our knowledge; however, no studies have examined these biomarkers in resilient seniors without clinical dementia and other age-associated morbidities. Assessing these biomarkers in a resilient senior population may enable identification of biological associations that might otherwise be masked by the presence of disease.

This study investigates the effect of *APOE* genotype, which encodes the apolipoprotein E (apoE) protein that regulates lipid metabolism and cholesterol transport, on plasma levels of $A\beta_{42}/40$, p-tau-181, NfL and GFAP. The *APOE* gene has three major alleles, *APOE2*, *APOE3*, and *APOE4*, which affect risk and age of onset of AD, with *APOE2* protective, *APOE3* neutral and *APOE4* detrimental²⁰. Many animal and human studies have demonstrated the pleiotropic

effects of ApoE4, including promoting amyloid plaque formation, decreasing Ab clearance from the brain, promoting phosphorylation of tau, increasing microglial mediated neuroinflammation, and increasing cerebrovascular damage²⁰. *APOE* genotype is also associated with other age-related morbidities, with the *APOE4* genotype being a risk factor for both diabetes and CVD³⁶².

The influence of *APOE4* on AD pathology in resilient populations is less clear. It is known that a proportion of seniors develop clinically silent AD pathology and that formation of amyloid plaques may not result in cognitive impairment even at the oldest ages^{83,363}. Understanding how *APOE4* genotype influences proxy markers of AD pathology regardless of clinical disease status may provide further insight on the relationship between *APOE* genotype and AD pathology, while also improving the interpretation of plasma biomarkers. Thus, this study aims to determine whether *APOE4* carriership modifies biomarker concentrations in healthy seniors who are resilient to dementia, cancer, diabetes, CVD, and major pulmonary disease.

6.2. Methods

6.2.1 Ethical Approval

This study was reviewed and approved by the University of British Columbia (UBC) Clinical Research Ethics Board (REB), Vancouver, Canada (H21-01783). The Super-Seniors Study, which provided banked plasma specimens for this study, was reviewed and approved by the joint Clinical Research Ethics Board of BC Cancer and UBC, with harmonized review by the REB of Simon Fraser University (H20-03117)

6.2.2 Participant Recruitment and Data Collection

Specimens for this secondary analysis study were acquired from phase 1 (collected 2004-2007) of the Super-Seniors study that recruited healthy individuals age 85 or older³²¹, who self-reportedly had never been diagnosed with cancer, CVD, diabetes, dementia, or major pulmonary disease. Health data and prescribed medications of enrolled participants were reviewed and participants were excluded if they had any of the above diseases, or were prescribed medications to treat one of these conditions. Demographic data included age,

biological sex, height, and weight. Functional tests included the Mini-Mental State Exam (MMSE), Instrumental Activities of Daily Living Scale (IADL), Timed Up and Go test (TUG), and Geriatric Depression Scale (GDS). Non-fasting blood samples were collected in EDTA vacutainers and centrifuged at 4° Celsius for 15 minutes at 2500g. Plasma was separated and stored in a -80° Celsius freezer within a maximum of 24 hours from collection. DNA was extracted using the PureGene DNA isolation kit (Gentra Systems, MN). *APOE* genotyping was previously reported by Tindale et al.³⁵²

6.2.3 Biomarker Analysis

Biomarker concentrations were quantified using the single-molecule array enzyme linked immunoassay (Simoa) HD-X platform from Quanterix (Billerica, MA) following the manufacturer's protocol. Plasma GFAP, NfL, A β 40, and A β 42 were measured using the Neurology-4-plex E advantage assay (catalogue #103670, lot #503105), and plasma p-tau-181 was measured using the ptau-181 V2 advantage assay (catalogue #103714, lot #502923). Each assay included a 7 or 8-point calibrator curve, two internal kit controls, and three plasma controls. All specimens were assayed in duplicate and the mean value was reported as the result. For all analytes, quality control information was documented across all runs. Standard curves had an average absolute error of 2.2-6.8% and an average recovery of 94-106%. Internal kit controls had an average absolute error of 8.0-12.0% and an average recovery of 106-110%. For plasma controls analyzed in duplicate across every plate, mean values had an average inter-plate CVs were between 3.7-18%. Average intra-plate CVs were between 2.3-4.3%.

6.2.4 Statistical Analysis

Groupwise comparisons were performed between participants dichotomized as *APOE4* carriers or non-carriers. Secondary analysis was also performed to repeat all analysis excluding participants with an E2E4 genotype. Biomarker concentrations were determined to be non-parametric by the Shapiro–Wilk test and group data is reported as the median. Group comparisons between *APOE4* carriers and non-carriers for demographics, study measures and raw biomarker concentrations were performed using a Mann-Whitney U test for continuous variables or a Fisher's exact test for categorical variables. To assess associations between

demographic variables and plasma biomarker concentrations, categorical variables were analyzed using Mann-Whitney U test, and continuous variables were analyzed using simple linear regression (SLR). To assess the association between plasma biomarker levels and *APOE4* carrier status while accounting for demographic variables, categorical variables were analyzed using a two-way analysis of variance (ANOVA), and continuous variables were analyzed using multivariable linear regression (MLR). SLR and MLR models utilized log transformed biomarker data to improve normality of the residuals. All statistical analyses were performed using GraphPad Prism version 9.5.1.

6.3. Results

6.3.1 Cohort Description

N=369 participants from of the Super-Seniors Study³²¹ were included. N=80 (22%) participants were *APOE4* carriers (E3E4 N=71, E4E4 N=3, E2E4 N=6) and N=289 (78%) were non-carriers (E2E2 N=3, E2E3 N=58, E3E3 N=228) (**Figure 6.1**). No significant differences were found between *APOE4* carriers and non-carriers for age, sex, BMI, physical fitness (TUG, IADL), mental health (GDS), or cognitive function (MMSE) (**Table 6.1**).

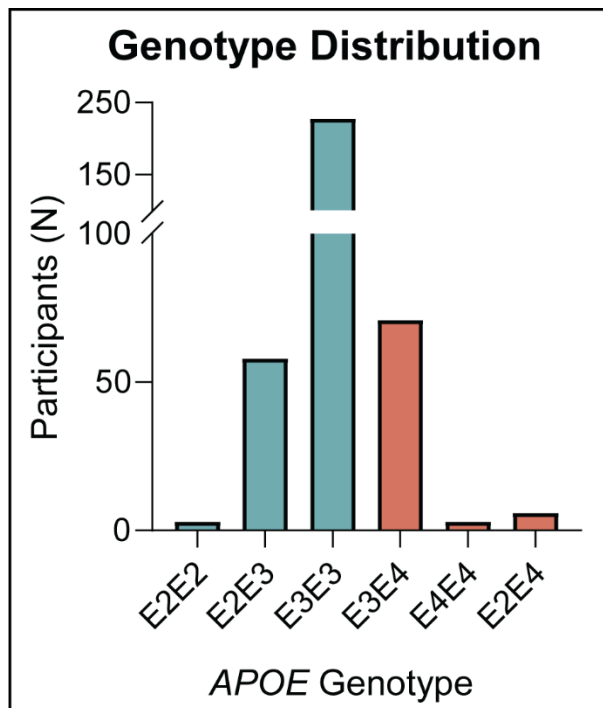


Figure 6.1 APOE genotype distribution of the Super Seniors Cohort. Participants without an *APOE4* allele, non-E4 carriers, are shown in blue (E2E2 N=3, E2E3 N=58, E3E3 N=228). Participants who are *APOE4* carriers are in red (E3E4 N=71, E4E4 N=3, E2E4 N=6).

Table 6.1 Demographics of the Super Seniors Cohort dichotomized by APOE4 carrier status. Comparisons were performed using a Mann-Whitney test for continuous variables and a Fisher's exact test for categorical variables. Abbreviations: BMI=body mass index, TUG=timed up and go, MMSE=mini mental state exam, IADL=instrumental activities of daily living, GDS=geriatric depression score.

<i>APOE</i> Genotype	Non-E4 Carriers	E4 Carriers	p-value
Demographics			
N (%)	289 (78)	80 (22)	
Age, y (median, IQR)	89 (87-91)	88 (87-90)	0.3262
Male Sex (N, %)	89 (31)	25 (31)	>0.9999
Female Sex (N, %)	200 (69)	55 (69)	
BMI (median, IQR)	24 (22-27)	24 (22-27)	0.9620
Study Measures			
MMSE (median, IQR)	29 (27-29)	28 (27-29)	0.1416
TUG (median, IQR)	11 (10-14)	11 (9-13)	0.5244
IADL (median, IQR)	23 (21-23)	23 (22-23)	0.9964
GDS (median, IQR)	1 (0-2)	1 (0-2)	0.3252

6.3.2 Plasma Biomarker Association with Demographic Variables and *APOE* Genotype

In *APOE4* carriers, A β 42/40 was lower (0.0596 vs 0.0618, $p=0.0430$), and both p-tau-181 (3.24 vs 2.77 pg/ml, $p=0.0068$) and GFAP (196 vs 172 pg/ml, $p=0.0457$) were higher compared to non-*APOE4* carriers. NfL was not significantly associated with *APOE4* carrier status (31.2 vs 33.5pg/ml, $p=0.1147$) (**Figure 6.2**). As the demographic variables of age, sex, and BMI have previously been reported to associate with these biomarkers^{246,271,361}, we explored these associations in this resilient population (**Table 6.2**). Both p-tau-181 and GFAP were associated with sex. P-tau-181 was higher in males vs females (3.52 vs 2.57 pg/ml, $p<0.0001$), and GFAP was lower in males vs females (168.6 vs 181.2 pg/ml, $p=0.0467$). Using log transformed biomarker concentrations analyzed by SLR, we observed that NfL was associated with age ($\beta=0.0067$, $p=0.0477$) and BMI ($\beta=-0.0082$, $p=0.0022$), and that p-tau-181 was associated with BMI ($\beta=-0.0072$, $p=0.0207$).

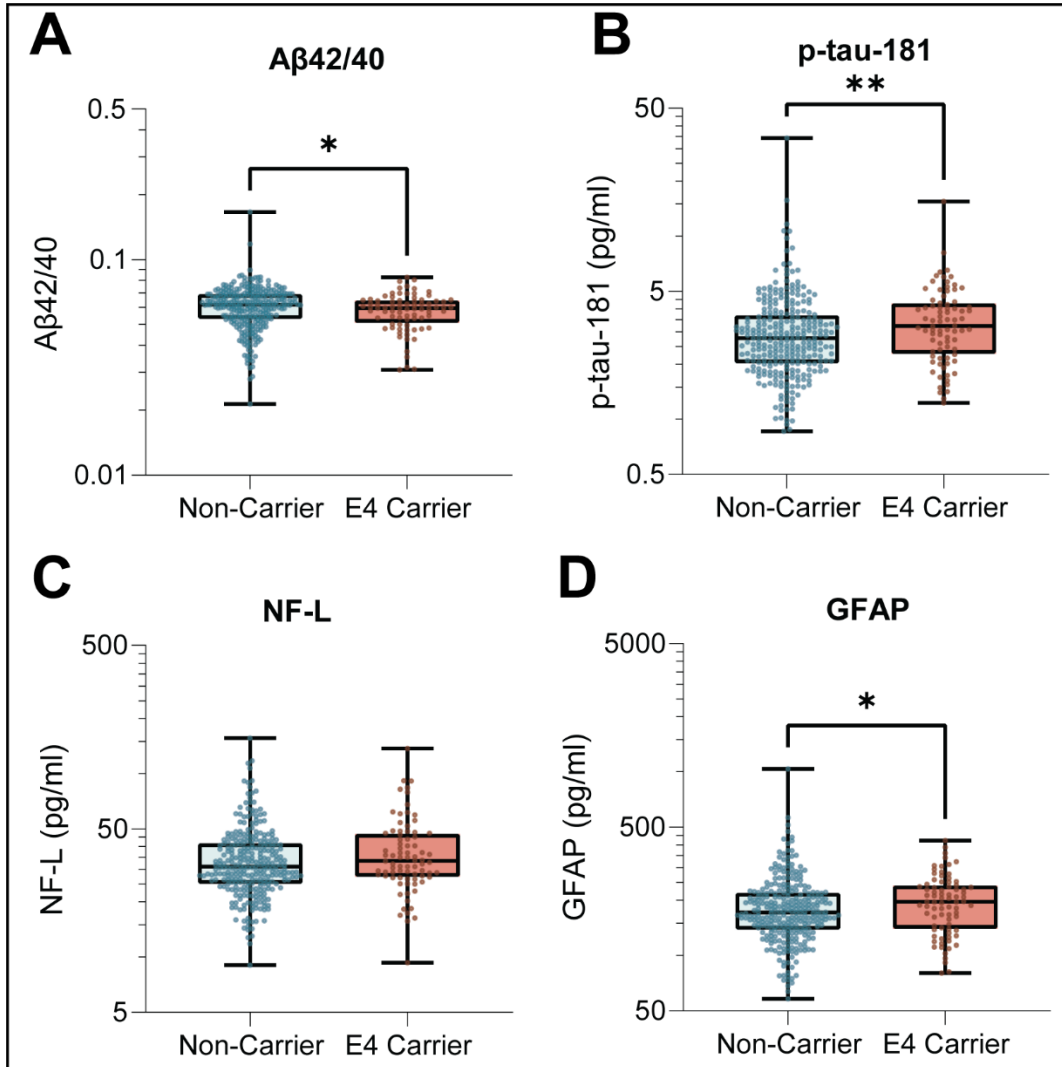


Figure 6.2 Concentrations of plasma Aβ 42/40 (A), p-tau-181 (B), NF-L (C), and GFAP (D) in the Super Seniors Cohort dichotomized by *APOE4* carrier status. E4 carriers N=80, non-E4 carriers N=289. Comparison done by Mann-Whitney test * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$.

Table 6.2 Associations of plasma biomarkers, demographic variables, and *APOE4* carrier status. Univariate analysis was first performed to assess associations between demographic variables and Aβ 42/40, p-tau-181, NF-L, and GFAP. Categorical variables (sex) were analyzed using Mann-Whitney U test, and continuous variables (age, BMI) were analyzed using simple linear regression. Further analysis was performed to assess the association between plasma biomarkers and *APOE4* carrier status while accounting for these demographic variables. Categorical variables (sex) were analyzed using a two-way ANOVA, and continuous variables (age, BMI) were analyzed using multivariable linear regression. Simple and multivariable linear regression models utilized log transformed biomarker data. Abbreviations: ANOVA=

two-way analysis of variance, SLR= simple linear regression, MLR= multivariable linear regression, BMI= body mass index.

Demographics		A β 42/40		p-tau-181		NfL		GFAP	
Sex		p-value		p-value		p-value		p-value	
Mann-Whitney: Sex		0.9757		****<0.0001		0.691		*0.0467	
		F	p-value	F	p-value	F	p-value	F	p-value
ANOVA : Sex + <i>APOE4</i>	Interaction	0.969	0.326	2.17	0.1451	1.015	0.314	0.355	0.552
	Sex	0.428	0.514	14.27	***0.0002	0.0435	0.835	4.555	*0.0335
	<i>APOE4</i>	1.079	0.300	2.93	0.0880	1.024	0.312	1.481	0.224
Age (y)		β	p-value	β	p-value	β	p-value	β	p-value
SLR: Age (y)		-2.14e-5	0.990	0.0062	0.108	0.0067	*0.0477	0.0043	0.168
MLR: Age + <i>APOE4</i>	Intercept	-1.207	<0.0001	-0.1678	0.625	0.8853	**0.0032	1.832	<0.0001
	Age	-2.07e-4	0.908	0.0069	0.0746	0.0070	*0.0365	0.0046	0.136
	<i>APOE4</i>	-0.0190	0.128	0.0673	*0.0131	0.0388	0.0993	0.0354	0.101
BMI		β	p-value	β	p-value	β	p-value	B	p-value
SLR: BMI		4.33e-4	0.763	-0.0072	*0.0207	-0.0082	**0.0022	-0.0035	0.157
MLR: BMI + <i>APOE4</i>	Intercept	-1.238	<0.0001	0.6232	<0.0001	1.712	<0.0001	2.329	<0.0001
	BMI	4.55e-4	0.751	-0.0073	*0.0188	-0.0082	**0.0020	-0.0036	0.152
	<i>APOE4</i>	-0.0191	0.126	0.0619	*0.0231	0.0330	0.158	0.0334	0.127
All Variables		β	p-value	β	p-value	β	p-value	B	p-value
MLR: <i>APOE4</i> + Sex + Age + BMI	Intercept	-1.23	<0.0001	0.121	0.335	1.126	***0.0002	1.912	<0.0001
	<i>APOE4</i>	-0.0194	0.127	0.0649	*0.0129	0.0362	0.1213	0.0356	0.103
	Sex	5.39e-4	0.962	0.135	<0.0001	0.0110	0.595	-0.0408	*0.0353
	Age	-1.20e-4	0.755	0.0056	0.131	0.0066	*0.0466	0.00472	0.127
	BMI	4.50e-4	0.755	-0.0087	**0.0034	-0.0084	**0.0017	-0.0032	0.201

We then used two-way ANOVA and MLR models to explore how sex, age and BMI influenced the associations between plasma biomarker levels and *APOE4* carrier status (**Table 6.2**). For A β 42/40, no model showed significant associations between sex, age, BMI or *APOE4* carrier status. For NfL, none of the models showed significant associations with *APOE4* carrier status. For GFAP, no significant associations were seen with *APOE4* carrier status when adjusting for variables independently associated with GFAP (i.e., sex), or variables that were not (i.e., age, BMI). P-tau-181 was associated with *APOE4* carrier status when adjusting for age and BMI separately. When accounting for sex, *APOE4* carrier status was no longer significantly

associated with p-tau-181. Of note, there was a significant difference between *APOE4* carrier status in females, with carriers having higher p-tau-181 concentrations (3.10 vs 2.49 pg/ml, $p=0.0036$), a pattern not observed in males. As p-tau-181 was shown to independently associate with both sex and BMI, MLR was performed to assess if the association with *APOE4* carrier status persisted when adjusting for the combination of variables. It was determined that *APOE4* carrier status associated with p-tau-181 concentrations ($\beta=0.06224$, $p=0.0170$) while accounting for sex ($\beta=0.1365$, $p<0.0001$) and BMI ($\beta=-0.0087$, $p=0.0036$), both of which contributed significantly to the MLR model. When performing MLR to account for all three covariates (sex, age, BMI), p-tau-181 again persisted as the only biomarker to associate with *APOE4* carriership ($\beta=0.0649$, $p=0.0129$) (**Table 6.2**).

As *APOE2* is considered protective in the context of AD²⁰, all analyses were redone excluding the N=6 participants with an E2E4 genotype. In this analysis, GFAP was no longer higher in *APOE4* carriers (196 vs 172 pg/ml, $p=0.0524$), and was also no longer associated with sex by Mann-Whitney test or two-way ANOVA. All other significant differences and associations remained the same.

6.4. Discussion

6.4.1 *APOE* in Super-Seniors

We identified associations between *APOE4* carriership and plasma biomarker concentrations in a group of Super-Seniors resilient to clinical dementia, cancer, diabetes, CVD, and major pulmonary disease. In the 369 participants studied here, 22% carried at least one *APOE4* allele. Tisdale et al. previously published an analysis of the same cohort, finding Super-Seniors had a lower prevalence of the *APOE4* allele compared to a cohort of 421 population-based middle-aged controls³⁵². This could reflect *APOE*'s risk for AD, CVD, and diabetes that were exclusions for the Super-Seniors study.

Analysis of how *APOE4* genotype associates with plasma biomarkers in a cohort without CVD and diabetes has the advantage of removing these variables that could act as potential mediators in the relationship between *APOE* genotype and AD biomarker concentrations^{249,271,323-325}.

CVD and diabetes are prevalent in adult populations^{364,365} and as such, are regularly screened for and have a wealth of diagnostic tools. However, with AD, in a recent study it was shown that in a population-based cohort, 41% of people between 80-89 years old without cognitive impairment were found to have underlying plaque pathology, based on amyloid proton emission tomography without symptoms required for a clinical diagnosis⁸³. As those with an *APOE4* genotype are more likely to develop amyloid pathology²⁰, it is plausible that *APOE4* also exacerbates amyloid accumulation in Super-Seniors, albeit without leading to cognitive decline.

6.4.2 *APOE4* Association with Plasma Biomarkers and AD Pathology

ApoE4 is associated with earlier onset and greater amyloid accumulation in the brain²⁰, which results in a decrease of A β 42/40 in plasma. Because of this, we hypothesized that plasma A β 42/40 levels would be lower in *APOE4*-carrying Super-Seniors. We found only a mild but significant median difference of 0.0022 between *APOE4* carriers and non-carriers. This association did not persist when adjusting for other demographic variables; however, we also found no independent associations between A β 42/40 and age, sex or BMI in this cohort. Previous literature has mixed findings as to whether *APOE* genotype associates with plasma A β concentrations. Some studies found no association of plasma A β levels and *APOE4*^{366,367} in cognitively healthy cohorts, however this trend is not always consistent³⁶⁸. Other studies found interactions between *APOE* genotype and the ability of plasma A β to detect amyloid pathology^{369,370}, indicating that the association of plasma A β and *APOE* may depend on the presence of increased amyloid pathology.

We found that p-tau-181 associated with *APOE4* carriership in Super-Seniors. This association persisted after adjusting for sex and BMI, which were independently associated with p-tau-181. This finding may be due to apoE4 mediating tangle pathology²⁰. However, it is less likely to find tau pathology, which is further along the AD continuum, in cognitively unimpaired individuals³⁴⁸. Recent studies show that p-tau-181 in plasma may be a better marker of amyloid plaque pathology than tau pathology¹⁶². Our findings also align with previous literature that consistently show higher concentrations of plasma p-tau-181 in *APOE4* carriers in seniors who are cognitively normal³⁷¹, from a population-based cohort³⁷², or are on the AD

continuum^{359,373}. Further investigation in younger healthy populations could help to clarify whether the association of *APOE4* with p-tau-181 exists before the typical age of neuropathological onset.

Additionally, Cai et al. found an interaction between *APOE4* carrier status and female sex that impacted the relationship between p-tau-181 and A β plaque formation³⁵⁹. This aligns with our findings that showing stronger association between *APOE4* carrier status and p-tau-181 in females than in males. This may point to the association between *APOE4* carrier status and p-tau-181 being female specific. This may be due sex specific differences in brain pathology, where the effect of *APOE4* carrier status on amyloid pathology formation is larger in females than males^{374,375}. Indicating that in this cohorts *APOE4* carriers, females may be more likely to have developed underlying pathology than males.

For GFAP, we may expect to see an association with *APOE* genotype as plasma GFAP increases due to astrocyte activation, which can be induced by the presence of amyloid plaques³⁶⁰. Here; however, we found only a mild association that does not persist when adjusting for demographic variables or excluding participants with an E2E4 genotype. This is aligned with previous literature on cognitively healthy seniors showing that although plasma GFAP trended higher in *APOE4* carriers, the two were not significantly associated^{367,371}.

The only plasma marker in this study we did not find to have an association with *APOE* genotype was NfL. This is not unexpected as NfL is not a specific marker of AD, but rather a general marker of neurodegeneration, which we would not expect to find in a cognitively healthy cohort. This finding is consistent with other studies conducted in cognitively normal seniors³⁶⁸ and a population-based cohort³⁷².

Though the Super-Senior *APOE4* carriers had higher p-tau-181, they were cognitively healthy. This observation implies that Super-Seniors are resilient to elevated p-tau-181. This could indicate either clinically-silent AD pathology with a resilience to associated cognitive decline, or an overall resistance to the formation of AD pathology, indicating other currently unknown factors causing increase p-tau-181 levels. Although this association between plasma p-tau-181

and *APOE4* carriership has been previously demonstrated, this is the first study to display this relationship in a cohort of advanced age (≥ 85 years), without other age-related comorbidities. Therefore, *APOE* genotype should be considered an important variable to account for in the interpretation of plasma biomarker concentrations in older adults, regardless of clinical disease status. It should also be noted that these associations only contribute to a minor amount of variability observed in plasma biomarker concentrations of healthy seniors. As the range of plasma biomarker concentrations in healthy seniors is wider than those observed in middle age³⁶¹, more research is needed to determine which other variables may contribute biomarker variability at older ages.

6.4.3 Limitations

As the Super Seniors study was originally designed to investigate the genetics of aging, no neuroimaging was performed nor was CSF collected in order to determine underlying AD pathology or neurodegeneration. This limits the authors from being able to determine how neuropathology may mediate the relationship between *APOE* genotype and plasma biomarker concentrations. Though the influence of important demographic variables were explored in this study, there are other physiological variables, such as other clinical lab results, that may influence the relationship between plasma biomarkers and *APOE4* carrier status that should be explored in future studies. Due to sample size, this study was limited to its assessment of *APOE* genotype dichotomized by *APOE4* carriership. Future studies should be conducted to determine if there is a dose response for *APOE4* homozygosity vs heterozygosity. This study is also limited in generalizability in terms of ethnicity, as all Super Senior participants analyzed here are Caucasian. Future studies should incorporate additional ethnicities as *APOE* genotype distribution can vary by race.

6.5 Conclusion

There is a strong association between plasma p-tau-181 concentrations and *APOE4* carrier status in cognitively healthy seniors over the age of 85 years. This association persists when accounting for sex and BMI, which independently associate with plasma p-tau-181 concentrations.

Chapter 7. Plasma Biomarkers to Detect and Grade Autopsy Confirmed Alzheimer's Disease Neuropathology

7.1 Introduction

Plasma biomarkers have recently emerged as excellent diagnostic tools for the presence of Alzheimer's disease (AD) pathology. Established biomarkers for AD include: amyloid-beta 42 over 40 ($A\beta_{42/40}$), components of amyloid plaques, which indicate the presence of amyloid plaque pathology^{144,254}; phosphorylated tau at residues 181 and 217 (p-tau-181, p-tau-217), a phosphorylated neuronal microtubule-associated protein that indicates the presence of neurofibrillary tangles and also tracks closely with amyloid burden in AD patients^{162,255}; neurofilament light chain (NfL), an axonal intermediate filament protein indicative of neurodegeneration^{191,376}; and glial fibrillary acidic protein (GFAP), a marker of astrocyte activation indicating neuroinflammation^{377,378}. Currently, the tests that are clinically used, and have the best performance in detecting amyloid-beta are p-tau-217 and $A\beta_{42/40}$. To date, most studies of candidate plasma biomarkers for AD diagnosis have been validated against positron emission tomography (PET) or cerebrospinal fluid (CSF) biomarkers as approved diagnostic tools, rather than the gold standard of autopsy confirmation. Although PET and CSF are very useful reference standards to assess pathology in living people, validation against these tools can give a skewed view of the true accuracy of these blood tests. PET has shown sensitivities and specificities of approximately 90% for the detection of $A\beta$ plaques^{114,379}. Consequently, biomarkers validated using PET as the reference standard are inherently constrained from exhibiting performance estimates that exceed this benchmark. Therefore, autopsy validation remains the gold standard to determine the true accuracy of tests. Investigating associations between AD plasma markers and autopsy measures allows for a more detailed investigation into the true accuracy of these biomarkers to grade the different AD neuropathologies.

Another limitation of many previous validation studies of these biomarkers is their inability to assess whether their concentrations might be modified by the presence of common age-related co-pathologies. It is known that mixed dementia accounts for 50-80% of dementia cases^{76,380,381}. Common co-pathologies in AD include: α -synuclein (α -syn) positive Lewy bodies (LB) and Lewy neurites⁶³, limbic predominant age-related transactive response DNA binding protein 43 kDa (TDP-43) encephalopathy neuropathologic change (LATE-NC)^{60,61},

and cerebrovascular disease (CVD)^{19,64}. As the ability to detect and grade these pathologies in living people is limited, the use of an autopsy cohort enables the potential modifying effects they have on AD plasma biomarkers to be assessed.

The primary goal of this study was to evaluate the current suite of AD diagnostic plasma biomarkers for their associations with neuropathological grading, to determine which measures of AD neuropathology these plasma biomarkers best inform on. We assessed associations of A β 42/40, p-tau-181, p-tau-217, NfL and GFAP with the levels of AD neuropathology, as graded by Thal phase, Braak stage, Consortium to Establish a Registry for Alzheimer's Disease (CERAD) neuritic plaque score, and National Institute on Aging (NIA) level of AD neuropathologic change (ADNC) rating (ABC score). From these results, we determined the primary pathological measure that each of these biomarkers can inform on and assessed the utility of each biomarker to predict that specific AD neuropathology at autopsy. As a secondary aim, we also investigated whether the presence of co-pathologies modified the ability of AD plasma biomarkers to detect AD neuropathology.

7.2 Methods

7.2.1 Ethical approval

This study was reviewed and approved by the University of British Columbia (UBC) Clinical Research Ethics Board (REB), Vancouver, Canada (H21-01783). The UBC Clinic for Alzheimer's Disease and Related Disorders (CARD) autopsy biobank was reviewed and approved by the UBC Clinical Research Ethics Board (CREB), Vancouver, Canada (H21-01931).

7.2.2 Participant and specimen selection

Specimens for this study were acquired from the UBC Hospital CARD, British Columbia's tertiary referral dementia clinic, which enrolls patients into their research biobank for future use. All patients were clinically diagnosed by neurologists and geriatricians with expertise in dementia assessments. Participants in this study included those diagnosed with dementia or mild cognitive impairment (MCI) at the time of blood collection. Demographic, clinical data,

and blood samples were collected during participants' clinical assessments. Clinical information was collected for diagnostic purposes according to physician's discretion. The Mini-Mental State Exam (MMSE) and Montreal Cognitive Exam (MoCA) were conducted on the majority of participants, with MMSE data available for 68% of participants and MoCA data being available for 59%. Blood samples were collected in EDTA vacutainers and centrifuged at 4° Celsius for 15 minutes at 2500g. Plasma was separated and stored in a -70° Celsius freezer within a maximum of 4 hours from collection.

7.2.3 Autopsy examinations

Autopsies were performed on participants referred from CARD with consent from legal next of kin, and examined by two neuropathologists. All participants underwent a standardized neuropathological examination³⁸². Brain tissue was formalin-fixed and paraffin-embedded, with blocks cut at 5 microns. Staining included hematoxylin and eosin (HE), HE combined with Luxol fast blue (HE/LFB), modified Bielschowsky silver, Gallyas silver, and Congo red stains. Using the Dako Omnis automated staining system, immunohistochemistry (IHC) was performed with primary antibodies against alpha-synuclein (α - syn; Thermo Scientific; 1:10,000 after microwave antigen retrieval), A β (DAKO; 1:100 with initial incubation for 3 hours at room temperature), hyperphosphorylated tau (clone AT-8; Innogenetics; 1:2000 after microwave antigen retrieval), phosphorylation-independent TDP-43 (ProteinTech; 1:1000 after microwave antigen retrieval), and ubiquitin (DAKO; 1:500 after microwave antigen retrieval).

7.2.4 Neuropathological classifications

AD pathologies were scored using multiple measures, including Thal phase for the anatomical distribution of A β peptide, Braak stage for the anatomical distribution neurofibrillary tangles, and CERAD neuritic plaque score for the maximal neocortical areal density of senile neuritic plaques. These measures were combined to generate a composite ABC score of ADNC . Scoring was determined by a neuropathologist following established guidelines^{25,26,29,30,383}. Primary tauopathies were coded as a binary present or absent, and included progressive supranuclear palsy (PSP), corticobasal degeneration (CBD), Pick's disease, chronic traumatic

encephalopathy (CTE), globular glial tauopathy (GGT), or suspected microtubule-associated protein tau (*MAPT*) mutations. Participants with primary tauopathies were excluded from the analysis of Braak staging. α -syn pathology was coded based on the presence or absence of any LB with additional staging based on the brain regions involved (1, amygdala only; 2, amygdala and hippocampus; 3 limbic structures and neocortex)³⁸⁴. LATE-NC was similarly coded as present or absent, with additional staging of amygdala only (1), amygdala and hippocampus (2), or limbic and frontal neocortex (3)^{385,386}. CVD was scored on a scale of 0-2, where a score of 1 indicated at least moderate arteriolosclerosis, previously shown to indicate cognitive impairment^{387,388}, but no hemorrhages or infarcts; whereas, a score of 2 was given to any participants with hemorrhages and/or infarcts or any size^{387,389}. CVD was considered present if a participant had a score of 1 or 2. Cerebral amyloid angiopathy (CAA) was scored by severity from 0-3 (0, none; 1, mild; 2, moderate; 3, severe).

7.2.4 Simoa biomarker analysis

Biomarker concentrations were quantified on the single-molecule array enzyme linked immunoassay (Simoa) HD-X platform from Quanterix (Billerica, MA). Plasma GFAP, NfL, A β 40, and A β 42 were measured using the Neurology-4-plex E advantage assay (catalogue #103670, lot #503811), and plasma p-tau-181 was measured using the ptau-181 V2 advantage assay (catalogue #103714, lot #503545) following the manufacturer's protocol. Each assay included a 7 or 8-point calibrator curve, two internal kit controls, and three plasma controls. All specimens were analyzed across two runs, assayed in duplicate and the mean value was reported as the result. For all analytes, quality control information was documented across all runs conducted using these specific lots. Standard curves had an average absolute error of 2.2% to 6.8% and an average recovery of 100% to 104%. Internal kit controls had an average absolute error of 8.0% to 12.0% and an average recovery of 106% to 110%. For plasma controls analyzed in duplicate across every plate, mean values had an average inter-plate coefficient of variation (CV) between 2.2% and 11.4%. Average intraplate CVs were between 4.8% and 7.1%.

7.2.5 Alamar ARGO biomarker analysis

We also measured plasma biomarkers using the Alamar (Fremont, CA) NULISaseq CNS Disease Panel (catalog# 801240, lot # 2415603), which measures 124 protein plasma biomarkers relevant to the CNS on the ARGO instrument. Each assay run contains 4 negative controls (NCs), 3 sample controls, and 3 inter-plate controls (IPCs). Each sample is spiked with the same concentration of an internal control (IC) used in the normalization procedure and as a method for assessing the uniformity of the assay run. IPC samples are pooled plasma controls used to normalize samples between experiments and to assess performance of the assay. NC wells are reactions where no sample input is provided and is used to assess limit of detection (LOD) for each target assay and to assess run quality. SC samples are pooled plasma controls from an independent source different from IPCs. SCs are used for calculating intra- and inter-plate coefficient of variation (CV). Specimens are assayed in singlicate in randomized order with no dilution required. Key quality control measures, including IC CV (coefficient of variation of internal control reads across all wells, expected threshold of 25%), IPC CV (coefficient of variation of reads across all IPCs, expected threshold of 25%), IPC target CV (median coefficient of variation of all IPC targets, expected threshold of 10%), target detectability (target is considered detectable if > 50% samples have signal > LOD), average LOD, defined as the mean plus 3 standard deviations (SD) from the 4 NCs, run detectability (percentage of targets that are detectable expected threshold of 90%), and total number of reads (expected threshold of 100,000,000) are recorded every run. NULISA Protein Quantification (NPQ) data is calculated from raw sequencing counts for each analyte in each well of the assay plate. To control well-to-well variation, the raw count value of each analyte for a given sample/well is divided by that well's IC raw count value. To control plate-to-plate variation, the IC-normalized values for each analyte are divided by the analyte-specific median IC-normalized counts from the 3 IPCs on the plate. Data is then rescaled and log₂-transformed to ensure distributions more amenable to statistical analysis. These normalization and data transformation steps yield data in NPQ units.

7.2.6 Statistical analysis

We assessed biomarker associations with the various measures of AD pathology using a Kruskal-Wallis test with Dunn's multiple comparisons. Associations were also assessed by simple linear regression with biomarkers as the dependent variable. To assess the influence of other neuropathologies on these associations, multivariable linear regression was performed to incorporate staging of other pathologies. To determine the utility of biomarkers to distinguish the presence or absence of significant AD pathology, we utilized the results of the association analyses to select our primary measure of AD pathology. We used the ABC score indicating intermediate or high AD pathology²⁶ to classify a participant as having significant ADNC. For participants who did not have Braak staging due to the presence of primary tauopathies, those with at least Thal phase 1 or 2 (A1) and CERAD moderate (C2), were considered to be positive for significant ADNC^{25,26}. Groupwise comparisons were performed between the AD pathology groups, for demographics, study measures and the presence of other neuropathologies, using a Mann-Whitney U test for continuous variables or a Fisher's exact test for categorical variables. Plasma biomarker concentrations were compared between the AD pathology groups using a Mann-Whitney U test and area under the receiver operator curve analysis (AUROC). To assess how co-pathology profiles influenced plasma biomarkers, a Kruskal-Wallis test with Dunn's multiple comparisons was performed within the group with significant ADNC across different co-pathology groups.

To assess how biomarkers from the NULISA 120-plex panel were associated with neuropathology staging, linear regression was performed using biomarkers as the dependent variables. Where possible, regressions were performed using pathological scores as continuous metrics or analyzed as binary if scoring was unavailable. A false discovery rate (FDR) correction was applied to the results for each neuropathology. Biomarkers were considered significant if they showed an effect size of at least 0.2 and maintained a p-value < 0.05 after FDR correction. All statistical analyses were performed using RStudio with R version 4.4.2 and GraphPad Prism version 10.5.0.

7.3 Results

7.3.1 Cohort Description

A total of n=94 participants were included in this study. The cohort was 53% female, and the median age of participants at the time of blood draw was 68.5 (57-77) years old, with a median of 4.6 (2.6-6.8) years between blood sampling and autopsy. Of the n=88 participants who had a clinical diagnosis at the time of blood draw, the majority had a diagnosis of dementia (89%), while the remaining participants were clinically diagnosed with mild cognitive impairment (MCI). The clinical diagnosis of participants largely corresponded to their underlying pathology, with only n=7 (8%) participants being misdiagnosed at the time of blood draw. N=64 (68%) participants met the threshold for having significant ADNC at autopsy, while n=30 (32%) did not and were designated as the non-AD group (**Table 7.1**). Those with significant ADNC were significantly older at the time of blood draw (72 vs 58 years, $p<0.0001$) and death (78 vs 62 years, $p<0.0001$) and had a longer time between blood draw and death (5.7 vs 3.0 years, $p=0.0004$). Those with significant ADNC also had a higher prevalence of α -syn (48% vs 13%, $p=0.0012$) and CVD (63% vs 37%, $p=0.0263$) co-pathology and a lower prevalence of non-AD related tau pathology (11% vs 50%, $p=0.0001$). There was no difference between the groups with and without significant ADNC for the proportion of participants with dementia vs MCI, or cognitive scores by MMSE or MoCA exam. However, a larger proportion of those with significant ADNC had MMSE score recorded vs those without AD (78% vs 47%, $p=0.0040$).

Table 7.1 Demographic and clinical descriptors of CARD Clinic participants dichotomized by presence of AD pathology on autopsy. Comparisons were performed using a Mann-Whitney test for continuous variables and a Fisher's exact test for categorical variables. Abbreviations: AD, Alzheimer's disease; TDP, TAR DNA-binding protein; CVD, cerebrovascular disease; MCI, mild cognitive impairment; MMSE, mini-mental state examination; MoCA, Montreal cognitive assessment.

	Full cohort	Non-AD	AD	p-value
N, %	94	30 (32)	64 (68)	
Demographics				
Age at Blood, years (median, IQR)	68.5 (57.0, 77.0)	58.0 (55.0, 68.0)	72.0 (66.0, 79.0)	<0.0001

Age at Death, years (median, IQR)	74.5 (65.0, 82.0)	61.5 (57.0, 74.0)	77.5 (70.5, 83.0)	<0.0001
Blood to Death, years (median, IQR)	4.6 (2.6, 6.8)	3.0 (1.2, 5.5)	5.7 (3.3, 7.9)	0.0004
Female sex (n, %)	50 (53)	18 (60)	32 (50)	0.3854
Male sex (n, %)	44 (47)	12 (40)	32 (50)	
Neuropathologies				
AD (n, %)	64 (68)	0 (0)	64 (100)	
1° tau (n, %)	22 (23)	15 (50)	7 (11)	0.0001
α-synuclien (n, %)	35 (37)	4 (13)	31 (48)	0.0012
TDP-43 (n, %)	44 (47)	14 (47)	30 (48)	>0.9999
CVD (n, %)	51 (54)	11 (37)	40 (63)	0.0263
Clinical Descriptors				
MCI	10 (11)	1 (3.4)	9 (15)	0.1556
Dementia	78 (89)	28 (97)	50 (85)	
MMSE (median, IQR)	25.0 (17.0, 26.0)	25.5 (16.0, 27.0)	25.0 (18.0, 26.0)	0.5029
MMSE (n, %)	64 (68)	14 (47)	50 (78)	0.0040
MMSE to Blood, years (median, IQR)	0.0 (0.0, 0.0)	0.0 (0.0, 0.2)	0.0 (0.0, 0.0)	0.5910
MoCA (median, IQR)	20.0 (16.0, 23.0)	21.0 (17.5, 23.0)	18.0 (15.0, 23.0)	0.2100
MoCA (n, %)	55 (59)	20 (67)	35 (55)	0.3695
MoCA to Blood, years (median, IQR)	0.0 (-0.3, 0.0)	0.0 (-0.2, 0.0)	0.0 (-0.6, 0.0)	0.8916

The majority of the participants were found to have multiple pathologies at autopsy. The upset plot in **Figure 7.1** displays the breakdown of neuropathologies, with bars indicating mixed proteinopathies and shading indicating vascular co-pathologies. Only six participants had pure ADNC, six had a primary tau proteinopathy alone, and eight had only FTLD-TDP. CVD and α -syn pathology rarely occurred in isolation (n=1 each). Across pathologies, various neuropathological stages were observed, as displayed in the pie charts of **Figure 7.1**. Of those with CVD, 67% had a milder score of 1. In our cohort, 51% of α -synuclein pathology cases had advanced to the neocortex, and 44% of TDP-43 pathology cases were present in the frontal lobe.

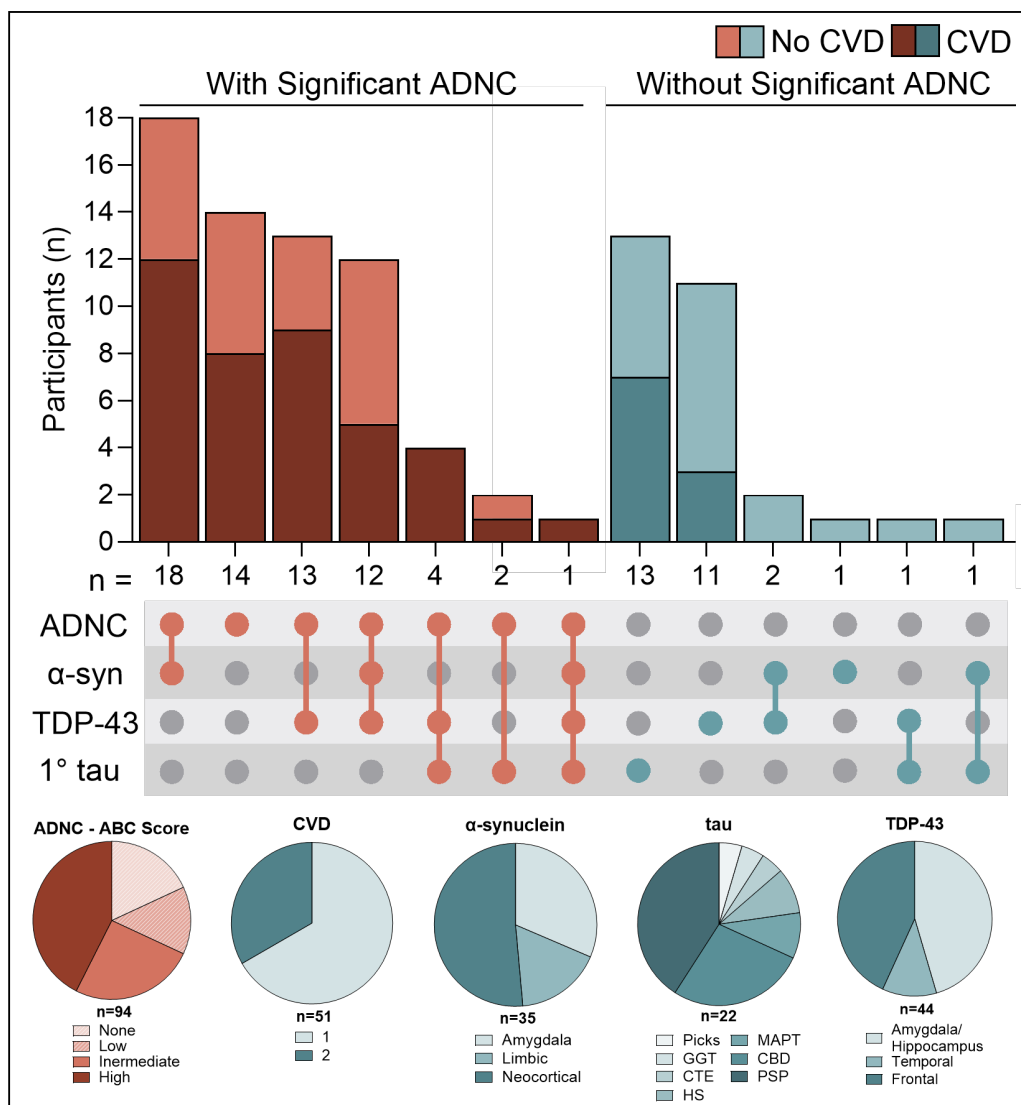


Figure 7.1 Neuropathological breakdown of CARD clinic participants. Upset plot in the top panel displays the overlapping pathologies present in a set of participants in the bar graph. Neuropathologies are indicated by the dot plot below the bar graph, and the presence of CVD is indicated by darker shading in the bar chart. Those in red were determined to have A β pathology based on an ABC score indicative of at least intermediate AD pathology (n=64). Those in blue did not have A β neuropathology (n=30). N=2 participants are not displayed in the upset plot due to only having CVD. Pie charts along the bottom panel indicate the breakdown of how individual pathologies were coded. CVD was classified by severity, α -synuclein and TDP-43 were classified by the brain region present, and tau was classified by the specific neuropathological diagnosis. Abbreviations: A β , amyloid beta; TDP, TAR DNA-binding protein; α -syn, alpha synuclein; CVD, cerebrovascular disease; CERAD, Consortium to Establish a Registry for Alzheimer's Disease; CTE, chronic traumatic encephalopathy; PSP, progressive supranuclear palsy; CBD, corticobasal degeneration; GGT, globular glial tauopathy; HS, hippocampal sclerosis; MAPT, suspected microtubule-associated protein tau mutation.

7.3.2 Simoa plasma biomarker associations with measures of AD pathology grading

Plasma A β 42/40, p-tau-181, p-tau-217, NfL and GFAP were significantly associated with Thal phase, Braak stage, CERAD score, and ABC score when assessed as continuous scores by linear regression (**Table 7.2**). Furthermore, some biomarkers were able to discriminate neuropathological stages when assessed as categorical groups (**Figure 7.2**). All biomarkers remained associated with all AD grading scores when assessed by Kruskal-Wallis, with the exception of NfL, which was not associated with CERAD score ($p=0.1353$). A β 42/40 was strongly associated with all scores ($p<0.0001$), except for Braak stage ($p=0.0039$), and consistently displayed significant differences between the lowest and the two highest pathology grades. Both p-tau markers showed excellent associations ($p<0.0001$) for all measures of pathology, with the largest differences in the highest pathological grade compared to low and intermediate grades. In addition, p-tau-217 was able to discriminate the second highest Thal phase and CERAD scores from the lowest. GFAP associated most strongly with Thal phase and CERAD score ($p<0.0001$), with significantly higher plasma concentrations in the highest pathological stage compared to the lowest and had a $p=0.0061$ for association with Braak stage and $p=0.0002$ for association with ABC score. NfL had the lowest associations of any marker and did not distinguish among neuropathological stages, with p-values for Thal phase $p=0.0420$, Braak stage $p=0.0921$, CERAD score $p=0.1353$, and ABC score $p=0.0039$.

Table 7.2 Associations of plasma biomarkers with measures of AD pathology staging and co-pathologies. Simple linear regression was first performed to assess associations between AD staging measures and plasma A β 42/40, p-tau-181, NF-L, and GFAP. Multivariable linear regression was then performed to incorporate measures of co-pathologies into the analysis. Tau pathology was considered a binary variable, while α -synuclein, TDP-43, and CVD were treated as continuous, using pathology staging scores.

			A β 42/40		p-tau-181		p-tau-217		NfL		GFAP	
			β	p-value	β	p-value	β	p-value	β	p-value	β	p-value
Thal Phase	AD Alone	Thal	-0.0024	<0.0001	0.6087	0.0003	0.2792	<0.0001	-8.47	0.0068	32.68	<0.0001
	α -synuclein	Thal	-0.0025	<0.0001	0.6722	0.0002	0.3026	<0.0001	-6.45	0.0477	35.20	<0.0001
		α -synuclein	0.0013	0.5609	-0.7240	0.2911	-0.2663	0.1798	-22.96	0.0693	-28.64	0.2974
	TDP-43	Thal	-0.0024	<0.0001	0.6167	0.0003	0.2758	<0.0001	-7.73	0.0104	33.42	<0.0001
		TDP-43	-0.0003	0.8926	0.0250	0.9682	-0.2881	0.1125	33.68	0.0034	26.99	0.2867
	tau	Thal	-0.0023	0.0001	0.5767	0.0014	0.2558	<0.0001	-10.09	0.0024	36.74	<0.0001
		tau	0.0005	0.8282	-0.4066	0.6025	-0.2782	0.2303	-21.46	0.1438	53.82	0.0890
	CVD	Thal	-0.0023	<0.0001	0.5981	0.0006	0.2892	<0.0001	-9.08	0.0048	30.66	<0.0001
		CVD	-0.0004	0.8323	0.1834	0.7742	-0.1782	0.3352	10.72	0.3686	35.77	0.1636
Braak Stage*	AD Alone	Braak	-0.0018	0.0074	0.4650	0.0084	0.2934	<0.0001	-13.70	0.0001	24.08	0.0003
	α -synuclein	Braak	-0.0018	0.0097	0.4793	0.0074	0.3023	<0.0001	-12.49	0.0003	24.46	0.0003
		α -synuclein	-0.0004	0.7049	-0.1743	0.5155	-0.0876	0.3203	-12.83	0.0123	-3.967	0.6836
	TDP-43	Braak	-0.0018	0.0390	0.2727	0.2041	0.1969	0.0057	-5.814	0.1483	22.65	0.0058
		TDP-43	0.0001	0.9117	-0.5177	0.1240	-0.2575	0.0196	20.41	0.0017	-4.186	0.7384
	CVD	Braak	-0.0017	0.0117	0.4486	0.0121	0.2979	<0.0001	-14.15	0.0001	22.49	0.0006
		CVD	-0.0015	0.3789	0.2935	0.5202	-0.0821	0.5887	7.875	0.3836	27.76	0.0957
CERAD Score	AD Alone	CERAD	-0.0031	0.0002	0.9881	0.0001	0.4758	<0.0001	-12.36	0.0086	48.86	<0.0001
	α -synuclein	CERAD	-0.0029	0.0006	1.0360	0.0001	0.4953	<0.0001	-9.906	0.0364	52.11	<0.0001
		α -synuclein	-0.0007	0.4493	-0.2139	0.4251	-0.0774	0.2926	-10.36	0.0368	-13.69	0.2045
	TDP-43	CERAD	-0.0031	0.0004	0.9820	0.0003	0.4401	<0.0001	-6.143	0.1795	52.54	<0.0001
		TDP-43	-0.0001	0.8737	-0.0343	0.9013	-0.1106	0.1576	19.39	0.0001	11.24	0.3226
	tau	CERAD	-0.0030	0.0008	0.9638	0.0005	0.4565	<0.0001	-15.63	0.0023	57.65	<0.0001
		tau	0.0004	0.8653	-0.1802	0.8191	-0.1346	0.5504	-24.66	0.1015	66.40	0.0401
	CVD	CERAD	-0.0031	0.0003	0.9752	0.0001	0.4792	<0.0001	-12.63	0.0078	47.40	<0.0001
		CVD	-0.0009	0.5135	0.2212	0.5895	-0.0596	0.5997	4.671	0.5488	25.80	0.1228
ABC Score	AD Alone	ABC	-0.0037	0.0001	0.9947	0.0005	0.4881	<0.0001	-14.14	0.0002	44.25	0.0002
	α -synuclein	ABC	-0.0038	0.0001	1.0898	0.0003	0.5341	<0.0001	-10.83	0.0449	46.99	0.0002
		α -synuclein	0.0010	0.6469	-0.6700	0.3283	-0.3059	0.1163	-22.97	0.0686	-19.04	0.5070
	TDP-43	ABC	-0.0037	0.0001	1.0301	0.0004	0.4869	<0.0001	-12.34	0.0145	46.49	0.0001
		TDP-43	-0.0005	0.7935	0.1276	0.8401	-0.2466	0.1651	32.78	0.0045	29.20	0.2706
	tau	ABC	-0.0037	0.0002	0.9705	0.0022	0.4651	<0.0001	-18.89	0.0011	54.74	<0.0001
		tau	-0.0005	0.8632	-0.1437	0.8611	-0.1361	0.5658	-28.88	0.0600	63.92	0.0652
	CVD	ABC	-0.0036	0.0001	0.9732	0.0008	0.5028	<0.0001	-15.01	0.0048	40.60	0.0007
		CVD	-0.0007	0.7328	0.2494	0.6959	-0.1712	0.3428	10.08	0.3956	42.41	0.1117

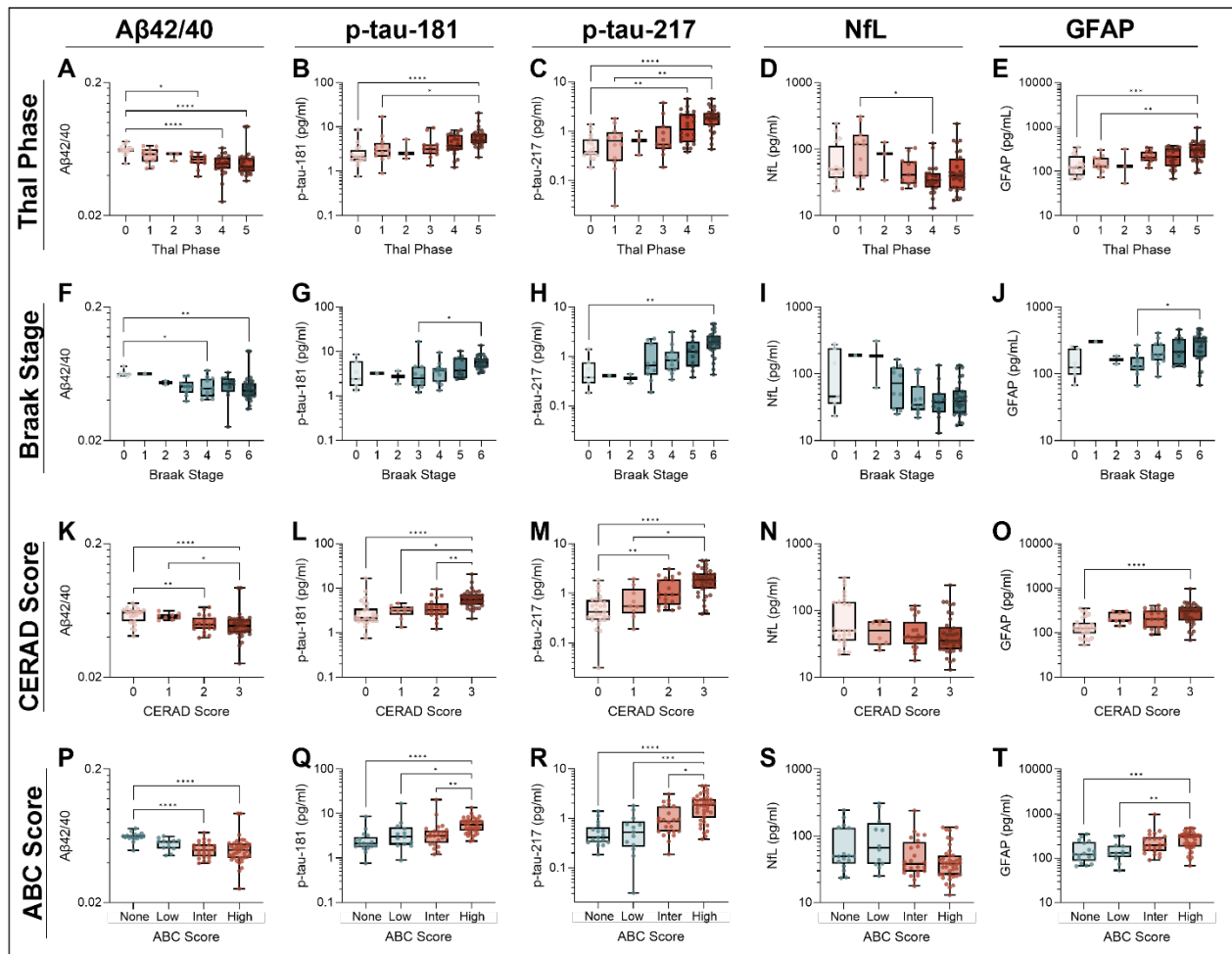


Figure 7.2 Associations of plasma biomarkers with measures of AD pathology staging. A-E Thal phase n=94, F-J Braak stage n=71 (all participants with primary tauopathies excluded from analysis), and K-O CERAD score n=94. P-T ABC score n=94. Analysis conducted by Kruskal-Wallis test with Dunn's multiple comparisons. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$. Abbreviations: CERAD, Consortium to Establish a Registry for Alzheimer's Disease.

To determine whether the presence of co-pathologies altered the association of AD plasma biomarkers with measures of AD pathology, multivariable linear regression was then performed to incorporate grading of other pathologies into the analysis (**Table 7.2**). Here, we found that A β 42/40 and p-tau-217 were not influenced by the presence of co-pathologies, consistently retaining their associations with AD pathology measures and showing no association with CVD scores, staging of α -syn and TDP-43, or presence of primary tauopathies. P-tau-181 maintained its association with Thal phase, CERAD score, and ABC

score regardless of any co-pathologies, but was no longer associated with Braak stage when accounting for the staging of TDP-43 pathology. GFAP consistently maintained association with all measures of AD pathology, and was found to significantly associated with the presence of primary tauopathies when accounting for CERAD score. NfL maintained most of its associations with AD pathology but was consistently associated with the scoring of TDP-43 pathology, which, when adjusted for in the analysis, removed the associations between NfL and Braak and CERAD. We also performed a separate multivariable regression analysis to determine if sex modified the association of AD plasma biomarkers with measures of AD pathology (**Appendix 6**). Here we found that while accounting for sex, all biomarkers remained significantly associated with all measures of AD pathology grading. Additionally, when accounting for any measure of AD pathology grade, no biomarkers were significantly associated with sex.

7.3.3 Simoa plasma biomarkers to detect AD pathology

We investigated plasma biomarkers' abilities to differentiate those with significant AD pathology burden from those without. We assessed each biomarker's ability to discriminate between a Thal phase of 0-2 vs 3-5, a CERAD score of 0-1 vs 2-3, and an ABC score of none/low vs intermediate/high (**Figure 7.3**). We found that AUROCs were comparable across measures of significant AD pathology for each biomarker, and within 0.01-0.05 of each other. A β 42/40, NfL and GFAP performed best with Thal phase, while both p-tau markers performed best with CERAD score. As ABC score represented a composite for all biomarkers, we utilized this as our determinant of participants having significant ADNC moving forward. When comparing biomarker concentrations between those with and without significant ADNC, all biomarkers showed a significant difference by Mann-Whitney test (**Figure 7.4**). P-tau-217 had the best ability to differentiate groups with an AUROC of 0.84, followed closely by A β 42/40 with an AUROC of 0.83. Combining p-tau-217 and A β 42/40 into one model improved the AUROC to 0.88, and a model containing A β 42/40, p-tau-217, p-tau-181, and GFAP showed even greater improvement with an AUROC of 0.91. Additionally, we determined if sex drove any biomarker differences between those with and without significant ADNC. In a two-way ANOVA analysis, no biomarkers differed by sex within either the non-AD or AD groups (**Appendix 6**). In females, biomarkers were still significantly different between those with and

without significant ADNC. However, for males only A β 42/40 and p-tau-217 remained significantly different between those with and without significant ADNC.

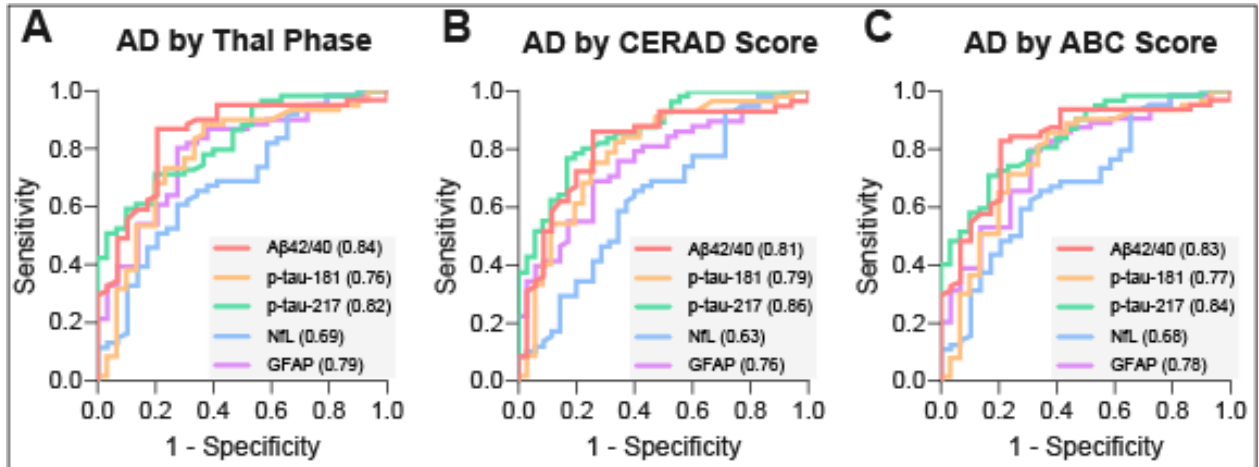


Figure 7.3 Plasma biomarkers to detect different measures of AD pathology. A Thal phase of 0-2 vs 3-5 (n=91), B CERAD score of 0-1 vs 2-3 (n=94), C ABC score of none/low vs intermediate/high (n=94). Area under the receiver operating curves for each biomarker's ability to detect AD pathology in the figure legend.

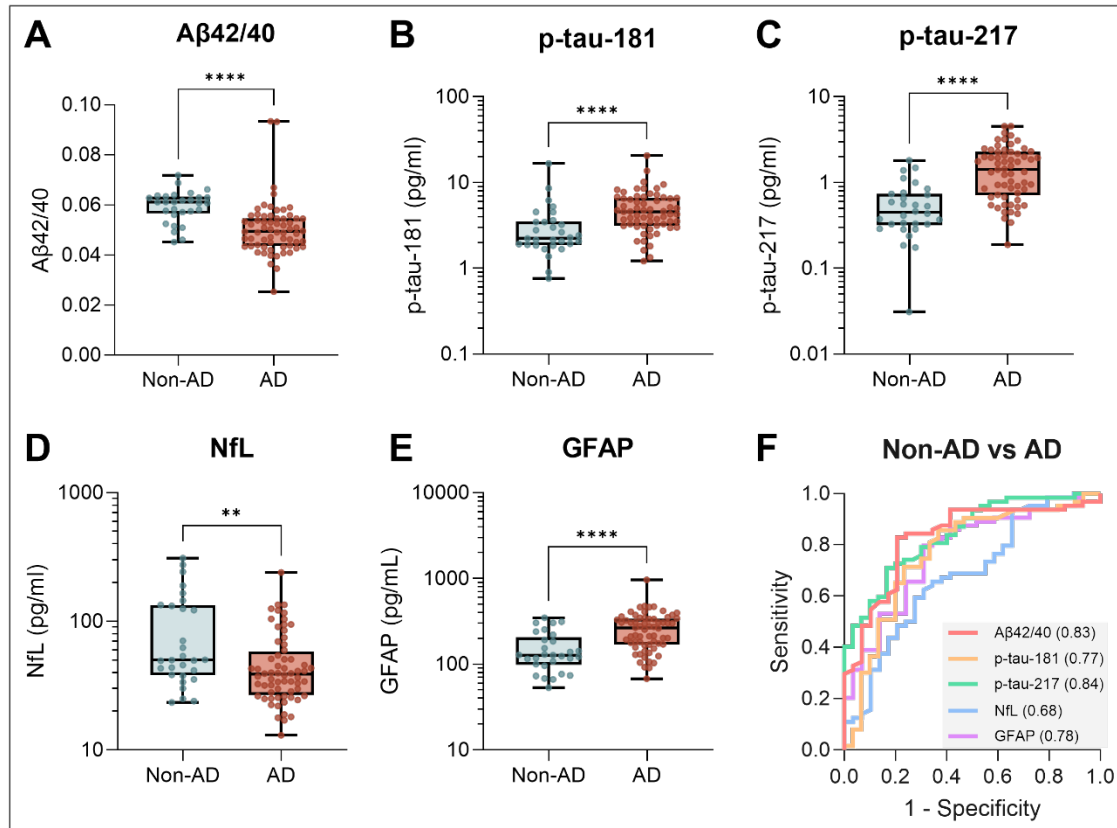


Figure 7.4 Plasma biomarkers to detect the presence of AD pathology at autopsy. AD pathology was determined based on an ABC score indicative of at least intermediate AD pathology (n=64). For those without AD n=30. Panels A-E display biomarkers dichotomized by AD status, with comparisons done by Mann-Whitney test * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$. Panel F displays receiver operating curves for each biomarker's ability to detect AD pathology. Area under the curve in the figure legend, with all $p < 0.0001$, except for NfL ($p = 0.0332$).

After establishing that staging of other neuropathologies did disrupt the association between most plasma biomarkers and grading of ADNC, we next aimed to investigate whether the presence of co-pathologies would alter biomarker profiles in those with significant ADNC (**Figure 7.5**). We assessed plasma biomarker concentrations in those with significant ADNC (n=64) across two categories of co-pathologies, proteinopathies and vascular. Proteinopathies were determined by the presence or absence of α -syn and TDP-43, as groupings with tau co-pathologies were $n \leq 4$ and were not included in this analysis and vascular co-pathologies were assessed using CVD scores of 0-2. The only biomarker that was significantly associated with the presence of any co-pathologies was NfL. NfL was significantly different across

proteinopathy groups, driven by the difference between those with TDP-43, having significantly higher NfL levels compared to those with α -syn co-pathologies. None of other biomarkers differed significantly across co-pathology profiles.

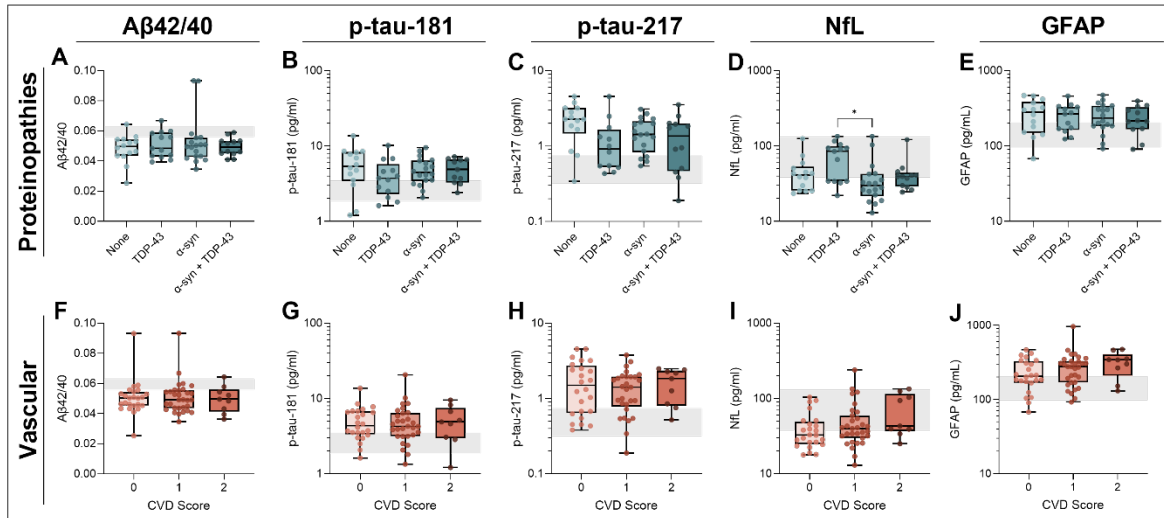


Figure 7.5 Influence of co-pathologies on AD plasma biomarkers in participants with AD pathology (n=64). A-E Groups divided by presence of proteinopathy co-pathologies in n=57 participants. Co-pathology groupings were excluded from analysis if $n \leq 4$. F-J Vascular copathologies scored by severity of 0-2 in n=64 participants. Grey bars in figures indicate the interquartile range of plasma biomarkers for the group without AD pathology (n=30). Analysis conducted by Kruskal-Wallis test with Dunn's multiple comparisons. * $p \leq 0.05$.

7.3.4 NULISA biomarkers for staging of AD pathology

In addition to assessing the established biomarkers of AD pathology, we performed an exploratory analysis on a subset of n=84 participants to determine if biomarkers from the Alamar NULISA CNS 120-plex could aid in the grading of ADNC. We assessed if biomarkers in the panel were associated with continuous measures of AD pathology (**Figure 7.6**). NULISA analysis validated our findings from the Simoa biomarker analysis, showing that p-tau-217, p-tau-181, GFAP, and NfL all associated with Thal phase, Braak stage, CERAD score, and ABC score. Interestingly, neither Aβ40 nor Aβ42 were significantly associated with any of the grading scores. In addition, we found that levels of acetylcholinesterase (ACHE), total tau (MAPT), and apolipoprotein E4 (apoE4) were positively associated with Thal phase, CERAD

score and ABC score. We also found that apoE4 carriership was positively associated with Braak stage, and that glial derived neurotrophic factor (GDNF) and S100 calcium-binding protein A12 (S100A12) were negatively associated with Braak stage.

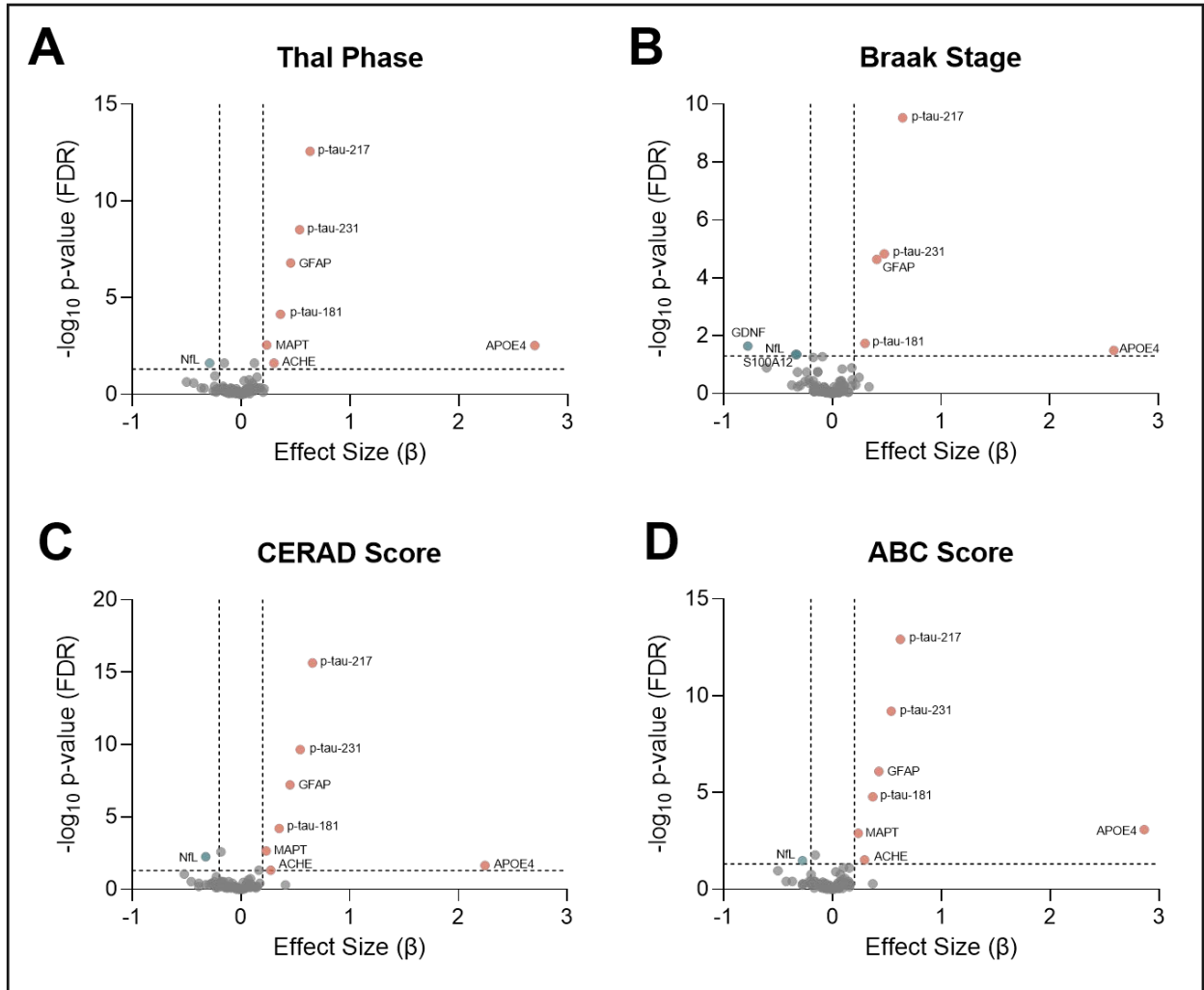


Figure 7.6 Volcano plots for association of NULISA 120-plex biomarkers with neuropathologies (n=84). A-D Biomarkers associated with staging of AD pathology by linear regression analysis. Analysis of Braak staging (B) (n=62) excludes participants with primary tauopathies. Biomarkers were considered significant if they showed a log2 fold change between categories of at least 0.5 or an effect size of at least 0.2 (vertical dashed lines), and maintained a p-value < 0.05 after FDR correction (horizontal dashed line).

7.4 Discussion

7.4.1 Grading AD neuropathology with plasma biomarkers

In this study, we assess the ability of multiple plasma biomarkers to detect and grade AD pathology. Although many groups have assessed the utility of plasma A β 42/40, p-tau-181, p-tau-217, NfL and GFAP as diagnostic biomarkers of AD¹⁰⁸, few studies have focused on using autopsy findings to evaluate plasma biomarker concentrations with neuropathological detection and grading³⁹⁰⁻³⁹². Here, we found that all markers were associated with AD pathology grading by Thal phase, Braak stage, CERAD score, and ABC score. Plasma A β 42/40, p-tau-181, p-tau-217 and GFAP could also detect the presence of significant ADNC, defined by an intermediate ABC score, with AUROC of 0.77 to 0.84.

By validating plasma biomarkers against autopsy-confirmed pathology, we can investigate if biomarkers inform on certain neuropathological measures better than others. It has been well established that A β 42/40, p-tau-181, and p-tau-217 all can be used to indicate presence of A β pathology^{144,162,254}. This is based on evidence of associations with overall pathological load defined by A β PET scans. Similarly, despite being an astrocyte marker, GFAP has also been shown to correlate with A β PET^{219,393}. Neuropathology measures allow us to investigate the nuances of biomarker correlations with pathology grading. Thal phase is used as an indicator of A β deposition that provides information on the anatomical spread of both diffuse and neuritic plaques throughout the brain³⁸³. CERAD score is another measure of A β plaques, but it focuses on the maximal focal density of neuritic plaques in the neocortex, rather than anatomical distribution³⁰. We found that most markers had similar associations with Thal phase and CERAD score, with lower levels of A β 42/40 and higher levels of p-tau-181, p-tau-217, NfL and GFAP associating with more severe pathology. Additionally, A β 42/40, p-tau-181, p-tau-217, and GFAP all consistently were able to distinguish the lowest from the highest grades of Thal phase and CERAD score, but could not discriminate the intermediate grades from the lower or higher grades. This indicates a dose-dependent response to A β pathology for all of these markers, rather than a strong response after a certain threshold of pathology. This aligns with previous findings showing similar associations with increasing A β PET and plasma biomarker concentrations^{144,162,219,254,255}. In addition to measures of A β pathology, we also investigated Braak stage, which, similar to Thal phase, indicates the regional distribution of

neurofibrillary tangles throughout the brain²⁹. Though biomarkers were still significantly associated with Braak stage, there was a less consistent ability to distinguish independent grades compared to our findings with Thal phase and CERAD score. This finding is surprising for p-tau-181 and p-tau-217 as tau markers have traditionally been thought of as indicators of neurofibrillary tangle burden. Studies using A β PET have shown that p-tau-181 and p-tau-217 both strongly associate with A β pathology, and some studies that include tau PET indicate that the association for plasma tau markers is actually stronger with A β than tau pathology^{162,164}. Here, our results show that p-tau-181 and p-tau-217 are stronger markers of A β pathology compared to neurofibrillary tangle pathology. This finding is also in line with other autopsy validated studies that assess the association of plasma p-tau markers with A β and tau pathology^{391,392}. To provide a composite score of Thal phase, Braak stage, CERAD score, the ABC score was developed. The ABC score grades overall ADNC as either none, low, intermediate, or high, with a score indicating intermediate or high ADNC indicating the AD pathology is likely sufficient to be a main driver of cognitive impairment^{25,26}. Assessing biomarker associations with ABC score revealed a pattern similar to what was observed for Thal phase and CERAD scores, which is unsurprising as the composite score is highly driven by these metrics.

7.4.2 Detecting significant ADNC with plasma biomarkers

We assessed biomarker ability to discriminate between a Thal phase of 0-2 vs 3-5, a CERAD score of 0-1 vs 2-3, and an ABC score of none/low vs intermediate/high. These staging cut points were based on previous evidence that higher scores indicated a pathological load that would result in cognitive impairment^{25,26}. AUROC analysis revealed that A β 42/40, NfL and GFAP performed best to discriminate AD by Thal phase, while both p-tau markers performed best to discriminate AD by CERAD score. However, the differences in AUROCs were marginal. As the ABC score represents a balance for all biomarkers, while also providing a composite score of both A β plaque distribution and density, we selected ABC score as our determinant of AD pathology moving forward. Here we found the best biomarker to indicate at least intermediate AD pathology was p-tau-217, with an AUROC of 0.84. The addition of multiple biomarkers increased the AUROC to 0.91 when using a model with A β 42/40, p-tau-

181, p-tau-217, and GFAP. This indicates the improved utility of looking at multiple biomarkers for the detection of significant ADNC.

Not unexpectedly, our results show p-tau-217 is the best marker to detect AD pathology, yet our AUROC is lower than many other published values that are typically 0.9 or greater^{170,224}. However, this is almost certainly due to the impact of comparing to a different reference standard. To date, the majority of studies assessing plasma biomarkers to detect AD have focused on their validation against either amyloid PET scans or CSF biomarkers. Although these methods are well-accepted reference standards to assess the presence of AD pathology in living people, they are still not the true gold standard of autopsy. PET scans to detect A β pathology were validated against autopsy findings; however, PET does not show perfect accuracy compared to the gold standard. Studies indicate sensitivities and specificities of approximately 90% for amyloid PET to detect A β plaques^{114,379}, indicating that there are still expected to be a proportion of false negatives and positives using amyloid PET. Therefore, studies that validate plasma biomarkers against PET can never exceed PET's accuracy threshold. Here, when validating plasma biomarkers against the true gold standard of autopsy, we can determine a more accurate estimate of how well these tests perform to detect AD pathology.

Notably, although we were able to produce a good AUROC to distinguish those with and without significant ADNC, we did find significant overlap in biomarker concentrations for those with low and intermediate ADNC. This is also true for the mild to moderate grades of Thal phase and CERAD score, indicating there is no measure of pathology suitable to develop a clean cut-off. This finding is aligned with recent recommendations to use a two-cut-off system for AD plasma biomarkers, in which an intermediate zone is created to denote further testing to discriminate the levels of AD pathology plasma markers cannot distinguish³⁹⁴⁻³⁹⁶. Using two cut-offs would likely provide better sensitivity for those with a high pathological load, and better specificity for those with a low pathological load, while recommending additional testing for those who fall in the low to intermediate ranges.

7.4.3 Cross-platform validation of AD biomarkers

In addition to assessing the utility of established biomarkers of AD neuropathology, we investigated novel biomarkers that may have utility in AD pathological grading and detection using the NULISA 120-plex assay. This analysis validated our findings from the Simoa platform as similar associations exist between Thal phase, Braak stage, CERAD score, and ABC score and NULISA plasma p-tau-217, p-tau-181, GFAP, and NfL, with p-tau-217 showing the strongest association with all AD neuropathology measures. Interestingly, neither A β 42 nor A β 40 was associated with any grading measure when using NULISA. However, this could be due to assessing these biomarkers as individual measures rather than the ratio. Additionally, we found that ACHE, MAPT, and being an *APOE4* carrier were positively associated with Thal phase, CERAD score and ABC score. ACHE is an enzyme responsible for breaking down acetylcholine, a neurotransmitter crucial for memory formation, in the brain. ACHE has additionally been shown to co-localize with A β plaques³⁹⁷, explaining why it may serve as a useful biomarker for plaque pathology. MAPT, or total tau, has shown utility in AD diagnosis previously, as it is utilized in CSF as an indicator of neurodegeneration, however its use in plasma is more limited as it is not entirely brain-specific^{108,398}. As being an *APOE4* carrier is the largest genetic risk factor for AD, and *APOE4* carriership is known to promote plaque formation, it is unsurprising *APOE4* also showed utility to grade A β pathology²⁰. Some NULISA biomarkers were uniquely associated with Braak staging, including GDNF and S100A12, which were both negatively associated with Braak stage. GDNF has previously been shown to have utility as a biomarker in AD in limited studies, but not specifically for tau pathology^{399,400}. S100A12 has not previously been considered an AD biomarker, but has been implicated in AD-associated neuroinflammatory pathways^{401,402}. Taken together, although the Alamar NULISA sub study identified several new biomarkers of interest to assist in neuropathological staging, they are outperformed by existing AD biomarkers.

7.4.3 Complex neuropathological profiles reveal robustness of established biomarkers

One of the key advantages of utilizing an autopsy cohort is the ability to assess whether co-pathologies influence AD plasma biomarker profiles. Here, we assessed the influence of non-AD tau, α -syn, TDP-43, and vascular pathologies on AD plasma biomarker concentrations.

Figure 7.2 displays the complexities of co-pathologies present in this cohort, where we see that pathologies very rarely occur independently. This is aligned with previous findings indicating that 50-80% of dementia is estimated to be associated with mixed pathology^{76,380,381}. We found that α -syn and CVD pathologies more commonly co-occurred with significant ADNC, while tauopathies occurred more often in its absence. TDP-43 was evenly distributed across those with and without significant ADNC. As many of these pathologies do not have reliable detection methods in living people, it is not yet possible to determine their presence as copathologies outside of an autopsy cohort. However, it is important to know if these co-pathologies may modify AD plasma biomarker concentrations in a way that may impact their ability to detect significant ADNC.

Therefore, we tested whether neuropathological grading and presence of co-pathologies can influence established AD plasma biomarkers measured by Simoa. With respect to grading, α -syn and TDP-43 were graded based on progression of pathology by brain region^{384,386}, CVD was graded based on severity of arteriolosclerosis and infarcts^{387,389}, and tauopathies were only assessed by presence or absence. Across the whole cohort, we found that AD biomarkers were highly robust in maintaining their associations with AD pathology as their associations with co-pathology grading was largely unaffected. There were few exceptions to this. First, when accounting for TDP-43 grade, p-tau-181 was no longer significantly associated with Braak stage, again adding to evidence that p-tau-181 is a more robust marker of A β than tangle pathology. Second, NfL consistently associated with TDP-43 pathology grading, which impacted the associations between NfL and Braak stage and CERAD scores. This finding is consistent with previous investigations showing NfL has potential as a diagnostic biomarker for frontotemporal dementia, where NfL shows a stronger response in frontotemporal dementia compared to other dementias⁴⁰³⁻⁴⁰⁵. NfL also showed associations with the presence of tauopathies and grading of α -syn, indicating its utility as a general marker of neurodegeneration severity rather being than a disease-specific marker^{191,376}.

In addition to understanding how associations may be modified by staging of co-pathologies across the whole cohort, we also investigated whether the presence of co-pathologies modified biomarkers specifically in those with significant ADNC. Here, we found very similar results,

where plasma A β 42/40, p-tau-181, p-tau-217, and GFAP levels were not significantly different when assessed in those with pure significant ADNC, with either TDP-43 or α -syn, or with multiple mixed pathologies of TDP-43 and α -syn. However, in those with significant ADNC, plasma NfL was significantly higher in those with TDP-43 compared to those with α -syn co-pathologies. No marker was affected by CVD grading. Taken together, our results show that plasma A β 42/40, p-tau-181, p-tau-217, and GFAP are highly robust markers of significant ADNC, regardless of the presence of co-pathologies. However, our findings reinforce the need to find additional biomarkers that can inform on additional neuropathologies, both independently and as co-pathologies in significant ADNC.

7.4.5 Limitations

The main limitation of this study was the relatively small sample size. As we investigate mixed pathologies in these participants, the sample size per grouping could be quite low, with some combinations of pathologies only existing in one participant. However, assessing pathological grading as opposed to pathology presence or absence ensured higher sample sizes for regression analyses. Another limitation of this study is the time between the blood sample and autopsy. Though all participants were clinically diagnosed with MCI or dementia at the time of blood draw, there was no guarantee that all the pathologies present at autopsy were already present at the time of blood draw. More studies with blood drawn closer to time of death, or repeated longitudinal measurements over time would be needed to validate our findings presented here. A final limitation of this study was the lack of a cognitively normal control group to investigate biomarkers in those with a lack of, or mild stages of pathology.

7.5 Conclusions

We found that established plasma biomarkers of AD pathology, including A β 42/40, p-tau-181, p-tau-217, and GFAP, associated with multiple staging measures of autopsy-confirmed AD neuropathology. These associations were robust to the presence of co-pathologies, making them optimal markers for AD diagnosis.

Chapter 8. Exploratory Investigation of Plasma Biomarkers to Detect Common Co-pathologies of Alzheimer's Disease in an Autopsy Cohort

8.1 Introduction

Plasma biomarkers are emerging as excellent tools for the diagnosis of neurological conditions. Alzheimer's disease (AD) has paved the way for much of this research, with a set of validated biomarkers, including amyloid beta ($A\beta$) and phosphorylated tau (p-tau), now moving towards clinical implementation^{108,224}. The diagnostic utility of $A\beta$ and p-tau biomarkers reflects their ability to detect the hallmark proteinopathies of AD. However, AD pathology rarely occurs independently, and dementia with mixed etiology is estimated to account for over 50% of cases⁷²⁻⁷⁵, with the true prevalence being unknown. Therefore, there is a great need to investigate plasma biomarkers that detect other dementia-associated neuropathologies.

Beyond AD, prevalent forms of dementia include Lewy body dementia (LBD), frontotemporal dementia (FTD), and vascular dementia. LBD is pathologically characterized by neuronal inclusion bodies of abnormally aggregated alpha synuclein (α -syn)⁶³, often present in the limbic and neocortical regions of the brain. Clinically, this may be reflected in shared symptoms, including balance and visuospatial problems⁶³. LBD is estimated to account for 5-10% of dementia cases⁴⁰⁶, and α -syn is a co-pathology in approximately 60% of AD cases⁴⁰⁷. FTD encompasses early onset dementias that often present with behavioural changes⁵⁹. One of the main frontotemporal lobar degeneration (FTLD) subtypes involves hyperphosphorylation and abnormal cytoplasmic localization of transactive response DNA binding protein 43 (TDP-43)^{60,61} in the frontal and/or temporal lobes. TDP-43 is also the proteinopathy responsible for limbic age-related TDP-43 encephalopathy (LATE), which is a common co-pathology in AD, occurring in approximately 40% of cases^{80,408}. Finally, vascular dementia is caused by a range of conditions that affect cerebrovascular physiology¹⁹ or disrupt blood flow to the brain, such as prior strokes or transient ischemic attacks⁶⁴. Cerebrovascular pathologies commonly occur with AD pathology with up to 80% of cases being expected to have vascular co-pathologies⁴⁰⁹. Different types of vascular co-pathologies have a range of prevalences, such as cerebral amyloid angiopathy (CAA) and small vessel disease (SVD) with arteriolosclerosis, microbleeds, lacunes, and microinfarcts being among the most common^{409,410}.

While we are able to detect amyloid pathology in living people via the use of amyloid position emission tomography (PET) or fluid biomarker tests, we do not yet have the techniques to accurately assess other proteinopathies in vivo. This has hindered our ability to develop accurate blood biomarker tests reflective of non-AD pathologic change as we lack tools against which to validate. Some advancements, such as a CSF assay to detect α -syn²²⁸, have been made. However, these newly developed tests are still under development and validation, and therefore not widely implemented even in research settings. Additionally, magnetic resonance imaging (MRI) sequences have been developed to visualize cerebrovascular anatomy, assess cerebral blood flow, brain perfusion, and cerebrovascular reactivity, and reveal evidence of vascular pathologies including white matter hyperintensities, microbleeds and cortical superficial siderosis^{411,412}. However, neuropathological examination at autopsy remains the diagnostic gold standard for these non-AD dementia-associated pathologies. Importantly, the vast majority of research on plasma biomarkers for both AD and non-AD dementia has been conducted in clinically diagnosed living participants who lack a confirmatory neuropathological diagnosis. Absence of pathological confirmation may confound results in participants with co-pathologies or those whose clinical diagnosis does not correspond to their underlying pathology. In this exploratory study, we leveraged a cohort of participants with autopsy-confirmed neuropathologies to identify novel plasma biomarkers with potential utility to detect autopsy-confirmed α -syn, TDP-43, and cerebrovascular disease (CVD). We utilized the Alamar NULISA CNS 120-plex biomarker panel to identify top biomarkers to detect these pathologies independently and as co-pathologies in AD to inform on study design for future validation efforts.

8.2. Methods

8.2.1 Ethical approval

This study was reviewed and approved by the University of British Columbia (UBC) Clinical Research Ethics Board (REB), Vancouver, Canada (H21-01783). The UBC Clinic for Alzheimer's Disease and Related Disorders (CARD) autopsy biobank was reviewed and approved by the UBC Clinical Research Ethics Board (CREB), Vancouver, Canada (H21-01931).

8.2.2 Participant and specimen selection

Plasma samples for this study were acquired from the UBC CARD clinic, which is British Columbia's tertiary referral dementia clinic that enrolls patient samples into their research biobank for future use upon informed consent. All CARD patients are clinically diagnosed by neurologists and geriatricians who have expertise in dementia assessments. Participants in this study included those diagnosed with dementia or mild cognitive impairment (MCI) at the time of blood collection. Demographic, clinical data, and blood samples were collected during participants' clinical assessments. Clinical information was collected according to physician's discretion for diagnostic purposes. Therefore, the Mini-Mental State Exam (MMSE) and Montreal Cognitive Exam (MoCA) were conducted on the majority of participants, with MMSE data available for 68% of participants and MoCA data being available for 59%. Blood samples were collected in EDTA vacutainers and centrifuged at 4° Celsius for 15 minutes at 2500g. Plasma was separated and stored in a -80° Celsius freezer within a maximum of 24 hours from collection.

8.2.3 Autopsy examinations

Autopsies were performed on participants referred from CARD with consent from legal next of kin, and examined by two neuropathologists. All participants underwent a standardized neuropathological examination³⁸². Brain tissue was formalin-fixed and paraffin-embedded, with blocks cut at 5 microns. Staining included hematoxylin and eosin (HE), HE combined with Luxol fast blue (HE/LFB), modified Bielschowsky silver, Gallyas silver, and Congo red stains. Using the Dako Omnis automated staining system, immunohistochemistry (IHC) was performed with primary antibodies against alpha-synuclein (α - syn; Thermo Scientific; 1:10,000 after microwave antigen retrieval), A β (DAKO; 1:100 with initial incubation for 3 hours at room temperature), hyperphosphorylated tau (clone AT-8; Innogenetics; 1:2000 after microwave antigen retrieval), phosphorylation-independent TDP-43 (ProteinTech; 1:1000 after microwave antigen retrieval), and ubiquitin (DAKO; 1:500 after microwave antigen retrieval).

8.2.4 Neuropathological classifications

The presence of AD pathology was determined using the National Institute on Aging (NIA) level of AD neuropathologic change (ADNC) rating (ABC score). Participants were determined to have significant ADNC if they had an intermediate or high ABC score, reflecting that AD pathology is likely to be a main driver of cognitive impairment^{25,26}. For participants who did not have Braak stage (B) scoring due to the presence of primary tauopathies, scoring was estimated using both Thal phase (A) and CERAD score (C), where a score of A1C2 or greater was considered intermediate AD pathology^{25,26}. Primary tauopathies were coded as absent or present, and included progressive supranuclear palsy (PSP), corticobasal degeneration (CBD), Pick's disease, chronic traumatic encephalopathy (CTE), globular glial tauopathy (GGT), hippocampal sclerosis (HS) with tau pathology, or suspected microtubule-associated protein tau (MAPT) mutations. Participants with primary tauopathies were excluded from the analysis of Braak staging. α -syn pathology was coded based on the presence or absence of any Lewy body pathology present in the amygdala, hippocampus, limbic structures and neocortex³⁸⁴. TDP-43 pathology was similarly coded by presence and absence in the amygdala, hippocampus, limbic, and frontal neocortex^{385,386}. CVD was determined to be present if participants had at least moderate arteriolosclerosis, previously shown to indicate cognitive impairment^{387,388}, or large, lacunar or microscopic infarcts^{387,389}. Cerebral amyloid angiopathy (CAA) was investigated as a co-pathology in AD and scored by severity from 0-3, with a score of 2 or 3 being used as the presence of at least moderate CAA.

8.2.5 Alamar ARGO biomarker analysis

We measured plasma biomarkers using the Alamar (Fremont, CA) NULISA CNS Disease Panel 120 (catalog# 801240, lot # 2415603), which includes 124 protein biomarkers relevant to CNS disease and injury, on the ARGOTM instrument, with sequencing performed on an Illumina NextSeq 2000 instrument. Each assay run contained 4 negative controls (NCs), 3 sample controls (SC), and 3 inter-plate controls (IPCs). Each sample was spiked with a fixed concentration of an internal control (IC), used both for normalization and to assess assay uniformity. IPC samples are pooled plasma controls used to normalize samples between experiments and to assess assay performance. NC wells contain no sample input and are used

to assess the limit of detection (LOD) for each target assay and to assess run quality. SC are pooled plasma controls from an independent source different from IPCs. SCs are used for calculating intra- and inter-plate coefficient of variation (CV). Samples were assayed in singlicate in randomized order with no dilution required. Key quality control measures were recorded for every run, including IC CV (coefficient of variation of IC reads across all wells, expected threshold of 25%), IPC CV (coefficient of variation of reads across all IPCs, expected threshold of 25%), IPC target CV (median coefficient of variation of all IPC targets, expected threshold of 10%), target detectability (target is considered detectable if > 50% samples have signal > LOD), average LOD (defined as the mean plus 3 standard deviations (SD) from the 4 NCs), run detectability (percentage of targets that are detectable expected threshold of 90%), and total number of reads (expected threshold of 100,000,000). NULISA Protein Quantification (NPQ) data is calculated from raw sequencing counts for each analyte in each well of the assay plate. To control for well-to-well variation, the raw count value of each analyte for a given sample/well is divided by that well's IC raw count value. To control for plate-to-plate variation, the IC-normalized values for each analyte are divided by the analyte-specific median IC-normalized counts from the 3 IPCs on the plate. Data are then rescaled and log2-transformed to ensure distributions are more amenable to statistical analysis. These normalization and data transformation steps yield data in NPQ units.

8.2.6 Statistical analysis

To assess the ability of biomarkers on the NULISA CNS Disease 120 panel to detect binary measures of neuropathologies and co-pathologies in AD, LASSO and elastic net regressions were performed, respectively. For each neuropathology or co-pathology, the cohort was redefined to create pathology-specific case and reference groups. Individuals meeting the criteria for the pathology of interest were coded as cases, and all remaining individuals served as the reference group. This reshuffling was performed separately for each pathology, resulting in a unique analytic dataset and outcome for every regression model. The mixing parameter α was selected manually, with $\alpha=1$ for neuropathologies to prioritize sparsity, and $\alpha=0.5$ for co-pathologies to allow inclusion of correlated biomarkers. We selected separate α values for analysis due to the large difference in sample sizes for each analysis; $\alpha=1$ was utilized in the full cohort to reduce overfitting, whereas $\alpha=0.5$ was used in the AD cohort to provide greater

model flexibility and retain informative but correlated biomarkers that might otherwise be excluded under a more restrictive penalty. The tuning parameter λ was selected by cross-validation using $\lambda_{\min}$, corresponding to the value that minimized the cross-validated deviance. Model stability was assessed by repeating cross-validation 1000 times, re-fitting models at the CV-chosen λ on each repeat, and recording the selection frequency of each biomarker. The final model was restricted to biomarkers selected in at least 75% of repeats, and utilized the largest λ that retained all stable biomarkers. Model discrimination ability was evaluated by using the area under the receiver operating characteristic curve (AUROC) computed from the model's predicted probabilities. From the results of the LASSO and elastic net regressions, top performing biomarkers were selected based on commonalities between ability to detect distinct neuropathologies and co-pathologies in AD. These biomarkers were further analyzed in a group-wise comparison looking at presence and absence of each pathology, using a Mann-Whitney or student's t-test depending on the normality of biomarker distribution. Top biomarkers were further analyzed by multivariable logistic regression to investigate their diagnostic utility in the presence of significant ADNC. All statistical analyses were performed using RStudio (R version 4.4.2), using the glmnet and pROC packages, and GraphPad Prism version 10.5.0.

8.3. Results

8.3.1 Cohort description

A total of n=84 CARD participants were included in this study. All participants had a diagnosis of either MCI or dementia at the time of their clinical visit, which included a blood draw. All participants underwent autopsy examination to determine the presence and staging of dementia-associated neuropathologies. At the time of blood draw, participants had a median age of 68 (58-78) years, and included n=42 (50%) females. **Table 8.1** describes demographic and clinical information for the cohort subdivided by the presence of α -syn, TDP-43, and CVD pathologies. The full neuropathological profile of CARD participants indicated that the vast majority have mixed pathology. The upset plot in **Figure 8.1** displays the breakdown of neuropathological diagnosis with bars indicating mixed pathologies and shading indicating the presence of significant ADNC. We found that only n=4 participants had pure AD pathology, n=5 had primary tauopathies alone, and n=7 had TDP-43 alone. CVD and α -syn rarely occurred

alone, with only n=1 participant each. Participants with α -syn most commonly had AD (87%) and CVD (50%) co-pathology. Participants with TDP-43 pathology most commonly had AD (67%) and CVD (49%) co-pathology. CVD was a common co-pathology across all classifications, but most commonly occurred with AD (77%). Across pathologies, various pathological gradings were observed, as displayed in the pie charts in **Figure 8.1**. Of those with CVD, 36% had a severe CVD denoted by the presence of infarcts. α -syn and TDP-43 were staged based on the brain region that the pathology had progressed furthest to. In our cohort, 50% of α -syn cases had advanced to the neocortex, and 44% of TDP-43 cases were present in the frontal lobe.

Table 8.1 Demographic and clinical descriptors of CARD Clinic participants based on the presence of neuropathology. Abbreviations: ADNC, significant Alzheimer’s disease neuropathologic change; α -syn, α -synuclein; TDP, TAR DNA-binding protein; CVD, cerebrovascular disease; MCI, mild cognitive impairment; MMSE, mini-mental state examination; MoCA, Montreal cognitive assessment.

	Full cohort	α -syn	TDP-43	CVD
N, %	84	30 (36)	39 (46)	47 (56)
Demographics				
Age at Blood, years (median, IQR)	68 (58-78)	69 (57-78)	71 (60-80)	72 (67-81)
Age at Death, years (median, IQR)	75 (64-82)	75 (66-82)	75 (64-84)	77 (71-85)
Blood to Death, years (median, IQR)	4.6 (2.7-6.8)	5.9 (3.5-8.0)	4.2 (2.4-6.8)	4.5 (2.7-6.8)
Female sex (n, %)	42 (50)	10 (33)	23 (59)	26 (55)
Male sex (n, %)	43 (50)	20 (67)	16 (41)	21 (45)
Co-pathologies				
ADNC (n, %)	56 (67)	26 (87)	26 (67)	36 (77)
α -syn (n, %)	30 (36)	30 (100)	12 (31)	15 (32)
TDP-43 (n, %)	39 (46)	12 (40)	39 (100)	19 (40)
1° tau (n, %)	21 (25)	2 (7)	6 (15)	13 (28)
CVD (n, %)	47 (56)	15 (50)	19 (49)	47 (100)
Clinical Descriptors				
MCI	10 (12)	1 (3)	2 (5)	5 (11)
Dementia	73 (87)	26 (87)	33 (85)	39 (83)

MMSE (median, IQR)	25 (16-26)	23 (16-26)	25 (22-26)	24 (16-26)
MMSE (n, %)	57 (68)	22 (73)	23 (59)	35 (74)
MMSE to Blood, years (median, IQR)	0 (-.03-0)	0 (-0.04-0.06)	0 (0-0)	0 (-0.14-0)
MoCA (median, IQR)	20 (16-23)	19 (15-24)	19 (15-22)	19 (15-24)
MoCA (n, %)	51 (60)	17 (57)	23 (59)	28 (60)
MoCA to Blood, years (median, IQR)	0 (-0.29-0)	-0.16 (-1.73-0)	0 (-0.36-0)	0 (-0.28-0)

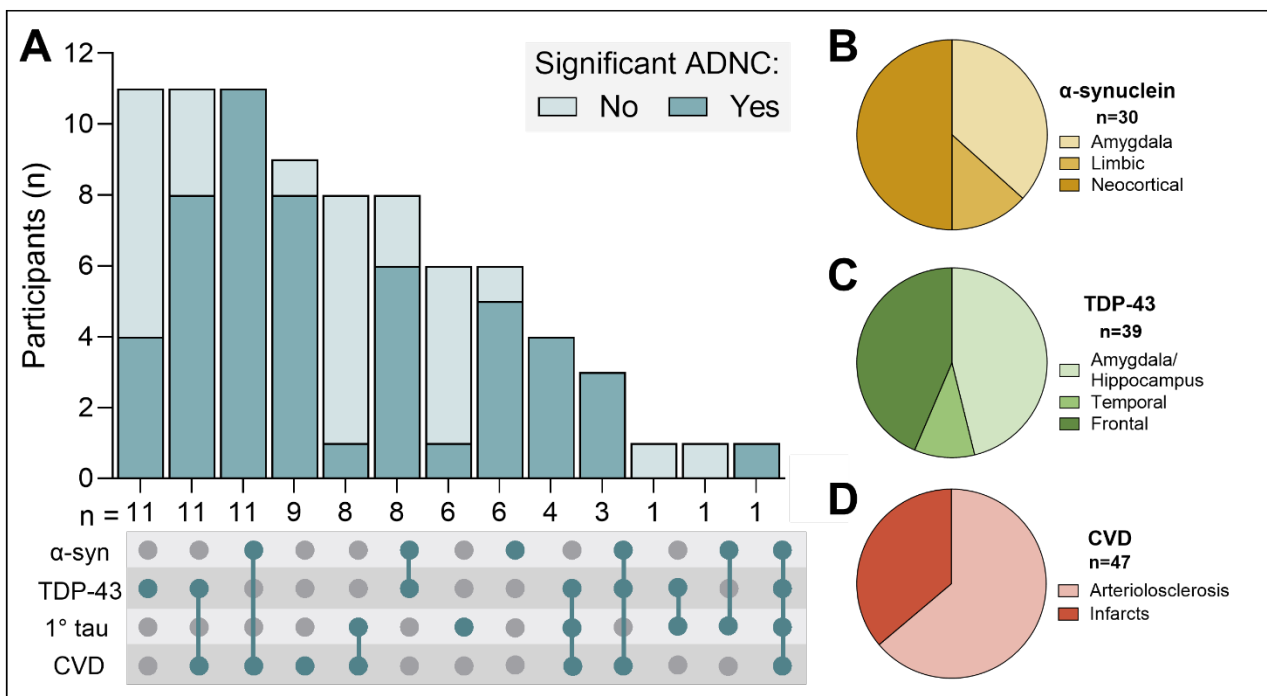


Figure 8.1 Neuropathological breakdown of CARD clinic participants. (A) Upset plot displaying the overlapping pathologies present in CARD participants. The bar graph depicts the frequency of neuropathological subtypes with (darker shading) and without (lighter shading) AD neuropathologic change. The dot plot below defines which co-pathologies, including combinations of co-pathologies, were detected for each bar. N=4 participants are not displayed in the upset plot due to only having AD. (B-D) Pie charts indicate the breakdown of how individual pathologies were coded. CVD was classified by severity, and α-synuclein and TDP-43 were classified by the brain region in which the pathology present. Abbreviations: TDP, TAR DNA-binding protein; α-syn, alpha synuclein; CVD, cerebrovascular disease.

8.3.2 Biomarker feature selection for detection of α -syn, TDP-43 and CVD

Using feature selection analysis, we tested whether biomarkers from the NULISA CNS Disease 120 panel could detect the presence of pathology in the full cohort (**Figure 8.2 A**). Using LASSO regression, we identified a set of 3-17 biomarkers that, when combined into a logistic regression model, could detect the presence of α -syn, TDP-43 and CVD (**Table 8.2**). A panel of 17 biomarkers was able to detect the presence of α -syn pathology with a combined AUROC of 0.95. The top NULISA 120-plex markers contributing to the detection of α -syn were elevated cystatin C (CST3; $\beta=2.09$) and acetylcholinesterase (ACHE; $\beta=0.302$), and lowered colony-stimulating factor 2 (CSF2; $\beta=-0.562$) and chitinase-3-like protein 1 (CHI3L1; $\beta=-0.240$). TDP-43 pathology was characterized by elevated interleukin-12 p70 (IL12p70; $\beta=0.272$), elevated neuronal pentraxin receptor (NPTXR; $\beta=0.081$), and lowered periostin (POSTN; $\beta=-0.085$) with a combined AUROC of 0.75. CVD pathology was detected by elevated C-C motif chemokine ligand 11 (CCL11; $\beta=0.641$), A β 40 ($\beta=0.213$), and ubiquitin C-terminal hydrolase L1 (UCHL1; $\beta=0.014$) with a combined AUROC of 0.86.

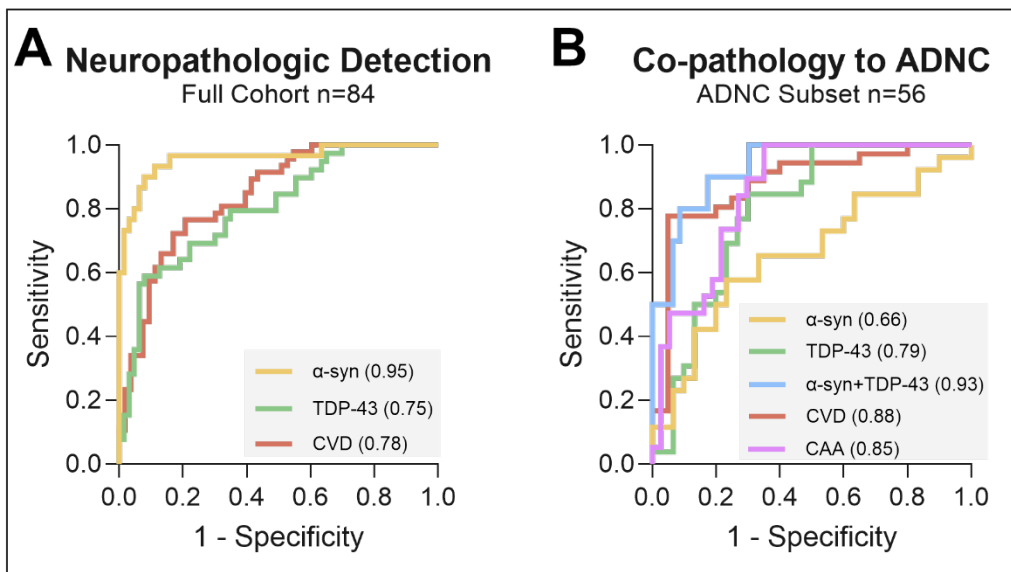


Figure 8.2 Receiver operating characteristic (ROC) curves for detection of α -syn, TDP-43 and CVD. (A) displays discrimination to detect each neuropathology based on presence and absence in the full analyzed cohort (n=84), using logistic LASSO regression. (B) displays discrimination to detect co-pathologies in a subset of participants with AD pathology (n=56), using logistic elastic-net regression ($\alpha = 0.5$). The λ for each regression was selected by cross-validation over 1000 repeats, with final models being restricted to biomarkers with non-zero coefficients in $\geq 75\%$ of repeated cross-validation runs. The AUROC values are displayed in the legend for each pathology. Biomarker predictors for each regression are outlined in **Tables 8.2 and 8.3**.

Table 8.2 Biomarkers retained in the final LASSO model to detect individual neuropathologies (n = 84). For each neuropathology, cases are defined by presence of the pathology, while the reference group is the remainder of the cohort without the pathology. Biomarkers are listed with their logistic lasso regression coefficient (β). Models were fit with logistic LASSO regression, with λ selected by cross-validation. Feature stability was assessed over 1000 repeated cross-validation runs, and only biomarkers selected (non-zero β) in $\geq 75\%$ of repeats were retained in the final model.

Biomarker	Coefficient (β)	Full Name	Marker Classification
α-synuclein (cases n=30; reference group n=54)			
CST3	2.09	Cystatin C	Protease inhibitor
ACHE	0.302	Acetylcholinesterase	Synaptic
S100B	0.203	S100 calcium-binding protein B	Neuroinflammation
p-tau-217	0.183	Phosphorylated tau (Thr217)	Pathological protein aggregation
CSF2	-0.562	Colony-stimulating factor 2	Inflammatory cytokine
CHI3L1	-0.240	Chitinase-3-like protein 1	Neuroinflammation
NfL	-0.222	Neurofilament light chain	Neuronal Injury
IFNG	-0.186	Interferon- γ	Inflammatory cytokine
CRP	-0.156	C-reactive protein	Inflammation
ICAM1	-0.087	Intercellular adhesion molecule 1	Endothelial
KDR	-0.087	Kinase Insert Domain Receptor	Endothelial
CHIT1	-0.080	Chitotriosidase-1	Neuroinflammation
MME	-0.080	Membrane metallo-endopeptidase	Protease
FCN2	-0.050	Ficolin-2	Inflammation
GDF15	-0.037	Growth differentiation factor 15	Inflammatory cytokine
TIMP3	-0.010	Tissue inhibitor of metalloproteinases-3	Extracellular Matrix
CXCL8	-0.009	C-X-C motif chemokine ligand 8	Inflammatory chemokine
TDP-43 (cases n=39; reference group n=45)			
IL12p70	0.272	Interleukin-12 p70	Inflammatory cytokine
NPTXR	0.081	Neuronal pentraxin receptor	Synaptic
POSTN	-0.085	Periostin	Extracellular Matrix
CVD (cases n=47; reference group n=37)			
CCL11	0.641	C-C motif chemokine ligand 11	Inflammatory chemokine
A β 40	0.213	Amyloid- β 1–40 peptide	Pathological protein aggregation
UCHL1	0.014	Ubiquitin C-terminal hydrolase L1	Neuronal Injury

8.3.3 Biomarker feature selection to detect α -syn, TDP-43 and CVD in cases with ADNC

To determine whether the presence of significant ADNC affects biomarker detection of α -syn, TDP-43, and CVD, we conducted further analysis on the subset of n=56 participants with AD using elastic net regression (**Figure 8.2 B, Table 8.3**). Additionally, we investigated detection of multiple proteinopathy co-pathologies including combined α -syn and TDP-43 co-pathologies, as well as the presence of CAA in AD. α -syn co-pathology was present in n=26 participants and was only detected by elevated ACHE ($\beta=0.688$) with an AUROC of 0.66. TDP-43 co-pathology was present in n=26 participants and detected by three markers including elevated interleukin-12 p70 (IL12p70; $\beta=0.023$), lowered RE1-silencing transcription factor (REST; $\beta=-0.236$) and lowered visinin-like protein 1 (VSNL1; $\beta=-0.090$) with a combined AUROC of 0.79. TDP-43 and α -syn co-pathologies co-existed in n=10 participants. This multiple mixed pathology was detected by n=7 markers with an AUROC of 0.93. The top markers contributing to mixed TDP-43 and α -syn detection included elevated IL12p70 ($\beta=0.570$), NPTXR ($\beta=0.524$), and CST3 ($\beta=0.044$), and lowered REST ($\beta=-0.535$). CVD copathologies were present in n=36 participants and detected by n=7 markers with an AUROC of 0.88. The top markers contributing to CVD detection included elevated A β 42 ($\beta=0.256$), C-X-C motif chemokine ligand (CXCL10; $\beta=0.185$), neuropeptide Y (NPY; $\beta=0.117$), and A β 40 ($\beta=0.111$). CAA was present in n=19 participants and detected by n=8 markers with an AUROC of 0.85. The top four markers contributing to CAA detection included lowered slit guidance ligand 2 (SLIT2; $\beta=-0.454$), and elevated p-tau-181 ($\beta=0.166$), p-tau-217 ($\beta=0.144$), and p-tau-231 ($\beta=0.074$).

Table 8.3 Biomarkers retained in the final elastic net model for the detection of co-pathologies in those with significant ADNC (n = 56). For each co-pathology, cases are defined by presence of the pathology, while the reference group is the remainder of the cohort without the pathology. biomarkers are listed with their logistic elastic net regression coefficient (β). Models were fit with logistic LASSO regression, with λ selected by cross-validation. Feature stability was assessed over 1000 repeated cross-validation runs, and only biomarkers selected (non-zero β) in $\geq 75\%$ of repeats were retained in the final model.

Biomarker	Coefficient (β)	Full Name	Marker Classification
α-synuclein (cases n=26; reference group n=30)			
ACHE	0.688	Acetylcholinesterase	Synaptic
TDP-43 (cases n=26; reference group n=30)			
IL12p70	0.023	Interleukin-12 p70	Inflammatory cytokine
REST	-0.236	RE1-silencing transcription factor	Transcription factor
VSNL1	-0.090	Visinin-like protein 1	Synaptic
α-synuclein + TDP-43 (cases n=10; reference group n=46)			
IL12p70	0.570	Interleukin-12 p70	Inflammatory cytokine
NPTXR	0.524	Neuronal pentraxin receptor	Synaptic
CST3	0.044	Cystatin C	Protease inhibitor
IL15	0.003	Interleukin-15	Inflammatory cytokine
REST	-0.535	RE1-silencing transcription factor	Transcription factor
S100A12	-0.057	S100 calcium-binding protein A12	Inflammation
SFRP1	-0.003	Secreted frizzled-related protein 1	Extracellular Matrix
CVD (cases n=36; reference group n=20)			
A β 42	0.256	Amyloid- β 1–42 peptide	Pathological protein aggregation
CXCL10	0.185	C-X-C motif chemokine ligand	Inflammatory chemokine
NPY	0.117	Neuropeptide Y	Synaptic
A β 40	0.111	Amyloid- β 1–40 peptide	Pathological protein aggregation
CCL11	0.109	C-C motif chemokine ligand 11	Inflammatory chemokine
NEFH	0.096	Neurofilament heavy chain	Neuronal Injury
IL5	-0.017	Interleukin-5	Inflammatory cytokine
CAA (n=19) (cases n=19; reference group n=37)			
p-tau-181	0.166	Phosphorylated tau (Thr181)	Pathological protein aggregation
p-tau-217	0.144	Phosphorylated tau (Thr217)	Pathological protein aggregation
p-tau-231	0.074	Phosphorylated tau (Thr231)	Pathological protein aggregation
MAPT	0.018	Microtubule-associated protein tau	Pathological protein aggregation

SLIT2	-0.454	Slit guidance ligand 2	Extracellular Matrix
BDNF	-0.031	Brain-derived neurotrophic factor	Synaptic
CRP	-0.028	C-reactive protein	Inflammation
TIMP3	-0.005	Tissue inhibitor of metalloproteinases-3	Extracellular Matrix

8.3.3 Biomarkers common across analyses

We next considered biomarkers that showed utility to detect a pathology both in the full cohort and, when controlling for the presence of significant ADNC as a sub-analysis, to define key markers of interest for future investigations. For α -syn these were ACHE and CST3, for TDP-43 these were IL12p70 and NPTXR, and for CVD these were CCL11 and A β 40. We extracted these biomarkers to conduct groupwise analysis comparing those with and without α -syn, TDP-43 or CVD in the full cohort of n=84 (**Figure 8.3**). In this analysis, all markers were found to be significantly higher in their pathology of interest, with the exception of CST3, which was not significantly different in those with α -syn pathology compared to those without. For the remaining markers, we performed individual multivariable logistic regression to determine if the individual biomarker could still predict the presence of pathology when adjusting for the presence of significant ADNC. We found that all models showed ability to detect the presence of pathology with AUROCs ranging from 0.63 to 0.78. All biomarkers significantly contributed to their respective models when accounting for the presence of significant ADNC. When detecting α -syn, ACHE had an odds ratio (OR) of 2.1 (1.2-3.7). When detecting TDP-43, IL12p70 had an OR of 2.9 (1.5-6.2) and NPTXR had an OR of 4.3 (1.5-14.7). When detecting CVD, CCL11 had an OR of 10.4 (3.0-42.8) and A β 40 had an OR of 5.9 (2.3-18.5).

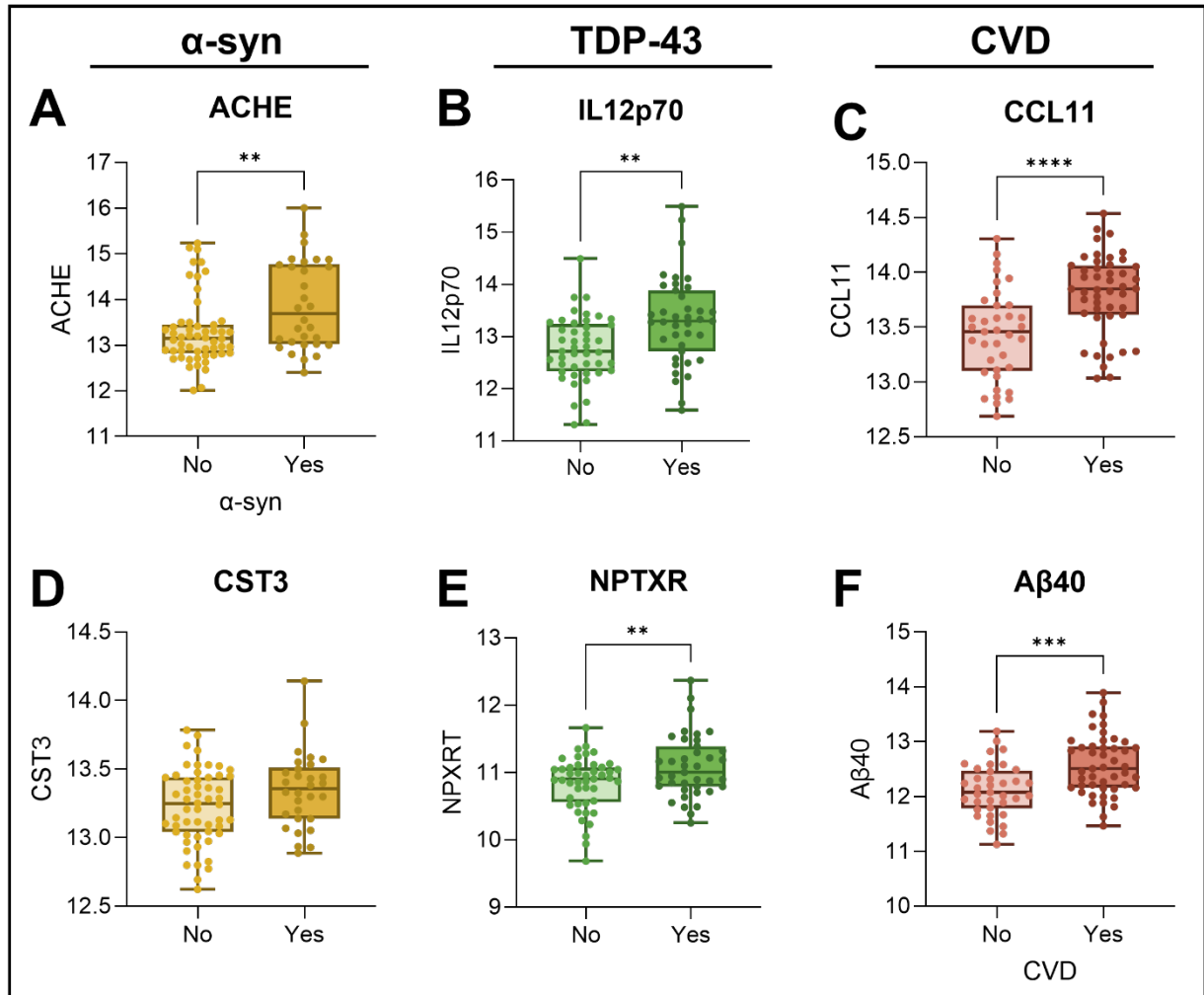


Figure 8.3 Groupwise comparison of plasma biomarkers to detect the presence of α -syn, TDP-43 and CVD. Plasma biomarkers, reported as NULISA Protein Quantification (NPQ) units, were dichotomized by the presence or absence of α -syn (yellow), TDP-43 (green), and CVD (red) in the full cohort of n=84. Comparisons done by Mann-Whitney (A) or Student's t-test (B-F) *p ≤ 0.05, **p ≤ 0.01, ***p ≤ 0.001, ****p ≤ 0.0001.

8.4. Discussion

8.4.1 Investigation of novel biomarkers in a mixed pathology cohort

In this study, we conducted an exploratory analysis using the NULISA CNS 120 assay to investigate novel biomarkers that may inform on α -syn, TDP-43, and CVD pathologies. Although the Alamar technology has been utilized to investigate differential diagnosis of clinical dementia subtypes⁴¹³⁻⁴¹⁸, this is the first study that, to our knowledge, investigates its use in detecting multiple autopsy-confirmed neuropathologies. In this exploratory analysis, we investigated the performance of this panel of biomarkers to predict the presence of α -syn, TDP-43, and CVD as distinct neuropathological diagnoses and as co-pathologies in those with significant ADNC. This analysis was further utilized to inform on common biomarkers across analyses that can be investigated in future studies to assess diagnostic utility of these biomarkers in larger cohorts. Using LASSO regression, we selected biomarkers that associated with the presence of neuropathologies and assessed the predictive power of these models using AUROC. The AUROCs for α -syn, TDP-43, and CVD all ranged from 0.75 to 0.95, showing good to excellent ability to detect these neuropathologies. We performed parallel biomarker analysis in both the full cohort and a sub-cohort composed of only participants with significant ADNC in order to determine biomarkers that associated with pathology only in the concomitant presence of AD pathology. Establishing methods to detect co-pathologies in AD is of high clinical importance as we now have highly reliable methods to detect AD pathology for the purpose of administering disease-modifying therapies¹⁰⁸. Little is known about how the presence of co-pathologies may affect the efficacy and safety of these new drugs. Therefore, this exploratory study is significant in that it highlights the importance of continued biomarker research for both clinical trial design and eventual guidance on treatment decisions.

8.4.2 Biomarkers for α -syn

α -syn is the hallmark underlying pathology in LBD. Clinical LBD diagnosis accounts for approximately 5-10% of dementia cases⁴⁰⁶. However, α -syn itself is a common co-pathology in AD⁴¹⁹. We identified a set of n=17 markers with the potential to detect any α -syn pathology in the overall cohort. However, only one biomarker was found to detect its presence as a co-pathology in those with significant ADNC. ACHE was the most promising biomarker across

both analyses to detect α -syn. There have been few studies on the potential utility of ACHE as a biomarker in LBD or PD, though ACHE activity on PET has been shown to help differentiate LBD from AD^{420,421}, and ACHE inhibitors have demonstrated better efficacy in LBD than in AD^{422,423}. We additionally looked at the detection of multiple mixed co-pathologies in those who had both α -syn and TDP-43 co-pathologies alongside AD pathology. Here, a set of n=7 markers detected this combination of pathologies. In this analysis, we identified CST3 as a marker for detection of α -syn and TDP-43 co-pathology in AD, which also emerged in the full cohort analysis for the detection of α -syn. CST3 is a cysteine protease inhibitor that has been previously genetically linked to AD⁴²⁴, but has had limited research in α -synucleinopathies. However, some studies have shown an association between CST3 and the development of dementia in those with LBD and PD^{425,426}. Interestingly, plasma α -syn itself was never selected to detect the presence of its pathology. This is a consistent finding across other studies using NULISA technology, where participants had been clinically determined to have LBD⁴¹³, or had the presence of pathology determined by the α -syn seed amplification assay^{414,415}. However, the NULISA plasma α -syn has been shown to be elevated in those with LBD compared to healthy controls⁴¹⁸, and in CSF of those with a positive α -syn seed amplification assay⁴¹⁶. This indicates that although plasma α -syn may still associate with the presence of pathology, it is unlikely to have utility in plasma to differentially diagnose against other forms of dementia.

8.4.3 Biomarkers for TDP-43

Hyperphosphorylation and abnormal cytoplasmic localization of TDP-43 occurs both in FTD and LATE. Although FTD often shows distinct clinical characteristics and has earlier onset than AD, LATE is a common co-pathology, likely due to the later onset of LATE compared to FTD⁴²⁷. To detect TDP-43 pathology, 3 biomarkers showed promise from the full cohort analysis and within the significant ADNC sub-cohort. The only consistent marker from both analyses was IL12p70, a pro-inflammatory cytokine. There is limited previous evidence supporting IL12p70 as a marker of TDP-43 pathology⁴²⁸. IL12p70 has been implicated in downstream neuroinflammatory pathways that can be triggered by TDP-43 pathology⁴²⁹. However, NPTXR identified TDP-43 in the full cohort, and detected TDP-43 and α -syn co-pathology in the AD group. NPTXR, and other synaptic biomarkers such as neuronal pentraxin 2 (NPTX2), have shown some previous associations with the progression of FTD and

amyotrophic lateral sclerosis (ALS), another TDP-43 related disorder⁴³⁰. Neuronal pentraxins play a crucial role in synaptic remodeling, which may be disrupted by TDP-43 mis-regulation, resulting in neuronal pentraxin accumulation⁴³¹. Similar to α -syn, no TDP-43 marker was found to reliably detect TDP-43 neuropathology. In other studies using NULISA technology, TDP-43 was also not found to be a marker that detected TDP-43 pathology based on the clinical definition of FTD^{413,417}. However, it did appear to have utility in detecting carrier status of common genetic mutations known to cause TDP-43 pathology⁴¹⁵. This indicates the need for further research into TDP-43 as a marker of its pathology.

8.4.4 Biomarkers for vascular neuropathologies

In this study, CVD was indicated by the presence of at least moderate arteriolosclerosis, which has been previously shown to indicate cognitive impairment^{387,388}, or presence of large, lacunar or microscopic infarcts^{387,389}. Three markers showed promise to detect CVD pathology in the full cohort, while seven markers were able to detect its presence as a co-pathology in those with significant ADNC. The overlapping markers included A β 40 and CCL11. While A β 40 was also found to inform on the staging of CVD, CCL11 was unique to its detection. A β markers have previously been shown to associate with CVD⁴³², which may be due to the contributions of CAA to the overall cerebrovascular burden⁴³³. Though it has not been investigated as a biomarker of CVD previously, CCL11 has shown utility for the prognosis of ischemic stroke⁴³⁴ and been linked to neurodegenerative processes⁴³⁵. Due to the consistent finding of A β markers to detect CVD, we also investigated biomarkers that could additionally detect at least moderate CAA in those with significant ADNC. The ability to detect CAA in those with AD pathology is also of high clinical importance due to its association with amyloid-related imaging abnormalities (ARIA) upon anti-amyloid therapy⁴³⁶. Here, we found a panel of 7 markers showed good ability to detect CAA with an AUROC of 0.85. Slit guidance ligand 2 (SLIT2), p-tau-181, and p-tau-217 displayed the strongest associations with CAA. This finding is backed by the recent findings that p-tau species associate strongly with A β pathology^{162,164}. Although no direct links have previously been established between SLIT2 and CAA, SLIT2 has a role in angiogenesis^{437,438} and has been shown to be a ligand for amyloid precursor protein⁴³⁹, indicating potential mechanistic links.

8.4.5 Considerations for future directions

This study serves as an exploratory analysis of plasma biomarkers that can aid in the detection of α -syn, TDP-43, and CVD pathology. We performed a simple binary analysis of to look at all-cause presence or absence of pathology. We found that ACHE may aid in the detection of α -syn, IL12p70 and NPTXR may aid in the detection of TDP-43, and A β 40 and CCL11 may can aid in the detection of CVD, all regardless of the presence of significant ADNC. However, our limited sample size constrains our ability to fully evaluate how overlapping co-pathologies influence these biomarker relationships, underscoring the need for replication in larger, well-characterized, autopsy-confirmed cohorts. There are many considerations that must be taken into account in the expansion of this work. Firstly, the overlapping co-pathologies make it highly difficult to derive which pathology is driving the difference in biomarker levels for highly associated pathologies. Cases with pure pathology make highly valuable additions to future research, however, as shown here, they can be incredibly rare. Therefore, future studies will require larger sample sizes that allow for more balanced representation of co-pathology combinations in both pathology-positive and pathology-negative reference groups. For example, we found ACHE to be associated with α -syn pathology even when adjusting for presence of significant ADNC, but we lack the sample size and statistical power to a) understand it's diagnostic utility in the absence of significant ADNC, and b) understand the complex nature of how multiple mixed pathology combinations of ADNC, TDP-43, and CVD collectively influence biomarker levels. A second consideration is the level of pathology we aim to detect. Here, we have evaluated detection of all-cause pathology. However, in previous work in Chapter 7, we explored markers to detect ADNC that was of a level significant enough to drive cognitive impairment. For example, investigation of TDP-43 pathology included all stages of LATE pathology, but not all LATE cases have cognitive impairment⁴⁰⁸. If the primary goal of the diagnostic biomarker is to detect pathology that contributes to cognitive impairment, sample size needs to be further increased to enrich the number of participants with severe pathology. However, if the goal is to detect all-cause co-pathologies that may impact efficacy of disease modifying therapies, this is a lesser consideration. Our results underscore a growing opportunity to leverage plasma biomarker panels for multi-pathology detection, but also reinforces the inherent limitations of small, heterogeneous cohorts. Strategic expansion of

sample size and co-pathology representation will be pivotal for advancing these markers toward meaningful clinical use.

8.4.6. Limitations

The main limitation of this study is sample size. As we investigated mixed pathologies in a total of n=84 CARD participants with autopsy data, the sample size per pathological group can be quite low, with some combinations of pathologies only existing in one participant. We acknowledge that applying the NULISA CNS Disease 120 panel across a range of neuropathologies is exploratory and will require future investigations and validation in larger autopsy cohorts. Another limitation of this study is the time between the blood sample and autopsy. Although all participants were clinically diagnosed with MCI or dementia at the time of blood draw, there is no guarantee that all the pathologies present at autopsy were present at the time of blood draw. More studies with blood drawn closer to death will be required to validate the findings presented here.

8.5. Conclusions

In this exploratory study, we investigated novel biomarkers for the detection of autopsy confirmed non-AD neuropathologies. We found that ACHE may aid in the detection of α -syn, IL12p70 and NPTXR may aid in the detection of TDP-43, A β 40 and CCL11 may aid in the detection of CVD, and p-tau and SLIT2 may have utility to detect CAA. Further analysis of these biomarkers in larger validation cohorts will be required to determine their clinical utility.

Chapter 9. Using Plasma Biomarkers to Estimate the Prevalence of Alzheimer's Disease Pathology in COMPASS-ND

9.1 Introduction

Many plasma biomarkers have recently been recognized as excellent tools to indicate multiple components of Alzheimer's disease (AD) neuropathology. The ratio of amyloid-beta 42 over 40 ($A\beta_{42}/40$) is an indicator of amyloid plaque pathology^{144,254}. Phosphorylated tau at residues 181 (p-tau-181), a phosphorylated neuronal microtubule-associated protein, can indicate neurofibrillary tangles and that tracks closely with amyloid burden in AD patients^{162,255}. Neurofilament light (NfL), is an axonal intermediate filament protein indicative of neurodegeneration^{191,376}. Glial fibrillary acidic protein (GFAP), a marker of astrocyte activation, indicates neuroinflammation^{377,378}. These biomarkers have been recognized for potential utility in AD diagnosis and staging in the recently updated ATN framework for AD¹⁰⁸. In the research context, plasma biomarkers have considerable potential to indicate AD pathology without the need for expensive positron emission tomography (PET) or invasive cerebrospinal fluid (CSF) testing. However, as many assays used in research studies do not have established cut-offs or guidelines for interpretation, researchers need to develop their own cut-offs, which requiring reference standards of autopsy, PET, or CSF to establish cut-offs and define test accuracy.

The Canadian Consortium on Neurodegeneration and Aging (CCNA) developed the Comprehensive Assessment of Neurodegeneration and Dementia (COMPASS-ND) study as a national observational cohort study⁴⁴⁰. COMPASS-ND is a rich cohort of participants ranging from those cognitively unimpaired to those diagnosed with dementia, including AD, frontotemporal dementia (FTD), Lewy body dementia (LBD), and Parkinson's disease dementia (PDD). However, CSF measures of pathology are only available on a limited number of COMPASS-ND participants and PET data are not available. In this study, we measured plasma $A\beta_{42}/40$, p-tau-181, NfL, and GFAP to provide a proxy measure of AD pathology in COMPASS-ND. We aimed to create a probability score using plasma biomarkers to determine the probability that a participant has AD pathology. We aim to create two different versions of the scoring system, one tailored for the pre-dementia stages and one tailored for those with clinical dementia.

Here we describe a robust procedure used to develop plasma probability scores (PPS), including initial validation in an external cohort with autopsy data and subsequent validation in a sub-cohort of COMPASS-ND participants with CSF AD testing. The overall objective of this study is to determine the prevalence of AD pathology in the clinical sub-groups of COMPASS-ND using PPS to provide COMPASS-ND researchers a proxy measure of AD pathology that can be utilized in future analyses of COMPASS-ND data.

9.2 Methods

9.2.1 Ethical approval

This study was reviewed and approved by the University of British Columbia (UBC) Clinical Research Ethics Board (REB), Vancouver, Canada (H21-01783). The COMPASS-ND study was reviewed and approved by the Research Ethics Board of the CIUSSS West-Central Montreal Board, Montreal, Canada (MP-05-2016-384).

9.2.2 COMPASS-ND Participants and samples

COMPASS-ND is an observational, multi-center, national study of participants living with or at risk of developing dementia. The full study protocol has been published by Chertkow et al.⁴⁴⁰. Participants are recruited into the study based on clinical diagnostic groups of cognitively unimpaired, subjective cognitive impairment (SCI), mild cognitive impairment (MCI), AD, LBD, FTD, and PD. Diagnosis is confirmed at the end of the clinical visit, where a diagnostic evaluation is made by the clinician (e.g., neurologist, geriatrician, physician) informed by the clinical test battery. After completion of baseline data collection, a second diagnostic evaluation was conducted to reappraise the baseline diagnosis by COMPASS-ND Diagnostic Committee members. Blood samples were collected in EDTA vacutainers and centrifuged at room temperature for 15 minutes at 3000g. Plasma was separated and stored in a -80 Celsius freezer within a maximum of 24 hours from collection. A sub-cohort of COMPASS-ND participants underwent lumbar puncture to collect CSF. CSF collected in polypropylene tubes

and stored in a -80 Celsius freezer. CSF A β 42 and p-tau-18 were analyzed on the Roche Elecsys at St. Paul's Hospital in Vancouver, Canada, following the manufacturer's protocol.

9.2.3 Biomarker Analysis

The ratio of p-tau-181/A β 42 was utilized as the CSF test to determine AD positivity, using a cut-off of <0.025 pg/mL for the first generation assay and <0.024 pg/mL for the second generation assay. Plasma biomarker concentrations in all cohorts were quantified on the single-molecule array enzyme linked immunoassay (Simoa) HD-X platform from Quanterix (Billerica, MA). Plasma GFAP, NfL, A β 40, and A β 42 were measured using the Neurology-4-plex E advantage assay (N4PE; catalogue #103670), and plasma p-tau-181 was measured using the ptau-181 V2 advantage assay (catalogue #103714) following the manufacturer's protocol. Each assay included a 7 or 8-point calibrator curve, two internal kit controls, and three plasma controls. All specimens were assayed in duplicate and the mean value was reported as the result. For all analytes, quality control information was documented across all runs. COMPASS-ND samples were analyzed using N4PE lot 503212, and p-tau-181 lot 503199. Standard curves had an average absolute error of 3.3-7.0%. Kit controls had an average absolute error of 4.8-13.2% and an average inter-run CV of 6.9-8.6%. For plasma controls analyzed in duplicate across every plate, mean values had an average inter-plate CVs were between 6.5-11.3%. Average intra-plate CVs were between 4.1-5.9%.

Biomarker results from the COMPASS-ND cohort were compared to results from two other cohorts that are described fully in previous Chapters. First, the Canadian Health Measures Survey (CHMS), used to create population-based reference curves, and second, the UBC Clinic for Alzheimer's and related disorders (CARD) which included participants with autopsy investigations. N=2,500 samples from CHMS were analyzed, with n=900 being analyzed using N4PE lot 503105 and p-tau-181 lot 502923, and n=1,600 being analyzed using N4PE lot 503212 and p-tau-181 lot 503199. N=78 CARD samples were analyzed using N4PE lot 503212 and p-tau-181 lot 503199. Cross lot analysis was conducted for across lots. To assess associations between assay results, Spearman correlation was performed. To assess the difference between assays, Bland-Altman analysis was performed to report the average bias. Comparing the first CHMS lot to the COMPASS-ND lot, all analytes had a rho >0.9, and the

average absolute bias was 11%, ranging from 3.7-22.1%. Comparing the second CHMS/CARD lot to the COMPASS-ND lot, all analytes had a $\rho > 0.9$, and the average absolute bias was 10%, and ranged from 0.1-13.5%. Detailed results can be found in **Chapter 4, Table 4.2**. No cross-lot harmonization was performed as the bias for all analytes comparing the second CHMS/CARD lot to the COMPASS-ND lot was under 15%. The one bias $> 15\%$ for p-tau-181 in the comparison between the first CHMS lot to the COMPASS-ND lot did not merit harmonization, due to lot variation being accounted for in combining lots for reference curve construction.

9.2.3 Canadian Health Measures Survey (CHMS): Age-specific reference curves

Reference curves were previously generated in Chapter 2 to encapsulate the normative range of plasma biomarkers from age 3 to 79 years old using $n=2,500$ CHMS samples²⁶². Here, we reported the 5th, 15th, 25th, 50th, 75th, 85th, and 95th percentiles of the population by performing a GAMLSS regression of biomarker concentration across the lifespan. We first extended the trajectory of the reference curves through interpolation to provide reference values for those 80 years and older. For this, a simple linear regression was performed using the concentrations that corresponded to each percentile at each age from 70 to 79. For each percentile, this regression formula was used to interpolate the extension of the curves and report expected concentration percentiles at each age from 80-95 years old. We investigated multiple percentiles as potential cut-offs. For A β 42/40 the 5th, 15th, and 25th percentiles were investigated as potential cut-offs, and for p-tau-181, NfL, and GFAP, the 75th, 85th, and 95th percentiles were investigated as potential cut-offs. Assessment of potential cutoffs was conducted visually and by assessing the sensitivity and specificity of individual cut-offs using the validation cohort with CSF as a reference standard. A maximum of two cut-offs per analyte were incorporated into the final scoring system, requiring at least 80% specificity or 70% sensitivity.

9.2.4 Clinic for Alzheimer's and Related Dementias (CARD): Autopsy-validated cohort

The CARD clinic is British Columbia's tertiary referral dementia clinic, which enrolls patients into their research biobank for future use. All patients are clinically diagnosed by neurologists and geriatricians who have expertise in dementia assessments. Participants in this study included those diagnosed with dementia at the time of blood collection. Demographic data and blood samples were collected during participants' clinical assessments. Blood samples were collected in EDTA vacutainers and centrifuged at 4 Celsius for 15 minutes at 2500g. Plasma was separated and stored in a -80 Celsius freezer within a maximum of 24 hours from collection. Autopsies were performed on participants referred from CARD with informed consent, and conducted by two neuropathologists. All participants underwent a standardized neuropathological examination³⁸². Brain tissue was formalin-fixed and paraffin-embedded, with blocks cut at 5 microns. Staining included hematoxylin and eosin (HE), HE combined with Luxol fast blue (HE/LFB), modified Bielschowsky silver, Gallyas silver, and Congo red stains. Using the Dako Omnis automated staining system, immunohistochemistry (IHC) was performed with primary antibodies against A β (DAKO; 1:100 with initial incubation for 3 hours at room temperature), and hyperphosphorylated tau (clone AT-8; Innogenetics; 1:2000 after microwave antigen retrieval). The presence of AD pathology was determined by an ABC score that indicates at least intermediate AD pathology. For participants who did not have Braak scoring due to the presence of primary tauopathies, scoring was estimated using both Thal (A) and CERAD (C), where a score of A1C2 or greater was considered intermediate AD pathology^{25,26}. Individual analyte cut-offs in this cohort were determined using a receiver operating curve (ROC) analysis and Youden's index.

9.2.5 Statistical analysis

Groupwise comparisons were performed between participants for demographics, study measures and biomarker concentrations using a Mann-Whitney U test for continuous variables or a Fisher's exact test for categorical variables. Biomarkers in the COMPASS-ND CSF sub-cohort were compared between those in pre-dementia and dementia stages and CSF test status using a two-way ANOVA with multiple comparisons. AD PPS were generated, validated, and

assessed separately for participants in pre-dementia stages and those clinically diagnosed with dementia. Pre-dementia stages utilized probability scores generated from age-specific reference curves, and dementia utilized probability scores derived from autopsy-confirmed cut-offs. To generate AD PPS, we applied scores to biomarker cut-offs in both reference curves and autopsy cut-offs, assigning biomarkers with single cut-offs a score of 0-1, and biomarkers with two cut-offs a score of 0-2. Composite scores were assessed to estimate the prevalence of AD pathology of the validation cohort per score. All scores with <30% were combined into the low AD probability category, scores with between 30-70% were combined into the intermediate AD probability category, and scores with >70% were combined into the high AD probability category. Scoring systems were applied to the COMPASS-ND cohort to report the estimated prevalence of AD across clinical diagnostic groups.

9.3 Results

9.3.1 COMPASS-ND cohort and analysis

A total of n=998 COMPASS-ND participants were included in this study. Of these, n=248 had a clinical diagnosis of dementia and n=750 were within the pre-dementia spectrum encompassing cognitively unimpaired to mild cognitive impairment (MCI). Sample sizes for individual clinical diagnostic groups can be found in **Figure 9.1**. The pre-dementia and dementia groups were analyzed separately using distinct approaches to determine the probability of having AD pathology. In the pre-dementia group, we tested whether plasma biomarkers deviated from previously generated population normative levels in a manner suggestive of AD pathological change. In the dementia group, we investigated whether plasma biomarkers could indicate whether AD pathology is contributing to dementia. A flowchart of the cohorts utilized and the analysis plan is shown in **Figure 9.1**.

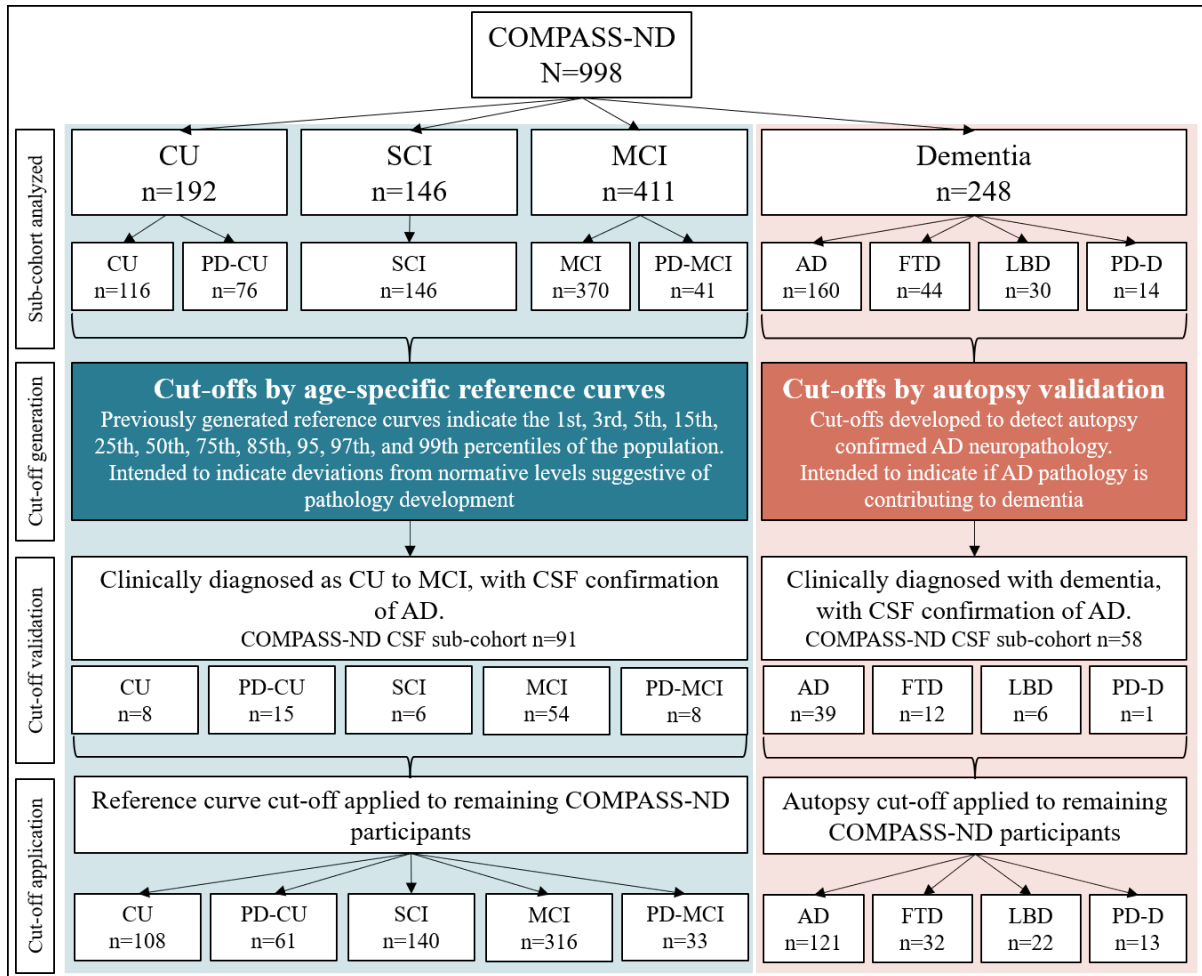


Figure 9.1 Flow chart of study design and cohorts utilized. Shading indicates the division of the COMPASS-ND cohort used to denote the different cut-offs used to develop and validate AD plasma probability scores. Dark blocks indicate the method of cut-off development used in score development, which were generated in cohorts external to COMPASS-ND. Abbreviations: CU, cognitively unimpaired; SCI, subjective cognitive impairment; MCI, mild cognitive impairment; PD, Parkinson's disease; AD, Alzheimer's disease; FTD, frontotemporal dementia; LBD, Lewy body dementia; PD-D, Parkinson's disease dementia.

9.3.2 COMPASS-ND CSF sub-cohort

Within COMPASS-ND, a sub-cohort of n=150 participants had CSF collected for biomarker analysis. CSF p-tau-181/A β 42 was analyzed and used to determine the presence of AD pathology. Of n=91 participants in the pre-dementia stages, n=39 (43%) were positive on their

AD CSF test. Of the n=59 with dementia, n=43 (73%) were positive on their AD CSF test. The breakdown by clinical diagnosis is displayed in **Figure 9.2**.

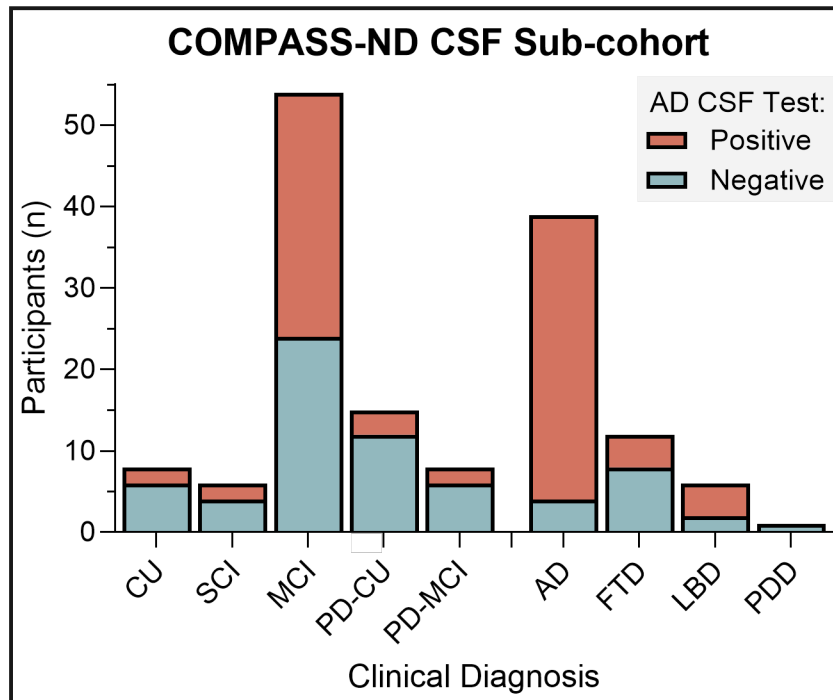


Figure 9.2 Histogram of COMPASS-ND participants with CSF biomarker testing. Those with a positive CSF p-tau-181/A β 42 test were considered to be positive for AD and are red, while the remaining participants are blue. Abbreviations: CU, cognitively unimpaired; SCI, subjective cognitive impairment; MCI, mild cognitive impairment; PD, Parkinson's disease; AD, Alzheimer's disease; FTD, frontotemporal dementia; LBD, Lewy body dementia.

As we planned to use the CSF sub-cohort as a validation cohort for PPS generation, we investigated how well the CSF sub-cohort represented the whole COMPASS-ND cohort in **Table 9.1**. Within pre-dementia and dementia groups, we compared the CSF sub-cohort to the remaining COMPASS-ND participants. In both groups, participants with CSF biomarker data were significantly older. In the pre-dementia group, there was a higher prevalence of those diagnosed with Parkinson's disease (PD) in the CSF sub-cohort. However, in the dementia group, there were significantly lower numbers of those diagnosed with PD in the CSF sub-cohort. All other demographic and clinical variables did not significantly differ between sub-cohorts. With respect to plasma biomarkers, p-tau-181 was slightly higher in COMPASS-ND

participants without CSF data, but there were no other significant differences for any of the other biomarkers. We further broke down these analyses by individual clinical diagnostic groups in **Appendix 7**.

Table 9.1 Demographics and clinical data for COMPASS-ND cohort. Cohort divided by those with a dementia diagnosis, including Alzheimer's disease, frontotemporal dementia, Lewy body dementia, and Parkinson's disease dementia, and those in pre-dementia stages of cognitively unimpaired, subjective cognitive impairment, and mild cognitive impairment. Each group is further divided based on the availability of CSF biomarkers to indicate how representative the CSF sub-cohorts of the remaining COMPASS-ND participants. Analysis of continuous variables was by the Mann-Whitney test, and categorical variables were analyzed by Fisher's exact test. Abbreviations: APOE4, apolipoprotein E4; BMI, body mass index; MoCA, Montreal Cognitive Assessment.

	Pre-dementia (n=749)			Dementia (n=248)		
CSF Sub-cohort	Yes	No	p-value	Yes	No	p-value
N, %	89 (11.9)	660 (88.1)		60 (24.2)	188 (75.8)	
Demographics						
Age, years (median, IQR)	72 (66–76)	69 (64–72)	0.0003	76 (68–80)	71 (65–77)	0.0353
Female Sex (n, %)	348 (52.9)	38 (42.2)	0.0716	71 (37.4)	20 (34.5)	0.7567
Male Sex (n, %)	310 (47.1)	52 (57.8)		119 (62.6)	38 (65.5)	
APOE4 Carrier (n, %)	204 (32.0)	35 (41.2)	0.1101	88 (48.4)	30 (53.6)	0.5426
Clinical						
BMI (median, IQR)	22.3 (19.7–25)	21.6 (19.2–24.3)	0.2330	21.7 (19.2–24.9)	22.7 (19.3–24.7)	0.6630
Parkinsons disease (n, %)	94 (14.3)	23 (25.3)	0.0127	1 (7.1)	13 (92.9)	
Hypertension (n, %)	246 (37.4)	30 (33.0)	0.4868	78 (41.3)	18 (31.0)	0.1700
Kidney disease (n, %)	14 (2.1)	2 (2.2)	>0.9999	3 (1.6)	1 (1.7)	>0.9999
MoCA (median, IQR)	26 (23–28)	25 (22–27)	0.3765	18 (15–22)	18.5 (16–22)	0.3489
Plasma Biomarkers						
Aβ42/40 (median, IQR)	0.0659 (0.0574–0.0728)	0.0642 (0.0563–0.0726)	0.5524	0.06 (0.0534–0.0662)	0.0609 (0.0532–0.068)	0.9441
p-tau-181, pg/mL (median, IQR)	2.14 (1.56–3.18)	2.14 (1.54–3.6)	0.6405	3.26 (2.26–4.42)	4.12 (2.52–5.24)	0.0455
NfL, pg/mL (median, IQR)	19.8 (15–26.3)	19.1 (14.3–25.5)	0.3808	28.3 (21.8–38.1)	30.3 (23.7–37.3)	0.4751

GFAP, pg/mL (median, IQR)	124 (85.8–178)	119 (80.8–174)	0.5723	179 (126–254)	205 (129–252)	0.5787
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To assess the performance of plasma biomarkers in the CSF sub-cohort, we investigated how biomarker concentrations differed between those with a positive versus negative AD CSF test in both pre-dementia and dementia groups (**Figure 9.3**). We found that in both pre-dementia ($p=0.0016$) and dementia ($p=0.0181$) groups, plasma A β 42/40 was lower in those with a positive AD CSF test. Plasma concentrations of both p-tau-181 and GFAP were higher in participants with positive CSF tests in both pre-dementia (p-tau-181 $p=0.0004$, GFAP $p<0.0001$) and dementia (p-tau-181 $p=0.0001$, GFAP $p<0.0001$) groups. Furthermore, plasma p-tau-181 and GFAP levels were higher in those with dementia than those in the pre-dementia stages with positive AD CSF tests (p-tau-181 $p=0.0036$, GFAP $p=0.0004$). These differences suggest all three plasma biomarkers perform well to detect AD pathology, with the elevations of p-tau-181 and GFAP in those with dementia validating the separation of groups for analysis. NfL, on the other hand, was only able to differentiate those with positive AD CSF tests in the dementia groups, and was higher in those with negative AD CSF tests ($p=0.0023$). NfL was also significantly higher in those with dementia compared to pre-dementia stages in those with negative AD CSF tests ($p<0.0001$), indicating potentially greater utility to detect non-AD pathology.

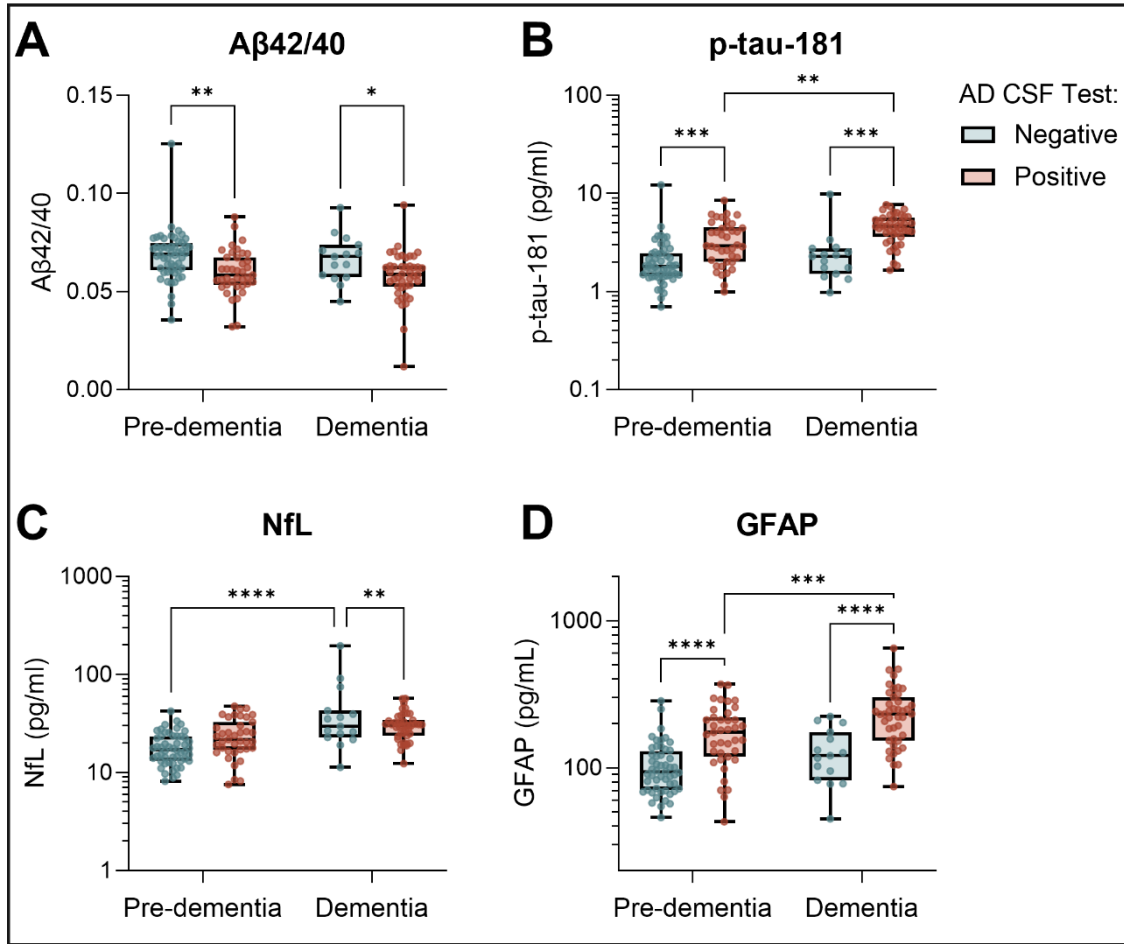


Figure 9.3 Plasma biomarkers in COMPASS-ND participants with CSF testing (n=151). CSF p-tau-181/Aβ42 was utilized as the reference standard to classify a negative or positive AD CSF test. Plasma biomarkers were compared across clinical grouping of those in pre-dementia stages of cognitively unimpaired, subjective cognitive impairment, and mild cognitive impairment and those with a dementia diagnosis, including Alzheimer's disease (AD), frontotemporal dementia, Lewy body dementia, and Parkinson's disease dementia. As well as being compared across AD CSF status. Analysis was conducted by 2-way ANOVA with multiple comparisons. *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

9.3.3 Normative reference curves to indicate pathology probability in pre-dementia

For participants in the pre-dementia stages, we aimed to determine if plasma biomarker levels that deviated from normative levels are suggestive of pathological development. To do this, we utilized reference curves, previously generated from Chapter 3 that describe representative age-specific percentiles of the Canadian population from ages 3-79, to create a reference curve

PPS (RC-PPS) (**Figure 9.4 A-D**). Here, we aimed to determine the deviation threshold for each biomarker to inform on the probability of AD pathology development. Plasma biomarker data from COMPASS-ND participants in the pre-dementia CSF sub-cohort were overlaid on the reference curves to visually assess where those with a positive AD CSF test fell on the curves relative to those with negative tests (**Figure 9.4 E-H**). This facilitated setting a percentile threshold for each biomarker. For A β 42/40 using the population 25th percentile as a cut-off provided a sensitivity of 28% and specificity of 88%. For p-tau-181, using the 95th percentile as a cut-off provided a sensitivity of 31% and specificity of 94%, and using the 75th percentile as a cut-off provided a sensitivity of 59% and specificity of 81%. For GFAP, using the 95th percentile as a cut-off provided a sensitivity of 36% and specificity of 90%, and using the 75th percentile as a cut-off provided a sensitivity of 74% and specificity of 67%. A composite biomarker score was then developed. The final scoring system for converting reference curve percentiles into the RC-PPS is displayed in **Figure 9.4 (I-L)**. Specifically, we assigned a score of 1 if A β 42/40 concentration was under the 25th percentile, and p-tau-181 and GFAP both received a score of 1 if over the 75th percentile and a score of 2 if over the 95th percentile. The final AD pathology probability score ranges from 0-5.

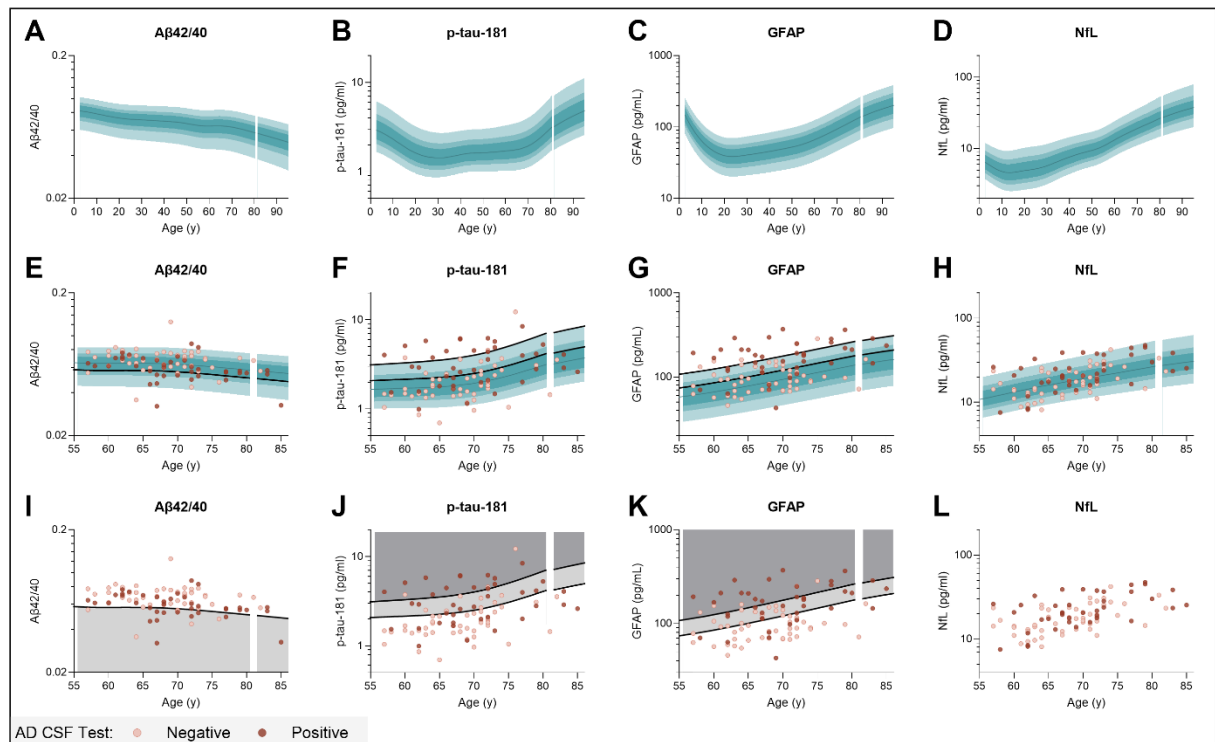


Figure 9.4 Deriving an AD plasma probability score from plasma biomarkers contextualized against normative age-specific reference curves. (A-D) Reference curves developed from n=2,500 normative participants from the Canadian Health Measures Study from ages 3-79 years old. Reference curves for age ≥ 80 were interpolated from curves of ages 70 to 79. Displayed percentiles include 5th (first shaded gradation from bottom), 15th (second shaded gradation from bottom), 25th (third shaded gradation from bottom), 50th (central solid line), 75th (fourth shaded gradation from bottom), 85th (fifth shaded gradation from bottom), 95th (end of shaded gradations). (E-H) COMPASS-ND CSF sub-cohort overlaid onto reference curves. Black dashed lines indicate percentiles utilized as cut-points for plasma biomarkers in the lower panel. (I-L) Conversion of percentiles into age-specific cut-offs for the plasma biomarker scoring system. Lightest grey shading indicates a score of 1 for that biomarker, and the darkest grey indicates a score of 2. Final AD plasma probability score ranges from 0-5. (I) A β 42/40 concentration receives a score of 1 if under the 25th percentile, (J-K) p-tau-181 and GFAP both receive a score of 1 if over the 75th percentile and a score of 2 if over the 9th percentile. (L) NfL concentrations were not utilized in the final score.

We then tested the utility of the RC-PPS to predict AD pathology in the COMPASS-ND CSF sub-cohort (**Figure 9.5**). Of the participants who received a score of 0, n=6 (20%) participants had a positive CSF test, and n=5 (22%) participants with a score of 1 had a positive AD CSF test. In those with a score of 2, n=13 (59%) had a positive AD CSF test. In participants with a score of 3, n=4 (67%) had a positive AD CSF test, and 100% of participants with a score of 4 or 5 had positive AD CSF tests. We then combined RC-PPS into groups that would describe low (scores of 0-1), intermediate (score of 2) and high (scores of 3-5) probability of AD based on a positive CSF test. We then assessed the sensitivity and specificity of low and high cut-offs in each group (**Table 9.2**), where a low cut-off differentiates those with low vs intermediate/high likelihood of AD pathology, and the high cut-off differentiates those with low/intermediate vs high likelihood of AD pathology. The low cut-off had a sensitivity of 72%, with n=11 false negative results. The high cut-off had a specificity of 90%, with only 2 false positives. Both cut-offs had a better specificity than sensitivity.

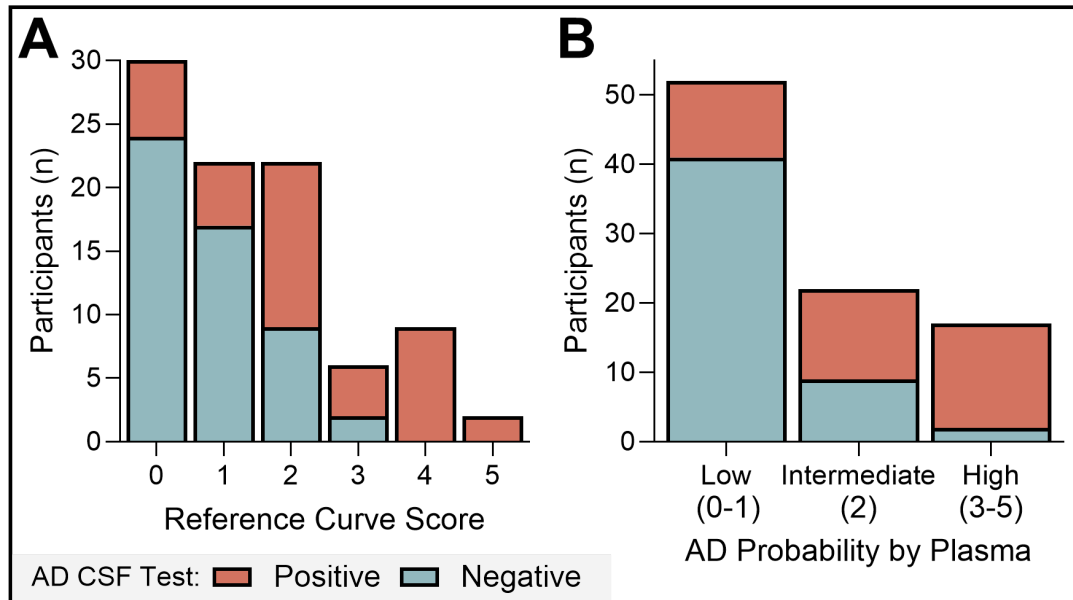


Figure 9.5 Conversion of normative reference curve scores into an reference curve plasma probability score (RC-PPS) using participants in the COMPASS-ND CSF sub-cohort without dementia (n=91). Participants in red have been determined to have AD pathology, using CSF p-tau-181/A β 42 as the reference standard. (A) Reference curve scores are denoted by the following metrics: A β 42/40 concentration receives a score of 1 if under the 25th percentile, and p-tau-181 and GFAP both receive a score of 1 if over the 75th percentile and a score of 2 if over the 9th percentile. (B) Those with low AD probability had a plasma score of 0-1, intermediate a score of 2 and high a score of 3-5

Table 9.2 Calculation of sensitivity and specificity for RC-PPS. Low cut-off indicates the ability to differentiate those with low vs intermediate/high likelihood of AD pathology. Those with low AD probability had a plasma score of 0-1, intermediate a score of 2 and high a score of 3-5. High cut-off indicates the ability to differentiate those with low/intermediate vs high likelihood of AD pathology.

Low Cut-off		CSF +	CSF -	High Cut-off		CSF +	CSF -
RC-PPS	Intermediate/High	28	11	RC-PPS	High	15	2
	Low	11	41		Low/Intermediate	24	50
Sensitivity		72%		Sensitivity		38%	
Specificity		79%		Specificity		96%	

Following validation of the RC-PPS in the CSF subcohort, we next applied it to the remaining participants in COMPASS-ND who were in the pre-dementia stages (**Figures 9.6 and 9.7**). We used this scoring system to determine the probability of AD pathologic change across all clinical diagnostic groups in the pre-dementia stages. The percentage of participants with a high probability of AD pathologic change increased across the spectrum of cognitively unimpaired to MCI, in both those with and without PD. Having an intermediate probability of AD pathologic change was more varied across groups, ranging from 11-24%. Participants with a low probability of AD pathologic change were the most prevalent in all groups, ranging from 61-64% in those with MCI to 74-77% in those who were cognitively unimpaired.

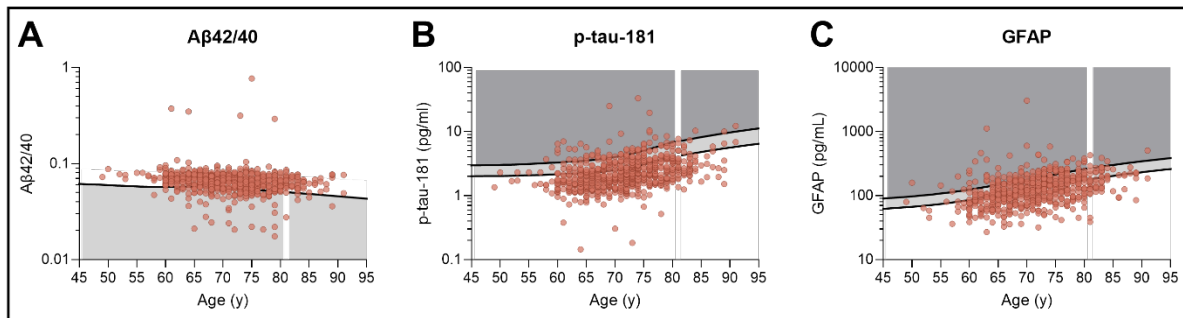


Figure 9.6 Plasma biomarkers in COMPASS-ND participants in pre-dementia stages of being cognitively unimpaired, subjective cognitive impairment, and mild cognitive impairment. Shading indicates numbers added to the RC-PPS. Light grey shading indicates a score of 1 for that biomarker, and dark grey indicates a score of 2. The final RC-PPS ranges from 0-5. (A) A β 42/40 concentration receives a score of 1 if under the 25th percentile, (B-C) p-tau-181 and GFAP both receive a score of 1 if over the 75th percentile and a score of 2 if over the 9th percentile.

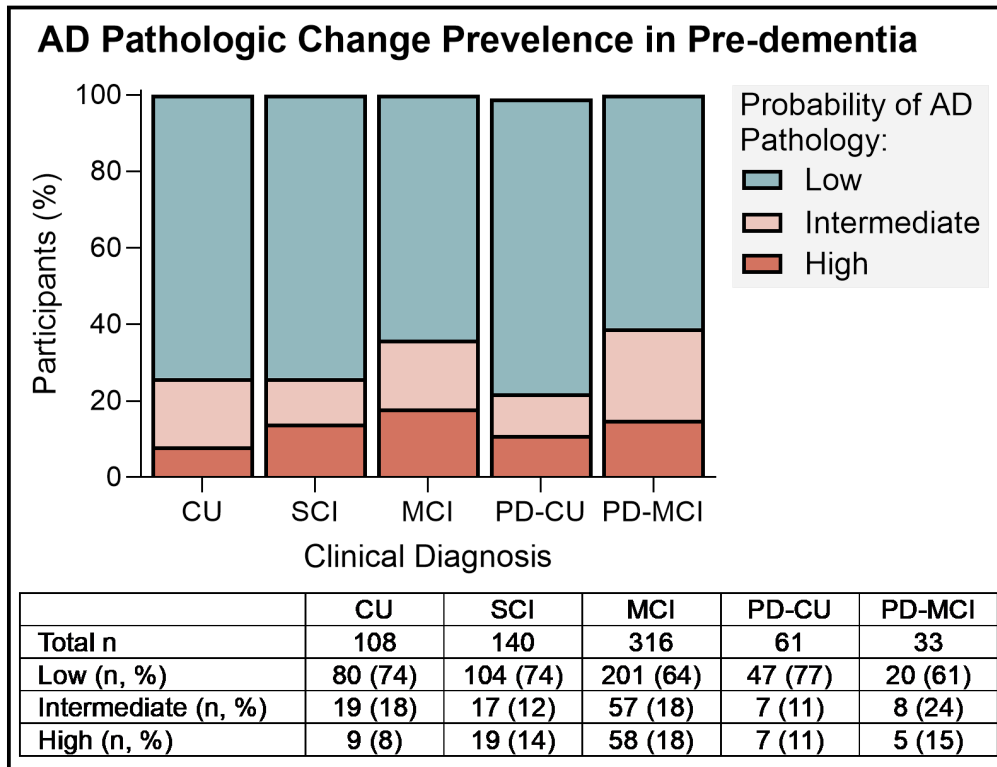


Figure 9.7 Estimated prevalence of AD pathologic change based on RC-PPS in COMPASS-ND participants in pre-dementia stages. Plasma probability score derived from normative reference curves. The histogram displays the percentage of participants in each clinical diagnostic group with low, intermediate, and high likelihood of underlying AD pathologic change. The table below indicates the exact sample size. Abbreviations: CU, cognitively unimpaired; SCI, subjective cognitive impairment; MCI, mild cognitive impairment; PD, Parkinson's disease.

9.3.4 Autopsy-derived cut-offs to indicate the probability of AD pathology

In COMPASS-ND participants who were clinically diagnosed with dementia, we aimed to determine whether plasma biomarkers could inform on the likelihood of AD neuropathologic change (ADNC) driving cognitive impairment. To do this, we utilized an independent autopsy cohort from the UBC CARD clinic to establish an ADNC-PPS for those diagnosed with clinical dementia. CARD participants were diagnosed with dementia and had plasma collected during life, followed by an autopsy examination of their neuropathology. We first evaluated comparability between CARD and COMPASS-ND demographic variables and plasma biomarker concentrations (**Table 9.3**). CARD participants were significantly younger than

COMPASS-ND participants, yet had lower A β 42/40 and higher p-tau-181 and NfL concentrations.

Table 9.3 Demographics and plasma biomarker levels in COMPASS-ND participants vs CARD participants clinically diagnosed with dementia. Analysis of continuous variables was by the Mann-Whitney test, and categorical variables were analyzed by Fisher's exact test.

	COMPASS-ND	CARD	p-value
N, %	248	78	
Demographics			
Age, years (median, IQR)	75 (68-79)	69 (58-76)	<0.0001
Female Sex (n, %)	91 (37)	35 (45)	0.2303
Male Sex (n, %)	157 (63)	43 (55)	
Plasma Biomarkers			
A β 42/40 (median, IQR)	0.0599 (0.0534-0.0669)	0.0528 (0.0458-0.0595)	<0.0001
p-tau-181, pg/mL (median, IQR)	3.42 (2.31-4.66)	3.86 (2.41-6.28)	0.0410
NfL, pg/mL (median, IQR)	29.2 (22.3-38.1)	43.0 (32.4-79.8)	<0.0001
GFAP, pg/mL (median, IQR)	185 (127-254)	211 (128-308)	0.1685

A full description of CARD participant neuropathological workup can be found in Chapters 7 and 8. Here, we focused on their AD neuropathological scores. We utilized CARD participant ABC scores to determine if they had none, low, intermediate, or high AD neuropathology, based on established neuropathological guidelines. Using the ABC system, intermediate and high levels of neuropathology are considered sufficient to contribute to cognitive impairment. Assessment of plasma biomarker concentrations across ABC scores in CARD participants revealed that plasma A β 42/40 was lower in those with intermediate ($p=0.0008$) and high ($p<0.0001$) levels of AD pathology compared to those with none (**Figure 9.8**). Plasma p-tau-181 was significantly higher in those with high levels of AD pathology compared to those with none ($p<0.0001$), low ($p=0.0265$) and intermediate ($p=0.0368$) pathology. Plasma GFAP was also significantly higher in those with high levels of AD pathology compared to those with none ($p=0.0007$) and low ($p=0.0059$) AD pathology. Plasma NfL did not significantly differ

across groups. Using these results, we can determine a threshold of pathology to develop cut-offs for plasma biomarkers. For A β 42/40 and GFAP, a cut-off was developed to differentiate those with no/low vs intermediate/high AD pathology. For p-tau-181, 2 cut-offs were developed, namely, a low cut-off to differentiate those with no/low vs intermediate/high AD pathology, and a high cut-off to differentiate those with no/low/intermediate vs high AD pathology. All cut-offs were generated using ROC analysis and Youden's index. Similar to our earlier reference curve analysis, ADNC-PPS was applied to indicate how biomarker data from participants compared to cut-offs. For plasma A β 42/40, participants with a concentration lower than 0.0559 received a score of 1. For plasma p-tau-181, participants with a concentration higher than 2.82 pg/mL received a score of 1 and participants with a concentration over 4.58 pg/mL received a score of 2. For plasma GFAP, participants with a concentration higher than 155.6 pg/mL received a score of 1. The composite scores ranged from 0-4, and were converted to an ADNC-PPS where those with a score of 0-1 were classified as having a low probability of AD pathology, a score of 2 having an intermediate probability, and those with a score of 3-4 having a high probability of having AD pathology.

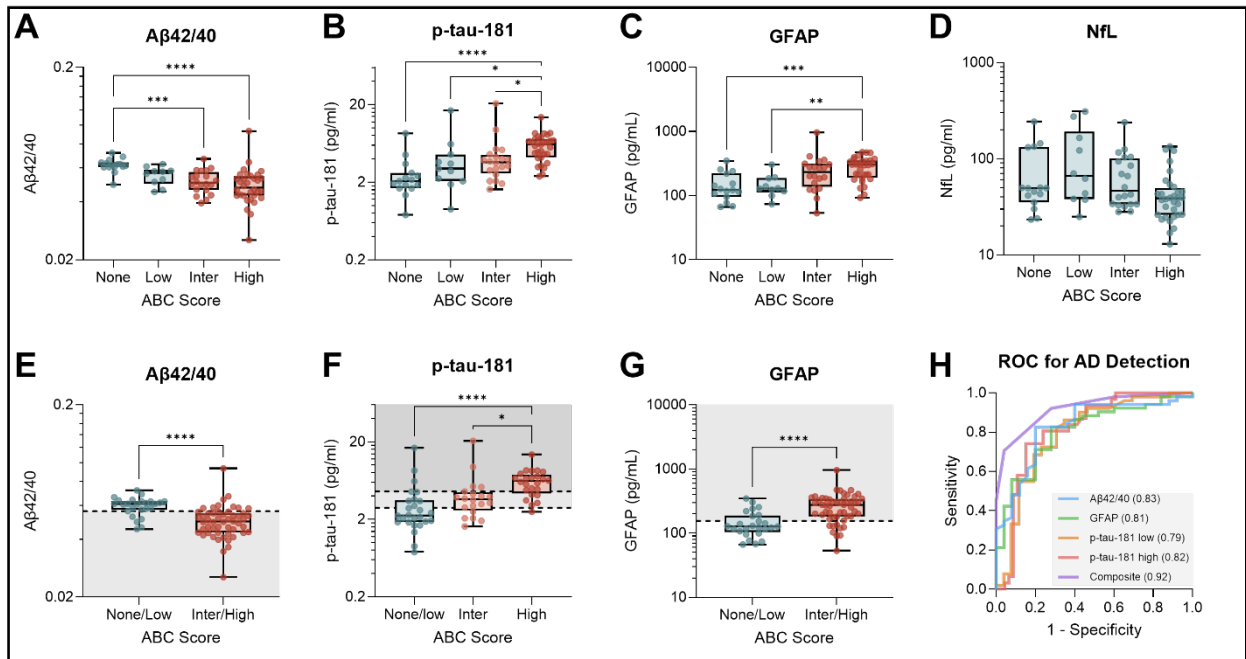


Figure 9.8 Development of plasma biomarker cut-offs to detect autopsy-confirmed AD pathology. (A-D) Plasma biomarkers across ABC scores for AD neuropathology to define each biomarker's ability to differentiate pathology levels (n=78). Colour coding is used to indicate how groups will be collapsed for cutoffs. (E-G) Plasma biomarker cut-off across combined groupings of ABC scores. Cut-offs developed by Youden's index are indicated by dashed lines. Conversion of cut-offs into plasma biomarker scoring system indicated by shading. Light grey shading indicates a score of 1 for that biomarker, and dark grey indicates a score of 2. Final AD PPS for symptomatic participants range from 0-4. Analysis for A to G by Mann Whitney for two groups of Kruskal Wallis with Dunns multiple comparisons for three or more groups, *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001. (H) Area under the receiver operating curve (AUROC) analysis for plasma biomarkers to differentiate ABC score groups displayed in E to G. p-tau-181 low indicates split of none/low vs intermediate/high, while p-tau-181 high indicates split of none/low/intermediate vs high. Composite score derived from giving 1 point for each cut-off for a score of 0-4. AUROC values are displayed in the legend of the figure.

This ADNC-PPS was applied to both participants from CARD and the COMPASS-ND CSF sub-cohort clinically diagnosed with dementia, utilizing their respective reference standards to assess the utility of the test (**Figure 9.9**). In CARD participants, n=4 (18%) participants who had a low ADNC-PPS had AD pathology on autopsy, n=11 (65%) participants who had an intermediate ADNC-PPS had AD pathology on autopsy, and n=36 (97%) participants who had a high ADNC-PPS had AD pathology on autopsy. In CSF COMPASS-ND participants with dementia, n=9 (41%) participants who had a low ADNC-PPS had a positive AD CSF test, n=10 (83%) participants who had an intermediate ADNC-PPS had a positive AD CSF test, and n=24

(100%) participants who had a high ADNC-PPS had a positive AD CSF test. The utility of the ADNC-PPS in participants with dementia was validated against autopsy confirmation and the CSF test by assessing the sensitivity and specificity of the low and high cut-offs (**Table 9.4**). The low cut-off indicates the ability to differentiate those with low vs intermediate/high likelihood of AD pathology, and the high cut-off indicates the ability to differentiate those with low/intermediate vs high likelihood of AD pathology. Compared to autopsy confirmation, the low cut-off had a sensitivity of 92%, with n=4 false negative results, and the high cut-off had a specificity of 90%, with only 1 false positive. Compared to CSF confirmation, the low cut-off had a sensitivity of 79%, with n=9 false negative results, and the high cut-off had a specificity of 100%. Overall, the ADNC-PPS showed better sensitivity when validated against autopsy confirmation, but better specificity when validated against CSF testing.

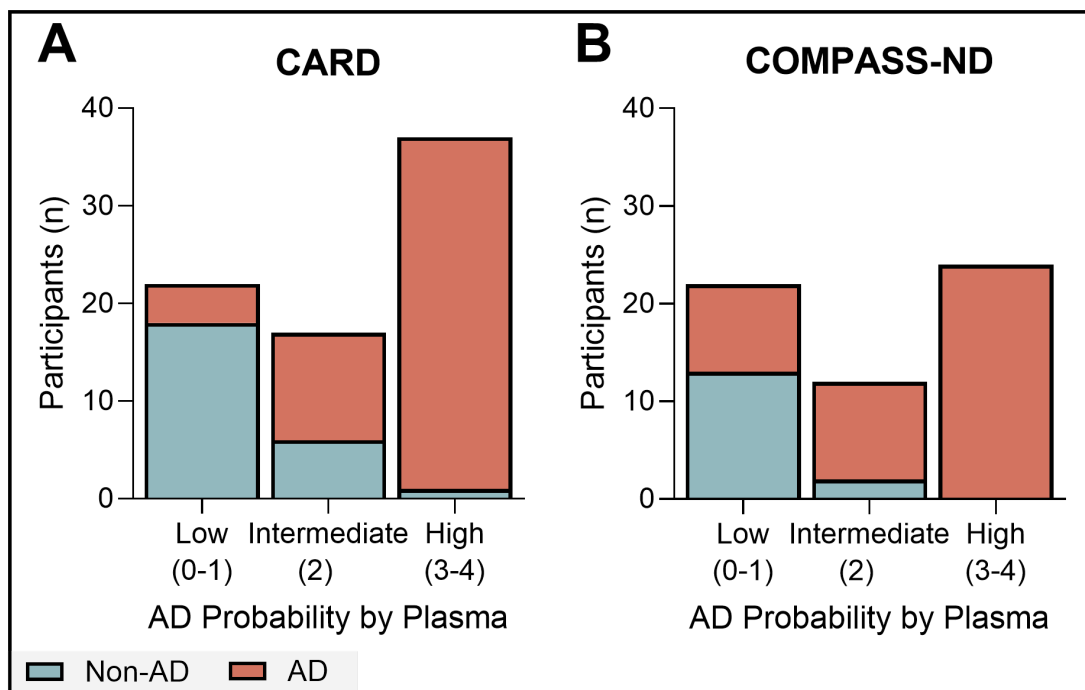


Figure 9.9 Validation of ADNC-PPS for the likelihood of AD pathology in those with dementia. Participants in red have been determined to have AD by the designated reference standard. **(A)** Participants from the CARD clinic, using autopsy-confirmed neuropathology as the reference standard (n=78). **(B)** Participants from the COMPASS-ND cohort, using CSF p-tau-181/A β 42 as the reference standard (n=58). Tables below histograms indicate the sensitivity and specificity of the cut-offs for both validation cohorts. Those with low AD probability had a plasma probability score of 0-1, intermediate a score of 2 and high a score of 3-4.

Table 9.4 Calculation of sensitivity and specificity for ADNC-PPS. Low cut-off indicates the ability to differentiate those with low vs intermediate/high likelihood of AD pathology. Those with low AD probability had a plasma probability score of 0-1, intermediate a score of 2 and high a score of 3-4. High cut-off indicates the ability to differentiate those with low/intermediate vs high likelihood of AD pathology.

CARD							
Low Cut-off		Autopsy +	Autopsy -	High Cut-off		Autopsy +	Autopsy -
ADNC-PPS	Intermediate/ High	47	7	ADNC-PPS	High	36	1
	Low	4	18		Low/ Intermediate	15	24
Sensitivity		92%		Sensitivity		71%	
Specificity		72%		Specificity		96%	
COMPASS-ND							
Low Cut-off		CSF +	CSF -	High Cut-off		CSF +	CSF -
AD Probability	Intermediate/ High	34	2	AD Probability	High	24	0
	Low	9	13		Low/ Intermediate	19	15
Sensitivity		79%		Sensitivity		56%	
Specificity		87%		Specificity		100%	

This ADNC-PPS was then applied to the remaining symptomatic participants in COMPASS-ND who were clinically diagnosed with dementia (**Figures 9.10 and 9.11**). We used this scoring system to determine the probability that AD pathology was present and contributing to cognitive impairment. Those clinically diagnosed with AD had the largest proportion of participants with a high probability of having AD pathology at 45%. The lowest probability of AD pathology was in those clinically diagnosed with FTD, and the likelihood of AD pathology increased in those diagnosed with LBD and PDD. An intermediate probability of AD pathology is most prevalent in those diagnosed with AD and LBD at 27-29%, and was much lower in those with FTD and PDD at 13-15%. Those clinically diagnosed with AD also had the lowest prevalence of participants with a low probability of AD pathology (28%), and those with FTD had the highest prevalence of low probability of AD pathology (78%). LBD and PDD were similar at 50-54%.

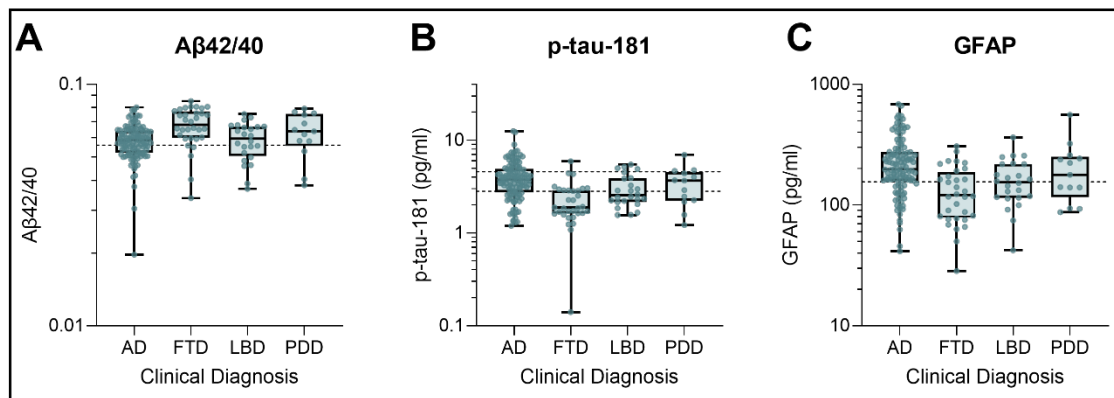


Figure 9.10 Plasma Aβ42/40 (A), p-tau-181 (B), and GFAP (C) in COMPASS-ND participants clinically diagnosed with dementia. Dashed lines indicate cut-offs for each plasma biomarker to incorporate into the plasma AD probability score.

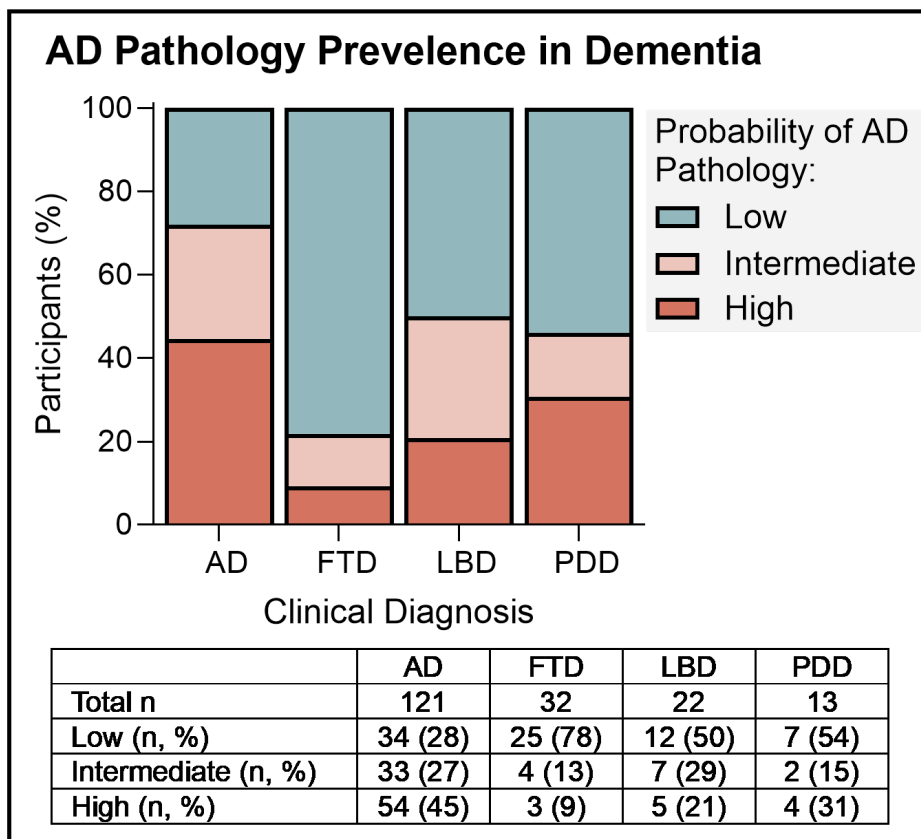


Figure 9.11 Estimated prevalence of AD pathology based on ADNC-PPS in COMPASS-ND participants diagnosed with dementia. Plasma probability score derived from autopsy-confirmed cut-offs for AD pathology. The histogram displays the percentage of participants in each clinical diagnostic group with low, intermediate, and high likelihood of AD pathology sufficiently severe to contribute to cognitive impairment. The table below indicates the exact sample size. Abbreviations: AD, Alzheimer's disease; FTD, frontotemporal dementia; LBD, Lewy body dementia.

9.4 Discussion

9.4.1 Creation of a plasma probability score for AD

In this study, we assessed the utility of plasma biomarkers to estimate the prevalence of AD pathology in the COMPASS-ND cohort through the use of an AD plasma probability score. The AD probability scoring systems we developed in this study integrates measures of plasma A β 42/40, p-tau-181, and GFAP to determine if a participant has a low, high or intermediate probability of having AD pathology. We created two different versions of the AD plasma probability score. First, we utilized age-specific reference curves to create a scoring system to flag deviations from normative levels suggestive of AD pathologic change in those in pre-dementia stages from cognitively unimpaired to MCI (RC-PPS). Second, we leveraged a cohort of autopsy-confirmed dementia patients to create plasma biomarker cut-offs indicative of AD pathology sufficient to contribute to dementia (ADNC-PPS). These two scoring systems were developed using a three-level grading system to improve confidence in the accuracy of those in the low and high probability groups. Similar to other studies that have used a two-cut-off system for AD plasma biomarkers³⁹⁴⁻³⁹⁶, we found that this approach improved sensitivity in distinguishing the low probability from the intermediate/high group, and improved specificity when distinguishing the high probability group from the low/intermediate group. When using the sub-cohort of COMPASS-ND that had CSF AD testing as our validation cohort, we found that both RC-PPS and ADNC-PPS had better specificity than sensitivity. When differentiating the low from intermediate/high probability groups, the PPS had sensitivities of 72% for RC-PPS in the pre-dementia group and 79% for ADNC-PPS in the dementia group compared to CSF testing. When differentiating the high from low/intermediate probability groups, the PPS had specificities of 96% for RC-PPS in the pre-dementia group and 100% for ADNC-PPS in the dementia group compared to CSF testing.

9.4.2 Utility of AD plasma probability scores

Recently published guidelines for the use of plasma biomarkers indicate that tests should have a sensitivity $\geq 90\%$ and a specificity between 75-90% depending on the context of use^{243,441}. These recommendations prioritize sensitivity over specificity to utilize the test as a rule-out measure, ensuring most participants with AD are identified and minimizing false negatives. In

this scenario, a negative test can rule out amyloid pathology in most cases, whereas a positive test could confirm AD pathology or indicate the need for further testing, depending on the level of clinical suspicion of AD⁴⁴¹. However, there is still utility in a high specificity test, which will have a low number of false positives. For those in pre-dementia stages, participants with a high RC-PPS could be prioritized for close follow-up to monitor for conversion to dementia. In contrast, pre-dementia participants with a low RC-PPS may not require further investigation, ultimately allowing those with subjective or mild memory complaints to be more easily triaged and managed by their primary care physician. In those clinically diagnosed with dementia, a high ADNC-PPS, alongside clinical suspicion of AD, could be very useful in setting inclusion/exclusion criteria for future AD clinical trials as well as determining eligibility to receive approved AD disease-modifying therapies. A high ADNC-PPS in those clinically diagnosed with other forms of dementia may indicate the need for follow-up testing using CSF or PET to confirm differential diagnosis.

9.4.3 Estimated prevalence of AD pathology development in pre-dementia stages

In participants who are in pre-dementia stages ranging from cognitively unimpaired to those with MCI, a screening test that indicates deviations from normative biomarker level trajectories and captures early AD pathologic change may be of interest for future primary prevention trials. In Chapter 3, we developed age-specific reference curves that described the major percentiles of biomarker concentrations for a normative Canadian population. These reference curves were developed from individuals without overt dementia but did not exclude any participants with potential underlying neuropathologies, as the CHMS did not include neurological measures to assess this. When developing the RC-PPS, we tested cut-offs at 95% of the normative population, as this is a standard reference interval cut-off. In addition, we also tested down to 75% of the normative population, as reports show a prevalence of approximately 25% of cognitively unpaired participants having AD pathology²³²⁻²³⁵. All biomarkers utilized the 95% cut-off, while p-tau-181 and GFAP utilized an additional cut-off for 75% of the population. The RC-PPS was validated against CSF testing in the COMPASS-ND sub-cohort, resulting in a sensitivity of 72% for the low-probability group and a specificity of 96% for the high-probability group. As the true AD pathology prevalence in the normative population is not known, and we utilized a small validation cohort, this RC-PPS requires

further validation in larger cohorts with more robust reference standards, such as PET. However, despite this scoring system being in the initial stages of development, there are multiple reasons to continue this line of investigation. One of the key advantages of using population-based reference curves as the basis for a screening measure is the incorporation of age into cut-offs, which is often overlooked by other AD biomarker studies despite clear evidence of biomarker changes with aging²⁸⁶. Given that AD pathologic change may initiate 15-20 years prior to the onset of cognitive symptoms, accounting for age is essential. Our reference curves demonstrate clear differences in normative ranges between individuals aged 50 and those aged 80, underscoring the importance of age-specific thresholds. The other key advantage of the percentile-based cut-off system is the ability to track biomarker changes over time based on age-related trajectories. Similar to how growth curves are used, these percentile-based reference curves for AD plasma biomarkers can be used to track deviations over time that are greater than the normative age-related changes, which may be cause for follow-up.

When applying the current RC-PPS to the remainder of the pre-dementia COMPASS-ND cohort, we found the expected association of increased prevalence of AD pathology from cognitively unimpaired to MCI. Previous studies using PET and CSF to detect AD pathology have indicated the prevalence of AD pathology in approximately 10-25% of cognitively unimpaired individuals, increasing up to approximately 25-50% in people with MCI²³²⁻²⁴⁰. Here, we found a high probability of AD pathology in 8% of cognitively unpaired participants and an intermediate probability of AD pathology in 18%. Thus, our estimation of the prevalence of AD pathology is on the higher end of reported studies, but well within range. The total prevalence of AD pathology in SCI participants was equivalent to the cognitively unpaired group, but a higher proportion of participants were observed to have a high probability of AD pathology development (14%). In the MCI group, 18% of participants had a high probability of AD pathology development, and 18% had an intermediate probability of pathology development, for a total of 36%. This is well within the range of the expected prevalence of AD pathology in those with MCI. Therefore, if we consider both the intermediate and high probability groups to be indicators of AD pathology, the RC-PPS we propose here produces the expected prevalence of AD pathology across clinical subgroups in the pre-dementia stages.

9.4.4 Estimated prevalence of AD pathology in dementia diagnoses

For participants who have already progressed to clinical dementia, we are interested in understanding the degree to which AD pathology contributes to their cognitive impairment. By utilizing an autopsy cohort to develop the ADNC-PPS, we can develop and validate cut-offs derived from the true gold standard of neuropathology in a manner that assesses its likelihood to contribute to cognitive impairment. Our previous work in Chapter 7 examined the association between plasma biomarkers and established AD staging frameworks and their ability to discriminate across AD neuropathological grades. We concluded that when using multiple plasma biomarkers, they are best able to differentiate AD staging by the composite ABC score rather than its component scores of Thal phase (A), Braak stage (B), and CERAD score (C). Here, we utilized the same autopsy cohort to distinguish participants with AD pathology from those without, where an intermediate or higher ABC score was indicative of pathology sufficient to cause cognitive impairment^{25,26}. Using autopsy-derived cut-offs, the resultant ADNC-PPS exhibited strong performance relative to autopsy findings, achieving a sensitivity of 94% for the low-probability group and a specificity of 93% for the high-probability group. Subsequent validation in the COMPASS-ND CSF sub-cohort revealed a modest reduction in sensitivity but an improvement in specificity.

When applying this ADNC-PPS to participants in the remainder of the COMPASS-ND cohort clinically diagnosed with dementia, we found an expected distribution of AD pathology prevalence. The highest prevalence of AD pathology was detected in those clinically diagnosed with AD, with 45% having a high probability and 27% having an intermediate probability of AD pathology. This also resulted in 28% of participants with clinical AD having a low probability of underlying AD pathology. This result is aligned with previous findings that approximately 10-30% of patients clinically diagnosed with AD are found not to have AD pathology^{232,237,241,242}. In participants clinically diagnosed with other forms of dementia, we also found expected patterns, with the FTD group having the lowest prevalence of AD pathology and the LBD and PDD groups falling in between. In those clinically diagnosed with FTD, 22% of participants had at least an intermediate probability of having AD pathology, with only 9% having a high probability. Previous reports found that approximately 10-30% of FTD cases had AD co-pathology, depending on the clinical subtypes⁴⁴². Both LBD and PDD groups

had a prevalence of approximately 50% of participants having at least an intermediate probability of having AD pathology. Previous reports indicate that somewhere between 50-80% of LBD patients and 20-60% of PDD patients have intermediate to high levels of AD co-pathology⁴⁴³⁻⁴⁴⁸. This puts our results on the lower end of what is expected in LBD and the higher end of what is expected for PDD. However, the higher prevalence in participants with PDD could be due to the limited sample size of only n=13 in this group. It is also important to note that, similar to the RC-PPS, the lower sensitivity and higher specificity of the ADNC-PPS may result in a potentially lower proportion of AD pathology reported compared to the true value. Therefore, these estimates should be considered the lowest estimate of AD pathology prevalence in these clinical dementia groups.

9.4.5 Limitations

One of the key limitations of this study is the limited sample size in the CSF sub-cohort we used for validation. This limitation precludes calculation of the positive and negative predictive values of these tests, which requires an accurate representation of the expected prevalence of pathology in each group. Although we have shown the CSF sub-cohort is representative of the overall COMPASS-ND cohort in terms of demographic and clinical descriptors, we can not confirm if the prevalence of AD is representative of the full cohort or the general population using this small sample size. Further validation of this AD probability scoring system in larger cohorts with additional reference standards, such as PET, will be important for future studies. Another limitation of this study is the lack of p-tau-217 data. As samples from the COMPASS-ND cohort were analyzed prior to the commercial release of p-tau-217 assays, we lack plasma p-tau-217 data for our primary cohort. This is a limitation due to the consensus that p-tau-217 is the top-performing biomarker to detect AD pathology^{171,449}.

9.5 Conclusions

We combined plasma A β 42/40, p-tau-181, and GFAP into two different AD plasma probability scores to indicate a low, intermediate and high probability of AD. The RC-PPS, to be used in pre-dementia stages, indicated deviations from normative levels suggestive of AD pathology development with a sensitivity of 72% and specificity of 96%. The ADNC-PPS, to be used in

those clinically diagnosed with dementia, indicated the probability of AD pathology contributing to cognitive impairment with a sensitivity of 79% and specificity of 100%.

Chapter 10. Discussion

10.1 Summary and significance of findings

In this thesis, we measured plasma amyloid beta (A β 42/40), phosphorylated tau-181 (p-tau-181), neurofilament light (NfL), and glial fibrillary acidic protein (GFAP) across four cohorts representing the spectrum of brain health. The results of this thesis provide information on important variables to consider when interpreting plasma biomarker results, and how plasma biomarkers can be used for the detection of AD pathology. The specific findings of each chapter, along with the broader implications of these findings, can be found in **Table 10.1**.

Table 10.1 Summary of thesis findings and their implication by chapter.

Chapter/ Cohorts	Content	
Chapter 2: CHMS	Title:	Age Specific Reference Intervals for Plasma Biomarkers of Neurodegeneration and Neurotrauma in a Canadian Population
	Question:	Do we need age-specific reference intervals for plasma biomarkers?
	Findings:	Plasma biomarkers display U-shaped curves across the life-span, requiring age specific reference intervals.
	Implication:	Age is an important variable to consider in the interpretation of plasma A β 42/40, p-tau-181, NfL, and GFAP.
Chapter 3: CHMS	Title:	Age-Specific Reference Curves for Plasma Biomarkers of Neurodegeneration
	Question:	Can we improve reference interval-based interpretation by increasing sample size to improve representation of the Canadian population?
	Findings:	Reference curves were produced to report the 1 st , 3 rd , 5 th , 15 th , 25 th , 50 th , 75 th , 85 th , 95, 97 th , and 99 th percentiles of normative biomarker levels.
	Implication:	Updated reference curves can be used as cut-offs or a reference population in other research, providing a useful tool for biomarker interpretation.
Chapter 4: CHMS, Super Seniors, CARD, COMPASS-ND	Title:	Investigating Reproducibility of Plasma Biomarker Results Through Cross Lot Analysis of Simoa Assays
	Question:	Can we compare biomarker results across cohorts analyzed on different assay lots?
	Findings:	The low bias between lots enabled direct comparison rather than requiring data harmonization for comparison between lots.
	Implication:	Different lots of commercial N4PE and p-tau-181 assays produce consistent data allowing for simple comparison of data produced using different lots.
Chapter 5: Super Seniors, CHMS	Title:	Cognitively Resilient Seniors Have Resilient Alzheimer's Disease Plasma Biomarker Profiles
	Question:	How do biomarker profiles of cognitively resilient seniors compare to population normative levels?

	Findings:	NfL and GFAP levels were reflective of a normative population, while A β 42/40 and p-tau 181 were less variable and concentrated in healthier ranges.
	Implication:	Instead of silent pathology, it may be resistance to pathological disease development that more closely relates to cognitive resilience.
Chapter 6: Super Seniors	Title:	<i>APOE4</i> Carrier Status Modifies Plasma p-tau-181 Concentrations in Cognitively Healthy Super-Seniors
	Question:	Does being an <i>APOE4</i> carrier modify biomarker concentrations in cognitively resilient seniors?
	Findings:	A β 42/40, p-tau-181, and GFAP associated with carrier status, but only p-tau-181 remained associated after accounting for age, sex, and BMI.
	Implication:	When available, <i>APOE</i> genotype may be important to consider when interpreting p-tau-181.
Chapter 7: CARD	Title:	Plasma Biomarkers to Detect and Grade Autopsy Confirmed Alzheimer's Disease Neuropathology
	Question:	What measures of AD neuropathology do plasma biomarkers best inform on for grading and detection of pathology?
	Findings:	A β 42/40, p-tau-181, NfL, and GFAP associated strongly with multiple measures of AD neuropathology, but informed best on overall AD neuropathologic change. These results were unaffected by co-pathologies.
	Implication:	Validation against autopsy can provide the true accuracy of plasma biomarkers. A β 42/40, p-tau-181, and GFAP are excellent indicators of AD neuropathology regardless of the presence of co-pathologies.
Chapter 8: CARD	Title:	Exploratory Investigation of Plasma Biomarkers to Detect Common Co-pathologies of Alzheimer's Disease in an Autopsy Cohort
	Question:	Can we discover novel biomarkers to detect the presence of neuropathologies of non-AD dementias?
	Findings:	We identified sets of biomarkers that have utility to inform on α -synuclein, TDP-43, and vascular neuropathologies.
	Implication:	This study demonstrates feasibility of conducting larger scale neuropathology studies to facilitate biomarker discovery of under studied forms of dementia.
Chapter 9: COMPASS-ND, CHMS, CARD	Title:	Using Plasma Biomarkers to Estimate the Prevalence of Alzheimer's Disease Pathology in COMPASS-ND
	Question:	Can we create cut-offs for plasma biomarkers to use them as a proxy measure of AD pathology?
	Findings:	Combinations of plasma biomarkers can be used to indicate if study participants have a low, intermediate or high probability of having AD pathology with excellent specificity and good sensitivity.
	Implication:	We created plasma probability scores provide a new method of interpreting plasma biomarkers, and have provided a biological measure of AD pathology for the COMPASS-ND study.

10.1.1 Gaining a deeper understanding of neurodegenerative plasma biomarkers

This thesis aimed to gain a deeper understanding of plasma biomarkers across the spectrum of health. In Chapters 2 and 3, we explored how plasma biomarkers are modified by age across the lifespan. In most cases, we observed U-shaped curves for biomarkers where concentrations decrease during childhood and adolescence, level out in early adulthood, and then begin to increase again with aging. This finding highlights the importance of considering age in the interpretation of plasma biomarker levels. Though this trajectory has also been established by other groups for NfL and GFAP^{188,248,250}, we were the first to identify age-related changes for p-tau-181, A β 40 and A β 42. The age-specific reference intervals and curves we produced here have the potential to be incredibly impactful on the field of neurological plasma biomarker research. The ability for other researchers to access this information and utilize this data as a reference group in their own work can not only reduce the burden and cost on other researchers to produce their own unique controls group, but also to allow for some standardization across studies using a common anchor point. Although true reference materials will ultimately be required for full inter-laboratory standardization, the dataset presented here represents an important early step toward that goal. While the broader development of this resource lies outside the scope of this thesis, these results provide valuable guidance for study design and interpretation and can be readily incorporated into external research efforts.

Beyond age, we also explored how other variables modify plasma biomarker concentrations in Chapters 3 and 6. Here, we explored the importance of sex, BMI, kidney function, and *APOE* genotype. When generating reference curves, we observed that sex was a mild modifier for plasma GFAP and p-tau-181 levels, which resulted in differing 95th percentiles of the curves beyond age 40 for GFAP and beyond age 20 for p-tau-181. Due to the age-specific nature of these associations, it is difficult to determine if sex differences in plasma biomarker levels are driven by sex specific biology or the development of underlying pathology that varies by sex. While future studies are required to delineate sex-specific pathways, for the purpose of this thesis we chose to utilize the combined reference intervals. Notably, the ability to test sex specific curves as a future direction will be contingent upon Statistics Canada enabling the release of more sex-specific percentiles and use of a larger validation cohort than we had access to for the duration of this thesis project.

In Chapter 5, we took a novel approach to understanding plasma biomarkers of neurodegeneration, by observing them in a cognitively resilient senior population. Here, we found that plasma NfL and GFAP levels in this population were largely reflective of the normative distribution described in Chapter 3. This observation indicates that cognitive resilience does not necessarily indicate resilience to the accumulation of neurodegenerative pathologies and inflammation typically seen with aging. However, we did find a significant proportion of cognitively resilient seniors with plasma A β 42/40 and p-tau-181 levels indicative of low AD pathological burden, which could contribute to their cognitive resilience. The results of this study are important for our understanding of whether cognitive resilience is driven by resistance to pathology development or resilience to the clinical manifestations of this pathology; our results suggest that both mechanisms may be at play in the cognitively resilient population we studied. We can additionally utilize this information in future studies to determine which biomarkers may have greater utility in screening for cognitive decline, as well as to further explore mechanisms that underly cognitive resilience in the presence of what appears to be a typical neuropathological burden. We would anticipate both genetic and lifestyle factors to be important.

10.1.2 Plasma biomarkers for Alzheimer's disease diagnosis

The goal of gaining a deeper understanding of the plasma biomarkers studied in this thesis is grounded in moving towards implementation of plasma tests to facilitate the diagnosis of AD. In Chapters 7 and 9, we began to explore the use of plasma A β 42/40, p-tau-181, p-tau-217, NfL and GFAP to detect AD pathology. To use these biomarkers for diagnostic purposes, we first had to understand how they informed on the various types of pathological burden in AD, leading to the investigation of associations with Thal phase (reflecting amyloid distribution), Braak stage (reflecting neurofibrillary tangle distribution), CERAD score (reflecting neuritic plaque density) and ABC score (composite of previous three) in Chapter 7. We found significant associations for A β 42/40, p-tau-181, p-tau-217, NfL and GFAP with all measures of AD neuropathology. However, we found that NfL showed the weakest associations of all the tested biomarkers with each pathological measure, and that all biomarkers had a stronger association with measures involving amyloid pathology rather than neurofibrillary tangle

pathology alone. We then determined that the ABC score, which encompasses overall AD neuropathologic change, displayed the best association with plasma biomarkers.

In Chapter 9, we leveraged findings from previous chapters to investigate use of these plasma biomarkers as proxy tools to indicate AD pathology. We created plasma probability scores (PPS) as practical tools to estimate the probability of study participants having underlying AD pathology. One of the main rationales behind this section of the thesis was to enable future research within the COMPASS-ND study. As COMPASS-ND did not include a proxy measure of AD pathology, other than a small subset of those with CSF biomarker analysis, and thus relied upon clinical diagnosis alone, we aimed to fill this gap this with plasma biomarker data. This study incorporated a variety of concepts that were established both within this thesis and by other researchers to create a simple but highly useful and relevant tool. Key considerations included in the development of PPS included: the ability to improve biomarker utility by using multiple markers, the need to handle biomarker results differently across the clinical spectrum, the importance of age-related changes, and the use of a multi-cutoff system. We were able to use the final PPS tools to estimate the prevalence of AD pathology across clinical diagnostic groups within the COMPASS-ND cohort, showing the tool estimated expected disease prevalences based on the literature. The PPS, similar to the reference curves themselves, is an important tool to be generated from this thesis that could indeed have wider applicability for researchers. With wider validation, we hope this could be a useful tool for estimating the presence of AD pathology in research studies, and form the basis of future studies aimed at earlier screening of AD pathology development.

A key future direction for neurodegenerative plasma biomarker research will be to investigate biomarkers to detect non-AD neuropathologies, we also conducted an exploratory investigation of novel biomarkers to detect α -syn, TDP-43, and CVD in Chapter 8. We found some novel biomarkers show promise to detect these pathologies both uniquely and as co-pathologies in AD. Though this work is highly exploratory due to the small sample size and mixed pathologies, it is highly significant as it is one of the first studies to investigate novel biomarkers for these pathologies using autopsy-confirmed samples and NULISA technology.

10.2 Past, present, and future: The landscape of neurological plasma biomarkers

10.2.1 The Past: Rapid progression of research

One of the important concepts to acknowledge in the context of this thesis is how much the landscape of neurological plasma biomarker research has changed since the inception of this project 5 years ago. Neurodegenerative plasma biomarker research has grown at an exponential pace over the past decade. Approximately 10 years ago, the technology to accurately measure biomarkers, like A β , p-tau, NfL, and GFAP in blood was limited. Platforms often lacked the sensitivity to detect these low-abundance analytes in blood. However, technology has since been developed that not only allows for greatly increased sensitivity, but also improved robustness and reproducibility. These technological advances have catalysed a tsunami of research in neurological plasma biomarkers across neurological diseases and context of use.

Evidence of growth and impact of biomarker studies can be appreciated when conducting a PubMed search using the terms:

("plasma"[Title/Abstract] OR "serum"[Title/Abstract]) AND ("amyloid beta"[Title/Abstract] OR "A β "[Title/Abstract] OR "phosphorylated tau"[Title/Abstract] OR "p-tau"[Title/Abstract] OR "pTau"[Title/Abstract] OR "neurofilament light"[Title/Abstract] OR "NfL"[Title/Abstract] OR "glial fibrillary acidic protein"[Title/Abstract] OR "GFAP"[Title/Abstract]) AND (biomarker[Title/Abstract] OR "blood biomarker"[Title/Abstract] OR "fluid biomarker"[Title/Abstract]),*

By 2015, 508 studies had been published, with an additional 772 studies being published between 2015 to 2019. In 2020 alone, 424 studies were published, and since then the number of papers per year has increased roughly 30% from the previous year. With 3,156 papers published from 2020 to 2024, there was a greater than 4-fold increase compared to the previous 5 years. During this time, a few key milestones occurred including the release of new biomarker tests and the establishment of core findings about these plasma biomarkers.

One of the first major milestones in the advancement of neurological plasma biomarker testing was the development of the Simoa HD-1 analyzer from Quanterix²²¹. The Simoa technology was the first widely adopted platform capable of analyzing neurological biomarkers in serum and plasma due to its improved sensitivity over traditional analog based systems. In the initial

years of expansion of blood biomarker work, Simoa technology was at the forefront, responsible for the development of some of the most widely used assays in plasma biomarker research to date. Simoa assays for NfL and GFAP were some of the most widely used in the early years of neurological plasma biomarker studies due to their utility across multiple neurological indications. Research for plasma biomarkers has often been split into two main areas of focus; neurotrauma and neurodegeneration, with NfL and GFAP often straddling this boundary due to their high sensitivity but lack of specificity towards any particular condition. From the perspective of neurotrauma, one central research objective was to establish the trajectories of biomarker concentrations following traumatic brain injury. GFAP peaks within hours to days followed by a more rapid normalization, while NfL peaks weeks post injury with a slower return to normal levels²³⁰. This understanding shaped many additional studies that aimed to define appropriate contexts of use for NfL and GFAP across various forms of neurotrauma.

Although NfL and GFAP are also often elevated in neurodegenerative conditions, AD specific biomarkers rapidly became the driving focus of the plasma biomarker world. A β has been a long-explored plasma biomarker for AD due to it directly representing the plaques forming in AD. A key advancement here was the understanding of the ability to improve diagnostic utility by utilizing the ratio of A β 42/40 as opposed to each biomarker alone. However, this advancement was also followed with the contingency that there is still large overlap between AD and controls, therefore limiting its accuracy. More recently, the rapid expansion of p-tau assays has shifted the landscape of AD biomarker research. Initially p-tau markers were investigated as measures of neurofibrillary tangle pathology, with researchers aiming to find phosphorylation sites most specific to AD. Up to the early 2020s, p-tau-181 was at the forefront as the top performing biomarker reflecting tau pathology^{173,320,450}. This led to its inclusion in this thesis project, with most experimental work being conducted between 2021 and 2023. However, in 2020, the first publications exploring p-tau-217 emerged. Over the next few years, p-tau-217 has rocketed to the forefront of p-tau biomarkers, with many proposing it as the top biomarker for AD pathologic change^{108,170,224,396}. This is a remarkable accomplishment considering the first publicly available plasma p-tau-217 assay was released in 2023. The other key factor in p-tau biomarkers moving to the forefront of AD research was the establishment

that, despite being tau markers, they actually tracked remarkably well with amyloid pathology and quickly became considered better markers of A β pathology than the A β markers themselves^{162,164}. Importantly, with these well-established findings for AD biomarkers, the research community at large has been enabled to produce a variety of consensus statements and guidelines for use. One of the first major milestones was the incorporation of plasma A β , p-tau, NfL and GFAP into the 2024 National Institute on Aging and the Alzheimer's Association (NIA-AA) revised criteria for diagnosis and staging of Alzheimer's disease¹⁰⁸. The Alzheimer's Association has also put out a variety of other guidelines such as appropriate use and clinical implementation recommendations, as well as clinical practice guidelines²²⁴⁻²²⁶.

10.2.2 *The Present: Bridging research to implementation*

Following consensus statements and recommendations from panels and associations, we have entered the phase of moving biomarkers from research to clinical practice. At the time of writing this thesis, the following biomarkers have been approved for clinical use:

1. NfL for detecting disease activity in those with relapsing-remitting MS or secondary progressive MS, first approved by the US FDA on the Roche Elecsys analyzer in 2023, and approved by Health Canada on the Siemens Atellica analyzer in 2024.
2. GFAP to rule out the need for CT scan following concussions, first approved by the US FDA on the Abbott iStat analyzer in 2021, and approved by Health Canada on the Abbott iStat analyzer in 2022.
3. Plasma p-tau-217/A β 42 to indicate the presence or absence of amyloid pathology in patients over 50 years old with symptoms of cognitive decline, first approved by the US FDA on the Fujirebio Lumipulse analyzer in 2025.
4. Plasma p-tau-181 to rule out AD pathology in those with cognitive decline in primary care settings, first approved by the US FDA on the Roche Elecsys analyzer in 2025.

The emergence of two blood tests for AD in the past few months is likely to be just the beginning of a wave of test approval, implementation, and uptake. As AD is a priority interest, it is likely these biomarkers will undergo approvals on a variety of platforms and expand upon their context of use. Currently, the approved tests are not often used to replace established diagnostic tools, but rather determine the need for further testing prior to treatment. Similar to

how GFAP is approved to rule out the need for a CT scan in adults after neurotrauma, a negative p-tau test could rule out more expensive PET or CSF testing to confirm the diagnosis and/or eligibility to receive anti-amyloid immunotherapy in a clinical or research setting. By contrast, positive p-tau testing is still often followed up with a PET or CSF testing prior to administering anti amyloid therapy, due to its potential adverse side effects. However, as approvals progress and more clinical data supports the use of these biomarkers as primary diagnostic tools, they will likely be able to be used as stand-alone tests for the diagnosis of AD and a determination for the use of anti amyloid therapy.

10.2.3 *The Future: Beyond Alzheimer's disease diagnosis*

With established biomarkers for AD diagnosis far along the pipeline for clinical implementation, much of the research community has turned their focus to the discovery of biomarkers for other neurological diseases and contexts of use. Within AD there are many other contexts of use to be explored. Now that we are able to quantify the degree of amyloid pathology, the logical progression is to be able to confirm the presence of the other key AD hallmarks, namely neurofibrillary tangles and neurodegeneration, to determine how far along the AD pathologic continuum a patient is. Though NfL has shown promise as a general marker to inform on the severity of neurodegeneration present¹⁰⁸, the investigation of biomarkers to inform on neurofibrillary tangles may be challenging. As the main p-tau markers are currently used to indicate amyloid pathology, it could be difficult to implement them clinically to also inform on tau. If the field stays focused on anti-amyloid treatments, this does not pose a problem. But as many anti-tau therapies are currently in clinical trials, there is a clear need for tau specific biomarkers that can inform on the presence and burden of neurofibrillary tangles. Potentially this will come from the current p-tau markers with use of additional cut-offs where certain thresholds will indicate the multiple pathologies. But more likely we will need to discover and validate new tau specific markers to accomplish this clinical goal.

Along with this approach to neuropathologic staging, many are interested in investigating biomarkers for AD screening and prognostication. The importance of these two will go hand in hand. Developing biomarkers for early screening will be important, as amyloid pathology can develop decades before the onset of clinical symptoms^{98,234,363}. If treatments or

interventions are eventually approved to treat amyloid pathology prior to clinical onset, we will need screening tools that can be deployed on a population wide scale to detect plaque pathology. There is potential this may come in the form of current biomarkers, with markers like p-tau-217 detecting even the earliest changes in amyloid plaque deposition^{164,224}. However, it is also possible that new biomarkers may be even more sensitive to early pathologic changes. Following a positive screening test, we will then need markers for clinical prognosis. The time between amyloid plaque development and onset of cognitive decline may vary greatly between individuals and be mediated by a variety of factors. Therefore, determining which biomarkers will work best for prognosis will be important for triaging patients in primary care settings as well as for clinical trial design, where maximizing the number of participants that progress along the timeframe of a trial is critically important to minimize participant burden, cost, and time. From an epidemiological perspective, that some people may have amyloid pathology for years into the oldest ages or even death without conversion to dementia^{235,236} raises many future questions that large-scale population wide cohort data could address. Thus, screening biomarkers have multiple highly important use cases that go far beyond use of a positive screening test for amyloid pathology to determine potential eligibility to receive anti-amyloid immunotherapy and to predict conversion to MCI or dementia stages.

With such success in AD, many researchers have turned to investigating plasma biomarkers for other forms of dementia. This is also a necessary next step as this knowledge will be important for care regardless of whether the AD test is positive or not. For example, if a patient tests positive on a plasma test for AD, one of the natural next questions is whether this patient has co-pathologies that could also be contributing to their cognitive impairment. This could easily affect the treatment plan for an individual if anti-amyloid therapies and future treatments are less effective in the presence of mixed pathologies. And of course, for those who test negative on a plasma AD test, but exhibit symptoms of cognitive decline, we need to investigate what the cause is. Although other forms of dementia such as FTD and LBD are less common than AD, they still result in similar detrimental effects on an individual's life and the lives of those around them. One of the things we have learned from advancements in AD diagnosis is how imperative the ability to accurately diagnosis pathology is for the

development of treatments. Therefore, larger scale studies similar to that conducted in Chapter 8 of this thesis, are crucially important to discover and develop biomarkers for the diagnosis of these other pathologies. In fact, the detection of these other pathologies can build off the success of the AD field. By leveraging large neuropathology cohorts, biomarkers for these other pathologies can be found without the need for intermediary reference standards like PET and CSF to be developed, therefore expediting the progression to a plasma-based diagnosis for non-AD dementias. This ultimately could work towards a standard plasma panel that will be administered for the differential diagnosis of dementias.

10.3 The Canadian perspective: Implementation and future directions

10.3.1 Canadian progress compared to the global landscape

Though the field of plasma biomarker research in AD has exploded internationally over the past decade, not all countries have experienced the same adoption of research and speed of progress towards implementation. Although Canada is a highly developed country and is considered a global competitor in producing high-quality research, there are some areas it lags behind comparable nations. With neurodegenerative plasma biomarkers, Canada was a slightly later adopter both from the research and clinical implementation perspectives. On the research side, much of the early stages of biomarker discovery before 2020 were published by groups from the United States and the European Union. Canadian contributions came after this phase when large Canadian cohort studies were being used to assess the utility of plasma biomarkers. Here, the TRIAD cohort from McGill University was a major contributor to this work, alongside large USA and EU cohorts, due to their collection of paired blood samples with PET scans^{162,172,394,396}. It is also important to note that despite being a large Canadian cohort, much of the work published out of TRIAD was done in collaboration with groups from the EU rather than Canada. Despite being a very large country with multiple world-class institutions, no other major studies from Canada were producing AD plasma biomarker work.

Though many Canadian groups are beginning to catch up now, it is evident that Canada has lagged behind advances pioneered elsewhere. There are a multitude of reasons for this that continue to limit quick adoption of new AD research in Canada. Starting with funding, Canada

has a much smaller national research funding pool. The Canadian Institutes of Health Research (CIHR) has provided relatively modest funding towards dementia research compared to the USA National Institute of Aging and EU equivalents due to its smaller grant sizes and shorter funding cycles. This in turn limits the ability to recruit large cohorts or sustain longitudinal studies for biomarker research. The limited funding also directly impacts the availability of resources. With the exception of the TRIAD cohort, very few Canadian dementia studies have PET data, and those that do are often limited to subset analyses. In Canada, CSF is more often used in research studies than PET, but again CSF analysis is often limited to a subset of a cohort. This has been a major limitation highlighted throughout this thesis, as despite working with what is arguably Canada's largest national study on dementia, COMPASS-ND, no PET scans were collected and only ~15% of the cohort had CSF measures taken. This in turn has greatly limited the ability to compare biomarkers against the regularly used reference standards.

Another infrastructure issue is the ability to obtain equipment required to conduct biomarker research. As equipment is not an expense allowed on many operating grants, such as those from the Canadian Institutes of Health Research (CIHR), Canadian researchers are limited to seeking infrastructure funded from limited highly competitive pools like the Canada Foundation for Innovation, or seeking donor funds to acquire equipment. As this biomarker research has relied heavily on the rapid development of new technologies, the inability to acquire new technology in a timely manner will often put Canadians behind on the large research landscape. All of this in turn has a tendency to cause fragmentation across Canada, with smaller pools of research funding limiting studies to often exist in provincial specific silos instead of generating national studies geared towards biomarker research.

On the clinical side of the implementation of AD blood tests, Canada also falls behind due to its complex pathway for the implementation of new test. Though some neurological blood tests have been approved for use in Canada, the AD blood tests have not yet been through any Health Canada approvals. Despite this, as of July 2025, Life Labs in British Columbia and Ontario have begun offering testing for A β 42, A β 40, p-tau-181 and 217, and NfL. This is due to Canada's support for lab developed tests (LDT) that does not require federal approval. However, it is

important to note, that the current tests are being performed outside of Canada, with samples being shipped to the USA for testing, and unlike the majority of blood tests in Canada, these are not publicly funded and will cost patients \$300 to \$600 per test. In order to get these tests covered, a more extensive approval process with health economics analysis will have to take place. As Lecanemab has just been approved by Health Canada as of October 2025, a clear need is to establish a national strategy and build testing capacity within Canada.

10.3.2 Balancing treatment availability and diagnosis

In 2021, the USA FDA approved the first disease-modifying therapy for AD. Aducanumab was a monoclonal antibody targeting amyloid plaques to break them down and remove them from the brain. It was discontinued in 2024 due to concerns over safety and limited efficacy, however since its approval, two additional anti-amyloid therapies have also been approved by the FDA: Lecanemab in 2023 and Donanemab in 2024. Both therapies are intended for patients in the early clinical stages of AD dementia, with Donanemab also being approved for use in those with MCI. Both drugs have also been approved for use in Japan, China, the United Kingdom UK, and the European Union (EU). However, the EU and UK restrict the use of both drugs to those who are not *APOE4* homozygotes, as being a carrier of two copies of the gene has shown to associate with increased risk of adverse effects, such as amyloid-related imaging abnormalities (ARIA)^{53,54,56}. As of October 25th, 2025, Health Canada has also approved the use of Lecanemab for patients with mild early-stage Alzheimer's who are not *APOE4* homozygotes. This recent development greatly expedites the need for blood tests that can detect A β pathology to be deployed in Canada. Currently, in other countries, the drug is priced around \$26,000 USD per year, limiting its accessibility for many Canadians. As it can often take multiple years after regulatory approval before a drug becomes publicly covered, it may take some time before Lecanemab and similar drugs could be widely used. However, it is still imperative to begin the process of implementing blood testing prior to this.

There are a multitude of additional reasons to introduce the AD plasma biomarker tests while Canada is launching access to approved disease-modifying therapies and other treatments are considered. The process of test approval and adoption can take quite a long time. The approvals through both the federal and provincial authorities are extensive and complex. However,

beyond governmental test approved, test adoption, uptake, and demand will be a long process. This involves capacity building in terms of both infrastructure and clinician education about the availability of the test, defining criteria for test eligibility, and how to interpret results. As this entire process is quite extensive, it is imperative that we do not wait until treatments are widely accessible to start the process. If we were to wait, we would inevitably be unable to deal with the demand from those who would like to be tested for AD to determine eligibility for treatment with the current limited infrastructure we have using CSF and PET. In addition to preparing for treatments, there is still great value in providing a pathological diagnosis, regardless of treatment availability. Having insight into a person's underlying pathology can still provide useful insights to clinicians about disease progression. Knowing this information earlier can allow for the patient and family members/caregivers to more accurately plan for the future which can greatly reduce stress and improve quality of life. And of course, respecting the patient's preference in learning the test result would be a priority.

10.3.3 Approval of new blood tests in Canada

Approval and implementation of new lab tests in Canada is a multistep process. There are two different pathways in which a test may be approved; one as an in-vitro-diagnostic device (IVD), and the second as a lab developed test (LDT). The process of approving an IVD for use in Canada starts with Health Canada's Medical Devices Directorate (MDD) approval of devices under the Food and Drug Act Medical Devices Regulation²⁵¹. This process would apply to both the platform and test kit used for analysis and assess safety, effectiveness, and quality. Approval from the MDD authorizes the sale of the medical device in Canada. Following this, Health Canada is also responsible for monitoring the safety of a device once on the market. For approved tests, the Canadian Agency for Drugs and Technologies in Health (CADTH) conducts health technology assessments (HTA)²⁵². The HTA will provide important information for implementation such as clinical, social, ethical, and economic effects of the test, which for laboratory tests will most likely involve patient indications and health economics. The results of the HTA are then utilized by individual provincial health authorities do determine which tests to implement. A large part of this process is assessing the health economic analysis to determine if the test will be designated for public funding. For example, in British Columbia, the Provincial Laboratory Medicine Services (PLMS) conducts a review

of tests, determines test coverage and adds tests to approved requisition forms under the Laboratory Service Act and Regulations⁴⁵¹. For LDTs, approvals are less regulated as Health Canada approvals are not required. LDTs are thus solely regulated by provincial and territorial authorities with approvals and accreditations being provided to single laboratory site. Therefore, these tests cannot be sold or distributed to other sites. Because of this, LDTs work well for tests that do not require large scale-up. However, a blood test for a highly prevalent condition such as AD would require large scale efforts to meet the demands of the general public and therefore may be better suited to follow the IVD route.

10.3.4 Public health and economic assessments

One of the key future directions this thesis work provides a foundation for is to be able to move AD blood biomarker tests into clinical practice in Canada. A key future objective is a large-scale assessment of both the utility of the test in the current system and the health economics associated with implementing it^{252,253}. When assessing the integration of the test into the health care system, the assessment will start with a review of the clinical evidence. This is focused on looking at evidence of the test's accuracy from both an analytical and clinical perspective. This will be based on research studies, like those conducted in this thesis. Once this evidence has been reviewed, if the test is deemed accurate enough, there is assessment of the test's clinical utility. This would compare the new test to the current standard of care. For blood tests, this would involve comparing to PET and CSF testing, as well as the more frequently used memory tests, to understand if it can lead to a more accurate or earlier diagnosis, if it can inform better on treatment decisions, or reduce unnecessary follow-up. When comparing to amyloid PET and CSF testing, it is likely that blood testing would rank fairly similarly on these points, but greatly outperform memory testing on all of these fronts. Where blood tests begin to outperform PET and CSF is assessing the tests impact on the healthcare system. This is where the delivery of the test is compared to current measures. Compared to all previous diagnostic techniques, blood testing requires far fewer and less specialized resources and can be conducted much more quickly. For a blood test, sample collection is quite simple, with resources already being in place and easily accessible, throughout Canada at facilities like Life Labs. By contrast, for CSF collection, a lumbar puncture is considered a medical procedure

that must be conducted in a clinical setting by a highly trained professional, which reduces accessibility.

Public health and health economic assessments also examine infrastructure requirements and test turnaround time, where blood testing also greatly outperforms PET imaging. PET scans require highly specialized equipment along with expensive and limited radioactive tracers, which not only makes it expensive, but also difficult to implement beyond specialized facilities. PET also requires multiple specialized personnel to administer testing and interpret scans, while blood tests have a lower barrier to being able to learn how to interpret the results. One of the final components of assessing the public health impact of new tests is assessing its health equity, which is another realm where blood tests are superior. With many remote and rural locations across Canada, PET and CSF testing is not available to many due to the specialized facilities required that are often limited to large urban centres.

The next component of this assessment is the economic evaluation, which determines what value the test provides to the healthcare system²⁵³. Similar to looking at the clinical utility of the new blood test, the economic analysis will compare against the current standards. For each diagnostic method, all direct and indirect costs associated are assessed. Direct costs include items like the testing materials and equipment, facility and staff fees, along with fees that may be associated with direct follow-up procedures. Indirect costs involve considerations such as the time patients lose to travel for testing, patient productivity gains and losses such as the ability to remain productive or avoid long-term care, caregiver time and burden based on the ability of the test to guide care, and the downstream societal effects related to hospitalizations, long term care and social services. These costs are then assessed against the outcome of quality adjusted life years (QALYs), which is a measure assessing both the length and quality of life gained from the healthcare intervention. For example, with blood tests enabling earlier diagnosis or avoiding invasive procedures, QALY could increase by extending the time patients spend in better health and by improving quality of life through reduced stress, discomfort, and delay in care. The costs and outcomes are then weighed against each other through economic modeling. In this, different models can simulate patient pathways between diagnostic methods, such as looking at the different pathways for a positive test on PET vs

blood, or assessing the pathway over disease progression. The results of these models are typically expressed as an incremental cost-effectiveness ratio (ICER), which can indicate when a test has a high economic value. Following this assessment is a budget impact analysis (BIA) which assesses the financial feasibility of scaling up the test. This looks at potential costs and savings to the healthcare system over multiple years, accounting for factors like the size of the population needing the test, and upgrading infrastructure. All of these assessments together are then considered when provincial authorities decide whether a test should be publicly funded.

10.4 Practical approaches to key future directions

10.4.1 Alzheimer's disease diagnosis and implementation

As plasma biomarkers for the diagnosis of AD have been implemented in other countries, it is likely just a matter of time before they become regularly used in Canada as well. As stated above, one of the first steps in the implementation of blood tests is reviewing the clinical evidence that describes a test's analytical and clinical accuracy. Though this thesis lays the foundation for assessing plasma biomarkers in a variety of prominent Canada's cohorts, additional work is required to validate and expand on these findings in order to support the translation into clinical use. Therefore, a key future direction from this thesis will be validation in larger, more diverse cohorts with robust reference standards. One of the key limitations of this thesis was the limited sample size for validation of cut-offs for plasma biomarkers based on autopsy and CSF as the reference standards. Validating the findings of this thesis will thus require utilizing a study of a similar scale to that of the CHMS study from Chapters 2 and 3, where participants are included from across Canada and are able to represent the population at a whole from the perspective of age, sex, ethnicity, and medical conditions. For the purpose of diagnosis, these participants should all be in the stages of MCI or dementia to gear the test towards understanding if AD pathology is contributing to their cognitive impairment. Most importantly, participants would be required to have either PET or CSF measures to compare the blood results against. Although PET would be more preferable as a more robust reference standard, it has limited feasibility due to the limited access restricted to large city centers and lower testing capacity. From there we can utilize results similar to methods explored in Chapter 9, refining cut-offs based on the test metrics generated from these new cohorts. This would also allow us to have insight into how to incorporate information such as age, sex, BMI, and

co-morbidities into test interpretation, something this thesis lacked the power to do beyond age due to limited sample sizes. Conducting this assessment on a larger scale with new cohorts would also present the opportunity to translate results onto additional platforms that are moving towards clinical implementation.

10.4.2 Moving towards midlife screening

In addition to using plasma biomarkers for the purpose of diagnosis, there is also considerable interest to consider these biomarkers for the context of use of screening. We started to explore this potential in Chapters 3 and 9, where we explored the development of reference curves that could indicate deviations from normal that may inform on the early stages of pathology development. However, this work requires further development. First, one of the key ways we need to expand this work is to develop reference curves for p-tau-217 and its associated ratios. This work has been initiated using the same methodology we have established in this thesis. This is an important addition as it is not only considered the top biomarker for the detection of A β pathology, but also anticipated to be one of the markers that shows early changes^{164,224}. In line with this, another study that should be conducted is a longitudinal assessment of plasma biomarker trajectories with pathological changes over time. This could be conducted by recruiting participants in midlife and following them periodically with paired PET scans and blood sampling to determine which biomarkers show alterations at the earliest signs of changes on amyloid PET. Ideally, the biomarkers we would want to develop into screening tests would be the markers that display the earliest changes in response to pathology. Once we have established the biomarkers that best inform on early pathology development, there is a need to conduct large-scale studies to establish what deviations from normal are cause for concern. Here, we can build of the design used in Chapter 9, which leverages normative reference curves. Using a large population-based sample, ranging from midlife to senior ages, of those without cognitive impairment, paired PET and blood sampling would be taken. Similar to our work in Chapter 9, blood biomarkers would be compared to the reference curves using amyloid PET positivity to denote what percentiles could indicate pathology development.

10.5 Concluding remarks

In this thesis, we measured plasma A β , p-tau-181, NfL, and GFAP in 4,072 samples across four Canadian cohorts to learn more about plasma biomarkers of neurodegeneration across the health spectrum. Spanning populations from cognitive resilience to normative populations, to those with dementia we established multiple key findings: 1) plasma biomarkers have significant age-specific trends across the lifespan; 2) In cognitive resilience, plasma NfL and GFAP appear normative, while levels of A β 42/40 and p-tau-181 appear more resilient; 3) These plasma biomarkers all associate with multiple measures of AD pathology confirmed on autopsy; 4) We can use these plasma biomarkers in research studies as a proxy tool to estimate the probability that participants have underlying AD pathology. Importantly, we created two key tools for use in future research, the age-specific reference curves and plasma probability scores, as well as laying the foundations for beginning the clinical implementation of these plasma biomarkers in Canada for multiple contexts of use.

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Appendix

Appendix 1. Tables of age-specific reference interval values for A β 42/40, p-tau-181, NfL, and GFAP from Chapter 2.

Appendix 1.1. A β 42/40 continuous reference interval values for by year of age. Point values pulled from quantile regression curves. Upper limit defined by the 95th percentile, lower limit defined by the 5th percentile, median defined by the 50th percentile.

Age (y)	Lower Limit	Lower Limit 95% CI	Median	Median 95% CI	Upper Limit	Lower Limit 95% CI
3	0.061	0.053, 0.069	0.079	0.075, 0.083	0.098	0.096, 0.100
4	0.061	0.055, 0.067	0.079	0.077, 0.081	0.098	0.096, 0.100
5	0.06	0.056, 0.064	0.079	0.077, 0.081	0.098	0.097, 0.099
6	0.06	0.056, 0.064	0.079	0.075, 0.083	0.098	0.097, 0.099
7	0.059	0.055, 0.063	0.079	0.075, 0.083	0.098	0.096, 0.100
8	0.059	0.055, 0.063	0.079	0.073, 0.085	0.098	0.096, 0.100
9	0.059	0.055, 0.063	0.079	0.073, 0.085	0.098	0.096, 0.100
10	0.058	0.054, 0.062	0.079	0.073, 0.085	0.098	0.096, 0.100
11	0.058	0.054, 0.062	0.079	0.075, 0.083	0.098	0.096, 0.100
12	0.057	0.053, 0.061	0.079	0.075, 0.083	0.098	0.096, 0.100
13	0.057	0.051, 0.063	0.079	0.075, 0.083	0.098	0.096, 0.100
14	0.057	0.051, 0.063	0.079	0.075, 0.083	0.097	0.095, 0.099
15	0.056	0.050, 0.062	0.079	0.075, 0.083	0.097	0.095, 0.099
16	0.056	0.050, 0.062	0.078	0.074, 0.082	0.097	0.095, 0.099
17	0.056	0.050, 0.062	0.078	0.072, 0.084	0.097	0.095, 0.099
18	0.056	0.050, 0.062	0.078	0.072, 0.084	0.096	0.094, 0.098
19	0.055	0.049, 0.061	0.078	0.070, 0.086	0.096	0.094, 0.098
20	0.055	0.047, 0.063	0.078	0.070, 0.086	0.096	0.093, 0.099
21	0.055	0.047, 0.063	0.078	0.070, 0.086	0.095	0.092, 0.098
22	0.055	0.047, 0.063	0.078	0.070, 0.086	0.095	0.092, 0.098
23	0.055	0.047, 0.063	0.078	0.070, 0.086	0.095	0.092, 0.098
24	0.055	0.047, 0.063	0.078	0.070, 0.086	0.094	0.091, 0.097
25	0.055	0.047, 0.063	0.078	0.070, 0.086	0.094	0.091, 0.097
26	0.054	0.046, 0.062	0.078	0.070, 0.086	0.094	0.091, 0.097
27	0.054	0.046, 0.062	0.077	0.069, 0.085	0.094	0.091, 0.097
28	0.054	0.046, 0.062	0.077	0.069, 0.085	0.093	0.090, 0.096
29	0.054	0.046, 0.062	0.077	0.069, 0.085	0.093	0.091, 0.095
30	0.054	0.046, 0.062	0.077	0.071, 0.083	0.093	0.091, 0.095
31	0.054	0.048, 0.060	0.077	0.071, 0.083	0.093	0.091, 0.095
32	0.054	0.048, 0.060	0.077	0.071, 0.083	0.092	0.090, 0.094

33	0.054	0.048, 0.060	0.077	0.071, 0.083	0.092	0.090, 0.094
34	0.054	0.048, 0.060	0.077	0.071, 0.083	0.092	0.090, 0.094
35	0.054	0.048, 0.060	0.077	0.071, 0.083	0.092	0.090, 0.094
36	0.054	0.048, 0.060	0.076	0.070, 0.082	0.092	0.090, 0.094
37	0.054	0.048, 0.060	0.076	0.070, 0.082	0.091	0.089, 0.093
38	0.054	0.048, 0.060	0.076	0.070, 0.082	0.091	0.089, 0.093
39	0.054	0.048, 0.060	0.076	0.070, 0.082	0.091	0.089, 0.093
40	0.054	0.048, 0.060	0.076	0.072, 0.080	0.091	0.089, 0.093
41	0.054	0.048, 0.060	0.076	0.072, 0.080	0.091	0.089, 0.093
42	0.054	0.048, 0.060	0.076	0.072, 0.080	0.09	0.088, 0.092
43	0.054	0.048, 0.060	0.075	0.071, 0.079	0.09	0.088, 0.092
44	0.054	0.048, 0.060	0.075	0.069, 0.081	0.09	0.088, 0.092
45	0.054	0.048, 0.060	0.075	0.069, 0.081	0.09	0.088, 0.092
46	0.053	0.047, 0.059	0.075	0.069, 0.081	0.09	0.088, 0.092
47	0.053	0.047, 0.059	0.075	0.069, 0.081	0.09	0.087, 0.093
48	0.053	0.045, 0.061	0.075	0.069, 0.081	0.089	0.086, 0.092
49	0.053	0.045, 0.061	0.074	0.068, 0.080	0.089	0.086, 0.092
50	0.053	0.045, 0.061	0.074	0.068, 0.080	0.089	0.086, 0.092
51	0.052	0.044, 0.060	0.074	0.068, 0.080	0.089	0.086, 0.092
52	0.052	0.044, 0.060	0.074	0.066, 0.082	0.089	0.086, 0.092
53	0.052	0.044, 0.060	0.074	0.066, 0.082	0.089	0.086, 0.092
54	0.051	0.043, 0.059	0.074	0.066, 0.082	0.089	0.086, 0.092
55	0.051	0.043, 0.059	0.073	0.065, 0.081	0.089	0.086, 0.092
56	0.051	0.043, 0.059	0.073	0.065, 0.081	0.089	0.086, 0.092
57	0.05	0.044, 0.056	0.073	0.065, 0.081	0.088	0.085, 0.091
58	0.05	0.044, 0.056	0.073	0.067, 0.079	0.088	0.085, 0.091
59	0.05	0.044, 0.056	0.073	0.067, 0.079	0.088	0.085, 0.091
60	0.049	0.043, 0.055	0.072	0.066, 0.078	0.088	0.086, 0.090
61	0.049	0.043, 0.055	0.072	0.066, 0.078	0.088	0.086, 0.090
62	0.048	0.042, 0.054	0.072	0.066, 0.078	0.088	0.086, 0.090
63	0.048	0.042, 0.054	0.072	0.066, 0.078	0.088	0.086, 0.090
64	0.048	0.042, 0.054	0.071	0.067, 0.075	0.088	0.086, 0.090
65	0.047	0.041, 0.053	0.071	0.067, 0.075	0.088	0.086, 0.090
66	0.047	0.041, 0.053	0.071	0.067, 0.075	0.088	0.086, 0.090
67	0.046	0.040, 0.052	0.07	0.064, 0.076	0.088	0.086, 0.090
68	0.046	0.040, 0.052	0.07	0.064, 0.076	0.088	0.086, 0.090
69	0.045	0.039, 0.051	0.069	0.063, 0.075	0.088	0.085, 0.091
70	0.044	0.038, 0.050	0.069	0.063, 0.075	0.088	0.085, 0.091
71	0.044	0.038, 0.050	0.068	0.062, 0.074	0.088	0.084, 0.092
72	0.043	0.037, 0.049	0.068	0.062, 0.074	0.088	0.084, 0.092
73	0.043	0.037, 0.049	0.067	0.061, 0.073	0.088	0.083, 0.093
74	0.042	0.034, 0.050	0.067	0.061, 0.073	0.088	0.082, 0.094

75	0.041	0.033, 0.049	0.066	0.062, 0.070	0.088	0.081, 0.095
76	0.041	0.029, 0.053	0.066	0.060, 0.072	0.088	0.080, 0.096
77	0.04	0.026, 0.054	0.065	0.057, 0.073	0.088	0.079, 0.097
78	0.039	0.021, 0.057	0.064	0.054, 0.074	0.088	0.077, 0.099
79	0.039	0.017, 0.061	0.064	0.048, 0.080	0.088	0.075, 0.101

Appendix 1.2. p-tau-181 continuous reference interval values for by year of age. Point values pulled from quantile regression curves. Upper limit defined by the 95th percentile, lower limit defined by the 5th percentile, median defined by the 50th percentile.

Age (y)	Lower Limit (pg/ml)	Lower Limit 95% CI	Median (pg/ml)	Median 95% CI	Upper Limit (pg/ml)	Lower Limit 95% CI
3	1.7	1.5, 1.9	2.7	2.5, 2.9	4.9	4.7, 5.1
4	1.6	1.4, 1.8	2.6	2.4, 2.8	4.8	4.6, 5.0
5	1.5	1.3, 1.7	2.5	2.3, 2.7	4.7	4.5, 4.9
6	1.4	1.2, 1.6	2.4	2.2, 2.6	4.5	4.3, 4.7
7	1.3	1.3, 1.3	2.3	2.1, 2.5	4.4	4.2, 4.6
8	1.3	1.3, 1.3	2.2	2.0, 2.4	4.3	4.1, 4.5
9	1.2	1.2, 1.2	2.2	2.0, 2.4	4.1	4.0, 4.2
10	1.2	1.2, 1.2	2.1	1.9, 2.3	4.0	3.9, 4.1
11	1.1	1.1, 1.1	2.0	1.8, 2.2	3.8	3.7, 3.9
12	1.1	1.1, 1.1	1.9	1.7, 2.1	3.7	3.6, 3.8
13	1.0	1.0, 1.0	1.9	1.7, 2.1	3.6	3.5, 3.7
14	1.0	1.0, 1.0	1.8	1.6, 2.0	3.5	3.4, 3.6
15	1.0	1.0, 1.0	1.7	1.5, 1.9	3.3	3.2, 3.4
16	0.9	0.9, 0.9	1.7	1.5, 1.9	3.2	3.1, 3.3
17	0.9	0.9, 0.9	1.6	1.4, 1.8	3.1	3.0, 3.2
18	0.9	0.9, 0.9	1.6	1.4, 1.8	3.0	2.9, 3.1
19	0.9	0.9, 0.9	1.5	1.3, 1.7	2.9	2.8, 3.0
20	0.8	0.8, 0.8	1.5	1.3, 1.7	2.9	2.8, 3.0
21	0.8	0.8, 0.8	1.4	1.2, 1.6	2.8	2.7, 2.9
22	0.8	0.8, 0.8	1.4	1.2, 1.6	2.7	2.6, 2.8
23	0.8	0.8, 0.8	1.4	1.2, 1.6	2.6	2.5, 2.7
24	0.8	0.8, 0.8	1.4	1.2, 1.6	2.6	2.5, 2.7
25	0.8	0.8, 0.8	1.3	1.1, 1.5	2.5	2.4, 2.6
26	0.8	0.8, 0.8	1.3	1.1, 1.5	2.5	2.4, 2.6
27	0.8	0.8, 0.8	1.3	1.1, 1.5	2.4	2.3, 2.5
28	0.8	0.8, 0.8	1.3	1.1, 1.5	2.4	2.3, 2.5
29	0.8	0.8, 0.8	1.3	1.1, 1.5	2.4	2.3, 2.5
30	0.8	0.8, 0.8	1.3	1.1, 1.5	2.4	2.3, 2.5
31	0.8	0.8, 0.8	1.3	1.1, 1.5	2.4	2.3, 2.5
32	0.8	0.8, 0.8	1.3	1.1, 1.5	2.3	2.2, 2.4

33	0.8	0.8, 0.8	1.3	1.3, 1.3	2.3	2.2, 2.4
34	0.8	0.8, 0.8	1.3	1.3, 1.3	2.3	2.2, 2.4
35	0.8	0.8, 0.8	1.3	1.3, 1.3	2.3	2.2, 2.4
36	0.8	0.8, 0.8	1.3	1.3, 1.3	2.3	2.2, 2.4
37	0.8	0.8, 0.8	1.3	1.3, 1.3	2.3	2.2, 2.4
38	0.8	0.8, 0.8	1.3	1.3, 1.3	2.3	2.2, 2.4
39	0.8	0.8, 0.8	1.3	1.3, 1.3	2.3	2.2, 2.4
40	0.8	0.8, 0.8	1.3	1.3, 1.3	2.3	2.2, 2.4
41	0.8	0.8, 0.8	1.3	1.3, 1.3	2.4	2.3, 2.5
42	0.8	0.8, 0.8	1.3	1.3, 1.3	2.4	2.3, 2.5
43	0.8	0.8, 0.8	1.3	1.3, 1.3	2.4	2.3, 2.5
44	0.8	0.8, 0.8	1.3	1.3, 1.3	2.4	2.3, 2.5
45	0.8	0.8, 0.8	1.3	1.3, 1.3	2.4	2.3, 2.5
46	0.8	0.8, 0.8	1.3	1.1, 1.5	2.4	2.3, 2.5
47	0.8	0.8, 0.8	1.3	1.1, 1.5	2.4	2.3, 2.5
48	0.8	0.8, 0.8	1.3	1.1, 1.5	2.5	2.4, 2.6
49	0.8	0.8, 0.8	1.3	1.1, 1.5	2.5	2.4, 2.6
50	0.8	0.8, 0.8	1.3	1.1, 1.5	2.5	2.4, 2.6
51	0.8	0.8, 0.8	1.3	1.1, 1.5	2.5	2.4, 2.6
52	0.8	0.8, 0.8	1.4	1.2, 1.6	2.6	2.5, 2.7
53	0.8	0.8, 0.8	1.4	1.2, 1.6	2.6	2.5, 2.7
54	0.9	0.9, 0.9	1.4	1.2, 1.6	2.6	2.5, 2.7
55	0.9	0.9, 0.9	1.4	1.2, 1.6	2.7	2.6, 2.8
56	0.9	0.9, 0.9	1.4	1.2, 1.6	2.7	2.6, 2.8
57	0.9	0.9, 0.9	1.4	1.2, 1.6	2.7	2.5, 2.9
58	0.9	0.9, 0.9	1.4	1.2, 1.6	2.8	2.6, 3.0
59	0.9	0.9, 0.9	1.4	1.2, 1.6	2.8	2.6, 3.0
60	0.9	0.9, 0.9	1.4	1.2, 1.6	2.8	2.6, 3.0
61	0.9	0.9, 0.9	1.4	1.2, 1.6	2.9	2.7, 3.1
62	1.0	1.0, 1.0	1.5	1.3, 1.7	2.9	2.7, 3.1
63	1.0	1.0, 1.0	1.5	1.3, 1.7	3.0	2.8, 3.2
64	1.0	1.0, 1.0	1.5	1.3, 1.7	3.0	2.8, 3.2
65	1.0	1.0, 1.0	1.6	1.4, 1.8	3.1	2.9, 3.3
66	1.0	1.0, 1.0	1.6	1.4, 1.8	3.1	2.9, 3.3
67	1.0	1.0, 1.0	1.6	1.4, 1.8	3.2	3.0, 3.4
68	1.1	0.9, 1.3	1.7	1.5, 1.9	3.3	3.1, 3.5
69	1.1	0.9, 1.3	1.7	1.5, 1.9	3.4	3.1, 3.7
70	1.1	0.9, 1.3	1.8	1.6, 2.0	3.6	3.3, 3.9
71	1.1	0.9, 1.3	1.9	1.7, 2.1	3.7	3.4, 4.0
72	1.1	0.9, 1.3	1.9	1.7, 2.1	3.9	3.6, 4.2
73	1.2	1.0, 1.4	2.0	1.8, 2.2	4.1	3.7, 4.5
74	1.2	1.0, 1.4	2.1	1.9, 2.3	4.4	4.0, 4.8

75	1.2	1.0, 1.4	2.2	2.0, 2.4	4.6	4.2, 5.0
76	1.2	1.0, 1.4	2.2	2.0, 2.4	4.9	4.5, 5.3
77	1.3	0.9, 1.7	2.3	2.1, 2.5	5.3	4.8, 5.8
78	1.3	0.9, 1.7	2.4	2.2, 2.6	5.6	5.1, 6.1
79	1.3	0.9, 1.7	2.5	2.1, 2.9	6.0	5.4, 6.6

Appendix 1.3. NFL continuous reference interval values for by year of age. Point values pulled from quantile regression curves. Upper limit defined by the 95th percentile, lower limit defined by the 5th percentile, median defined by the 50th percentile.

Age (y)	Lower Limit (pg/ml)	Lower Limit 95% CI	Median (pg/ml)	Median 95% CI	Upper Limit (pg/ml)	Lower Limit 95% CI
3	3.7	3.3, 4.1	6.2	5.8, 6.6	10.3	9.6, 11.0
4	3.4	3.0, 3.8	5.8	5.4, 6.2	10.2	9.6, 10.8
5	3.2	3.0, 3.4	5.5	5.1, 5.9	10.1	9.5, 10.7
6	3.0	2.8, 3.2	5.2	4.8, 5.6	10.0	9.4, 10.6
7	2.9	2.7, 3.1	4.9	4.7, 5.1	10.0	9.5, 10.5
8	2.7	2.5, 2.9	4.7	4.5, 4.9	9.9	9.4, 10.4
9	2.7	2.5, 2.9	4.6	4.4, 4.8	9.8	9.3, 10.3
10	2.6	2.4, 2.8	4.4	4.0, 4.8	9.8	9.3, 10.3
11	2.5	2.3, 2.7	4.3	3.9, 4.7	9.8	9.2, 10.4
12	2.5	2.3, 2.7	4.3	3.9, 4.7	9.7	9.1, 10.3
13	2.5	2.3, 2.7	4.3	3.9, 4.7	9.7	9.1, 10.3
14	2.4	2.2, 2.6	4.3	3.9, 4.7	9.7	9.1, 10.3
15	2.4	2.2, 2.6	4.3	3.9, 4.7	9.7	9.1, 10.3
16	2.4	2.2, 2.6	4.4	4.0, 4.8	9.7	9.1, 10.3
17	2.4	2.2, 2.6	4.4	4.0, 4.8	9.7	9.1, 10.3
18	2.4	2.2, 2.6	4.5	4.1, 4.9	9.7	9.1, 10.3
19	2.4	2.0, 2.8	4.5	4.1, 4.9	9.7	9.1, 10.3
20	2.4	2.0, 2.8	4.6	4.2, 5.0	9.8	9.2, 10.4
21	2.4	2.0, 2.8	4.7	4.3, 5.1	9.8	9.2, 10.4
22	2.4	2.0, 2.8	4.7	4.3, 5.1	9.8	9.2, 10.4
23	2.5	2.1, 2.9	4.8	4.4, 5.2	9.9	9.3, 10.5
24	2.5	2.1, 2.9	4.9	4.5, 5.3	9.9	9.3, 10.5
25	2.5	2.1, 2.9	4.9	4.5, 5.3	10.0	9.4, 10.6
26	2.6	2.2, 3.0	5.0	4.6, 5.4	10.1	9.5, 10.7
27	2.6	2.2, 3.0	5.1	4.7, 5.5	10.2	9.6, 10.8
28	2.7	2.3, 3.1	5.2	4.8, 5.6	10.3	9.7, 10.9
29	2.8	2.4, 3.2	5.3	4.9, 5.7	10.4	9.8, 11.0
30	2.9	2.5, 3.3	5.4	5.0, 5.8	10.5	10.0, 11.0
31	2.9	2.5, 3.3	5.5	5.1, 5.9	10.6	10.1, 11.1
32	3.0	2.6, 3.4	5.6	5.2, 6.0	10.8	10.3, 11.3

33	3.1	2.7, 3.5	5.8	5.4, 6.2	10.9	10.4, 11.4
34	3.2	2.8, 3.6	5.9	5.5, 6.3	11.1	10.6, 11.6
35	3.3	2.9, 3.7	6.1	5.7, 6.5	11.3	10.8, 11.8
36	3.5	3.1, 3.9	6.3	5.9, 6.7	11.5	11.0, 12.0
37	3.6	3.2, 4.0	6.5	6.1, 6.9	11.6	11.1, 12.1
38	3.7	3.3, 4.1	6.7	6.3, 7.1	11.8	11.3, 12.3
39	3.8	3.4, 4.2	6.9	6.5, 7.3	12.1	11.6, 12.6
40	3.9	3.5, 4.3	7.1	6.7, 7.5	12.3	11.8, 12.8
41	4.1	3.7, 4.5	7.3	6.9, 7.7	12.5	12.0, 13.0
42	4.2	3.8, 4.6	7.5	7.1, 7.9	12.8	12.3, 13.3
43	4.4	4.0, 4.8	7.8	7.4, 8.2	13.0	12.5, 13.5
44	4.5	4.1, 4.9	8.0	7.4, 8.6	13.3	12.8, 13.8
45	4.7	4.3, 5.1	8.1	7.5, 8.7	13.6	13.1, 14.1
46	4.8	4.4, 5.2	8.3	7.7, 8.9	13.8	13.3, 14.3
47	5.0	4.6, 5.4	8.5	7.9, 9.1	14.1	13.5, 14.7
48	5.1	4.7, 5.5	8.6	8.0, 9.2	14.5	13.9, 15.1
49	5.3	4.9, 5.7	8.8	8.2, 9.4	14.8	14.2, 15.4
50	5.5	5.1, 5.9	9.0	8.4, 9.6	15.2	14.5, 15.9
51	5.7	5.1, 6.3	9.1	8.5, 9.7	15.7	15.0, 16.4
52	5.9	5.3, 6.5	9.3	8.7, 9.9	16.1	15.3, 16.9
53	6.1	5.5, 6.7	9.5	8.9, 10.1	16.7	15.9, 17.5
54	6.2	5.6, 6.8	9.8	9.0, 10.6	17.3	16.4, 18.2
55	6.4	5.8, 7.0	10.0	9.2, 10.8	18.0	17.1, 18.9
56	6.7	6.1, 7.3	10.3	9.5, 11.1	18.7	17.7, 19.7
57	6.9	6.3, 7.5	10.6	9.8, 11.4	19.6	18.6, 20.6
58	7.1	6.5, 7.7	10.9	10.1, 11.7	20.5	19.4, 21.6
59	7.3	6.7, 7.9	11.3	10.5, 12.1	21.5	20.4, 22.6
60	7.5	6.9, 8.1	11.7	10.9, 12.5	22.5	21.3, 23.7
61	7.8	7.2, 8.4	12.1	11.3, 12.9	23.6	22.4, 24.8
62	8.0	7.4, 8.6	12.5	11.7, 13.3	24.8	23.6, 26.0
63	8.2	7.6, 8.8	13.0	12.2, 13.8	26.1	24.8, 27.4
64	8.5	7.9, 9.1	13.5	12.7, 14.3	27.5	26.2, 28.8
65	8.7	8.1, 9.3	14.0	13.2, 14.8	28.9	27.5, 30.3
66	9.0	8.2, 9.8	14.5	13.7, 15.3	30.4	29.0, 31.8
67	9.3	8.5, 10.1	15.1	14.1, 16.1	32.0	30.5, 33.5
68	9.5	8.7, 10.3	15.7	14.7, 16.7	33.7	32.2, 35.2
69	9.8	8.8, 10.8	16.3	15.3, 17.3	35.4	33.8, 37.0
70	10.1	8.9, 11.3	16.9	15.9, 17.9	37.3	35.6, 39.0
71	10.4	9.0, 11.8	17.6	16.4, 18.8	39.2	37.4, 41.0
72	10.7	9.1, 12.3	18.3	17.1, 19.5	41.2	39.3, 43.1
73	11.0	9.2, 12.8	19.0	17.6, 20.4	43.2	41.2, 45.2
74	11.3	9.3, 13.3	19.8	18.2, 21.4	45.4	43.2, 47.6

75	11.6	9.2, 14.0	20.5	18.7, 22.3	47.6	45.3, 49.9
76	11.9	9.2, 14.6	21.3	19.3, 23.3	49.9	47.4, 52.4
77	12.2	9.3, 15.1	22.2	19.8, 24.6	52.3	49.6, 55.0
78	12.5	9.2, 15.8	23.0	20.5, 25.5	54.7	51.7, 57.7
79	12.9	9.0, 16.8	23.9	21.0, 26.8	57.2	54.0, 60.4

Appendix 1.4. GFAP continuous reference interval values for by year of age. Point values pulled from quantile regression curves. Upper limit defined by the 95th percentile, lower limit defined by the 5th percentile, median defined by the 50th percentile.

Age (y)	Lower Limit (pg/ml)	Lower Limit 95% CI	Median (pg/ml)	Median 95% CI	Upper Limit (pg/ml)	Lower Limit 95% CI
3	76.6	71.5, 81.7	116.6	111.5, 121.7	181.4	175.3, 187.5
4	67.8	62.9, 72.7	107.9	103.0, 112.8	174.9	169.0, 180.8
5	60.1	55.6, 64.6	99.7	95.2, 104.2	168.6	163.0, 174.2
6	53.3	49.0, 57.6	92.0	87.7, 96.3	162.4	157.0, 167.8
7	47.4	43.3, 51.5	84.8	80.7, 88.9	156.4	151.2, 161.6
8	42.4	38.7, 46.1	78.1	74.4, 81.8	150.6	145.6, 155.6
9	38.1	34.6, 41.6	71.9	68.4, 75.4	145.0	140.2, 149.8
10	34.6	31.3, 37.9	66.2	62.9, 69.5	139.5	134.8, 144.2
11	31.6	28.5, 34.7	61.0	57.9, 64.1	134.3	129.8, 138.8
12	29.2	26.3, 32.1	56.4	53.5, 59.3	129.3	124.9, 133.7
13	27.2	24.5, 29.9	52.2	49.3, 55.1	124.5	120.3, 128.7
14	25.7	23.2, 28.2	48.5	45.8, 51.2	119.9	115.8, 124.0
15	24.5	22.0, 27.0	45.2	42.5, 47.9	115.5	111.5, 119.5
16	23.6	21.2, 26.0	42.5	40.0, 45.0	111.3	107.4, 115.2
17	22.9	20.5, 25.3	40.1	37.6, 42.6	107.3	103.5, 111.1
18	22.4	20.0, 24.8	38.3	35.8, 40.8	103.6	99.9, 107.3
19	22.0	19.8, 24.2	36.8	34.3, 39.3	100.1	96.4, 103.8
20	21.7	19.5, 23.9	35.8	33.3, 38.3	96.7	93.1, 100.3
21	21.4	19.2, 23.6	35.2	32.7, 37.7	93.6	90.1, 97.1
22	21.1	18.9, 23.3	34.9	32.4, 37.4	90.7	87.3, 94.1
23	20.9	18.7, 23.1	34.8	32.3, 37.3	88.0	84.6, 91.4
24	20.7	18.5, 22.9	35.0	32.5, 37.5	85.6	82.3, 88.9
25	20.5	18.3, 22.7	35.4	32.9, 37.9	83.3	80.1, 86.5
26	20.4	18.0, 22.8	36.0	33.5, 38.5	81.2	78.1, 84.3
27	20.3	17.9, 22.7	36.6	33.9, 39.3	79.4	76.3, 82.5
28	20.2	17.8, 22.6	37.3	34.6, 40.0	77.8	74.8, 80.8
29	20.2	17.8, 22.6	37.9	35.2, 40.6	76.4	73.5, 79.3
30	20.2	17.8, 22.6	38.6	35.9, 41.3	75.1	72.3, 77.9
31	20.3	17.9, 22.7	39.3	36.6, 42.0	74.2	71.5, 76.9
32	20.3	17.9, 22.7	40.0	37.1, 42.9	73.4	70.8, 76.0
33	20.5	18.0, 23.0	40.7	37.8, 43.6	72.8	70.2, 75.4
34	20.6	18.1, 23.1	41.4	38.5, 44.3	72.5	70.0, 75.0
35	20.8	18.3, 23.3	42.1	39.2, 45.0	72.3	69.9, 74.7
36	21.0	18.5, 23.5	42.8	39.9, 45.7	72.4	70.1, 74.7

37	21.3	18.8, 23.8	43.6	40.7, 46.5	72.7	70.5, 74.9
38	21.6	19.1, 24.1	44.3	41.4, 47.2	73.1	70.9, 75.3
39	21.9	19.4, 24.4	45.0	42.1, 47.9	73.9	71.7, 76.1
40	22.3	19.8, 24.8	45.8	42.9, 48.7	74.8	72.7, 76.9
41	22.7	20.2, 25.2	46.5	43.6, 49.4	75.9	73.8, 78.0
42	23.2	20.7, 25.7	47.3	44.4, 50.2	77.2	75.0, 79.4
43	23.6	21.1, 26.1	48.0	45.1, 50.9	78.8	76.6, 81.0
44	24.2	21.8, 26.6	48.8	45.9, 51.7	80.6	78.3, 82.9
45	24.7	22.3, 27.1	49.5	46.6, 52.4	82.5	80.0, 85.0
46	25.3	22.9, 27.7	50.3	47.4, 53.2	84.7	82.1, 87.3
47	25.9	23.5, 28.3	51.1	48.2, 54.0	87.1	84.3, 89.9
48	26.6	24.2, 29.0	52.0	49.1, 54.9	89.7	86.7, 92.7
49	27.3	24.9, 29.7	52.8	49.9, 55.7	92.6	89.3, 95.9
50	28.0	25.6, 30.4	53.8	50.9, 56.7	95.6	92.0, 99.2
51	28.8	26.3, 31.3	54.8	51.7, 57.9	98.8	94.9, 102.7
52	29.6	27.1, 32.1	55.9	52.8, 59.0	102.3	98.1, 106.5
53	30.4	27.9, 32.9	57.1	54.0, 60.2	106.0	101.5, 110.5
54	31.3	28.8, 33.8	58.4	55.1, 61.7	109.9	105.0, 114.8
55	32.2	29.5, 34.9	59.8	56.5, 63.1	114.0	108.7, 119.3
56	33.1	30.4, 35.8	61.4	57.9, 64.9	118.3	112.6, 124.0
57	34.1	31.2, 37.0	63.0	59.3, 66.7	122.8	116.7, 128.9
58	35.1	32.0, 38.2	64.8	60.9, 68.7	127.5	121.0, 134.0
59	36.2	33.1, 39.3	66.8	62.7, 70.9	132.5	125.5, 139.5
60	37.3	34.0, 40.6	68.8	64.5, 73.1	137.6	130.2, 145.0
61	38.4	34.9, 41.9	71.0	66.5, 75.5	143.0	135.1, 150.9
62	39.6	35.9, 43.3	73.3	68.6, 78.0	148.6	140.2, 157.0
63	40.8	36.7, 44.9	75.7	70.8, 80.6	154.4	145.5, 163.3
64	42.0	37.7, 46.3	78.3	73.0, 83.6	160.4	150.9, 169.9
65	43.3	38.8, 47.8	81.0	75.5, 86.5	166.6	156.6, 176.6
66	44.6	39.7, 49.5	83.8	77.9, 89.7	173.0	162.4, 183.6
67	45.9	40.8, 51.0	86.7	80.6, 92.8	179.7	168.5, 190.9
68	47.3	41.8, 52.8	89.8	83.3, 96.3	186.5	174.7, 198.3
69	48.7	43.0, 54.4	92.9	86.0, 99.8	193.6	181.2, 206.0
70	50.2	44.1, 56.3	96.2	88.9, 103.5	200.9	187.9, 213.9
71	51.6	45.1, 58.1	99.7	92.1, 107.3	208.4	194.8, 222.0
72	53.2	46.3, 60.1	103.2	95.2, 111.2	216.1	201.8, 230.4
73	54.7	47.4, 62.0	106.9	98.5, 115.3	224.0	209.1, 238.9
74	56.3	48.7, 63.9	110.7	101.9, 119.5	232.1	216.5, 247.7
75	57.9	49.9, 65.9	114.6	105.2, 124.0	240.4	224.1, 256.7
76	59.6	51.2, 68.0	118.7	108.9, 128.5	249.0	232.0, 266.0
77	61.3	52.5, 70.1	122.8	112.4, 133.2	257.8	240.0, 275.6
78	63.0	53.8, 72.2	127.1	116.3, 137.9	266.7	248.2, 285.2
79	64.8	55.2, 74.4	131.6	120.2, 143.0	275.9	256.6, 295.2

Appendix 2. Variables from the Canadian Health Measures Survey analyzed in Chapter 3.

Appendix 2.1. Variables from the Canadian Health Measures Survey analyzed The table indicates the origin of the survey variable and the original coding, with any modifications made for analysis. Sample sizes for variables analyzed in the lower section of the table. Reference groups were females for sex, Caucasians for ethnicity, and not fasted for fasted.

CHMS Variable	Simple Name	CHMS Data Dictionary	Survey Description	Other Code Descriptions	Variable Cleaning (make binary)							
PGDCGT	Ethnicity	Household	Cultural or racial group - (D)	01 = White, 02 = Black, 03 = Korean, 04 = Filipino, 05 = Japanese, 06 = Chinese, 07 = South Asian, 08 = Southeast Asian, 09 = Arab, 10 = West Asian, 11 = Latin America, 12 = Other racial or cultural origin, 13 = Multiple racial or cultural origins	Combine 2,3,4,5,6,7,8,9,10,11,12,13 to be Non-Caucasian, coded 2							
CLC SEX	Sex	Clinic		1 = Male, 2 = Female								
CLC AGE	Age	Clinic										
PHID12	Fasted	Clinic	Fasting status - (D)	1 = Selected for subsample, fasted, 2 = Selected for subsample, not fasted, 3 = Not selected for subsample, fasted, 4 = Not selected for subsample, not fasted	Yes/Fasted (1) = 1, 3 No/Not fasted (2) = 2, 4							
HWMDBMI	BMI	Clinic	Body Mass Index measured	N/A								
LAB_BCRE	Creatinine	Lab	Blood creatinine (mol/L)	N/A								
LAB_UCRE	Creatinine (urine)	Lab	Creatinine (mmol/L)	N/A								
Sample sizes												
	Age 3-19			Age 20-39			Age 40-59			Age 60-79		
	All	Ref	Test	All	Ref	Test	All	Ref	Test	All	Ref	Test
Ethnicity	670	470	210	530	380	150	520	400	120	700	620	80
Sex	700	350	350	550	270	270	550	270	270	720	360	360
Age	700			550			550			720		
Fasted	700	450	240	550	270	280	540	260	280	720	270	450
BMI	700			530			540			710		
Creatinine	700			540			550			720		
Creatinine (urine)	670			540			540			700		

Appendix 3. Tables of age-specific reference curve values for A β 42/40, p-tau-181, NfL, and GFAP from Chapter 3.

Appendix 3.1. Age-specific reference curves percentiles for A β 42/40. Points pulled from the GAMLSS model, with percentiles extracted at each age, where the midpoint of an age is used to represent the range of that year of age.

	A β 42/40 Percentiles										
Age	1	3	5	15	25	50	75	85	95	97	99
3	0.0449	0.0555	0.0600	0.0694	0.0740	0.0816	0.0890	0.0933	0.1017	0.1055	0.1144
4	0.0446	0.0551	0.0596	0.0690	0.0736	0.0811	0.0885	0.0928	0.1011	0.1050	0.1138
5	0.0443	0.0548	0.0593	0.0686	0.0731	0.0807	0.0880	0.0922	0.1006	0.1044	0.1132
6	0.0439	0.0544	0.0588	0.0681	0.0726	0.0801	0.0874	0.0917	0.0999	0.1038	0.1125
7	0.0436	0.0540	0.0584	0.0676	0.0721	0.0796	0.0869	0.0911	0.0993	0.1031	0.1118
8	0.0432	0.0535	0.0580	0.0671	0.0716	0.0791	0.0863	0.0905	0.0987	0.1025	0.1111
9	0.0428	0.0531	0.0575	0.0666	0.0711	0.0785	0.0857	0.0899	0.0980	0.1018	0.1104
10	0.0424	0.0527	0.0570	0.0661	0.0706	0.0779	0.0851	0.0892	0.0973	0.1011	0.1096
11	0.0421	0.0522	0.0566	0.0656	0.0700	0.0773	0.0845	0.0886	0.0966	0.1004	0.1089
12	0.0417	0.0518	0.0561	0.0651	0.0695	0.0768	0.0838	0.0879	0.0960	0.0997	0.1081
13	0.0413	0.0513	0.0556	0.0646	0.0689	0.0762	0.0832	0.0873	0.0953	0.0990	0.1074
14	0.0409	0.0509	0.0552	0.0641	0.0684	0.0756	0.0826	0.0867	0.0946	0.0983	0.1067
15	0.0405	0.0505	0.0547	0.0636	0.0679	0.0751	0.0821	0.0861	0.0940	0.0976	0.1060
16	0.0401	0.0501	0.0543	0.0631	0.0674	0.0746	0.0815	0.0855	0.0934	0.0970	0.1053
17	0.0398	0.0497	0.0539	0.0627	0.0670	0.0741	0.0810	0.0850	0.0928	0.0964	0.1047
18	0.0395	0.0493	0.0535	0.0623	0.0665	0.0736	0.0805	0.0845	0.0923	0.0959	0.1041
19	0.0392	0.0490	0.0532	0.0619	0.0661	0.0732	0.0801	0.0840	0.0918	0.0954	0.1036
20	0.0389	0.0486	0.0528	0.0615	0.0658	0.0728	0.0797	0.0836	0.0914	0.0950	0.1031
21	0.0386	0.0483	0.0525	0.0612	0.0655	0.0725	0.0793	0.0833	0.0910	0.0946	0.1027
22	0.0383	0.0481	0.0523	0.0609	0.0652	0.0722	0.0790	0.0830	0.0907	0.0943	0.1024
23	0.0381	0.0478	0.0520	0.0607	0.0649	0.0719	0.0788	0.0827	0.0904	0.0940	0.1021
24	0.0379	0.0476	0.0518	0.0605	0.0647	0.0717	0.0785	0.0825	0.0902	0.0938	0.1019
25	0.0377	0.0474	0.0516	0.0603	0.0645	0.0715	0.0784	0.0823	0.0900	0.0936	0.1017
26	0.0375	0.0473	0.0514	0.0601	0.0644	0.0714	0.0782	0.0822	0.0899	0.0935	0.1016
27	0.0374	0.0471	0.0513	0.0600	0.0642	0.0712	0.0781	0.0820	0.0898	0.0933	0.1015
28	0.0372	0.0469	0.0511	0.0598	0.0641	0.0711	0.0780	0.0819	0.0897	0.0932	0.1014
29	0.0370	0.0468	0.0510	0.0597	0.0639	0.0710	0.0778	0.0818	0.0895	0.0931	0.1013
30	0.0369	0.0466	0.0508	0.0595	0.0638	0.0708	0.0777	0.0817	0.0894	0.0930	0.1011
31	0.0367	0.0464	0.0506	0.0594	0.0636	0.0707	0.0776	0.0815	0.0893	0.0929	0.1010
32	0.0365	0.0463	0.0505	0.0592	0.0635	0.0705	0.0774	0.0814	0.0891	0.0927	0.1009
33	0.0363	0.0461	0.0503	0.0590	0.0633	0.0704	0.0773	0.0812	0.0890	0.0926	0.1008
34	0.0361	0.0459	0.0501	0.0589	0.0631	0.0702	0.0771	0.0811	0.0888	0.0924	0.1006
35	0.0359	0.0457	0.0499	0.0587	0.0630	0.0701	0.0770	0.0809	0.0887	0.0923	0.1005
36	0.0358	0.0455	0.0498	0.0585	0.0628	0.0699	0.0768	0.0808	0.0886	0.0922	0.1004

37	0.0356	0.0454	0.0496	0.0584	0.0627	0.0698	0.0767	0.0806	0.0884	0.0920	0.1002
38	0.0354	0.0452	0.0494	0.0582	0.0625	0.0696	0.0765	0.0805	0.0883	0.0919	0.1001
39	0.0352	0.0450	0.0493	0.0581	0.0624	0.0695	0.0764	0.0804	0.0882	0.0918	0.1000
40	0.0350	0.0448	0.0491	0.0579	0.0622	0.0693	0.0763	0.0803	0.0881	0.0917	0.0999
41	0.0348	0.0447	0.0489	0.0577	0.0620	0.0692	0.0761	0.0801	0.0879	0.0916	0.0998
42	0.0346	0.0445	0.0487	0.0576	0.0619	0.0690	0.0760	0.0800	0.0878	0.0914	0.0996
43	0.0344	0.0443	0.0485	0.0574	0.0617	0.0688	0.0758	0.0798	0.0876	0.0912	0.0995
44	0.0342	0.0440	0.0483	0.0571	0.0615	0.0686	0.0756	0.0796	0.0874	0.0910	0.0993
45	0.0340	0.0438	0.0480	0.0569	0.0612	0.0684	0.0753	0.0793	0.0872	0.0908	0.0990
46	0.0337	0.0435	0.0478	0.0566	0.0609	0.0681	0.0750	0.0790	0.0869	0.0905	0.0987
47	0.0334	0.0432	0.0475	0.0563	0.0606	0.0678	0.0747	0.0787	0.0865	0.0901	0.0983
48	0.0332	0.0429	0.0472	0.0560	0.0603	0.0674	0.0743	0.0783	0.0861	0.0897	0.0979
49	0.0328	0.0426	0.0468	0.0556	0.0599	0.0670	0.0739	0.0779	0.0857	0.0893	0.0975
50	0.0325	0.0422	0.0465	0.0552	0.0595	0.0666	0.0735	0.0775	0.0852	0.0888	0.0970
51	0.0322	0.0419	0.0461	0.0548	0.0591	0.0662	0.0730	0.0770	0.0847	0.0883	0.0964
52	0.0319	0.0415	0.0457	0.0544	0.0587	0.0658	0.0726	0.0765	0.0843	0.0878	0.0959
53	0.0316	0.0412	0.0453	0.0541	0.0583	0.0653	0.0722	0.0761	0.0838	0.0874	0.0954
54	0.0313	0.0408	0.0450	0.0537	0.0580	0.0650	0.0718	0.0757	0.0834	0.0869	0.0950
55	0.0310	0.0406	0.0447	0.0534	0.0577	0.0647	0.0715	0.0754	0.0831	0.0866	0.0947
56	0.0308	0.0403	0.0445	0.0532	0.0574	0.0644	0.0712	0.0752	0.0828	0.0864	0.0944
57	0.0306	0.0401	0.0443	0.0530	0.0572	0.0643	0.0711	0.0750	0.0827	0.0862	0.0943
58	0.0304	0.0400	0.0442	0.0529	0.0571	0.0642	0.0710	0.0750	0.0826	0.0862	0.0943
59	0.0303	0.0399	0.0441	0.0528	0.0571	0.0642	0.0710	0.0750	0.0827	0.0863	0.0944
60	0.0302	0.0398	0.0440	0.0528	0.0571	0.0642	0.0711	0.0751	0.0828	0.0864	0.0945
61	0.0301	0.0398	0.0440	0.0528	0.0571	0.0643	0.0712	0.0752	0.0830	0.0866	0.0947
62	0.0300	0.0397	0.0440	0.0528	0.0572	0.0643	0.0713	0.0753	0.0831	0.0867	0.0949
63	0.0299	0.0396	0.0439	0.0528	0.0571	0.0643	0.0713	0.0753	0.0832	0.0868	0.0951
64	0.0297	0.0395	0.0438	0.0527	0.0571	0.0643	0.0713	0.0754	0.0832	0.0869	0.0951
65	0.0296	0.0394	0.0437	0.0526	0.0570	0.0643	0.0713	0.0753	0.0832	0.0869	0.0952
66	0.0294	0.0392	0.0435	0.0525	0.0569	0.0641	0.0712	0.0752	0.0831	0.0868	0.0951
67	0.0292	0.0390	0.0433	0.0523	0.0567	0.0639	0.0710	0.0750	0.0829	0.0866	0.0949
68	0.0289	0.0387	0.0430	0.0520	0.0564	0.0636	0.0707	0.0747	0.0827	0.0863	0.0946
69	0.0286	0.0384	0.0427	0.0517	0.0561	0.0633	0.0703	0.0744	0.0823	0.0859	0.0942
70	0.0283	0.0381	0.0423	0.0513	0.0557	0.0629	0.0699	0.0740	0.0818	0.0855	0.0937
71	0.0280	0.0377	0.0420	0.0509	0.0553	0.0625	0.0695	0.0735	0.0813	0.0850	0.0932
72	0.0277	0.0373	0.0416	0.0504	0.0548	0.0620	0.0689	0.0730	0.0808	0.0844	0.0926
73	0.0273	0.0369	0.0411	0.0500	0.0543	0.0615	0.0684	0.0724	0.0802	0.0838	0.0919
74	0.0269	0.0365	0.0407	0.0495	0.0538	0.0609	0.0678	0.0718	0.0795	0.0831	0.0912
75	0.0266	0.0360	0.0402	0.0490	0.0533	0.0604	0.0672	0.0712	0.0789	0.0824	0.0905
76	0.0262	0.0356	0.0398	0.0485	0.0527	0.0598	0.0666	0.0705	0.0782	0.0818	0.0898
77	0.0258	0.0352	0.0393	0.0480	0.0522	0.0592	0.0660	0.0699	0.0775	0.0810	0.0890
78	0.0254	0.0347	0.0388	0.0474	0.0517	0.0586	0.0654	0.0693	0.0768	0.0803	0.0883
79	0.0251	0.0343	0.0383	0.0469	0.0511	0.0580	0.0647	0.0686	0.0761	0.0796	0.0875

Appendix 3.2. Age-specific reference curves percentiles for p-tau-181. Points pulled from the GAMLSS model, with percentiles extracted at each age, where the midpoint of an age is used to represent the range of that year of age.

	p-tau-181 Percentiles										
Age	1	3	5	15	25	50	75	85	95	97	99
3	1.28	1.50	1.63	2.00	2.26	2.85	3.70	4.34	5.98	6.95	9.83
4	1.25	1.46	1.59	1.95	2.20	2.78	3.60	4.22	5.81	6.75	9.54
5	1.21	1.42	1.55	1.89	2.14	2.70	3.50	4.10	5.63	6.54	9.22
6	1.18	1.38	1.50	1.84	2.07	2.62	3.39	3.97	5.45	6.32	8.90
7	1.14	1.34	1.45	1.78	2.01	2.53	3.27	3.83	5.25	6.09	8.56
8	1.11	1.30	1.41	1.72	1.94	2.44	3.16	3.69	5.06	5.86	8.22
9	1.07	1.25	1.36	1.66	1.87	2.36	3.04	3.55	4.86	5.63	7.88
10	1.03	1.21	1.31	1.60	1.80	2.27	2.92	3.42	4.67	5.40	7.54
11	1.00	1.17	1.26	1.54	1.74	2.18	2.81	3.28	4.47	5.17	7.21
12	0.96	1.12	1.22	1.49	1.67	2.10	2.70	3.15	4.29	4.95	6.89
13	0.93	1.08	1.17	1.43	1.61	2.02	2.59	3.02	4.11	4.74	6.58
14	0.89	1.04	1.13	1.38	1.55	1.94	2.49	2.90	3.94	4.54	6.29
15	0.86	1.01	1.09	1.33	1.49	1.87	2.40	2.79	3.78	4.35	6.01
16	0.83	0.97	1.05	1.28	1.44	1.80	2.31	2.68	3.63	4.18	5.76
17	0.81	0.94	1.02	1.24	1.39	1.74	2.23	2.59	3.49	4.02	5.53
18	0.79	0.92	0.99	1.20	1.35	1.69	2.16	2.50	3.37	3.87	5.32
19	0.77	0.89	0.97	1.17	1.31	1.64	2.09	2.43	3.26	3.74	5.13
20	0.75	0.87	0.94	1.14	1.28	1.60	2.03	2.36	3.16	3.63	4.96
21	0.73	0.85	0.92	1.12	1.25	1.56	1.99	2.30	3.08	3.53	4.81
22	0.72	0.84	0.91	1.10	1.23	1.53	1.94	2.25	3.01	3.44	4.68
23	0.71	0.83	0.89	1.08	1.21	1.50	1.91	2.20	2.94	3.36	4.56
24	0.70	0.82	0.88	1.07	1.19	1.48	1.88	2.17	2.89	3.30	4.46
25	0.70	0.81	0.87	1.06	1.18	1.46	1.85	2.14	2.84	3.24	4.38
26	0.69	0.80	0.87	1.05	1.17	1.45	1.83	2.11	2.80	3.20	4.30
27	0.69	0.80	0.86	1.04	1.16	1.44	1.82	2.09	2.77	3.16	4.24
28	0.69	0.80	0.86	1.04	1.16	1.43	1.80	2.08	2.75	3.12	4.19
29	0.69	0.80	0.86	1.04	1.16	1.43	1.80	2.07	2.73	3.10	4.15
30	0.69	0.80	0.86	1.04	1.16	1.43	1.79	2.06	2.71	3.08	4.11
31	0.69	0.80	0.87	1.04	1.16	1.43	1.79	2.06	2.71	3.08	4.09
32	0.70	0.81	0.87	1.05	1.17	1.43	1.80	2.07	2.71	3.08	4.09
33	0.71	0.82	0.88	1.06	1.18	1.44	1.81	2.08	2.72	3.08	4.09
34	0.71	0.83	0.89	1.07	1.19	1.46	1.82	2.09	2.73	3.10	4.10
35	0.72	0.84	0.90	1.08	1.20	1.47	1.84	2.11	2.75	3.11	4.12
36	0.73	0.85	0.91	1.09	1.21	1.49	1.86	2.13	2.78	3.14	4.14
37	0.74	0.86	0.92	1.11	1.23	1.50	1.88	2.15	2.80	3.16	4.17
38	0.75	0.87	0.94	1.12	1.24	1.52	1.90	2.17	2.83	3.19	4.20
39	0.76	0.88	0.95	1.13	1.26	1.54	1.92	2.19	2.85	3.22	4.23
40	0.77	0.89	0.96	1.15	1.27	1.56	1.94	2.21	2.88	3.25	4.26

41	0.78	0.90	0.97	1.16	1.29	1.57	1.96	2.23	2.90	3.27	4.29
42	0.79	0.91	0.98	1.17	1.30	1.59	1.97	2.25	2.92	3.30	4.32
43	0.80	0.92	0.99	1.18	1.31	1.60	1.99	2.27	2.94	3.31	4.34
44	0.80	0.92	0.99	1.18	1.32	1.61	2.00	2.28	2.96	3.33	4.36
45	0.80	0.93	1.00	1.19	1.32	1.61	2.01	2.29	2.97	3.34	4.38
46	0.81	0.93	1.00	1.19	1.32	1.62	2.01	2.30	2.98	3.35	4.39
47	0.81	0.93	1.00	1.20	1.33	1.62	2.02	2.30	2.99	3.37	4.41
48	0.81	0.93	1.00	1.20	1.33	1.62	2.02	2.31	2.99	3.38	4.42
49	0.81	0.93	1.00	1.20	1.33	1.63	2.02	2.31	3.00	3.39	4.44
50	0.81	0.93	1.00	1.20	1.33	1.63	2.03	2.32	3.02	3.40	4.47
51	0.81	0.93	1.00	1.20	1.33	1.63	2.04	2.33	3.03	3.42	4.50
52	0.81	0.93	1.00	1.20	1.34	1.64	2.05	2.34	3.05	3.45	4.55
53	0.81	0.94	1.01	1.21	1.34	1.65	2.06	2.35	3.08	3.48	4.60
54	0.81	0.94	1.01	1.21	1.35	1.65	2.07	2.37	3.10	3.51	4.65
55	0.81	0.94	1.01	1.21	1.35	1.66	2.08	2.39	3.13	3.55	4.71
56	0.81	0.94	1.01	1.22	1.36	1.67	2.10	2.41	3.17	3.59	4.78
57	0.81	0.94	1.02	1.22	1.36	1.68	2.11	2.43	3.20	3.64	4.86
58	0.82	0.94	1.02	1.23	1.37	1.69	2.13	2.45	3.24	3.68	4.93
59	0.82	0.95	1.02	1.23	1.37	1.70	2.14	2.47	3.27	3.73	5.01
60	0.82	0.95	1.02	1.23	1.38	1.71	2.16	2.49	3.31	3.78	5.09
61	0.82	0.95	1.02	1.24	1.38	1.72	2.18	2.51	3.35	3.82	5.17
62	0.82	0.95	1.03	1.24	1.39	1.73	2.19	2.54	3.39	3.88	5.27
63	0.82	0.95	1.03	1.25	1.40	1.74	2.21	2.56	3.44	3.94	5.37
64	0.82	0.96	1.03	1.26	1.41	1.76	2.24	2.60	3.49	4.01	5.48
65	0.83	0.96	1.04	1.27	1.42	1.77	2.27	2.64	3.55	4.08	5.61
66	0.83	0.97	1.05	1.28	1.44	1.80	2.31	2.68	3.63	4.18	5.76
67	0.84	0.98	1.07	1.30	1.46	1.83	2.35	2.74	3.72	4.28	5.94
68	0.85	1.00	1.08	1.32	1.49	1.87	2.40	2.80	3.82	4.41	6.14
69	0.87	1.02	1.10	1.35	1.52	1.91	2.47	2.88	3.94	4.56	6.37
70	0.89	1.04	1.13	1.38	1.56	1.97	2.54	2.98	4.08	4.73	6.64
71	0.91	1.07	1.16	1.43	1.61	2.03	2.63	3.08	4.24	4.93	6.95
72	0.94	1.11	1.20	1.48	1.67	2.11	2.74	3.21	4.43	5.16	7.31
73	0.98	1.15	1.25	1.53	1.73	2.20	2.86	3.36	4.66	5.43	7.73
74	1.02	1.20	1.30	1.60	1.81	2.30	3.00	3.53	4.91	5.73	8.20
75	1.06	1.25	1.36	1.67	1.90	2.41	3.15	3.72	5.19	6.07	8.73
76	1.11	1.31	1.42	1.76	1.99	2.54	3.33	3.93	5.51	6.46	9.33
77	1.16	1.37	1.49	1.85	2.10	2.68	3.52	4.16	5.85	6.88	9.99
78	1.22	1.44	1.57	1.94	2.21	2.83	3.72	4.42	6.23	7.34	10.71
79	1.28	1.52	1.65	2.05	2.33	2.99	3.95	4.69	6.64	7.84	11.50

Appendix 3.3. Age-specific reference curves percentiles for NfL. Points pulled from the GAMLSS model, with percentiles extracted at each age, where the midpoint of an age is used to represent the range of that year of age.

	NfL Percentiles										
Age	1	3	5	15	25	50	75	85	95	97	99
3	2.86	3.29	3.55	4.32	4.86	6.12	7.80	8.96	11.49	12.73	15.63
4	2.71	3.13	3.37	4.11	4.63	5.84	7.47	8.59	11.05	12.26	15.10
5	2.57	2.97	3.21	3.91	4.42	5.58	7.16	8.25	10.64	11.82	14.60
6	2.45	2.83	3.06	3.74	4.22	5.35	6.88	7.94	10.27	11.42	14.16
7	2.33	2.70	2.92	3.58	4.05	5.14	6.63	7.66	9.94	11.08	13.78
8	2.24	2.59	2.80	3.44	3.90	4.96	6.42	7.43	9.67	10.79	13.46
9	2.15	2.50	2.71	3.33	3.78	4.82	6.24	7.24	9.46	10.57	13.23
10	2.09	2.43	2.63	3.24	3.68	4.71	6.11	7.10	9.31	10.42	13.09
11	2.04	2.37	2.57	3.18	3.61	4.63	6.03	7.02	9.23	10.35	13.05
12	2.01	2.34	2.54	3.14	3.58	4.60	6.00	6.99	9.23	10.36	13.10
13	1.99	2.32	2.52	3.13	3.57	4.59	6.01	7.02	9.29	10.44	13.25
14	1.98	2.32	2.52	3.13	3.57	4.61	6.05	7.07	9.39	10.58	13.46
15	1.99	2.33	2.53	3.15	3.60	4.65	6.12	7.16	9.53	10.74	13.72
16	2.00	2.34	2.55	3.18	3.63	4.70	6.19	7.26	9.68	10.93	14.00
17	2.01	2.36	2.57	3.21	3.67	4.76	6.28	7.36	9.85	11.13	14.28
18	2.03	2.38	2.60	3.24	3.71	4.82	6.36	7.47	10.01	11.32	14.56
19	2.05	2.41	2.62	3.28	3.76	4.88	6.45	7.57	10.16	11.50	14.82
20	2.07	2.43	2.65	3.32	3.80	4.93	6.52	7.66	10.29	11.66	15.04
21	2.09	2.46	2.68	3.35	3.84	4.98	6.59	7.75	10.40	11.79	15.22
22	2.11	2.48	2.71	3.39	3.88	5.04	6.66	7.82	10.50	11.90	15.38
23	2.13	2.51	2.74	3.43	3.93	5.09	6.73	7.90	10.60	12.01	15.51
24	2.16	2.55	2.78	3.48	3.98	5.15	6.80	7.98	10.70	12.11	15.65
25	2.20	2.59	2.83	3.53	4.04	5.23	6.88	8.07	10.80	12.23	15.79
26	2.24	2.64	2.88	3.60	4.11	5.31	6.98	8.17	10.93	12.36	15.95
27	2.29	2.70	2.94	3.67	4.19	5.41	7.09	8.30	11.07	12.52	16.13
28	2.35	2.77	3.02	3.76	4.29	5.52	7.23	8.44	11.24	12.70	16.35
29	2.42	2.85	3.10	3.86	4.40	5.65	7.38	8.61	11.44	12.92	16.60
30	2.50	2.94	3.20	3.97	4.52	5.80	7.55	8.80	11.67	13.16	16.89
31	2.58	3.04	3.30	4.10	4.66	5.96	7.74	9.01	11.92	13.43	17.21
32	2.67	3.14	3.42	4.23	4.81	6.14	7.95	9.24	12.19	13.73	17.56
33	2.77	3.26	3.54	4.38	4.97	6.33	8.18	9.49	12.48	14.04	17.94
34	2.87	3.37	3.67	4.53	5.14	6.52	8.41	9.74	12.79	14.37	18.33
35	2.98	3.50	3.80	4.69	5.31	6.72	8.65	10.00	13.09	14.70	18.72
36	3.08	3.62	3.93	4.84	5.48	6.92	8.88	10.25	13.39	15.02	19.11
37	3.19	3.74	4.06	5.00	5.65	7.12	9.11	10.50	13.68	15.34	19.48
38	3.30	3.86	4.19	5.15	5.82	7.32	9.33	10.75	13.97	15.65	19.84
39	3.40	3.99	4.32	5.31	5.98	7.51	9.56	10.99	14.25	15.95	20.21
40	3.51	4.11	4.46	5.46	6.15	7.70	9.78	11.23	14.53	16.25	20.57
41	3.61	4.23	4.59	5.61	6.32	7.90	10.00	11.47	14.82	16.56	20.94

42	3.71	4.35	4.72	5.77	6.49	8.09	10.22	11.71	15.10	16.86	21.31
43	3.81	4.47	4.84	5.92	6.65	8.28	10.44	11.95	15.38	17.17	21.69
44	3.91	4.59	4.97	6.07	6.81	8.47	10.66	12.18	15.66	17.48	22.07
45	4.01	4.70	5.09	6.21	6.97	8.65	10.87	12.41	15.93	17.78	22.46
46	4.10	4.81	5.21	6.35	7.12	8.83	11.07	12.63	16.20	18.08	22.84
47	4.18	4.91	5.32	6.49	7.27	9.00	11.27	12.85	16.47	18.37	23.23
48	4.26	5.01	5.43	6.62	7.41	9.16	11.46	13.06	16.73	18.67	23.62
49	4.34	5.11	5.54	6.75	7.56	9.33	11.66	13.28	17.01	18.98	24.05
50	4.43	5.22	5.66	6.89	7.71	9.52	11.88	13.52	17.32	19.33	24.53
51	4.53	5.34	5.79	7.05	7.90	9.73	12.14	13.81	17.69	19.75	25.11
52	4.64	5.48	5.94	7.25	8.11	9.99	12.44	14.16	18.14	20.26	25.81
53	4.77	5.64	6.12	7.47	8.35	10.28	12.81	14.57	18.67	20.87	26.64
54	4.92	5.83	6.33	7.72	8.64	10.63	13.22	15.04	19.28	21.57	27.61
55	5.09	6.04	6.56	8.01	8.96	11.02	13.70	15.58	19.99	22.38	28.72
56	5.28	6.27	6.82	8.33	9.32	11.45	14.24	16.19	20.79	23.29	29.98
57	5.48	6.52	7.10	8.68	9.71	11.93	14.82	16.86	21.66	24.29	31.38
58	5.69	6.79	7.40	9.05	10.12	12.43	15.45	17.57	22.60	25.37	32.89
59	5.91	7.07	7.70	9.43	10.56	12.96	16.10	18.31	23.59	26.51	34.51
60	6.13	7.35	8.02	9.83	11.00	13.51	16.77	19.08	24.62	27.70	36.21
61	6.35	7.64	8.34	10.23	11.46	14.06	17.46	19.86	25.67	28.93	37.99
62	6.57	7.92	8.66	10.64	11.91	14.62	18.15	20.66	26.74	30.18	39.84
63	6.78	8.20	8.97	11.04	12.37	15.18	18.84	21.45	27.83	31.46	41.76
64	6.98	8.47	9.28	11.44	12.82	15.74	19.54	22.26	28.93	32.76	43.75
65	7.18	8.75	9.59	11.85	13.28	16.30	20.24	23.07	30.07	34.11	45.85
66	7.38	9.02	9.91	12.26	13.75	16.88	20.97	23.91	31.24	35.51	48.07
67	7.57	9.30	10.23	12.68	14.23	17.48	21.71	24.77	32.46	36.98	50.44
68	7.76	9.58	10.55	13.11	14.72	18.09	22.48	25.66	33.72	38.52	52.96
69	7.95	9.86	10.88	13.55	15.22	18.71	23.26	26.57	35.03	40.12	55.64
70	8.13	10.14	11.20	13.99	15.72	19.33	24.05	27.50	36.38	41.77	58.50
71	8.31	10.41	11.53	14.43	16.23	19.97	24.85	28.44	37.77	43.50	61.55
72	8.48	10.69	11.86	14.89	16.75	20.63	25.68	29.42	39.22	45.32	64.83
73	8.64	10.97	12.19	15.35	17.29	21.30	26.54	30.43	40.75	47.23	68.38
74	8.81	11.25	12.54	15.84	17.85	22.00	27.43	31.48	42.35	49.27	72.26
75	8.97	11.54	12.89	16.34	18.43	22.73	28.36	32.59	44.05	51.45	76.50
76	9.12	11.84	13.25	16.86	19.04	23.49	29.34	33.74	45.84	53.77	81.16
77	9.28	12.13	13.62	17.40	19.66	24.29	30.35	34.95	47.74	56.25	86.28
78	9.43	12.44	14.01	17.96	20.32	25.11	31.41	36.22	49.76	58.90	91.94
79	9.57	12.75	14.40	18.54	20.99	25.97	32.51	37.54	51.89	61.73	98.19

Appendix 3.4. Age-specific reference curves percentiles for GFAP. Points pulled from the GAMLSS model, with percentiles extracted at each age, where the midpoint of an age is used to represent the range of that year of age.

	GFAP Percentiles										
Age	1	3	5	15	25	50	75	85	95	97	99
3	48.7	61.8	69.0	88.5	101.0	126.9	158.2	178.6	221.6	242.6	292.2
4	42.4	53.8	60.1	77.2	88.2	110.9	138.4	156.3	194.1	212.6	256.2
5	37.3	47.4	53.0	68.2	77.9	98.1	122.5	138.4	172.0	188.4	227.3
6	33.2	42.2	47.2	60.8	69.5	87.6	109.5	123.7	153.9	168.7	203.6
7	29.8	38.0	42.5	54.8	62.6	79.0	98.8	111.7	139.1	152.4	184.1
8	27.0	34.4	38.5	49.7	56.9	71.8	89.9	101.7	126.7	138.9	167.8
9	24.7	31.5	35.3	45.5	52.1	65.9	82.5	93.3	116.4	127.6	154.3
10	22.7	29.0	32.5	42.0	48.2	60.9	76.3	86.4	107.8	118.2	143.0
11	21.1	27.0	30.2	39.1	44.8	56.7	71.1	80.5	100.5	110.3	133.5
12	19.7	25.2	28.3	36.6	42.0	53.2	66.7	75.6	94.4	103.6	125.5
13	18.6	23.8	26.7	34.6	39.6	50.2	63.0	71.4	89.3	98.0	118.7
14	17.6	22.5	25.3	32.8	37.6	47.7	59.9	67.9	84.9	93.3	113.0
15	16.7	21.5	24.1	31.3	35.9	45.5	57.3	64.9	81.2	89.2	108.2
16	16.0	20.6	23.1	30.0	34.5	43.7	55.0	62.4	78.2	85.9	104.2
17	15.5	19.9	22.3	29.0	33.3	42.3	53.2	60.4	75.6	83.1	100.9
18	15.0	19.3	21.6	28.1	32.3	41.1	51.7	58.7	73.6	80.9	98.2
19	14.6	18.8	21.1	27.5	31.6	40.1	50.6	57.4	72.0	79.2	96.2
20	14.3	18.4	20.7	27.0	31.0	39.5	49.8	56.5	70.9	78.0	94.7
21	14.1	18.2	20.4	26.6	30.7	39.0	49.2	55.9	70.2	77.2	93.9
22	14.0	18.0	20.3	26.5	30.4	38.8	49.0	55.6	69.9	76.8	93.5
23	13.9	18.0	20.2	26.4	30.4	38.7	48.9	55.6	69.8	76.8	93.5
24	13.9	18.0	20.2	26.4	30.4	38.8	49.0	55.7	70.0	77.1	93.8
25	14.0	18.0	20.3	26.5	30.5	39.0	49.3	56.0	70.5	77.5	94.4
26	14.0	18.1	20.4	26.7	30.7	39.2	49.6	56.5	71.0	78.2	95.2
27	14.1	18.2	20.5	26.9	31.0	39.6	50.1	57.0	71.7	79.0	96.2
28	14.2	18.4	20.7	27.1	31.3	39.9	50.6	57.6	72.5	79.8	97.3
29	14.3	18.5	20.9	27.4	31.6	40.3	51.1	58.2	73.3	80.8	98.5
30	14.4	18.7	21.1	27.6	31.9	40.8	51.7	58.8	74.2	81.7	99.7
31	14.5	18.9	21.3	27.9	32.2	41.2	52.3	59.5	75.0	82.7	100.9
32	14.7	19.0	21.5	28.2	32.5	41.6	52.8	60.2	75.9	83.7	102.1
33	14.8	19.2	21.7	28.4	32.8	42.1	53.4	60.9	76.8	84.7	103.4
34	14.9	19.4	21.9	28.7	33.2	42.5	54.0	61.6	77.7	85.7	104.7
35	15.0	19.6	22.1	29.0	33.5	43.0	54.7	62.3	78.7	86.8	106.0
36	15.2	19.8	22.3	29.3	33.9	43.5	55.3	63.1	79.7	87.9	107.5
37	15.3	20.0	22.5	29.7	34.3	44.0	56.0	63.9	80.8	89.1	108.9
38	15.5	20.2	22.8	30.0	34.7	44.6	56.7	64.7	81.9	90.3	110.5
39	15.7	20.4	23.1	30.4	35.2	45.2	57.5	65.6	83.0	91.6	112.1
40	15.8	20.6	23.3	30.8	35.6	45.8	58.3	66.5	84.2	92.9	113.8

41	16.0	20.9	23.6	31.2	36.1	46.4	59.1	67.5	85.5	94.3	115.5
42	16.2	21.2	23.9	31.6	36.6	47.0	60.0	68.5	86.7	95.8	117.3
43	16.4	21.4	24.2	32.0	37.1	47.7	60.8	69.5	88.0	97.2	119.1
44	16.6	21.7	24.5	32.4	37.6	48.4	61.7	70.5	89.4	98.7	121.0
45	16.8	21.9	24.8	32.8	38.1	49.0	62.6	71.5	90.7	100.2	122.8
46	17.0	22.2	25.2	33.3	38.6	49.7	63.5	72.6	92.1	101.7	124.7
47	17.2	22.5	25.5	33.7	39.1	50.4	64.4	73.6	93.5	103.3	126.7
48	17.4	22.8	25.8	34.2	39.7	51.2	65.4	74.8	94.9	104.9	128.7
49	17.6	23.1	26.2	34.7	40.2	52.0	66.4	76.0	96.5	106.6	130.9
50	17.9	23.5	26.6	35.2	40.9	52.8	67.5	77.2	98.2	108.5	133.2
51	18.2	23.8	27.0	35.8	41.6	53.7	68.7	78.6	100.0	110.5	135.8
52	18.5	24.2	27.5	36.5	42.3	54.7	70.1	80.2	102.0	112.7	138.5
53	18.8	24.7	28.0	37.2	43.2	55.8	71.5	81.9	104.1	115.2	141.5
54	19.2	25.2	28.6	38.0	44.1	57.1	73.1	83.7	106.6	117.9	144.9
55	19.6	25.8	29.3	38.9	45.2	58.5	74.9	85.8	109.2	120.8	148.6
56	20.1	26.4	30.0	39.9	46.3	60.0	76.9	88.1	112.2	124.1	152.7
57	20.6	27.1	30.8	40.9	47.6	61.6	79.0	90.6	115.4	127.7	157.1
58	21.2	27.8	31.6	42.1	48.9	63.4	81.4	93.2	118.8	131.5	161.9
59	21.8	28.6	32.5	43.3	50.4	65.3	83.8	96.1	122.5	135.6	167.0
60	22.4	29.5	33.5	44.6	51.9	67.4	86.5	99.1	126.5	140.0	172.4
61	23.1	30.4	34.5	46.0	53.6	69.5	89.3	102.4	130.7	144.7	178.2
62	23.8	31.4	35.6	47.5	55.3	71.8	92.3	105.9	135.1	149.6	184.4
63	24.5	32.4	36.8	49.1	57.2	74.3	95.5	109.5	139.8	154.9	190.9
64	25.3	33.5	38.0	50.8	59.1	76.8	98.8	113.4	144.8	160.4	197.8
65	26.2	34.6	39.3	52.5	61.2	79.6	102.4	117.5	150.1	166.3	205.1
66	27.1	35.8	40.7	54.4	63.4	82.4	106.1	121.7	155.6	172.4	212.7
67	28.0	37.0	42.1	56.3	65.7	85.4	110.0	126.3	161.4	178.9	220.8
68	29.0	38.4	43.6	58.4	68.1	88.6	114.1	131.0	167.6	185.7	229.2
69	30.0	39.7	45.2	60.5	70.6	91.9	118.4	136.0	174.0	192.8	238.1
70	31.1	41.2	46.9	62.8	73.2	95.3	122.9	141.2	180.7	200.3	247.4
71	32.2	42.7	48.6	65.1	75.9	99.0	127.6	146.6	187.7	208.1	257.2
72	33.4	44.3	50.4	67.6	78.8	102.7	132.5	152.3	195.1	216.3	267.3
73	34.6	45.9	52.3	70.1	81.8	106.7	137.7	158.2	202.7	224.8	277.9
74	35.9	47.6	54.2	72.7	84.9	110.8	143.0	164.4	210.7	233.7	289.0
75	37.2	49.4	56.2	75.5	88.1	115.0	148.6	170.8	219.0	243.0	300.5
76	38.5	51.2	58.3	78.3	91.5	119.5	154.3	177.5	227.6	252.6	312.5
77	39.9	53.1	60.5	81.3	95.0	124.0	160.3	184.4	236.6	262.5	324.9
78	41.4	55.0	62.8	84.4	98.6	128.8	166.5	191.6	245.9	272.9	337.8
79	42.9	57.1	65.1	87.5	102.3	133.7	172.9	199.0	255.5	283.6	351.1

Appendix 4. Regression analysis of CHMS variables in Chapter 3.

Appendix 4.1 Results of each regression were reported using standardized betas and confidence intervals. Univariate regression analysis with biomarker as the response variable and health variables as the predictors was performed using svyglm function in each age bin.

Age Bin	Age 3-19			Age 20-39			Age 40-59			Age 60-80		
Variable	β	95% CI		β	95% CI		β	95% CI		β	95% CI	
A β 42/40												
Age	-0.199	-0.298	-0.100	-0.074	-0.221	0.074	-0.184	-0.308	-0.059	-0.137	-0.233	-0.040
Sex	0.030	-0.084	0.143	0.122	-0.025	0.269	-0.090	-0.244	0.064	-0.072	-0.177	0.034
Ethnicity	0.122	0.018	0.226	0.028	-0.123	0.180	-0.018	-0.153	0.116	0.065	-0.042	0.173
Fasted	-0.026	-0.104	0.053	0.042	-0.063	0.147	-0.060	-0.174	0.054	0.061	-0.007	0.128
BMI	-0.222	-0.327	-0.117	-0.072	-0.218	0.075	-0.235	-0.392	-0.079	-0.032	-0.143	0.079
Creatinine	-0.212	-0.325	-0.100	0.248	0.109	0.386	-0.063	-0.215	0.090	0.012	-0.074	0.098
Creatinine (urine)	-0.115	-0.208	-0.022	0.154	-0.003	0.310	-0.197	-0.329	-0.065	-0.091	-0.213	0.032
p-tau-181												
Age	-0.500	-0.594	-0.405	-0.069	-0.202	0.065	0.073	-0.077	0.222	0.275	0.165	0.386
Sex	0.058	-0.056	0.172	0.240	0.073	0.408	0.195	0.063	0.326	0.247	0.137	0.357
Ethnicity	-0.078	-0.187	0.031	-0.173	-0.325	-0.020	-0.052	-0.170	0.065	-0.067	-0.162	0.028
Fasted	-0.046	-0.129	0.037	-0.071	-0.189	0.046	0.012	-0.083	0.108	-0.028	-0.107	0.051
BMI	-0.366	-0.473	-0.260	-0.051	-0.157	0.055	0.056	-0.109	0.222	0.042	-0.081	0.165
Creatinine	-0.347	-0.440	-0.254	0.091	-0.233	0.415	0.286	0.158	0.414	0.187	0.088	0.286
Creatinine (urine)	-0.270	-0.364	-0.176	-0.010	-0.132	0.112	0.105	-0.045	0.254	0.014	-0.104	0.131
NfL												
Age	-0.126	-0.253	0.002	0.264	0.098	0.430	0.362	0.239	0.486	0.417	0.337	0.498
Sex	0.045	-0.069	0.159	0.110	-0.054	0.274	-0.072	-0.220	0.077	-0.001	-0.096	0.095
Ethnicity	-0.028	-0.139	0.083	0.153	0.003	0.304	-0.032	-0.211	0.147	-0.002	-0.126	0.122
Fasted	0.012	-0.075	0.099	0.021	-0.095	0.138	0.119	0.020	0.218	0.081	0.019	0.142
BMI	-0.221	-0.350	-0.092	-0.293	-0.456	-0.130	-0.273	-0.392	-0.153	-0.203	-0.292	-0.115
Creatinine	-0.043	-0.181	0.096	0.143	0.014	0.272	0.154	-0.006	0.315	0.224	0.133	0.314
Creatinine (urine)	-0.010	-0.141	0.121	0.043	-0.128	0.214	-0.108	-0.244	0.029	-0.017	-0.091	0.057
GFAP												
Age	-0.652	-0.742	-0.562	0.116	-0.039	0.271	0.198	0.058	0.339	0.451	0.355	0.547
Sex	0.007	-0.103	0.117	-0.124	-0.284	0.036	-0.321	-0.463	-0.180	-0.165	-0.261	-0.070
Ethnicity	-0.097	-0.208	0.014	0.050	-0.121	0.221	0.137	-0.016	0.290	0.036	-0.061	0.134
Fasted	-0.133	-0.212	-0.053	0.005	-0.108	0.118	0.021	-0.091	0.133	0.021	-0.042	0.085
BMI	-0.508	-0.627	-0.388	-0.292	-0.430	-0.155	-0.237	-0.402	-0.072	-0.195	-0.276	-0.113
Creatinine	-0.534	-0.618	-0.450	-0.162	-0.309	-0.015	-0.135	-0.287	0.018	0.002	-0.097	0.100
Creatinine (urine)	-0.300	-0.388	-0.211	-0.214	-0.388	-0.041	-0.114	-0.251	0.022	-0.127	-0.226	-0.027

Appendix 5. Web application for Chapter 3 reference curve interpretation.

A draft copy of the reference curve application can be found at <https://age-specific-reference-intervals-for-plasma-biomarkers.shinyapps.io/ri-biomarker-calculator/> using the password:

wellingtonabcdef2025

Please note that the z-score calculator is still under development pending further release of reference curve models from Statistics Canada. Below is a screenshot showing an example use of the calculator using the Super Senior cohort from Chapter 5.

Neurological Plasma Biomarker Reference Interval Calculator

[Home](#)[Percentile Calculator](#)[Group Comparisons](#)[Reference Curve Visualization](#)[Z-Score Calculator](#)

Download Input Template

Clear data

Upload CSV or Excel File

Browse... SS_for RI calc.csv

Upload complete

Process Data

Summary View Type

All Counts and %

Summary Table

Download Summary CSV

Show 13 entries

Search:

Bin	AB4240	ptau181	NfL	GFAP
Below 1st	0 (0.0%)	65 (13.5%)	3 (0.6%)	1 (0.2%)
Between 1st and 3rd	5 (1.0%)	59 (12.3%)	12 (2.5%)	4 (0.8%)
Between 3rd and 5th	8 (1.7%)	22 (4.6%)	12 (2.5%)	12 (2.5%)
Between 5th and 15th	29 (6.0%)	92 (19.2%)	66 (13.8%)	44 (9.2%)
Between 15th and 25th	17 (3.5%)	46 (9.6%)	60 (12.5%)	56 (11.7%)
Between 25th and 50th	58 (12.1%)	92 (19.2%)	116 (24.2%)	124 (25.8%)
Between 50th and 75th	79 (16.5%)	71 (14.8%)	90 (18.8%)	122 (25.4%)
Between 75th and 85th	68 (14.2%)	15 (3.1%)	51 (10.6%)	49 (10.2%)
Between 85th and 95th	108 (22.5%)	12 (2.5%)	45 (9.4%)	47 (9.8%)
Between 95th to 97th	40 (8.3%)	2 (0.4%)	12 (2.5%)	6 (1.2%)
Between 97th to 99th	48 (10.0%)	3 (0.6%)	11 (2.3%)	10 (2.1%)
Above 99th	20 (4.2%)	1 (0.2%)	2 (0.4%)	5 (1.0%)
Total	480 (100%)	480 (100%)	480 (100%)	480 (100%)

Showing 1 to 13 of 13 entries

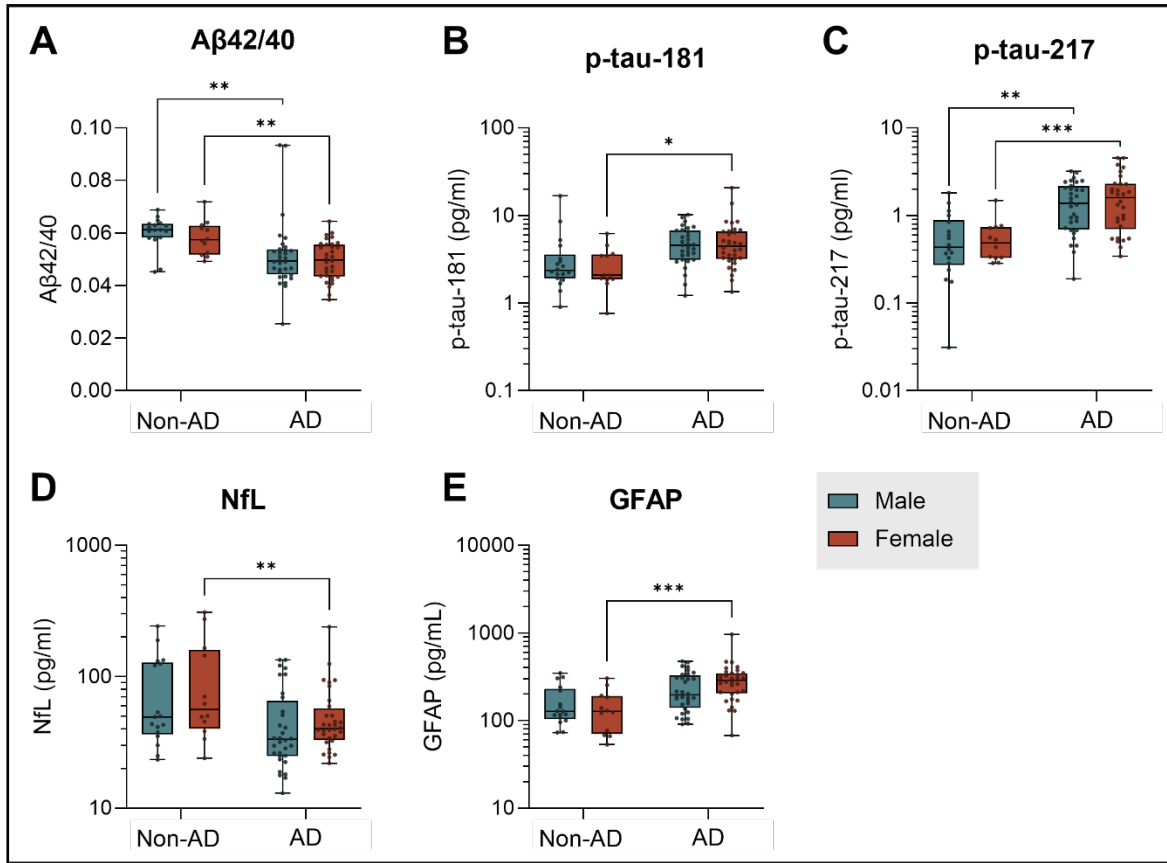
Previous1Next

Detailed Participant Output

Appendix 6. Sex stratified analysis of plasma biomarkers to detect and grade autopsy confirmed Alzheimer's disease neuropathology

Appendix 6.1. Associations of plasma biomarkers with measures of AD pathology staging and sex. Simple linear regression was first performed to assess associations between AD staging measures and plasma A β 42/40, p-tau-181, NF-L, and GFAP. Multivariable linear regression was then performed to incorporate sex into the analysis, using male sex as the reference group.

			A β 42/40		p-tau-181		p-tau-217		NfL		GFAP	
			β	P-value	β	P-value	β	P-value	β	P-value	β	P-value
Thal Phase	AD Alone	Thal	-0.0024	<0.0001	0.6087	0.0003	0.2792	<0.0001	-8.47	0.0068	32.68	<0.0001
	AD + Sex	Thal	-0.0023	<0.0001	0.6100	0.0004	0.2756	<0.0001	-8.75	0.0055	31.88	<0.0001
		Female Sex	-0.0019	0.3455	-0.0431	0.9452	0.1529	0.4004	11.16	0.3381	32.36	0.1979
Braak Stage*	AD Alone	Braak	-0.0018	0.0074	0.4650	0.0084	0.2934	<0.0001	-13.70	0.0001	24.08	0.0003
	AD + Sex	Braak	-0.0018	0.0102	0.4895	0.0062	0.2898	<0.0001	-14.19	0.0001	23.92	0.0004
		Female Sex	-0.0017	0.5045	-0.6708	0.3137	0.1205	0.5864	14.96	0.2529	4.89	0.8414
CERAD Score	AD Alone	CERAD	-0.0031	0.0002	0.9881	0.0001	0.4758	<0.0001	-12.36	0.0086	48.86	<0.0001
	AD + Sex	CERAD	-0.0030	0.0003	0.9947	0.0001	0.4707	<0.0001	-12.92	0.0065	47.48	<0.0001
		Female Sex	-0.0018	0.3779	-0.1232	0.8428	0.1258	0.4659	11.87	0.3108	29.67	0.2397
ABC Score	AD Alone	ABC	-0.0037	0.0001	0.9947	0.0005	0.4881	<0.0001	-14.14	0.0002	44.25	0.0002
	AD + Sex	ABC	-0.0036	0.0001	1.0074	0.0005	0.4828	<0.0001	-15.01	0.0043	42.32	0.0004
		Female Sex	-0.0015	0.4657	-0.1713	0.7869	0.0832	0.6429	13.14	0.2623	29.35	0.2671



Appendix 6.2. Sex stratified plasma biomarkers to detect the presence of AD pathology at autopsy. AD pathology was determined based on an ABC score indicative of at least intermediate AD pathology. Analysis of biomarkers stratified by AD status and sex conducted by two-way ANOVA with multiple comparisons * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$. Non-AD males $n=18$, Non-AD females $n=12$, AD males $n=31$, AD females $n=32$.

Appendix 7. Demographics and clinical data for COMPASS-ND participants sub-divided by diagnosis,

Appendix 7.1. Demographics and clinical data for COMPASS-ND participants in pre-dementia stages. The group was divided by clinical diagnosis of being cognitively unimpaired, subjective cognitive impairment, and mild cognitive impairment. Each group is further divided based on the availability of CSF biomarkers to indicate how representative the CSF sub-cohorts of the remaining COMPASS-ND participants. Analysis of continuous variables was by the Mann-Whitney test, and categorical variables were analyzed by Fisher's exact test. Abbreviations: APOE4, apolipoprotein E4; BMI, body mass index; MoCA, Montreal Cognitive Assessment.

	All Pre-dementia (n=749)			CU (n=192)			SCI (n=146)			MCI (n=411)		
CSF Sub-cohort	Yes	No	p-value	Yes	No	p-value	Yes	No	p-value	Yes	No	p-value
N, %	89 (11.9)	660 (88.1)		21 (10.9)	171 (89.1)		6 (4.1)	140 (95.9)		62 (15.1)	349 (84.9)	
Demographics												
Age, years (median, IQR)	72 (66–76)	69 (64–72)	0.0003	69 (64–73)	64 (60–69)	0.0074	70 (66–74)	66.5 (64–72)	0.2098	74 (68–79)	70 (65–73)	0.0011
Female Sex (n, %)	348 (52.9)	38 (42.2)	0.0716	111 (65.7)	12 (52.2)	0.2481	102 (72.9)	4 (66.7)	0.6656	135 (38.7)	22 (36.1)	0.7759
Male Sex (n, %)	310 (47.1)	52 (57.8)		58 (34.3)	11 (47.8)		38 (27.1)	2 (33.3)		214 (61.3)	39 (63.9)	
APOE4 Carrier (n, %)	204 (32.0)	35 (41.2)	0.1101	45 (27.6)	5 (21.7)	0.6254	34 (25.8)	4 (66.7)	0.0487	125 (36.4)	26 (46.4)	0.1811
Clinical												
BMI (median, IQR)	22.3 (19.7–25)	21.6 (19.2–24.3)	0.2330	21.6 (18.9–24.3)	21.5 (18.4–25.2)	0.8950	22.4 (19.4–25)	19.9 (17.8–21.8)	0.1070	22.3 (20.3–25.4)	21.8 (19.7–24.3)	0.2805
Parkinsons disease (n, %)	94 (14.3)	23 (25.3)	0.0127	61 (36.1)	15 (65.2)	0.0113	N/A	N/A	N/A	33 (9.5)	8 (12.9)	0.3665
Vascular contributions (n, %)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	25 (7.9)	5 (9.3)	0.7862
Hypertension (n, %)	246 (37.4)	30 (33.0)	0.4868	38 (22.5)	8 (34.8)	0.2006	52 (37.1)	0 (0.0)	0.0890	156 (44.7)	22 (35.5)	0.2109
Kidney disease (n, %)	14 (2.1)	2 (2.2)	>0.9999	2 (1.2)	1 (4.3)	0.3195	2 (1.4)	0 (0.0)	>0.9999	10 (2.9)	1 (1.6)	>0.9999
MoCA (median, IQR)	26 (23–28)	25 (22–27)	0.3765	28 (26–29)	27 (27–29)	0.5432	27 (26–28)	29 (28–30)	0.0673	23 (21–25)	23.5 (21–26)	0.6177
Plasma Biomarkers												
Aβ42/40 (median, IQR)	0.0659 (0.0574–0.0728)	0.0642 (0.0563–0.0726)	0.5524	0.0685 (0.0614–0.0739)	0.0711 (0.0601–0.0771)	0.5379	0.0669 (0.0575–0.0762)	0.0661 (0.0523–0.0751)	0.5979	0.0627 (0.0559–0.0715)	0.0633 (0.0561–0.0695)	0.8215
p-tau-181, pg/mL (median, IQR)	2.14 (1.56–3.18)	2.14 (1.54–3.6)	0.6405	1.8 (1.45–2.49)	1.68 (1.43–2.52)	0.6427	1.81 (1.35–2.56)	1.98 (1.57–2.46)	0.4995	2.53 (1.81–3.61)	2.52 (1.69–3.98)	0.7829
NfL, pg/mL (median, IQR)	19.8 (15–26.3)	19.1 (14.3–25.5)	0.3808	17.6 (12.7–21.6)	16.8 (13–23.4)	0.6658	18.9 (14.6–23.5)	17.4 (11.2–23.9)	0.4627	22 (16.8–29.9)	20.5 (15.9–26.9)	0.1737
GFAP, pg/mL (median, IQR)	124 (85.8–178)	119 (80.8–174)	0.5723	109 (79.3–154)	83.7 (68.5–132)	0.0745	114 (81–157)	147 (126–180)	0.2124	136 (94.3–193)	126 (89.5–192)	0.6004

Appendix 7.2. Demographics and clinical data for COMPASS-ND participants diagnosed with dementia. The group was divided by clinical diagnosis of Alzheimer's disease, frontotemporal dementia, Lewy body dementia, and Parkinson's disease dementia. Each group is further divided based on the availability of CSF biomarkers to indicate how representative the CSF sub-cohorts of the remaining COMPASS-ND participants. Analysis of continuous variables was by the Mann-Whitney test, and categorical variables were analyzed by Fisher's exact test. Abbreviations: APOE4, apolipoprotein E4; BMI, body mass index; MoCA, Montreal Cognitive Assessment.

	All Dementia (n=248)			AD (n=160)			FTD (n=44)			LBD (n=30)			PDD (n=14)		
CSF Sub-cohort	Yes	No	p-value	Yes	No	p-value	Yes	No	p-value	Yes	No	p-value	Yes	No	p-value
N, %	60 (24.2)	188 (75.8)		39 (24.4)	121 (75.6)		12 (27.3)	32 (72.7)		8 (26.7)	22 (73.3)		1 (7.1)	13 (92.9)	
Demographics															
Age, years (median, IQR)	76 (68–80)	71 (65–77)	0.0353	77 (71–81)	73 (68–78)	0.0473	66.5 (57.5–71)	64 (60.5–74)	0.7717	75 (68.5–78)	70 (69–77)	0.4830	77 (76–78)	62 (62–62)	0.1319
Female Sex (n, %)	71 (37.4)	20 (34.5)	0.7567	50 (41.3)	15 (38.5)	0.8519	15 (46.9)	5 (41.7)	>0.9999	5 (20.8)	0 (0.0)	0.5526	1 (7.7)	0 (0.0)	>0.9999
Male Sex (n, %)	119 (62.6)	38 (65.5)		71 (58.7)	24 (61.5)		17 (53.1)	7 (58.3)		19 (79.2)	6 (100.0)		12 (92.3)	1 (100.0)	
APOE4 Carrier (n, %)	88 (48.4)	30 (53.6)	0.5426	63 (54.3)	25 (65.8)	0.2589	7 (23.3)	2 (18.2)	>0.9999	16 (69.6)	3 (50.0)	0.6328	2 (15.4)	0 (0.0)	>0.9999
Clinical															
BMI (median, IQR)	21.7 (19.2–24.9)	22.7 (19.3–24.7)	0.6630	21.5 (18.7–24.9)	22.1 (19.1–24.5)	0.8973	22.6 (19.4–26.2)	21.8 (18.7–24.8)	0.3682	22.2 (20.1–24)	24 (22.4–25.7)	0.1139	20.9 (19.8–22.4)	30 (30–30)	0.1429
Vascular contributions (n, %)	N/A	N/A	N/A	36 (29.8)	9 (23.1)	0.5398	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Hypertension (n, %)	78 (41.3)	18 (31.0)	0.1700	54 (45.0)	12 (30.8)	0.1368	12 (37.5)	4 (33.3)	>0.9999	9 (37.5)	1 (16.7)	0.6328	3 (23.1)	1 (100.0)	0.285
Kidney disease (n, %)	3 (1.6)	1 (1.7)	>0.9999	2 (1.7)	1 (2.6)	0.5728	0 (0.0)	0 (0.0)	N/A	1 (4.2)	0 (0.0)	>0.9999	0 (0.0)	0 (0.0)	N/A
MoCA (median, IQR)	18 (15–22)	18.5 (16–22)	0.3489	18 (15–21)	18 (15–22)	0.7406	20 (16–23)	20 (17–22.5)	0.7037	17.5 (15–22)	18 (17–22)	0.4802	19 (11–20)	19 (19–19)	0.8999
Plasma Biomarkers															
Aβ42/40 (median, IQR)	0.06 (0.0534–0.0662)	0.0609 (0.0532–0.068)	0.9441	0.0586 (0.0522–0.0642)	0.0597 (0.0528–0.063)	0.5971	0.068 (0.0601–0.0768)	0.0685 (0.0611–0.0705)	0.5581	0.0596 (0.051–0.0668)	0.0549 (0.0307–0.0579)	0.1286	0.0639 (0.0582–0.0748)	0.0772 (0.0772–0.0772)	0.2857
p-tau-181, pg/mL (median, IQR)	3.26 (2.26–4.42)	4.12 (2.52–5.24)	0.0455	3.76 (2.76–4.89)	4.53 (3.3–5.62)	0.0373	1.88 (1.62–2.88)	1.84 (1.46–2.97)	0.9275	2.54 (2.18–3.88)	4.37 (2.76–5.82)	0.0647	3.66 (2.26–4.39)	2.39 (2.39–2.39)	0.7143
NfL, pg/mL (median, IQR)	28.3 (21.8–38.1)	30.3 (23.7–37.3)	0.4751	28.4 (21.8–38.6)	30.2 (21.8–35.2)	0.9398	29.8 (22.5–39.3)	35.4 (30.4–60.2)	0.0946	24.7 (19.8–33.2)	26.8 (23.7–29.3)	0.5953	29.7 (24.6–37.5)	11.3 (11.3–11.3)	0.1429
GFAP, pg/mL (median, IQR)	179 (126–254)	205 (129–252)	0.5787	199 (159–276)	230 (142–317)	0.6794	121 (79.1–182)	169 (99.5–206)	0.2782	155 (116–218)	129 (103–241)	0.7426	178 (140–243)	45 (45–45)	0.1429